Microelectronic Technologies

Lecture Notes

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Preface

Welcome to the course of "Microelectronic technologies" by professor Marcello Mariani. This course aims to explore the ins and outs of modern semiconductor fabrication. The semiconductor cleanroom is arguably the most complex manufacturing environment in human history, featuring technological marvels deeply rooted in condensed matter physics and chemistry. By taking an orderly approach, the course will teach you the fundamental steps required to create the devices powering our global infrastructure.

About this Course Material

First of all, I should mention that I wrote this material coming from a background in Engineering Physics. However, this course can be of great interest to many different kinds of engineering. As such, not every reader will have the same background in solid state physics, quantum physics, chemistry and so on. Despite trying to make this resource as accessible as possible, some topics are to be assumed as known and hopefully treated in previous engineering courses (such as the workings of a MOSFET).

I decided to write these notes to create a textbook-style material that can be consulted as a study guide. Over the years, I've learned how helpful a straightforward text-based reference can be while studying. However, the usual disclaimers: this text is supposed to complement the content present in the professor's slides and in your notes. I may not have included all the information your notes have. On the other hand, some content may have been amplified for clarity or interest. Consequently, content not present in the professor's slides may be skipped at the reader's discretion.

Again, this text is not a summary. It aims to be complete in what it treats, but may lack some information I forgot to mention or decided to omit for whatever reason. While these notes were prepared with care, they represent an independent student effort and were not overseen by the professor. If anything feels off, I encourage you to cross-reference with your own notes.

The sources for this material were mostly the professor's slides, my personal notes and, obviously, the internet.

One distinctive feature of these notes is *insights*. These are coloured boxes in which I talk about a **physical insight** or a **process insight**. The purpose of these boxes is to give you a momentary, focused glimpse into the true nature of the physics or the manufacturing process. Sometimes, when seeking to give the most instructive mathematical models and process explanations, one may skip over physical details or the actual tricks/issues that the industry faces. The idea of these boxes is to give you an "Aha!" moment that ties everything together, getting into the weeds of a specific topic.

To aid the explanation, I decided to include pictures (which definitely contributed to the length of these notes). As this is an unofficial document, pictures don't include proper references. If a picture does not specifically mention a source, it was taken from the professor's slides. The pictures that do mention a source include a hyperlink to the paper or site I found them on.

Furthermore, I mention a couple of times external sources or videos that can help in understanding a certain matter. These sources were chosen purely for personal liking. There might be better sources out there, they might lack some information, but they looked alright to me.

Contents

Prologue: The CMOS Inverter Traveller **8**

1 Silicon Substrates **18**
1.1 Silicon Crystallography . . . 18
1.1.1 Bravais Lattices . . . 18
1.1.2 The Diamond Structure . . . 18
1.1.3 Miller Indices and Wafer Orientation . . . 19
1.2 Crystallographic Defects . . . 19
1.2.1 Point Defects . . . 20
1.2.2 Line Defects: Dislocations . . . 21
1.2.3 Area and Volume Defects . . . 22
1.3 Silicon Manufacturing: From Sand to Ingot . . . 22
1.3.1 Purification to Electronic Grade Silicon . . . 22
1.3.2 Czochralski (CZ) Crystal Growth and Pull Rate . . . 23
1.3.3 Doping and the Segregation Coefficient . . . 25
1.4 Wafer Preparation and Metrology . . . 26
1.4.1 Mechanical Preparation . . . 26
1.4.2 Electrical Metrology . . . 26
1.4.3 Defect Analysis and Microscopy . . . 27

2 Silicon Oxidation **29**
2.1 Introduction to Silicon Oxidation . . . 29
2.2 The Structure of Silicon Dioxide . . . 29
2.2.1 Impurities and Network Modifiers . . . 30
2.3 Oxidation Kinetics and the Deal-Grove Model . . . 30
2.3.1 The Deal-Grove Flux Equations . . . 31
2.3.2 Solving for the Interfacial Concentration . . . 33
2.3.3 Growth Rate and the Linear-Parabolic Equation . . . 34
2.3.4 Corrections to Deal-Grove . . . 35
2.4 Factors Influencing the Oxidation Rate . . . 35
2.4.1 Substrate Doping and Impurity Segregation . . . 35
2.4.2 Crystal Orientation and 2D Topography . . . 36
2.4.3 The Microscopic Model: Vacancies and Interstitials . . . 36
2.5 Charges in the Oxide and C-V Characterisation . . . 36
2.5.1 The Four Oxide Charges . . . 37
2.5.2 Capacitance-Voltage (C-V) Characterisation . . . 38
2.5.3 Extracting Charge Data from C-V Curves . . . 38
2.6 Advanced Dielectrics and Oxidation Equipment . . . 39
2.6.1 Nitridation . . . 39
2.6.2 Oxidation Equipment . . . 40
2.6.3 The Scaling Limit and High-κ Dielectrics . . . 40

3 Dopant Diffusion **43**
3.1 Introduction to Doping and Scaling . . . 43
3.1.1 Technology Scaling and Junction Depth . . . 43
3.1.2 The Two-Step Doping Process . . . 43
3.1.3 Solid vs. Electrical Solubility . . . 44
3.2 Macroscopic Diffusion: Fick's Laws and Analytical Solutions . . . 45
3.2.1 Fick's Laws and the Diffusion Equation . . . 45
3.2.2 The Pre-deposition Step: The Error Function Solution . . . 46
3.2.3 The Drive-In Step: The Gaussian Solution . . . 46
3.2.4 Temperature Dependence of Diffusivity . . . 47
3.2.5 Thermal Budget and Multiple Diffusion Steps . . . 47
3.3 Corrections to Fick's Law: Extrinsic Effects . . . 48
3.3.1 Electric Field Enhancement . . . 48
3.3.2 Concentration-Dependent Diffusion . . . 48
3.3.3 Recovering the correction to Fick's First Law . . . 49
3.3.4 Segregation and Interfacial Pile-Up . . . 50
3.4 Atomic-Scale Microscopic Diffusion . . . 50
3.4.1 Vacancy and Interstitial Mechanisms . . . 50
3.4.2 Fractional Diffusivity . . . 51
3.4.3 Modelling Interstitial-Assisted Diffusion . . . 51
3.5 Metrology of Doped Profiles . . . 53
3.5.1 Secondary Ion Mass Spectrometry (SIMS) . . . 53
3.5.2 Electrical Profiling: SRP and SSRM . . . 54
3.5.3 Electron Holography . . . 55

4 Ion Implantation **57**
4.1 The Physics and Architecture of Ion Implantation . . . 57
4.1.1 Ion Generation and Extraction . . . 57
4.1.2 Mass Analysis and Filtering . . . 58
4.1.3 Acceleration and Neutral Filtering . . . 58
4.1.4 Dosimetry and Beam Heating . . . 59
4.2 Ion Stopping Mechanisms and Profile Modelling . . . 60
4.2.1 Nuclear and Electronic Stopping . . . 61
4.2.2 The Gaussian Approximation . . . 62
4.2.3 Masking Rules . . . 62
4.2.4 Higher-Order Moments and Subsequent Diffusion . . . 63
4.3 Channelling and Crystallographic Damage . . . 64
4.3.1 The Channelling Effect . . . 64
4.3.2 Crystallographic Damage . . . 64
4.3.3 Solid Phase Epitaxy (SPE) and Activation . . . 65
4.4 Defect Evolution and Transient Enhanced Diffusion . . . 66
4.4.1 The "+1" Model and 311 Clusters . . . 66
4.4.2 Transient Enhanced Diffusion (TED) . . . 67
4.5 Advanced Doping for Ultra-Shallow Junctions . . . 68
4.5.1 Millisecond and Microsecond Annealing . . . 68
4.5.2 Defect Engineering and Surface Treatments . . . 69
4.5.3 3D Doping: PLAD and Monolayer Doping . . . 69

5 Thin Film Deposition **71**
5.1 Introduction to Thin Film Deposition . . . 71
5.1.1 Conformality and Gap Filling . . . 71
5.1.2 Deposition Methods: PVD vs. CVD . . . 72
5.2 CVD Kinetics and Fluid Dynamics . . . 73
5.2.1 The Seven Steps of CVD . . . 73
5.2.2 Fluid Dynamics and the Stagnant Layer . . . 73
5.2.3 Flux Equations for CVD . . . 75
5.2.4 Mass Transfer vs. Surface Reaction Regimes . . . 75
5.3 Low-Pressure CVD (LPCVD) . . . 76
5.3.1 Reactant Gas Diffusivity . . . 76
5.3.2 Advantages of LPCVD . . . 77
5.4 Plasma Physics and PE-CVD . . . 77
5.4.1 Definition of a Plasma . . . 77
5.4.2 Weakly Ionised Plasmas . . . 77
5.4.3 Plasma-Enhanced CVD (PE-CVD) . . . 78
5.4.4 High Density Plasma CVD (HDP-CVD) . . . 79
5.5 Atomic Layer Deposition (ALD) . . . 79
5.6 Common CVD Films and Reactions . . . 80
5.7 Physical Vapour Deposition (PVD): Evaporation . . . 81
5.7.1 The Evaporation Mechanism and Langmuir-Knudsen Theory . . . 81
5.7.2 Deposition Rate and Geometric Approximations . . . 82
5.7.3 Molecular Beam Epitaxy (MBE) . . . 84
5.8 Sputter Deposition . . . 84
5.8.1 DC Sputtering and Arrival Dynamics . . . 85
5.8.2 Sputtering Yield and Alloy Deposition . . . 85
5.8.3 Advanced Sputtering Processes . . . 86
5.9 Topographical Modelling of Deposition . . . 87
5.9.1 Arrival Flux and Sticking Coefficients . . . 87
5.9.2 Redeposition Mechanics . . . 88
5.9.3 Diffusion mechanics . . . 88
5.9.4 Process-Specific Modelling . . . 89

6 Photolithography **91**
6.1 Introduction to Lithography . . . 91
6.2 Illumination Sources and Photomasks . . . 91
6.2.1 Illumination Sources . . . 92
6.2.2 Photomasks (Reticles) . . . 92
6.3 Wafer Exposure Systems . . . 93
6.3.1 Contact and Proximity Printing . . . 93
6.3.2 Projection Printing . . . 94
6.3.3 Steppers and Scanners . . . 94
6.3.4 The Pellicle . . . 96
6.4 The Photoresist Chemistry . . . 96
6.4.1 g-line and i-line Chemistry (DQN) . . . 97
6.4.2 Deep UV (DUV) Chemistry and Chemical Amplification . . . 97
6.4.3 Resist Contrast . . . 98
6.5 The Lithographic Process Flow . . . 99
6.5.1 The Track Processing Steps . . . 99
6.5.2 Thermal Reflow and Optical Challenges . . . 100
6.6 Registration and Overlay . . . 100
6.6.1 Alignment Marks and Measurement . . . 100

6.6.2 Overlay Models . . . 101
6.7 The Sub-Wavelength Gap and Diffraction Physics . . . 101
6.7.1 Diffraction at the Reticle . . . 101
6.7.2 The Objective Lens and Pattern Fidelity . . . 102
6.7.3 Optical Parameters: NA and σ . . . 104
6.8 Fourier Optics and Image Formation . . . 104
6.8.1 The Mask Spectrum . . . 104
6.8.2 The Objective Lens (Coherent Imaging $\sigma = 0$) . . . 105
6.8.3 Aerial Image Formation . . . 106
6.8.4 Resolution Limits in 1D Coherent Imaging . . . 106
6.8.5 Partially Coherent Imaging and Off-Axis Illumination . . . 107
6.9 Quantification of Image Quality and the Rayleigh Criteria . . . 108
6.9.1 Modulation Transfer Function and Image Contrast . . . 108
6.9.2 Depth of Focus (DoF) . . . 109
6.9.3 The Rayleigh Criterion for Depth of Focus . . . 109
6.9.4 The Rayleigh Equations: The Fundamental Trade-off . . . 110
6.9.5 Numerical Aperture and Wavelength Evolution . . . 110
6.10 Resolution Enhancement Techniques (RET) . . . 111
6.10.1 Wavefront Direction: Off-Axis Illumination (OAI) . . . 111
6.10.2 Wavefront Phase: Attenuated Phase Shift Masks (Att-PSM) . . . 111
6.10.3 Wavefront Amplitude: Optical Proximity Corrections (OPC) . . . 112
6.11 Immersion Lithography . . . 113
6.11.1 The Resolution Advantage . . . 114
6.11.2 The Depth of Focus Advantage . . . 115
6.12 Pushing the Limits: Multiple Patterning . . . 115
6.12.1 Double Exposure: Litho-Etch-Litho-Etch (LELE) . . . 115
6.12.2 Self-Aligned Double Patterning (SADP) . . . 116
6.13 Extreme Ultraviolet (EUV) Lithography . . . 118
6.13.1 The Paradigm Shift: Reflective Optics . . . 119
6.13.2 Primary Challenges of EUV . . . 119
6.14 Alternative Next-Generation Lithography (NGL) . . . 120
6.14.1 Electron Beam Lithography (EBL) . . . 120
6.14.2 Nano Imprint Lithography (NIL) . . . 122
6.14.3 Directed Self-Assembly (DSA) . . . 122

7 Etching and Pattern Transfer **124**
7.1 Introduction to Etch Metrics . . . 124
7.2 Wet Chemical Etching . . . 124
7.2.1 Wet Etching Examples . . . 125
7.2.2 Advanced Wet Processes . . . 126
7.3 Dry Plasma Etching Mechanisms . . . 126
7.3.1 Plasma Physics and Chamber Parameters . . . 126
7.3.2 The Three Etching Regimes . . . 127
7.4 Etching Tools and Metrology . . . 129
7.4.1 Evolution of Etching Tools . . . 129
7.4.2 End Point Detection . . . 131
7.5 Process “Knobs” and Integration Issues . . . 131
7.5.1 Plasma Etching Process Parameters . . . 131
7.5.2 Micro-Loading and Topographical Issues . . . 132
7.5.3 Radiation Damage . . . 133
7.6 Specific Etch Chemistries . . . 134
7.6.1 Silicon Oxide (SiO_2) . . . 134

7.6.2 Silicon and Poly-Silicon . . . 134
7.6.3 Aluminium (Al) . . . 135
7.7 Etching Models . . . 135
7.7.1 The Simple Linear Model . . . 136
7.7.2 The Ion-Enhanced Etching Model . . . 136

8 Interconnects and Metallisation 138
8.1 Introduction to the Back End of the Line (BEOL) . . . 138
8.1.1 Local Interconnects and Silicides . . . 139
8.1.2 Silicide Formation: The Salicide Process . . . 139
8.1.3 Evolution of Silicide Materials . . . 140
8.2 Contacts and Plugs . . . 140
8.2.1 Metal-Semiconductor Junction Physics . . . 141
8.2.2 Contact Resistance and Scaling Issues . . . 143
8.2.3 The Evolution of the Contact: Aluminium Spiking . . . 143
8.2.4 Modern Tungsten (W) Plugs . . . 143
8.3 Global Interconnects and the Aluminium Era . . . 145
8.3.1 Thermal Mismatch: Hillocks and Voids . . . 145
8.4 Electromigration and Reliability . . . 145
8.4.1 The Electron Wind and Flux Divergence . . . 146
8.4.2 Microstructure and the Bamboo Effect . . . 147
8.4.3 A Mathematical Model of Electromigration Stress . . . 148
8.4.4 Mitigation Strategies in Aluminium Interconnects . . . 149
8.5 Back-End Dielectrics and the RC Bottleneck . . . 150
8.5.1 Technology Scaling and the RC Delay Crisis . . . 150
8.5.2 The Transition to Low-k Dielectrics . . . 152
8.6 The Copper Damascene Revolution . . . 152
8.6.1 Phase 1: Line 1 Single Damascene . . . 152
8.6.2 Phase 2: Vias and Line 2 Dual Damascene . . . 153
8.6.3 Nanoscale Copper Challenges . . . 155
8.7 Future Perspectives and Conflict Minerals . . . 155
8.7.1 Beyond Copper: Radical Interconnect Technologies . . . 156
8.7.2 The Human Cost of Metallisation . . . 156

9 3D Integration and Advanced Packaging 158
9.1 The Purpose and Evolution of Packaging . . . 158
9.1.1 The Evolution of Interconnects: Wire Bonding to Flip-Chip . . . 158
9.1.2 The Drive for Interconnect Density . . . 159
9.2 Through-Silicon Vias (TSVs) and HBM Architecture . . . 160
9.2.1 TSV Integration Schemes and Scaling . . . 160
9.2.2 Case Study: High Bandwidth Memory (HBM) . . . 161
9.2.3 2.5D Integration and the Silicon Interposer . . . 161
9.3 Advanced Bonding Techniques . . . 162
9.3.1 Fusion Bonding . . . 162
9.3.2 Hybrid Bonding . . . 163
9.4 Current Applications of 3D Integration . . . 164
9.4.1 Fusion Bonding in SOI and MEMS . . . 164
9.4.2 Hybrid Bonding in High-Density Silicon . . . 165
9.4.3 The Future of Memory: VCT DRAM and 16H HBM . . . 166
9.5 The 3D Interconnect Roadmap . . . 166
9.5.1 TSMC SoIC and Heterogeneous Stacking . . . 166
9.5.2 The IMEC Interconnect Density Roadmap . . . 167

9.5.3 Backside Power Delivery Networks (BSPDN) 167

Prologue: The CMOS Inverter Traveller

Before diving into the rigorous physics and chemistry of individual semiconductor manufacturing tools, it is crucial to establish a macroscopic roadmap. In a semiconductor fabrication plant (fab), the sequence of operations required to transform a bare silicon wafer into a functional integrated circuit is known as a "traveller" .

Studying isolated physical mechanisms, such as the plasma dynamics of an etcher or the fluid dynamics of a chemical vapour deposition chamber, can easily cause one to lose sight of the final objective. To prevent this, we will first walk through a simplified process flow to create a basic CMOS (Complementary Metal-Oxide-Semiconductor) inverter. Even in this simplified overview, the creation of this fundamental logic gate requires approximately 80 distinct process steps and 13 individual photolithographic masks.

How to Read This Prologue

The slide deck provided by the professor (*02.CMOS process flow*) uses step-by-step 3D renders to aid the explanation. I highly recommend keeping it on the side, as the pictures provide great help to imagine the structure. Nevertheless, because replicating the exhaustive infographic in a written text is impractical, this traveller is condensed into major manufacturing **Milestones**. Each milestone represents a phase of construction that brings the device significantly closer to its final architecture.

As you read through these milestones, pay attention to the following structural elements:

- **The Mask Tracker:** Lithography is the heartbeat of semiconductor manufacturing. Every time a new layer of the circuit must be defined, a new mask is required. You will notice a counter in the margins of these notes tracking the current mask number (from 0 up to 13). This provides a persistent sense of progression.
- **Cross-Sectional Perspective:** The final structure at the end of each milestone is reported as per the slides. Rather than looking at the circuit from a top-down layout view, the steps are best understood through 2D cross-sections. This perspective reveals how layers are sequentially deposited, patterned, and etched to build the device vertically.
- **Insights:** Just as in the main chapters, *Process Insight* and *Physical Insight* boxes are used here. The insights in this prologue serve to explain physical or process-related reasons for the techniques used across the traveller.

The pedagogical purpose of this prologue is to build your intuition. You do not need to memorize every technique used or the exact sequence of the 80 steps right now. The goal is to observe how deposition, lithography, etching, and doping interact cyclically. Once you understand the *purpose* of each milestone in building the CMOS inverter, the dense physics of the tools described in the subsequent chapters will have a clear, practical context.

Milestone 1: Substrate Selection and Shallow Trench Isolation (STI)

Mask: 1/13

The creation of an integrated circuit begins with the selection of the substrate. For standard CMOS processes, a p-type (boron-doped) silicon wafer is chosen. The doping concentration is relatively light (typically around $3-5\times10^{14}$ cm^{-3}, yielding a resistivity of $25-50\ \Omega\cdot$cm). The crystal orientation is almost exclusively chosen as $\langle100\rangle$. The surfaces can be seen in Figure 1.

Physical Insight

The choice of the $\langle100\rangle$ crystalline orientation over $\langle111\rangle$ is driven by solid-state physics at the interface. Silicon atoms on a $\langle100\rangle$ surface have fewer dangling bonds per unit area than on a $\langle111\rangle$ surface. When the silicon is oxidised, fewer dangling bonds result in a lower density of interface-trapped charges (D_{it}). This interface quality is paramount, as trapped charges would otherwise scatter channel electrons, severely degrading the mobility and shifting the threshold voltage of the final transistors.

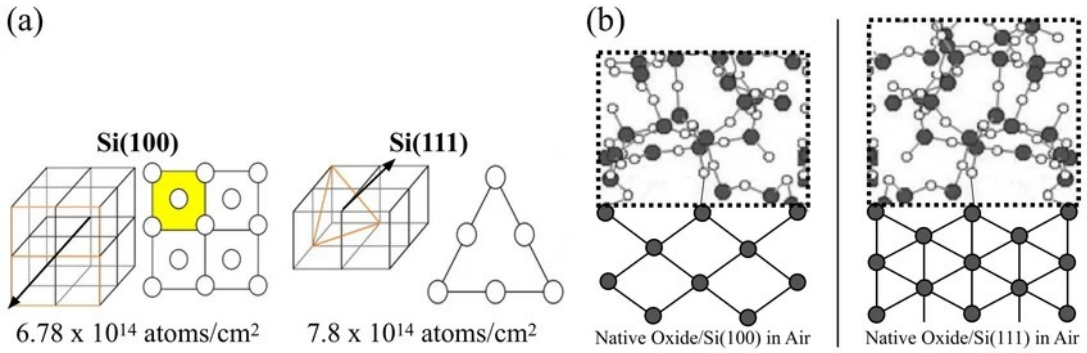


Figure 1: a) The bulk[1] structure of Si(100) and Si(111) is shown. b) A side view of the bulk structure of the two orientations with a Monte Carlo simulation of the oxidation at the surface. Image taken from ResearchGate.

Before any structures can be built, the individual transistors must be electrically isolated from one another to prevent short circuits and parasitic leakage. This is achieved through **Shallow Trench Isolation (STI)**.

The STI process begins with a rigorous surface cleaning (such as the standard RCA clean) to strip away native oxides and particulate contaminants. Following this, a thin layer of pad oxide is thermally grown at approximately 1000 °C, followed by the deposition of a thicker silicon nitride (Si_3N_4) layer via Low Pressure Chemical Vapour Deposition (LPCVD) at around 800 °C.

At this stage, the first lithographic step occurs. Photoresist is spun onto the wafer, and **Mask 1 (Active Area)** is exposed. The resist is developed, washing away the polymer in the regions where the isolation trenches will be located, whilst protecting the regions that will eventually become the active transistors.

The exposed nitride, oxide, and underlying silicon are then removed using an anisotropic plasma dry etch, creating trenches hundreds of nanometres deep into the silicon substrate. The trench sidewalls are often deliberately tapered (etched in a V shape, wider at the top) to

[1]When left alone, surfaces undergo structural changes due to surface thermodynamics. While it is not strictly necessary for this course, if interested one can look up "Si surface reconstruction". Professor Bussetti's course on "Quantum Phenomena in Low Dimensional Systems" is a great source for understanding surface physics.

reduce stress and facilitate the subsequent filling step. Once the etch is complete, the remaining photoresist is stripped away using an oxygen downstream plasma (dry ashing) and wet chemical etching (e.g., $H_2SO_4 + H_2O_2$).

Because the intense plasma etch damages the silicon crystalline lattice at the trench walls, a brief thermal oxidation is performed to grow a so-called *liner* oxide, effectively repairing the surface damage. The trenches are then overfilled with a thick layer of silicon dioxide (SiO_2), typically using High-Density Plasma CVD (HDP-CVD) or Atomic Layer Deposition (ALD) to ensure void-free filling of the high-aspect-ratio gaps. The reason we etch so much is to make sure parasitic transistors are long enough to be irrelevant.

Finally, the excess oxide must be removed to restore a flat surface. This is achieved via Chemical Mechanical Polishing (CMP). The CMP process grinds down the silicon dioxide but is highly selective, stopping abruptly when it reaches the underlying silicon nitride layer. The nitride is then stripped away using a wet etch (such as hot phosphoric acid, H_3PO_4), leaving behind perfectly flush oxide-filled trenches separating pristine silicon islands. The outcome can be seen in Figure 2.

Process Insight

The STI sequence perfectly highlights the concept of a *sacrificial layer*. The silicon nitride deposited at the beginning of the flow never makes it into the final device. Instead, it serves a dual technological purpose. Firstly, it acts as a hard mask during the deep silicon trench etch. Secondly, it acts as a robust "stop layer" for the CMP process.
CMP is a remarkably brute-force step in an otherwise delicate nanoscale environment, relying on an abrasive slurry to mechanically grind away material. Without the nitride acting as a mechanical buffer, the CMP pad would gouge into the delicate active silicon areas, completely destroying the wafer yield. Once its mechanical job is done, the nitride is stripped away, leaving a planarised surface ready for the next milestone.

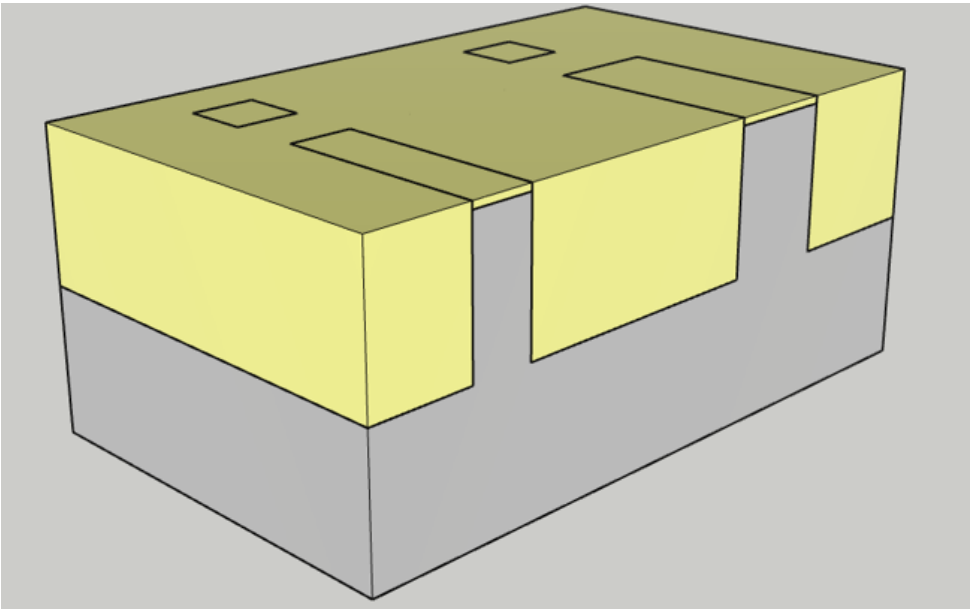


Figure 2: The picture is a 3D render taken from the slides which shows the result at the end of milestone 1. The grey part is the silicon substrate in which the trenches were etched. The yellow part is the silicon oxide.

Milestone 2: Well Formation and Dopant Activation

Mask: 3/13

With the active silicon islands electrically isolated by the STI oxide, the next phase dictates the electrical polarity of the future transistors. Because a CMOS architecture requires both n-channel and p-channel devices on the same substrate, local regions of specific background doping, known as "wells", must be created. Although the starting wafer is already p-type, modern processes typically employ a "twin-well" approach, implanting both p-wells and n-wells to precisely control the background doping concentrations.

The sequence begins with the application of photoresist and the exposure of **Mask 2 (p-well)**. The resist is developed to open the areas[2] corresponding to the future n-channel transistors, whilst the rest of the wafer remains covered. Boron, a group III element, is then introduced into the exposed silicon via **ion implantation**. This is not a single implant; typically, a high-energy (hundreds of keV), higher-dose implant forms the deep well body, while a lower-energy (tens of keV), lower-dose implant is performed closer to the surface to finely tailor the threshold voltage (V_{th}) of the transistor. The resist is subsequently stripped using an oxygen downstream plasma and a wet chemical etch.

The exact same photolithographic cycle is repeated for the p-channel devices using **Mask 3 (n-well)**. The photoresist protects the newly formed p-wells, and the future p-channel regions are left open. Phosphorus, a group V element, is implanted using similar multi-energy steps to define the n-well body and its specific threshold voltage. Following the implant, the resist is stripped.

At this stage, the wafer undergoes an activation annealing. This is a high-temperature thermal treatment (often around 1000 °C) that drives the dopants deeper into the substrate to establish the final well depths. Finally, the original pad oxide, which has collected metallic contaminants and debris during the violent implantation steps, is stripped away using a wet etch (such as dilute hydrofluoric acid), exposing the pristine silicon surface in preparation for the critical gate stack formation. The outcome can be seen in Figure 3.

Physical Insight

Ion implantation is a highly energetic and destructive process. When boron or phosphorus ions are accelerated into the substrate, they crash into the silicon lattice, knocking host atoms out of place and amorphising the crystal structure. Furthermore, the implanted ions come to rest in random, interstitial positions. In these interstitial sites, the dopants do not form the four covalent bonds required to donate or accept free charge carriers, effectively rendering them electrically inactive.
The activation annealing step resolves both of these issues. The intense thermal energy allows the damaged silicon lattice to spontaneously recrystallise, repairing the crystalline defects. Simultaneously, the dopant atoms acquire enough mobility to migrate into regular, substitutional lattice sites. Only once a dopant atom successfully replaces a silicon atom in the continuous lattice does it become electrically active and functionally dictate the n-type or p-type nature of the semiconductor well.

[2] Actually, you can see from the 3D renders in the slides that some of the SiO_2 remains exposed too. This means some ions will be implanted in the silicon oxide as well, but it is not a big deal.

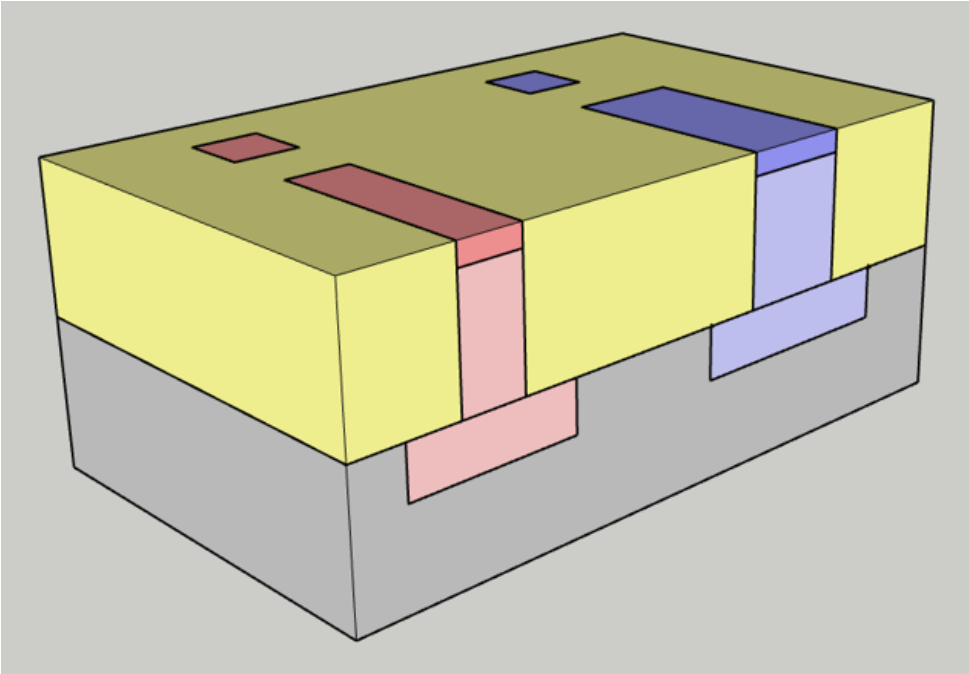


Figure 3: In this 3D render, the coloured areas represent the doped silicon (red for boron doping and blue for phosphorus doping). Darker-coloured areas represent more highly-doped sections of the silicon due to the different energies of the ion implantation steps.

Milestone 3: The Gate Stack

Mask: 4/13

Following the removal of the sacrificial pad oxide, the exposed active silicon regions are ready to receive the most critical structure of the transistor: the gate stack. The electrical performance of the device is fundamentally governed by the quality and precise dimensions of these layers.

The process begins with the growth of the gate dielectric. Exposed silicon is oxidised using wet, dry, radical, or plasma oxidation. The thickness of this oxide can range from a few nanometres to hundreds of nanometres, dictated entirely by the target operating voltage of the final transistor. Immediately following oxidation, a nitridation step is performed to grow a few atomic layers of silicon nitride at the SiO_2/silicon interface using an NH_3 or N_2O plasma.

Physical Insight

The nitridation of the gate oxide solves a severe solid-state diffusion problem. In modern CMOS processes, the poly-silicon gate of a p-channel transistor is heavily doped with boron. During subsequent high-temperature annealing steps, boron atoms can easily diffuse through pure SiO_2 and penetrate the underlying silicon channel. This unwanted doping would drastically shift the threshold voltage of the transistor. The ultra-thin layer of silicon nitride acts as an impermeable physical barrier, blocking boron diffusion and significantly improving the overall reliability of the gate dielectric.

Next, the gate electrode material is deposited. Typically, this is polycrystalline[3] silicon (poly-silicon), deposited via LPCVD using silane gas ($SiH_4 \rightarrow Si + 2H_2$). The deposited layer

[3]Polycrystalline materials are halfway between crystals and amorphous compositions. They are made of randomly-oriented grains, which are ordinary portions of crystal-like compositions. This is the best thing we can achieve growing on an oxide as there is no "template" for orientation.

is generally a few hundred nanometres thick, and careful control over the grain texture is maintained to ensure long-term transistor reliability.

To pattern these continuous layers into individual transistor gates, photoresist is spun onto the wafer and **Mask 4 (Gate)** is exposed. Once developed, the poly-silicon undergoes a highly specialised dry plasma etch. The remaining photoresist is then stripped using an oxygen downstream plasma (which avoids ion bombardment damage) and a wet chemical etch (such as $H_2SO_4 + H_2O_2$). The result can be seen in Figure 4.

Process Insight

The patterning of the gate electrode is one of the most demanding dry etching steps in the entire fabrication process. The plasma etch must be perfectly anisotropic to guarantee strictly vertical sidewalls, ensuring the final physical width of the gate perfectly matches the layout dimensions. Furthermore, the etch chemistry must possess an extreme selectivity to silicon dioxide. The plasma must rapidly blast through hundreds of nanometres of poly-silicon but stop instantaneously upon reaching the gate oxide. If the selectivity is poor, the plasma will dig through the ultra-thin oxide and severely damage the active silicon channel below, destroying the transistor.

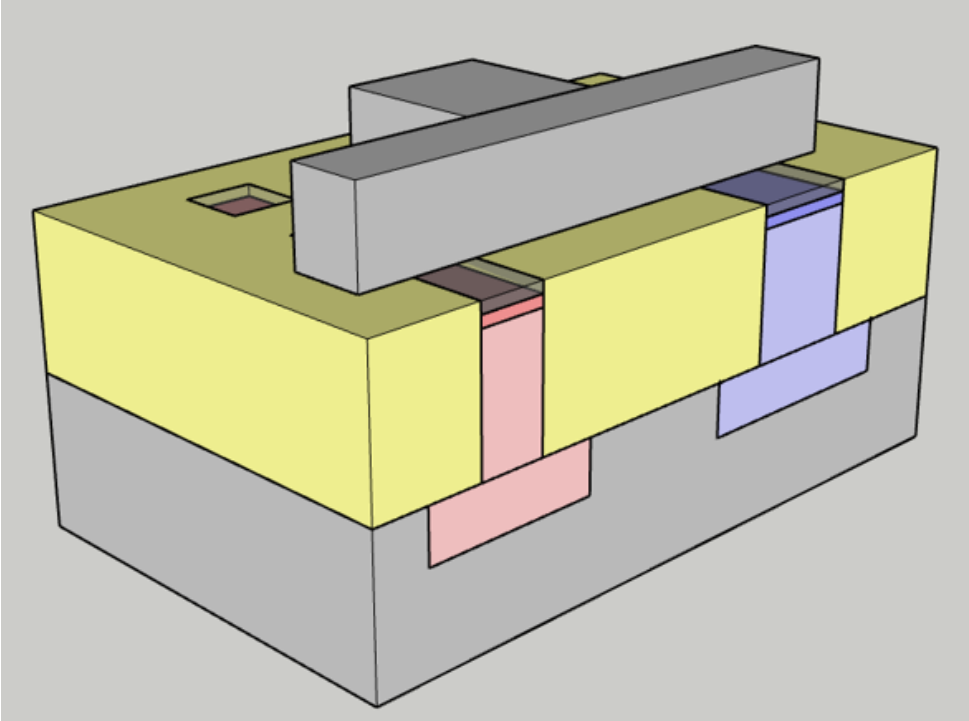


Figure 4: The grey bridge that can be seen on top is the poly line which now connects the gates created in the previous milestone.

Milestone 4: Junctions and Local Interconnections

Mask: 8/13

With the gate stack patterned, the source and drain junctions must be formed. This is done in a two-stage implantation process to create a graded junction profile, beginning with the Lightly Doped Drain (LDD) implants.

Photoresist is spun onto the wafer, and **Mask 5 (n-LDD)** is exposed. The resist is removed from the n-channel transistor regions and n-well contact areas. A shallow, low-dose implantation of n-type dopants (typically arsenic or phosphorus) is performed. The resist is then stripped. This process is mirrored using **Mask 6 (p-LDD)**, where boron or indium is implanted into the exposed p-channel regions. It is important to note that these implants also introduce dopants into the exposed poly-silicon gate electrodes.

Physical Insight

The necessity of the LDD structure arises from the scaling laws of semiconductor physics. Historically, transistor operating voltages have not scaled down as rapidly as their physical dimensions. For instance, applying 5 V across an older 2 µm channel yields an electric field of 2.5×10^4 V $\cdot$ cm^{-1}, whereas the same voltage across a modern 45 nm channel generates an electric field approaching 1 MV $\cdot$ cm^{-1}.
These extreme electric fields accelerate charge carriers to highly energetic states. These can cause severe reliability issues, such as impact ionisation or charge injection directly into the gate oxide. To mitigate these short-channel effects, a lightly doped, shallow junction (the LDD) is placed immediately adjacent to the channel, gradually dropping the electric field and preventing the carriers from gaining excessive kinetic energy.

Following the LDD implants, a dielectric layer (either silicon dioxide or silicon nitride) is deposited using a highly conformal CVD or ALD process. For example, a typical LPCVD nitride process reacts silane and ammonia ($3SiH_4 + 4NH_3 \rightarrow Si_3N_4 + 12H_2$) at approximately 800 °C. This conformal layer undergoes a highly anisotropic dry etch. Because the plasma only etches vertically, the dielectric is completely cleared from the flat surfaces but is left intact on the vertical sidewalls of the gate stacks, forming "spacers".

With the spacers in place, the deep, high-dose source and drain implants are performed. **Mask 7 (n+)** is used to define the heavy arsenic or phosphorus implants for the n-channel devices, and **Mask 8 (p+)** is used for the heavy boron implants in the p-channel devices. Following these implants, a highly controlled Rapid Thermal Process (RTP) or spike/flash anneal at around 1000 °C is performed to activate the junction dopants. The duration must be extremely short to avoid anomalous diffusion effects, such as Transient Enhanced Diffusion (TED).

The final step in this milestone is the creation of local interconnections via a self-aligned silicide (often termed "salicide"). Any remaining thin oxide over the active silicon areas is removed using a wet etch (e.g., dilute HF). A thin layer of titanium (tens of nanometres) is deposited across the entire wafer at room temperature, so as not to destroy the structures built so far, using Physical Vapour Deposition (PVD). An RTP at approximately 700 °C causes the titanium to chemically react with the underlying silicon to form titanium silicide ($TiSi_2$). The unreacted titanium is then stripped away using a wet chemical etch ($NH_4OH : H_2O_2 : H_2O$). The structure can be seen in Figure 5.

Process Insight

Sidewall spacers and the salicide process are known as self-aligned manufacturing.
Firstly, the spacers physically block the heavy n+ and p+ implants from reaching the silicon immediately adjacent to the gate. This preserves the LDD profile near the channel while allowing the creation of deep, highly doped regions further away, which are strictly required to form low-resistance ohmic contacts.
Secondly, during the silicide formation, the titanium reacts only where it is in direct physical contact with silicon (the source/drain active areas and the poly-silicon gate). It does not react with the dielectric spacers or the STI oxide. The subsequent wet etch is also highly selective, leaving the $TiSi_2$ completely intact. This brilliant sequence allows the simultaneous creation of low-resistance metallic contacts on the gate, source, and drain without requiring an additional, impossibly precise lithography mask to prevent them from short-circuiting together.

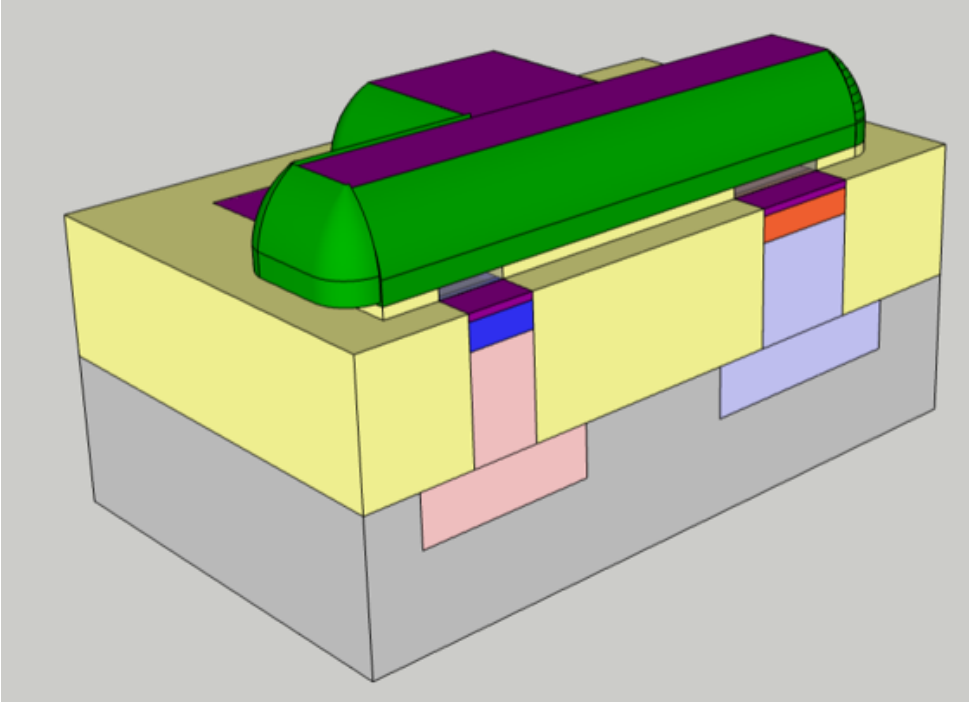


Figure 5: The green parts are the insulating nitride forming the spacers. The purple represents $TiSi_2$, which indeed is only present where there was an exposed Si contact, leaving all the oxidised area intact.

Milestone 5: Metallisation and Passivation

Mask: 13/13

With the transistors fully formed and their terminals capped with low-resistance silicide, the final phase of manufacturing is the Back End of Line (BEOL). This phase constructs the intricate multi-level wiring network required to link billions of isolated transistors into a functional integrated circuit.

The sequence begins with the deposition of a conformal silicon nitride layer via CVD to act as an etch stop. A thick layer of Pre-Metal Dielectric (PMD), typically silicon dioxide, is then

deposited using Plasma-Enhanced CVD (PECVD) and planarised to a perfectly flat surface using CMP.

To connect the transistors to the wiring layers, vertical contact holes must be patterned. Photoresist is spun onto the wafer, and **Mask 9 (Contact)** is exposed. The contact holes are etched in a critical two-step dry plasma process. The PMD oxide is first etched selectively to the underlying silicon nitride. Once the nitride is exposed, the chemistry is switched to etch the nitride selectively to the underlying titanium silicide ($TiSi_2$).

Process Insight

This two-step contact etch elegantly solves a major geometrical challenge. The gate electrodes physically sit higher than the source and drain active areas. If the contact holes were etched directly through the oxide without the conformal nitride stop layer, the plasma would reach the elevated gate silicide much earlier than the active area silicide. The plasma would heavily over-etch and destroy the gate whilst waiting for the deeper active area holes to finish clearing. The nitride layer halts the oxide etch across all depths simultaneously, ensuring that the final, highly selective punch-through to the silicide occurs safely and uniformly.

The contact holes are then filled with metal. A thin titanium/titanium nitride (Ti/TiN) barrier is deposited via CVD. This barrier promotes adhesion and protects the surrounding silicon dioxide from the fluorine-heavy chemistry of the next step. Tungsten (W) is then deposited via CVD (e.g., using tungsten hexafluoride, $WF_6 + 3H_2 \rightarrow W + 6HF$) to completely fill the holes. A metal CMP step grinds away the excess tungsten and barrier material from the wafer surface, leaving perfectly flush, low-resistance tungsten plugs.

With the plugs in place, the first horizontal wiring layer is formed. A metal stack consisting of TiN and aluminium-copper (Al(Cu)) is deposited via Physical Vapour Deposition (PVD). Photoresist is applied and **Mask 10 (Metal 1)** is exposed. The metal stack is dry etched to define the wires, followed immediately by a wet surface clean. This clean is absolutely mandatory to remove residual chlorine (Cl_2) from the etch chemistry, which would otherwise rapidly corrode the aluminium.

Physical Insight

The addition of copper and a TiN underlayer to the aluminium wires addresses the physical phenomenon of electromigration. In highly scaled microelectronics, the current density inside the narrow metal lines is immense. The conducting electrons transfer significant momentum to the metal lattice, physically dislodging aluminium atoms over time. This mass transport creates voids (open circuits) and hillocks (short circuits). Doping the aluminium with a small percentage of copper impedes the migration of aluminium atoms along the grain boundaries, significantly increasing the wire's lifespan. Furthermore, the underlying TiN layer acts as a resilient conductive shunt; if an electromigration void does form in the aluminium, the current simply bypasses the crack by flowing through the TiN, preventing immediate device failure.

This plug-and-line sequence is repeated to build the upper interconnect layers. An Inter-Metal Dielectric (IMD) oxide is deposited and planarised. **Mask 11 (Via 1)** defines the vertical "vias" connecting Metal 1 to Metal 2. The vias are etched, lined with Ti/TiN, filled with tungsten, and planarised via CMP. This looping sequence continues until the topmost, thickest metal layer is patterned, such as **Mask 12 (Bonding Pads)**.

To finalise the wafer, a passivation layer (often silicon oxide, silicon nitride, or an organic polymer) is deposited across the entire surface to shield the circuitry from environmental contamination and mechanical handling. **Mask 13 (Passivation Opening)** is exposed to etch

windows exclusively over the large metal bonding pads, leaving them exposed for future packaging and wire bonding.

The entire traveller concludes with a final alloy annealing step. The wafer is heated to approximately 400 °C in a forming gas ambient (H_2/N_2). This thermal treatment alloys the metal-to-metal contacts to ensure low resistance and allows the hydrogen to diffuse through the layers, passivating dangling bonds and repairing residual defects in the active oxides. The final structure can be seen in Figure 6.

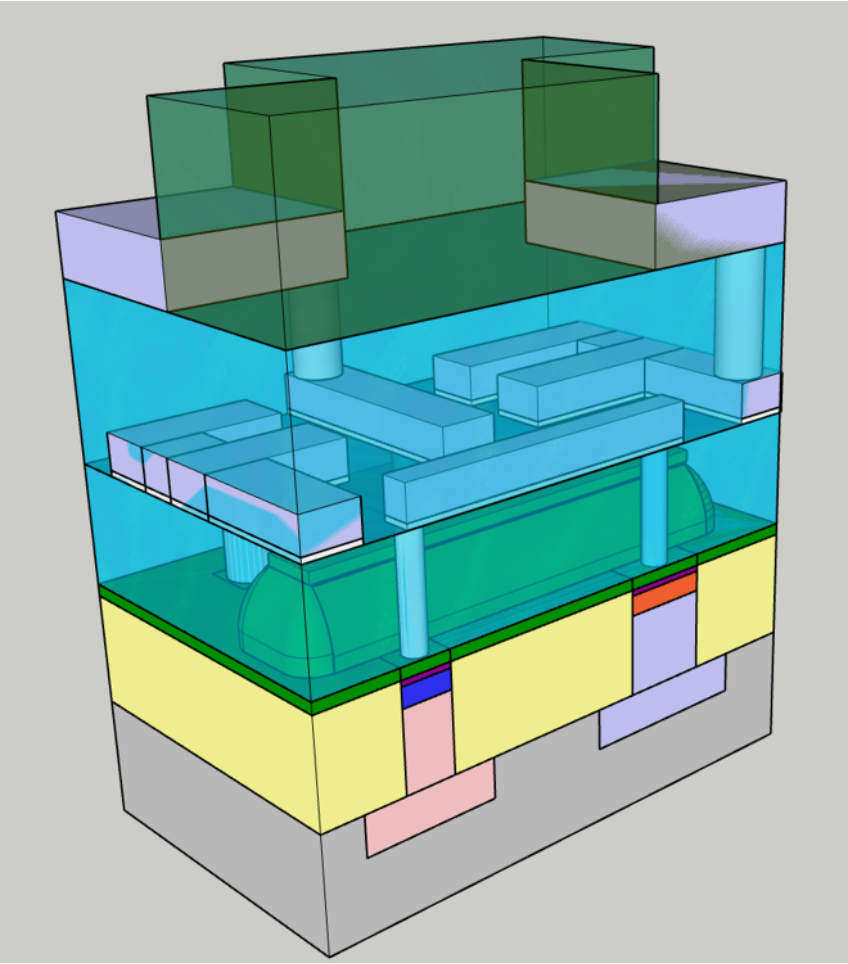


Figure 6: This picture shows the final CMOS unit. The light blue bulk is still SiO_2 but was made transparent to see the inner connections.

With the final alloy annealing complete, our simplified CMOS traveller comes to an end. This prologue has provided the *what* and the *when* of semiconductor manufacturing; we'll now delve into the *how* and the *why*.

The following chapters will dismantle this traveller and examine its constituent parts in deep physical detail, starting with a study of silicon substrates.

Chapter 1

Silicon Substrates

1.1 Silicon Crystallography

To understand the mechanical and electrical properties of silicon, we must first establish its atomic arrangement. Crystals are fundamentally defined by their underlying lattice structure, which dictates how atoms are positioned in space.

1.1.1 Bravais Lattices

A Bravais lattice is defined as an infinite array of discrete points with an arrangement and orientation that appears exactly the same from whichever of the points the array is viewed. Mathematically, it consists of all points with position vectors $\mathbf{R}$ of the form:

$$\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$$

where $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are any three linearly independent vectors, and n_1, n_2, n_3 range through all integer values.

Common examples of unit cells include:

- **Simple Cubic (SC):** Each atom has 6 nearest neighbours. α-Polonium is the only element known to crystallise in a simple cubic form under normal conditions.
- **Body-Centred Cubic (BCC):** Each atom has 8 nearest neighbours. Examples include Na, Cr, and Fe.
- **Face-Centred Cubic (FCC):** Each atom has 12 nearest neighbours. Examples include Al, Cu, Au, and Ag.

1.1.2 The Diamond Structure

Silicon, alongside germanium, α-Tin, and carbon (in its diamond form), crystallises in the diamond structure. It is important to note that the diamond structure itself is not a Bravais lattice. Rather, it is obtained by two interpenetrating FCC lattices, with the origin of the second lattice offset by $\frac{a}{4}$ in every direction (where a is the lattice constant). Alternatively, it can be considered a single FCC Bravais lattice with a two-point basis.

For a silicon crystal, in the cubic cell, which can be seen in Figure 1.1, the side is 5.4 Å, the distance between nearest neighbours is 2.36 Å, and the tetrahedral radius is 1.18 Å. The primary cleavage planes for this structure, that are the planes along which the crystal preferentially splits, are the (111) planes.

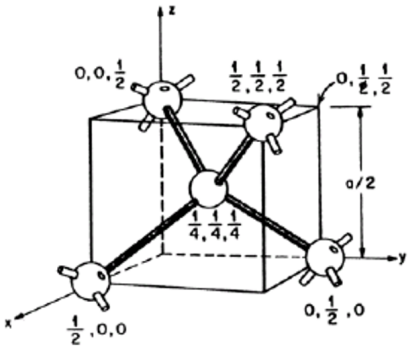


Figure 1.1: The 4 bonds of a silicon atom are enclosed in the cubic cell. It is easy to spot the tetrahedral configuration of the atoms.

Physical Insight

The physical origin of the diamond structure in silicon is rooted in quantum mechanics. Silicon is a group IV element, meaning it has four valence electrons. Through sp^3 hybridisation, these electrons form four highly directional, equally spaced covalent bonds. This specific tetrahedral bonding configuration naturally dictates the spatial arrangement of the atoms, forcing the crystal to adopt the interpenetrating FCC structure to maximise orbital overlap and minimise the electrostatic energy of the atomic bonds.

1.1.3 Miller Indices and Wafer Orientation

Directions within a crystal are described in terms of integers. Due to the cubic symmetry of the lattice, many directions are physically equivalent to one another. For instance, the [100], [010], and [001] directions are identical in their properties and are collectively referred to as the $\langle 100 \rangle$ family of directions.

Planes are determined using Miller indices, which are defined as the reciprocals of the intercepts that the plane makes with the three crystallographic axes. A specific plane is denoted by parentheses, such as (100), while a family of equivalent planes is denoted by braces, such as $\{100\}$.

As mentioned in the Physical Insight of Page 9, the $\langle 100 \rangle$ orientation is the industry standard[4]. The surface of Si(100) has the lowest number of atoms per square centimetre (6.78×10^{14} atoms $\cdot$ cm^{-2} compared to 7.8×10^{14} atoms $\cdot$ cm^{-2} for the (111) plane). Figure 1.2 shows the surface of different silicon orientations.

In terms of processing, this lower atomic density leads to lower oxidation rates and, most importantly, a lower density of interfacial defects (dangling bonds) when growing the gate oxide. This prevents charge trapping and maintains high electron mobility in the transistor channel.

1.2 Crystallographic Defects

In reality, a perfect periodic crystal does not exist. Any disruption in the regular arrangement of the silicon lattice is considered a defect. These irregularities profoundly affect the mechanical

[4]Si(111) used to be used for older BJT technologies. However, it no longer meets the required interface quality standards.

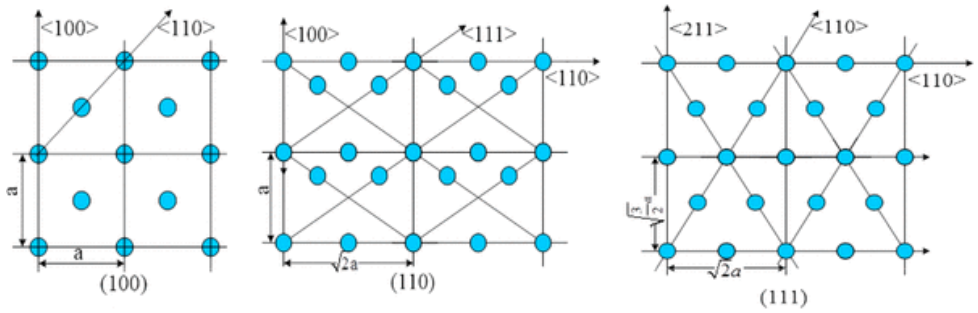


Figure 1.2: This picture shows the surface of the three main orientations of silicon. The centre cubic and hexagonal patterns are easily recognised in the (100) and (111) surfaces. Image taken from ResearchGate.

and electrical properties of the material and are generally classified by their dimensionality into point defects (0D), line defects (1D), area defects (2D), and volume defects (3D).

1.2.1 Point Defects

Point defects are zero-dimensional disruptions localised around a single lattice site. They can be intrinsic (involving only silicon atoms) or extrinsic (involving foreign atoms). Some examples can be seen in Figure 1.3.

- **Vacancy (V):** A lattice site where a silicon atom is missing.
- **Interstitial (I):** An extra silicon atom that resides in a non-lattice site. Interstitials can take various forms, from a simple atom in an empty void to an *interstitialcy* (where the defect perturbs nearest neighbours only) or an *extended interstitial* (causing longer-range perturbation).
- **Frenkel Pair:** A specific defect combination formed when an atom is displaced from its normal lattice site into a nearby interstitial site, leaving a vacancy behind.

Dopants and impurities are also technically considered point defects, acting substitutionally (replacing a silicon atom) or interstitially.

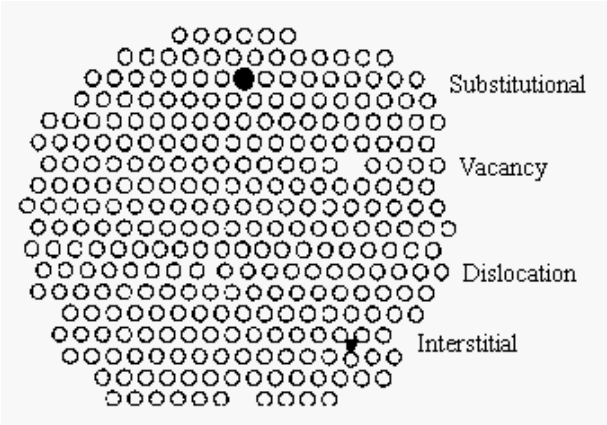


Figure 1.3: A few examples of 0D and 1D defects. Image taken from ChemEd.

The equilibrium density of neutral intrinsic point defects is governed by thermodynamics. For a given defect, its concentration C can be described by:

$$C = N_S \exp\left(\frac{S_f}{k_B}\right) \exp\left(-\frac{H_f}{k_B T}\right)$$

where N_S is the density of silicon lattice sites (5.02×10^{22} cm^{-3}), S_f is the formation entropy, H_f is the formation enthalpy[5], k_B is the Boltzmann constant, and T is the absolute temperature.

It is important to note that there is no thermodynamic requirement for the concentrations of vacancies and interstitials to be equal. Furthermore, even at high processing temperatures like 1000 °C, the intrinsic defect concentration is exceedingly small (rough values are 10^{12} cm^{-3} for interstitials and 5×10^{13} cm^{-3} for vacancies) compared to typical dopant concentrations (which range from 10^{15} to 10^{20} cm^{-3}).

Native point defects can also exist in a charged state. For example, if an electron is "left behind" when a vacancy is created, the result is a negatively charged vacancy[6]. These charged defects introduce deep energy levels within the bandgap, acting as deep donors or acceptors.

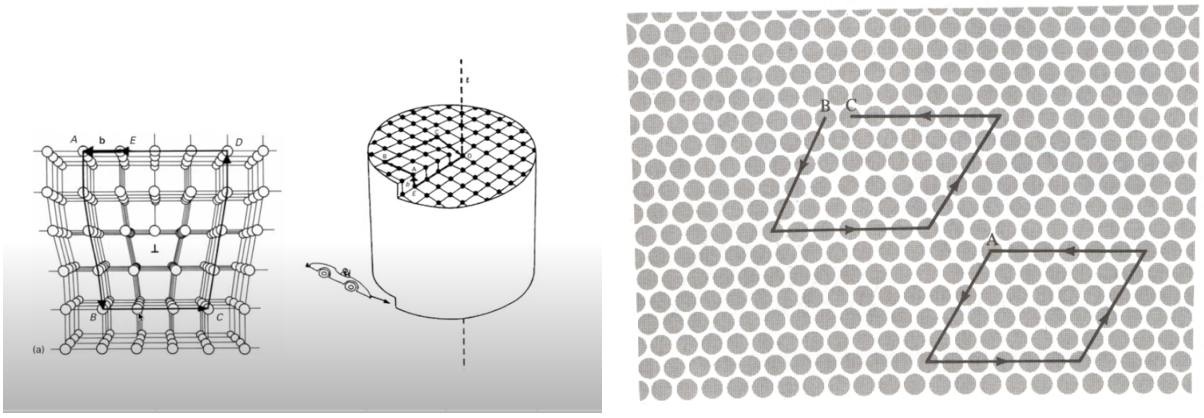
Physical Insight

The fact that point defects can be charged creates a direct coupling between the macroscopic doping of the wafer and its microscopic defect density. When the silicon is heavily doped (for example, highly n-type), the Fermi level shifts towards the conduction band. This shift lowers the energy required to form negatively charged vacancies. Consequently, altering the doping not only changes the dominant charged defect population but actively increases or decreases the *total* number of defects in the crystal at thermal equilibrium.

1.2.2 Line Defects: Dislocations

Line defects, or dislocations, represent a local 1D deformation of the silicon lattice. A classic example is the **edge dislocation**, which can be visualised as an extra half-plane of atoms inserted into the crystal, terminating abruptly within the lattice. Another fundamental type is the **screw dislocation**, which resembles a spiral ramp twisting through the crystal planes. These defects are shown in Figure 1.4a.

A dislocation is formally characterised by its **Burgers vector** (**b**). If one traces a closed loop path from atom to atom in a perfect crystal, tracing the exact same sequence of steps around a dislocated region will fail to close the loop. The vector required to close this gap is the Burgers vector, which defines the magnitude and direction of the lattice distortion. Figure 1.4b shows how to construct it.



(a) Line Defects (b) Burgers vector construction

Figure 1.4: a) A representation of edge dislocation and screw dislocation. b) How to define the Burgers vector in a lattice with a dislocation.

[5]Exact values of the formation entropy and enthalpy are still under debate.

[6]Charged vacancies and charged interstitials are experimentally detected using optical methods such as optical fluorescence, which you can read an example of in Nature Communications.

Dislocations can move through the crystal lattice via two primary mechanisms:

- **Glide:** The movement of the dislocation along its slip plane under the influence of shear stress.
- **Climb:** The movement of the dislocation perpendicular to its slip plane. This process requires the absorption or emission of point defects (vacancies or interstitials), effectively "eating" or "adding" to the extra half-plane of atoms.

During manufacturing, dislocations can be severely detrimental. They are typically induced by mechanical stress, the agglomeration of point defects, or the massive structural damage caused by ion implantation.

1.2.3 Area and Volume Defects

Area defects (2D) primarily take the form of stacking faults. These are planar disruptions in the regular stacking sequence of the crystal planes, usually occurring on the {111} planes in silicon. An *extrinsic* stacking fault occurs when an extra partial plane of atoms is inserted into the lattice, whereas an *intrinsic* stacking fault occurs when a partial plane is removed. These faults must physically terminate either at the wafer surface or at a bounding dislocation. Figure 1.5a shows an example of an area defect in Si(111).

Volume defects (3D) are macroscopic flaws, the most common being precipitates. If impurities are incorporated into the silicon at the extremely high temperatures of crystal growth, their concentration may exceed the solid solubility limit as the crystal cools. Unable to remain dissolved in the silicon lattice, these impurities clump together to form solid precipitates or clusters.

Process Insight

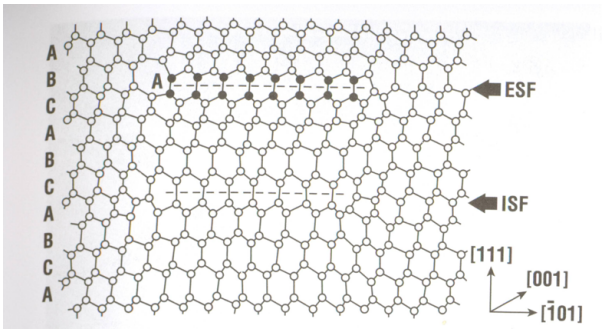
While most defects are carefully avoided, volume defects are sometimes engineered on purpose. During Czochralski crystal growth, a significant amount of oxygen from the quartz crucible dissolves into the liquid silicon. As the wafer cools, this oxygen exceeds its solid solubility and forms SiO_2 precipitates.
Through a highly controlled thermal sequence, fab engineers cause the oxygen near the wafer's surface to out-diffuse, leaving behind a pristine, defect-free region known as the **denuded zone**. The active transistors are built safely within this top layer. Deep in the bulk of the wafer, however, the oxygen precipitates are deliberately left intact to create a "buried" damaged zone. Throughout the rigorous processing in the fab, this damaged bulk acts as a sink, trapping (or "gettering") harmful metallic impurities and preventing them from migrating up into the delicate active devices. Figure 1.5b shows an example of this.

1.3 Silicon Manufacturing: From Sand to Ingot

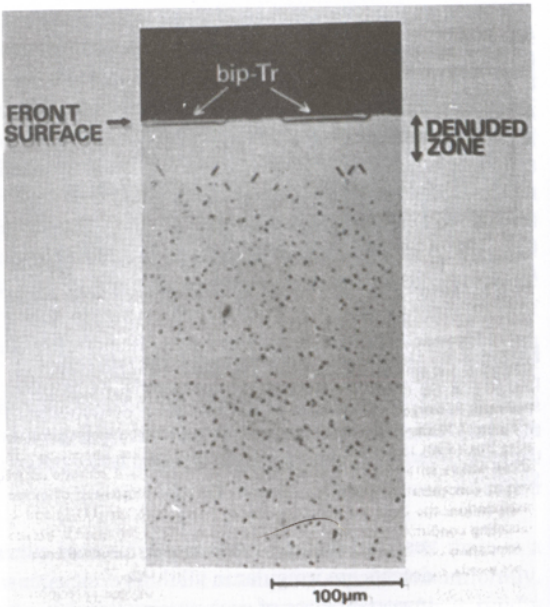
Having established the theoretical structure of the silicon lattice and the defects that can plague it, we now turn to the practical challenge of building this perfect crystal from raw materials. Silicon is an exceedingly abundant element, making up approximately 27% of the Earth's crust. However, it does not naturally occur in the ultra-pure, single-crystal form required for microelectronics.

1.3.1 Purification to Electronic Grade Silicon

The manufacturing journey begins with quartzite (sand, predominantly SiO_2). This raw material is placed in a furnace with a carbon source and heated to approximately 2000 °C. The resulting



(a) Area Defects



(b) Volume Defects

Figure 1.5: a) A side view of an area defect that breaks the lattice periodicity of Si(111). b) A picture showing the creation of the denuded zone. The black dots are the diffused defects.

carbo-thermic reduction strips the oxygen away as carbon monoxide gas, yielding Metallurgical Grade Silicon (MGS):

$$2C_{(s)} + SiO_{2(s)} \rightarrow Si_{(l)} + 2CO_{(g)}$$

MGS is roughly 98% pure, which is sufficient for steel-making but completely useless for semiconductors.

To achieve electronic-grade purity, the MGS undergoes a rigorous chemical transformation. It is reacted with anhydrous hydrogen chloride (HCl) and a catalyst to form a volatile liquid called trichlorosilane ($SiHCl_3$). Because it is a liquid at room temperature, $SiHCl_3$ can be purified using fractional distillation[7], separating the silicon-bearing molecules from the remaining trace metallic impurities.

Once ultra-pure, the trichlorosilane is vaporised and introduced into a Chemical Vapour Deposition (CVD) reactor alongside hydrogen gas. At high temperatures, the reaction reverses, depositing solid silicon onto a heated filament:

$$2SiHCl_{3(g)} + 2H_{2(g)} \rightarrow 2Si_{(s)} + 6HCl_{(g)}$$

The result of this CVD process is Electronic Grade Silicon (EGS). EGS boasts a purity of 99.9999999% (often referred to as "nine nines" purity), but it is physically deposited as a highly fractured, polycrystalline rod.

1.3.2 Czochralski (CZ) Crystal Growth and Pull Rate

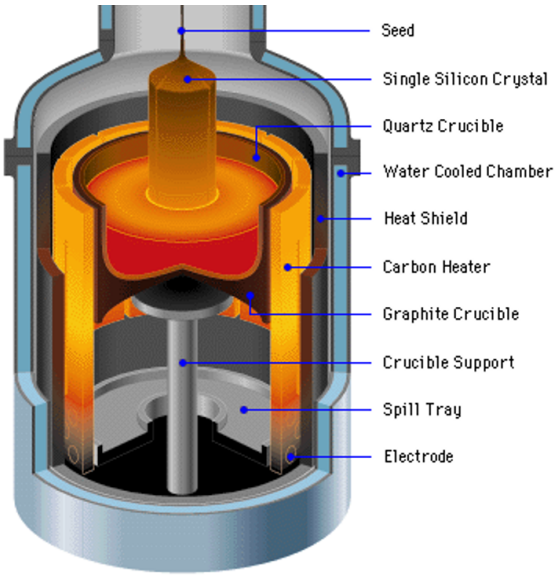
To transform the fragmented poly-silicon rod into a single, continuous crystal, the industry relies on the Czochralski (CZ) growth technique.

The EGS is placed inside a quartz (SiO_2) crucible and heated beyond its melting point. A "seed", that is a small, chemically etched[8] piece of single-crystal silicon with the precise desired

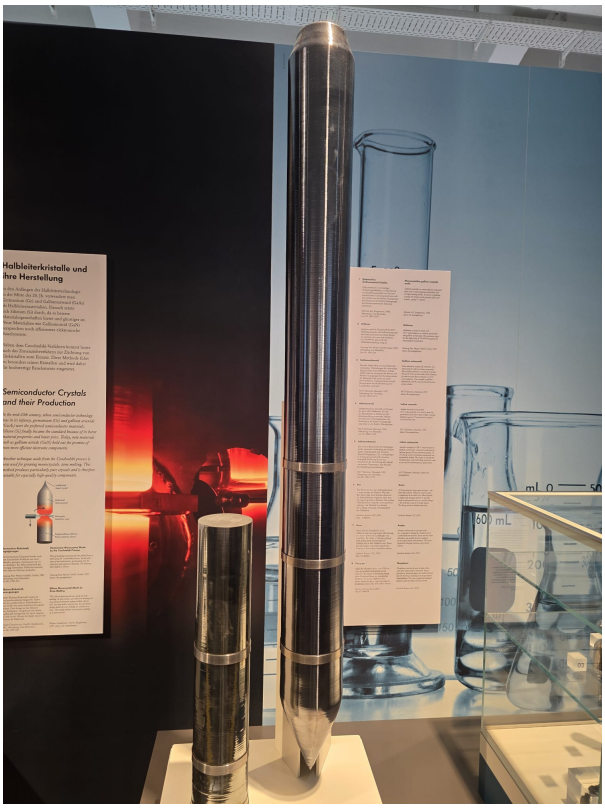
[7]This is the same process commonly used to separate ethanol from water (and other contaminants) in the creation of vodka.

[8]To create a seed, special chemical etchants are used on a high-purity single-crystal ingot. These etchants have different etch rates depending on the crystal's orientation.

crystallographic orientation (e.g., ⟨100⟩ or ⟨111⟩), is lowered until it just touches the surface of the melt. The melt naturally conforms to the atomic template of the seed. Figure 1.6a shows a drawing of what the furnace looks like.



(a) Czochralski furnace



(b) Boule

Figure 1.6: a) A scheme detailing all components of the furnace used in CZ growth. b) An example of a boule. The conical ends are usually cut in industrial processes while some fab workers keep them as knick-knacks.

As the seed is slowly pulled upwards, the molten silicon drawn up with it cools and solidifies, forming a massive, continuous single-crystal ingot known as a "boule". To ensure thermal symmetry and chemical uniformity, both the seed and the crucible are continuously rotated in opposite directions during the pulling process. Modern CZ boules can achieve immense dimensions, reaching diameters up to 450 mm (though 300 mm remains the dominant standard today) and lengths of 1 to 2 metres. Figure 1.6b is a picture of a boule I took at the Deutsches Museum in Munich.

The speed at which the boule is pulled from the melt is a fine balance between thermodynamics and practical efficiency. The melt has to be given enough time to turn into a crystal, otherwise it would just turn back into liquid and collapse. On the other hand, the manufacturing throughput must be kept in mind; if this process is too slow it becomes economically unviable.

Physical Insight

This way of growing the silicon is a freezing event. When the liquid silicon solidifies at the interface, it needs to release its latent heat of fusion (L). If this heat is not removed rapidly enough, the interface will melt back, and the crystal will fail to grow. Realistically, in the CZ process, most of this latent heat will be conducted away vertically through the solid silicon boule itself. Assuming that *all* the heat is conducted through the solid and that the absolute minimum heat is transported away from the surface, the maximum pull rate is:

$$v_{max} = \frac{k}{\rho L}\frac{dT}{dx_s} \sim \frac{1}{\rho L}\sqrt{\frac{1}{r}}$$

Where k is the thermal conductivity of solid Si, ρ is silicon's density, L is the latent heat of fusion and $\frac{dT}{dx_s}$ is the temperature gradient at the solid surface.
To prove of the last proportionality, one must find $\frac{dT}{dx_s}$. The derivation can be a good exercise for the reader. Hints: neglecting convection, write $\frac{dQ}{dx}$ via its radiation formula and equate it to the heat from the conduction formula (assume thermal conductivity as constant with respect to space). Then, solve the differential equation using $k \approx k_{melt}\frac{T_{melt}}{T}$ and $T = \frac{A}{\sqrt{x+B}}$.

1.3.3 Doping and the Segregation Coefficient

A completely pure silicon crystal is highly resistive and virtually useless for building transistors. To set the background electrical conductivity of the wafer (making it n-type or p-type), precise amounts of dopant atoms, such as boron (B), phosphorus (P), arsenic (As), or antimony (Sb), are added directly into the molten silicon before the CZ pull begins.

However, incorporating these dopants uniformly into the solidifying crystal presents a major physical challenge. Because of the difference in chemical potential between the solid and liquid phases, impurities typically exhibit a different solid solubility than liquid solubility. This is quantified by the coefficient (k):

$$k = \frac{C_s}{C_l}$$

where C_s is the equilibrium concentration of the impurity in the solid and C_l is the concentration in the liquid at the interface.

Process Insight

For almost all common silicon dopants, the segregation coefficient is strictly less than 1. For example, boron has a $k \approx 0.8$, phosphorus has $k \approx 0.35$, and antimony has an extremely low $k \approx 0.023$.
Physically, a $k < 1$ means that the solidifying silicon lattice actively "rejects" the dopant atoms, preferring to push them back into the remaining liquid melt. As the boule is continuously pulled, the remaining melt becomes increasingly enriched with the rejected dopants.
Consequently, the dopant concentration will not be uniform across the length of the silicon boule. If C_0 is the initial impurity concentration in the melt and X is the fraction of the melt that has solidified, the dopant concentration along the boule follows the normal freezing equation:

$$C_s(X) = kC_0(1-X)^{k-1}$$

This physical phenomenon forces manufacturers to discard the heavily doped tail-end of the boule, as its characteristics deviate too far from the target specifications.

1.4 Wafer Preparation and Metrology

The output of the Czochralski process is a massive, heavy silicon boule. While this ingot is a single, nearly perfect crystal, it is far from being ready for the fabrication plant. It must undergo a rigorous sequence of mechanical and chemical steps to be sliced into individual, mirror-flat wafers, followed by stringent metrology to guarantee its electrical and structural integrity.

1.4.1 Mechanical Preparation

The transformation from boule to wafer follows a strict mechanical sequence:

1. **Cropping and Trimming:** The tapered top and bottom of the boule are removed, and portions of the ingot are selected based on their specific resistivity requirements. The ingot is then ground down to its precise final diameter (e.g., 300 mm).
2. **Notch Grinding:** A small notch is ground into the side of the ingot. This serves as a permanent crystallographic alignment mark, allowing automated fab equipment to orient the wafer precisely. It is also cut in a specific direction, usually [110] in a (100) surface.
3. **Slicing:** The solid ingot is sliced into individual wafers using a highly tensioned diamond wire saw.
4. **Edge Grinding:** The sharp, fragile edges of the newly cut wafers are mechanically shaped and rounded according to the specific requirements of the Integrated Circuit (IC) manufacturer.
5. **Lapping and Polishing:** The wafer is lapped (ground with an abrasive slurry) to achieve the desired uniform thickness and perfectly flat macroscopic profile. Finally, chemical-mechanical polishing (CMP) is performed to ensure the surface is completely haze-free and mirror-like. Typically the top is mirror-polished while the bottom is not.

It is crucial to note that mega-sonic cleaning and rinsing are performed after every single mechanical step to remove abrasive particles and silicon dust. Following the mechanical preparation, thermal treatments (such as those required to form the denuded zone) and the growth of epitaxial silicon layers are performed if needed by the specific technology.

Process Insight

The mechanical steps of slicing and lapping inflict immense physical trauma on the silicon crystal, leaving a layer of micro-cracks and structural damage on the wafer surfaces. If left untreated, this damage would completely ruin any devices built upon it.
The final polishing step is therefore not merely cosmetic; it is an absolute necessity for modern photolithography. Advanced lithographic scanners have incredibly shallow depths of focus (often measured in tens of nanometres). If the wafer is not polished to atomic-level flatness, the projected circuit patterns will be out of focus and blurred, destroying the yield of the entire wafer.

1.4.2 Electrical Metrology

Before a wafer can be used, its background electrical properties must be validated.

To determine whether a silicon wafer is n-type or p-type, a **hot point probe** is used. This technique involves placing two probes, one hot and one cold, onto the silicon surface. Due to the temperature gradient, the majority charge carriers (electrons in n-type, holes in p-type) acquire thermal kinetic energy and migrate away from the hot probe towards the cold probe. This

migration builds up a Seebeck voltage between the two probes, which is measured as positive if the majority carriers are electrons, and negative if the majority carriers are holes.

To measure the wafer's resistivity (which directly correlates to the doping concentration), a **four-point probe** is utilised. Four equally spaced probes are brought into contact with the wafer. A known current (I) is forced between the two outer probes, while the resulting voltage drop (V) is measured across the two inner probes.

Physical Insight

Why use four probes instead of two? If we were to inject current and measure voltage using the same two probes, the voltmeter would measure not only the resistance of the silicon, but also the contact resistance between the metal probes and the semiconductor, as well as the resistance of the wiring.
By separating the current injection from the voltage measurement, the four-point probe elegantly avoids contact resistance problems. Because voltmeters have near-infinite internal impedance, practically zero current flows through the two inner probes, meaning there is no voltage drop across their contacts. This allows for a highly accurate measurement of the silicon's true sheet resistance.

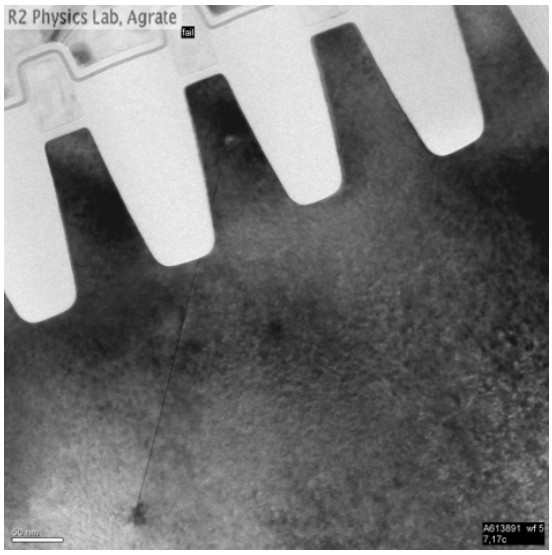
1.4.3 Defect Analysis and Microscopy

To inspect the wafer for the crystallographic defects discussed in previous sections, advanced metrology is required.

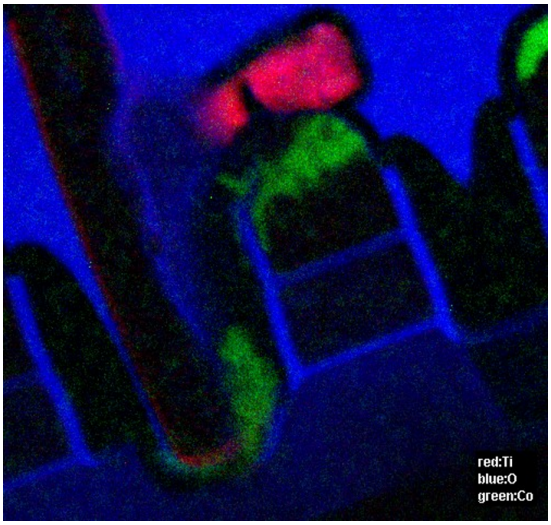
Optical microscopes are fundamentally limited by the diffraction of visible light. However, the resolution of an optical system depends directly on the wavelength of the probing medium. By accelerating electrons, we can leverage their De Broglie wavelength, which is 10^4 to 10^5 times smaller than that of visible photons.

- **Scanning Electron Microscopy (SEM)** scans a focused beam of electrons across the surface, collecting secondary or backscattered electrons to build a 3D-like topological map with a resolution of approximately 5 nm.
- **Transmission Electron Microscopy (TEM)** fires a beam of electrons *through* an ultra-thin sample of the wafer, detecting the transmitted electrons to observe the internal atomic lattice directly, achieving resolutions down to about 0.2 nm. Some TEM images can be seen in Figure 1.7.

Finally, destructive chemical methods can also be used to evaluate structural integrity. The **Secco etch** utilises a solution of potassium dichromate and hydrofluoric acid ($K_2Cr_2O_7$+ HF, very toxic). The etch rate of this chemical mixture is highly sensitive to the stress fields within the crystal; it etches damaged or strained silicon much faster than a perfect, unstrained lattice. Consequently, if the wafer contains line defects, the Secco etch will rapidly dig into the strained regions around them, revealing the otherwise invisible dislocations as distinct, macroscopic etch pits.



(a) TEM image of an STI structure



(b) Coloured TEM image of residues

Figure 1.7: a) There is a (somewhat) clear straight line spanning diagonally in the bottom-left of this picture. It is a dislocation, visible to the TEM microscope. b) The electrons of TEM are highly energetic and can create characteristic X-rays when passing through the target. By collecting these X-rays it is essentially possible to perform spectroscopy on the sample and find out what the residues are made of.

With the completion of the metrology and inspection phase, the journey from raw quartzite sand to a polished, crystallographically perfect, and precisely doped silicon wafer is complete. We now possess the pristine, mirror-flat canvas required for microelectronic fabrication.

However, a bare silicon substrate is only the beginning. As established in the macroscopic roadmap of our CMOS traveller, one of the very first steps in transforming this blank wafer into a functional integrated circuit is the creation of insulating layers, whether for Shallow Trench Isolation (STI) or the ultra-thin gate dielectric. To achieve this, the bare silicon must be exposed to oxygen to grow a high-quality layer of silicon dioxide (SiO_2).

In the next chapter, we will transition from the macroscopic preparation of the wafer to the atomic-level thermodynamics and reaction kinetics of **Silicon Oxidation**. Be prepared for an analytical exploration of how oxygen diffuses into the silicon lattice and reacts to build the fundamental dielectric structures of modern field-effect transistors.

Chapter 2

Silicon Oxidation

2.1 Introduction to Silicon Oxidation

In microelectronic manufacturing, the formation of silicon dioxide (SiO_2) on a silicon surface is typically achieved by exposing the wafer to an oxidant ambient, such as oxygen (O_2) or water vapour (H_2O). While SiO_2 can also be deposited using various techniques, thermal oxidation physically consumes the silicon substrate to grow the oxide layer. The two primary reactions are:

- **Dry Oxidation:** $Si + O_2 \rightarrow SiO_2$
- **Wet Oxidation:** $Si + 2H_2O \rightarrow SiO_2 + 2H_2$

Silicon is universally used in integrated circuits primarily because it forms this excellent native oxide, creating a nearly defect-free interface with the underlying semiconductor. However, while bare silicon will spontaneously oxidise in air at room temperature, this native oxide will not grow thicker than approximately 2 nm. At room temperature, there is simply not enough thermal energy for oxygen molecules to diffuse through the initial thin oxide layer to reach the unreacted silicon below. Consequently, industrial thermal oxidation is performed at high temperatures, typically ranging from 700 °C to 1100 °C.

2.2 The Structure of Silicon Dioxide

Thermally grown SiO_2 forms as fused silica, which is an amorphous solid characterised by short-range order. Although silicon dioxide can exist in crystalline forms (like quartz), the rate of crystallisation at typical processing temperatures is so low that it can be completely neglected.

The fundamental building block of this amorphous network is the SiO_4 tetrahedron, where a central silicon atom is surrounded by four oxygen atoms. Figure 2.1 shows what the structure looks like.

Within the continuous oxide network, oxygen atoms can be classified into two categories:

- **Bridging Oxygen:** Oxygen atoms that are bonded to two adjacent silicon atoms, linking the tetrahedra together.
- **Non-Bridging Oxygen:** Oxygen atoms that are bonded to only a single silicon atom.

A higher fraction of bridging oxygen directly correlates to higher cohesion and structural strength within the SiO_2 film. Dry oxidation generally produces an oxide with a higher fraction of bridging oxygen compared to wet oxidation.

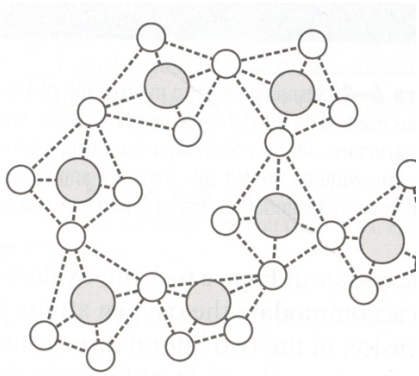


Figure 2.1: A representation of how the building blocks of silicon dioxide interact.

2.2.1 Impurities and Network Modifiers

The presence of impurities can significantly alter the structural integrity of the oxide network:

- **Substitutional Impurities (Network Formers):** Elements like boron (B) and phosphorus (P) replace silicon atoms within the tetrahedral structure. This can lead to the elimination of bridging oxygen, thereby weakening the structure, or the formation of new bridging bonds.
- **Interstitial Impurities (Network Modifiers):** Alkali metals such as sodium (Na) and potassium (K) do not substitute silicon. Instead, they reside in the interstitial voids, increasing the amount of non-bridging oxygen. For example, one could reside in the empty area in the centre of the structure of Figure 2.1. This makes the oxide structure weaker and more porous, which dramatically increases the diffusivity of other species through the glass.

Physical Insight

The differing qualities of dry and wet oxides can be understood through the atomic interactions of water within the silica network. During wet oxidation, water molecules react with the established oxide structure:

$$\text{Si-O-Si} + \text{H}_2\text{O} \rightarrow \text{Si-OH} + \text{HO-Si}$$

This reaction generates hydroxyl (-OH) groups that effectively substitute the strong bridging oxygen bonds with non-bridging terminations. Because the O-H bond within the hydroxyl group is relatively weak and can be easily removed, these sites often generate charged states or traps within the oxide. Consequently, even though wet oxidation proceeds much faster than dry oxidation, it produces a less dense, mechanically weaker dielectric with inferior electrical properties.

2.3 Oxidation Kinetics and the Deal-Grove Model

To model the growth of a thermal oxide, we must understand the physical mechanism driving it. Several experiments, including oxygen isotope profiling[9], have definitively demonstrated that the oxidation process occurs strictly at the Si/SiO_2 interface. This means that for the oxide

[9] In this kind of experiment, silicon was first oxidised using ^{18}O, then using ^{16}O. Using Secondary Ion Mass Spectrometry (SIMS), it was shown that the top layer was composed of ^{18}O while the inner oxide was made of ^{16}O.

layer to grow thicker, the oxidant species (either O_2 or H_2O) must diffuse completely through the already existing silicon oxide to react with the pristine silicon surface below.

Process Insight

Because the reaction occurs at the interface and incorporates oxygen into the silicon lattice, the process involves a massive volume expansion. Specifically, approximately 46% of the final oxide thickness grows *into* the silicon. In other words, the original silicon surface is "pushed" away, and a thickness of silicon equal to 46% of the new oxide is consumed.
This expansion generates significant mechanical stress, which plays a determinant role in oxidation kinetics. This is particularly problematic in 2D structures, such as LOCOS (Local Oxidation of Silicon). Because the diffusivity of oxidants is extremely low in silicon nitride compared to silicon dioxide, the nitride acts as a hard mask. However, oxidants diffuse laterally underneath the edge of the mask. The subsequent volume expansion pushes up against the hard nitride layer, creating a stressed, tapered region known as a "bird's beak", which can be seen in Figure 2.2.

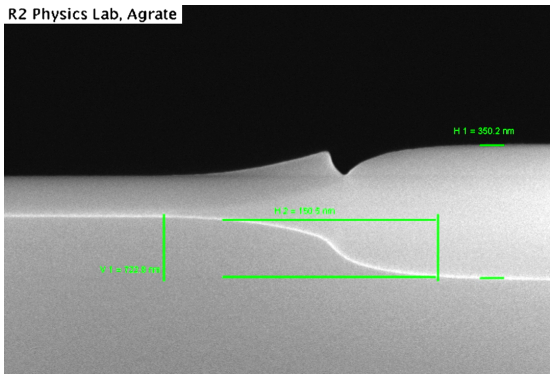


Figure 2.2: In this SEM image a LOCOS can be spotted. The darker grey area in the top-left is silicon nitride (Si_3N_4), under which an oxide is growing.

2.3.1 The Deal-Grove Flux Equations

The standard macroscopic mathematical framework used to describe this process is the **Deal-Grove model**. The model divides the physical system into three distinct regions: the gas phase (including a stagnant boundary layer), the growing SiO_2 film, and the solid silicon substrate. This division can be seen in figure 2.3.

The process is modelled using three continuous fluxes of the oxidant species:

1. F_1 **(Mass transport in the gas phase):** The flux of oxidant from the bulk gas across the stagnant boundary layer to the surface of the oxide.
2. F_2 **(Solid-state diffusion):** The flux of oxidant diffusing through the existing oxide layer.
3. F_3 **(Chemical reaction):** The flux of oxidant being consumed by the chemical reaction at the Si/SiO_2 interface.

To mathematically describe these fluxes, we define several oxidant concentrations, measured in $[\text{mol} \cdot \text{m}^{-3}]$:

- C_g: Concentration in the bulk gas.

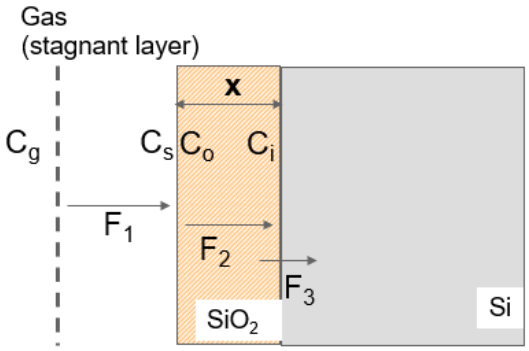


Figure 2.3: A scheme for the Deal-Grove model. To the left of the dotted line, the gas is free. Between it and the SiO_2 layer there is the stagnant layer.

- C_s: Concentration in the gas immediately adjacent to the oxide surface.
- C_0: Concentration inside the oxide, exactly at the gas/oxide surface.
- C_i: Concentration inside the oxide, exactly at the oxide/silicon interface.

Physical Insight

The concept of a "stagnant layer" in the Deal-Grove model is not merely a mathematical convenience; it is a direct consequence of the fluid dynamics inside an oxidation furnace. During thermal oxidation, oxidant gases flow through a heated cylindrical quartz tube and pass over the silicon wafers. To determine the nature of this gas flow, we evaluate the Reynolds number ($Re = \frac{\rho v D}{\mu}$). In a typical oxidation furnace, the bulk gas velocity (v) is extremely low (often on the order of 1 $\mathrm{cm \cdot s^{-1}}$) while the high temperatures ($\approx 1000\,^{\circ}\mathrm{C}$) significantly increase the kinematic viscosity ($\frac{\mu}{\rho}$) of the gas. Consequently, the Reynolds number in an oxidation tube is typically on the order of 10 to 20.
Because this value is vastly below the threshold for turbulence ($Re \approx 2000$), the gas flow is strictly laminar. In laminar flow, viscous forces dominate. According to the "no-slip condition" in fluid mechanics, the velocity of a viscous fluid exactly at the boundary of a solid surface (the wafer) must be zero.
This creates a velocity boundary layer adjacent to the wafer where the gas flow smoothly transitions from zero to the bulk velocity. Within this region, the gas is effectively "stagnant". Because there is no turbulent mixing to forcefully sweep the oxidant molecules directly onto the oxide surface, the oxidant must physically *diffuse* across this stagnant boundary layer. This fluid dynamic reality is exactly what the flux term $F_1 = h_g(C_g - C_s)$ models: the diffusion of oxidant from the bulk gas (C_g) through a motionless layer to reach the surface (C_s).

Deriving the Individual Fluxes

The gas-phase flux F_1 is linearly proportional to the concentration gradient across the stagnant layer, governed by a mass transport coefficient h_g:

$$F_1 = h_g(C_g - C_s)$$

To relate the gas concentration to the concentration directly inside the solid oxide surface, we use Henry's Law, which states that C_0 is proportional to the partial pressure at the surface (p_s): $C_0 = Hp_s$. This partial pressure is an experimental parameter. We now introduce an equilibrium concentration $C^* = Hp_g$. Using the perfect gas law, we can rewrite C_g and C_s as a

function of p_g and p_s, which we'll use to introduce C_0 and C^* into the flux equation. We can thus absorb the constants and rewrite F_1 as:

$$F_1 = h(C^* - C_0)$$

where h is a modified transport coefficient with value $h = \frac{h_g}{Hk_BT}$.

The diffusion flux F_2 is described by Fick's First Law. Assuming a steady-state linear concentration gradient across an oxide of thickness x, we have:

$$F_2 = D\frac{C_0 - C_i}{x}$$

where D is the diffusion coefficient of the oxidant in SiO_2.

The reaction flux F_3 assumes that no oxidant is generated within the oxide and that the reaction rate is strictly proportional to the concentration of the oxidant at the interface:

$$F_3 = k_s C_i$$

where k_s is the interface reaction rate constant.

2.3.2 Solving for the Interfacial Concentration

Under steady-state conditions, the fluxes must be equal so that no oxidant dynamically accumulates anywhere in the system:

$$F_1 = F_2 = F_3 = F$$

We can use this equality to find the unknown concentrations C_0 and C_i. First, equating F_2 and F_3:

$$D\frac{C_0 - C_i}{x} = k_s C_i$$

Solving for C_0 yields:

$$C_0 = C_i + \frac{k_s x}{D}C_i = C_i\left(1 + \frac{k_s x}{D}\right)$$

Next, we equate F_1 and F_3:

$$h(C^* - C_0) = k_s C_i$$

Substitute our expression for C_0 into this equation:

$$h\left(C^* - C_i\left(1 + \frac{k_s x}{D}\right)\right) = k_s C_i$$

$$hC^* - hC_i - \frac{hk_s x}{D}C_i = k_s C_i$$

Isolating C_i, we obtain:

$$hC^* = C_i\left(k_s + h + \frac{hk_s x}{D}\right)$$

Dividing both sides by h, we arrive at the final expression for the interface concentration:

$$C_i = \frac{C^*}{1 + \frac{k_s}{h} + \frac{k_s x}{D}} \approx \frac{C^*}{1 + \frac{k_s x}{D}}$$

where the last approximation is validated experimentally by noticing h is big compared to D. Physically, this means that travel is faster in the stagnant layer, as one might intuitively think.

This equation brilliantly reveals the two limiting regimes of oxidation:

- **Reaction-Controlled Regime:** If $k_s x/D \ll 1$ (which occurs when the oxide x is very thin or diffusion D is very fast), then $C_i \approx C^*$. The bottleneck is how fast the silicon can react with the abundant oxidant.
- **Diffusion-Controlled Regime:** If $k_s x/D \gg 1$ (which occurs when the oxide x is very thick or diffusion D is very slow), then $C_i \approx 0$. The interface is starved of oxygen, and the bottleneck is how fast the oxidant can diffuse through the thick glass layer.

2.3.3 Growth Rate and the Linear-Parabolic Equation

To find how the oxide thickness x evolves over time t, we relate the growth rate to the flux F_3. The rate of growth is simply the flux divided by N_1, which is the number of oxidant molecules incorporated per unit volume of oxide grown:

$$\frac{dx}{dt} = \frac{F_3}{N_1} = \frac{k_s C_i}{N_1}$$

Substituting our expression for C_i:

$$\frac{dx}{dt} = \frac{k_s C^*/N_1}{1 + \frac{k_s}{h} + \frac{k_s x}{D}}$$

This is a separable differential equation. Rearranging to separate x and t:

$$\left(1 + \frac{k_s}{h} + \frac{k_s x}{D}\right) dx = \frac{k_s C^*}{N_1} dt$$

We integrate this from the start of the oxidation process ($t = 0$, where an initial oxide thickness $x = x_i$ might already exist) to a future time t with thickness x:

$$\int_{x_i}^{x} \left(1 + \frac{k_s}{h} + \frac{k_s x}{D}\right) dx = \int_0^t \frac{k_s C^*}{N_1} dt$$

$$\left(1 + \frac{k_s}{h}\right)(x - x_i) + \frac{k_s}{2D}(x^2 - x_i^2) = \frac{k_s C^*}{N_1} t$$

To standardise this into the classic Deal-Grove form, we divide the entire equation by $\frac{k_s}{2D}$ and rearrange. By defining two new constants, A and B:

$$A = 2D\left(\frac{1}{k_s} + \frac{1}{h}\right) \quad \text{and} \quad B = \frac{2DC^*}{N_1}$$

we arrive at the fundamental Linear-Parabolic Deal-Grove equation:

$$x^2 + Ax = B(t + \tau)$$

where $\tau = \frac{x_i^2 + Ax_i}{B}$ is a time-shift parameter accounting for the initial oxide thickness x_i.

Linear and Parabolic Rate Constants

The constants B and B/A have profound physical meanings and are governed by Arrhenius temperature dependencies:

- **Parabolic Rate Constant (B):** Dominates for thick oxides. It is strongly related to the solid-state diffusion of the oxidant (D). Consequently, the activation energies differ significantly depending on the diffusing species (e.g., 1.23 eV for dry O_2 vs. 0.78 eV for wet H_2O).
- **Linear Rate Constant (B/A):** Dominates for thin oxides. It is strictly related to the chemical reaction at the silicon surface (k_s). Its activation energy (≈ 2.0 eV) is remarkably consistent regardless of the oxidant, as it represents the energy required to physically break Si-Si bonds. Because it depends on surface bonds, $\frac{B}{A}$ is highly dependent on crystal orientation (e.g., (111) oxidises faster than (100)).

2.3.4 Corrections to Deal-Grove

While the Deal-Grove model is a triumph of macroscopic kinetics, it notoriously fails to predict the growth rate of very thin oxides (typically < 20 nm) in dry oxygen. In this ultra-thin regime, the initial oxidation rate is significantly faster than the linear-parabolic model predicts.

To account for this, empirical models, such as the Massoud correction, append an exponentially decaying term to the growth rate equation. While a single definitive physical explanation for this thin-film anomaly is still debated, proposed mechanisms include coupled diffusion of O_2^- ions, thermionic emission of electrons from the silicon into the SiO_2, the presence of micro-pores, and blocking layers.

2.4 Factors Influencing the Oxidation Rate

The Deal-Grove model provides an excellent macroscopic framework, but it assumes an idealised, uniformly flat, and pure silicon lattice. In practical microelectronic fabrication, the silicon substrate is heavily doped, possesses specific crystal orientations, and contains complex 2D topographies. All these factors dramatically influence the oxidation kinetics.

2.4.1 Substrate Doping and Impurity Segregation

During thermal oxidation, the substrate doping profile does not remain static. As the silicon is consumed to form SiO_2, the dopant atoms previously residing in the consumed silicon must either be incorporated into the growing oxide glass or rejected back into the silicon substrate. This redistribution depends on the impurity's diffusivity and its segregation coefficient, that is the ratio of the impurity's concentration in the silicon layer to its concentration in the silicon dioxide layer during oxidation ($m = \frac{C_{Si}}{C_{SiO_2}}$).

We observe two primary behaviours at the moving Si/SiO_2 interface:

- **Oxide takes up the impurity ($m < 1$):** Dopants such as boron exhibit an $m < 1$. The growing oxide actively absorbs the boron, leading to a depletion of the dopant concentration in the silicon immediately adjacent to the interface.
- **Oxide rejects the impurity ($m > 1$):** Dopants such as phosphorus, arsenic, and antimony exhibit an $m \approx 10$. The oxide rejects these atoms, forcing them ahead of the advancing interface and causing a significant "pile-up" (accumulation) of dopants in the silicon substrate.

Crucially, all kinds of heavy substrate doping result in an increased overall oxidation rate, but they affect the Deal-Grove parameters differently:

- **Effect on the Parabolic Rate (B):** Because boron is incorporated into the SiO_2 network, it acts as a network modifier/former. It weakens the structure of the amorphous glass, enhancing the diffusivity (D) of the oxidant. As a result, heavy boron doping significantly increases the parabolic rate constant, B.
- **Effect on the Linear Rate (B/A):** Phosphorus doping exhibits a massive increase in the linear rate constant, B/A. Because phosphorus piles up at the silicon surface, it heavily n-dopes the interfacial region. This massive dopant concentration shifts the Fermi level (E_f) towards the conduction band, which thermodynamically increases the concentration of vacancies at the interface. More vacancies provide more available reaction sites for the incoming oxygen, dramatically accelerating the interface reaction rate (k_s).

2.4.2 Crystal Orientation and 2D Topography

As established in the derivation of the Deal-Grove equations, the two rate constants have distinctly different physical origins:

- The parabolic constant (B) is governed by the diffusion of the oxidant through the amorphous oxide. Because the bulk SiO_2 is amorphous and lacks long-range order, B does not depend on the crystal orientation of the underlying silicon.
- The linear constant (B/A) is strictly dependent on the chemical reaction at the silicon surface. Consequently, B/A strongly depends on the crystal orientation, primarily because different planes present a different number of available Si-Si bonds per cm^2. As mentioned before, the (111) plane presents more available bonds than the (100) plane, leading to a proportionally higher linear oxidation rate.

Furthermore, when growing oxide over non-planar 2D structures (such as trenches or patterned lines), the oxidation rate becomes highly non-uniform. Extensive studies (e.g., by Kao et al.) have demonstrated that the oxidation rate is a strong function of the local radius of curvature. This is due to a combination of 2D oxidant diffusion effects, viscous flow of the oxide, and massive orientation-dependent stress generated by the volume expansion at convex and concave corners.

2.4.3 The Microscopic Model: Vacancies and Interstitials

At a purely microscopic level, the mechanism of oxidation is more violent than a simple atomic substitution. Because the newly formed SiO_2 requires vastly more volume than the original silicon lattice, the interface undergoes extreme compressive stress.

To accommodate a portion of this immense volume expansion, the oxidation process actively "consumes" vacancies from the silicon lattice and concurrently "produces" (or injects) silicon interstitials deep into the underlying substrate.

Process Insight

The injection of silicon interstitials during oxidation has profound secondary consequences on the electrical design of the integrated circuit. Many common dopants (such as Boron and Phosphorus) diffuse through the silicon lattice primarily via an "interstitialcy" mechanism, meaning they require an adjacent interstitial silicon atom to move from one lattice site to another.
When thermal oxidation injects a massive flood of extra interstitials into the bulk silicon, the local concentration of interstitials skyrockets past its thermal equilibrium value. Consequently, the diffusion rate of boron and phosphorus suddenly and dramatically increases. This phenomenon is known as **Oxidation-Enhanced Diffusion (OED)**. If process engineers fail to account for OED in their thermal budget calculations, the transistor junctions will diffuse much deeper than intended, destroying the device.
Furthermore, if these excess interstitials agglomerate around existing nucleation sites or damage in the crystal, they can precipitate to form extended 2D area defects known as **Oxidation-induced Stacking Faults (OSF)**.

2.5 Charges in the Oxide and C-V Characterisation

Although the Si/SiO_2 interface is celebrated for its perfection, it is never truly flawless. A pristine silicon surface possesses approximately 10^{15} atoms per cm^2. Following thermal oxidation, the interface typically contains between 10^9 and 10^{11} defects per cm^2. While this represents a

remarkably low defect density (less than one defect per ten thousand surface atoms), these non-idealities profoundly impact the electrical performance of MOS devices.

2.5.1 The Four Oxide Charges

Electrical defects within the SiO_2 system are formally categorised into four distinct types based on their physical origin and location. Figure 2.4 shows where the following defects are found.

1. **Fixed Oxide Charge (Q_f):** This is a sheet of positive charge located in the oxide, very close to the interface (within approximately 2 nm). It is believed to be formed by incompletely oxidised silicon atoms. Q_f is strictly fixed; it does not change its charge state during device operation, nor does it move. It can, however, be reduced by performing a high-temperature anneal after the oxidation step.

2. **Interface-Trapped Charge (Q_{it}):** Located exactly at the physical interface between the silicon and the oxide. These are primarily dangling silicon bonds or incompletely oxidised atoms. Unlike fixed charges, interface traps introduce energy levels directly within the forbidden bandgap of the silicon[10] (usually near the valence and conduction bands). Consequently, they can be positive, negative, or neutral depending on whether they are filled or empty. A 450 °C anneal in a hydrogen ambient (H_2) is highly effective in passivating these dangling bonds and reducing Q_{it} density.

3. **Mobile Ionic Charge (Q_m):** These are physically mobile charges attributed to alkali metal ions (such as Na^+, K^+, Li^+), as well as negative ions or heavy metals. They are incorporated into the oxide as contamination. Because they are mobile, they can drift through the oxide under the influence of an electric field, causing severe instability in the transistor's threshold voltage. This issue is mitigated by strict cleanroom contamination controls and by cleaning furnace tubes with HCl.

4. **Oxide Trapped Charge (Q_{ot}):** These traps can be located anywhere in the bulk of the oxide and are caused by broken Si-O bonds. They are typically induced by damaging process steps (such as ion implantation or plasma etching) or by exposure to ionising radiation.

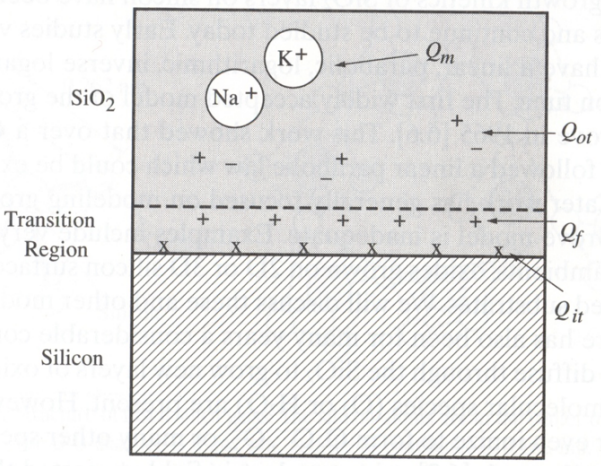


Figure 2.4: Positions of charges in oxides.

[10]This introduces substantial Random Telegraph Noise. As such, interface-trapped charges are where most of the focus in dealing with oxide charges is poured in.

2.5.2 Capacitance-Voltage (C-V) Characterisation

To measure and quantify these charges, engineers use the Capacitance-Voltage (C-V) measurement technique on a simple MOS capacitor structure[11]. As a DC gate voltage (V_G) is swept across the device, the silicon surface beneath the oxide transitions through three electrostatic regimes:

- **Accumulation:** Majority carriers are attracted to the interface. The capacitance measured is simply the oxide capacitance (C_{ox}).
- **Depletion:** Majority carriers are repelled, leaving behind a depleted region of fixed dopant ions with depth x_D. This adds a depletion capacitance (C_D) in series with C_{ox}, causing the total measured capacitance to drop.
- **Inversion:** Minority carriers are pulled to the interface, forming an inversion layer (a conductive channel) of charge Q_I. The depletion width reaches a maximum (x_{DMax}).

Physical Insight

When the MOS capacitor enters strong inversion, the C-V curve diverges depending on the frequency of the small AC signal used to measure the capacitance.
In a **Low-Frequency (LF)** measurement, the alternating signal is slow enough that the thermal generation and recombination of minority carriers can keep pace. The inversion layer rapidly absorbs and releases charge in sync with the AC signal. Because this conductive inversion layer sits immediately beneath the oxide, it effectively shields the depletion region from the AC signal. The measured capacitance therefore climbs back up to equal C_{ox}.
In a **High-Frequency (HF)** measurement, the AC signal oscillates so rapidly that the minority carriers simply cannot generate or recombine fast enough to follow the signal. The inversion layer cannot respond. Instead, the AC signal variations must be accommodated by minuscule expansions and contractions of the depletion region edge deep in the bulk. Consequently, the series combination of C_{ox} and C_{Dmin} is maintained, and the HF capacitance stays flat at a minimum value (C_{Min}).
If the DC bias itself is swept too fast, minority carriers do not even have time to form the DC inversion layer, driving the device into a state called **Deep Depletion**, where the capacitance drops even further below C_{Min}.

2.5.3 Extracting Charge Data from C-V Curves

The non-ideal charges present in the oxide alter the shape and position of the C-V curves, as seen in Figure 2.5.

- **Effect of Q_f and Q_m:** The presence of a fixed positive charge (Q_f) induces a negative image charge in the underlying silicon. This makes inversion easier to achieve (for a p-type substrate) since electrons are naturally attracted to the surface, but requires a more negative gate voltage to achieve flatband. As a result, the entire C-V characteristic undergoes a parallel shift to the left. The mobile charge (Q_m) produces a similar shifting effect; however, because the ions can physically move within the oxide, the magnitude of the shift changes depending on the applied voltage and temperature history.

[11] I will not go into detail on how an MOS capacitor physically works (band-bending, charge motion etc.) as it is likely you have seen it in another course already. If not, there are plenty of resources and YouTube videos online you can use.

- **Effect of Q_{it}:** interface-trapped charges do not cause a simple parallel shift. Instead, as the DC gate voltage sweeps the Fermi level across the bandgap, interface traps are progressively filled or emptied. This charging and discharging behaviour has a dynamic-shifting effect, which essentially "stretches" the C-V curve along the voltage axis.

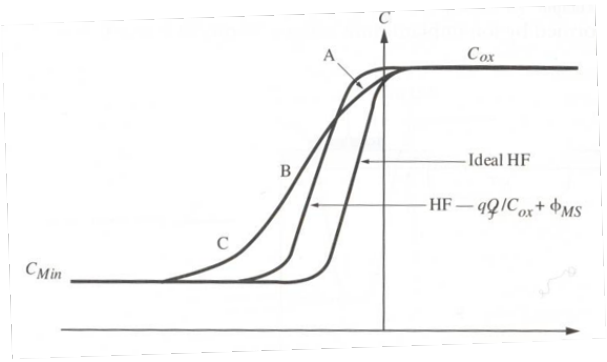


Figure 2.5: C-V curve effects due to charge defects. First, the shift due to the presence of Q_f can be seen. The stretched curve is divided into 3 zones depending on where the traps energetically sit in the band: near the conduction band, states require little band-bending to be filled, which translates to zone A in the graph. Similar reasoning can be applied to zone B and C the deeper in the gap the states are.

The density of interface states (D_{it}, measured in $\text{cm}^{-2} \cdot \text{eV}^{-1}$), aka interface traps, can be precisely mapped across the bandgap using the **Quasi-static method**. This method calculates D_{it} by comparing the physical distortion of the measured LF curve with respect to the HF curve. D_{it} is typically highest near the band edges and lowest near mid-gap.

Beyond C-V analysis, other standard characterisation metrics for SiO_2 include the dielectric breakdown voltage, the Time-Dependent Dielectric Breakdown (TDDB) to determine charge-to-breakdown, and Bias Temperature Stress (BTS) testing specifically used to estimate mobile ion contamination (Q_m).

2.6 Advanced Dielectrics and Oxidation Equipment

As transistor dimensions are scaled down to improve performance and packing density, the demands placed on the gate dielectric become increasingly severe. To meet these demands, both the chemical composition of the oxide and the equipment used to grow it have evolved significantly.

2.6.1 Nitridation

A standard, pure SiO_2 gate dielectric faces severe reliability issues in modern CMOS processes. Specifically, the heavily boron-doped polycrystalline silicon used for p-channel control gates tends to allow boron to diffuse straight through the thin oxide and into the underlying silicon substrate during high-temperature annealing steps.

To combat this, the oxide is subjected to a process called nitridation. Nitridation incorporates nitrogen into the dielectric, which acts as a robust physical barrier to boron diffusion from the control gate into the silicon substrate. Furthermore, the addition of nitrogen significantly improves the electrical quality of the thin oxide, increasing the charge-to-breakdown (Q_{bd}) and enhancing its resistance to trapping and the generation of interface states after hot-electron stress.

The most common nitridation techniques involve annealing the wafer immediately after the thermal oxidation step at high or medium temperatures using nitrogen-rich gases such as

ammonia (NH_3), nitrous oxide (N_2O), or nitric oxide (NO). These processes can be performed in standard furnaces or via Rapid Thermal Processing (RTP). Currently, nitridation by NO is becoming the leading technology. For optimal device performance, the process is carefully tuned to form a nitrogen-rich layer exclusively at the Si/SiO_2 interface, deliberately avoiding the incorporation of nitrogen into the bulk of the oxide.

2.6.2 Oxidation Equipment

The total amount of thermal energy transferred to the wafer during processing, also known as thermal budget, must be strictly minimised in advanced technology nodes to prevent the unwanted diffusion of shallow junctions. Consequently, oxidation equipment has shifted from slow, high-temperature batch processes to rapid, highly controlled techniques:

- **Vertical Furnaces:** The traditional method. Wafers are loaded in batches (up to 100 wafers at a time) into a quartz tube. Reactant gases are flowed into the furnace, and the desired temperature is maintained via resistive heating elements. While excellent for uniformity, furnaces impose a massive thermal budget due to long ramp-up and cool-down times.
- **Rapid Thermal Oxidation (RTO):** A single-wafer process increasingly used to grow thin, high-quality dielectric layers while drastically reducing the thermal budget. The wafer is heated rapidly using high-intensity lamps. The oxidation kinetics in RTO are influenced by both thermally-induced and photon-induced processes, leading to activation energies that differ from conventional furnace growth.
- **Radical Oxidation (e.g., ISSG):** In processes like In-Situ Steam Generation (ISSG), H_2 and O_2 gases are injected directly into an RTO chamber at very low pressures. This generates water and highly reactive oxygen and hydroxyl radicals (O^*, OH^*). The presence of these highly reactive species at low pressures guarantees the growth of ultra-high-quality oxides at impressive rates, even for thicker layers.
- **Plasma Oxidation:** Techniques such as Slot Plane Antenna (SPA) oxidation utilise a plasma to drive (supply energy to) the reaction, allowing for an exceptionally low thermal budget, sustained by operating temperatures between 300 °C and 500 °C. This is ideal for growing ultra-thin, high-quality oxides without disturbing previously established dopant profiles.

2.6.3 The Scaling Limit and High-κ Dielectrics

For decades, the semiconductor industry followed Moore's Law by simply growing thinner and thinner layers of SiO_2 to increase the gate capacitance and drive current. However, this scaling eventually hit a fundamental wall. The practical minimum thickness for an SiO_2 gate dielectric to avoid unacceptable gate leakage is approximately 0.8 nm.

If the physical thickness of the oxide (x_{ox}) can no longer be scaled down, the only way to continue increasing the gate capacitance is to replace silicon dioxide with a material that possesses a higher dielectric constant (κ).

Physical Insight

Why is 0.8 nm the absolute limit for SiO_2? At this thickness, the oxide is only about three atomic layers thick. According to classical physics, this dielectric should still act as a perfect insulator. However, as is known, at the nanoscale the wave-like nature of electrons dominates.
Through the phenomenon of quantum tunnelling, electrons possess a non-zero probability of passing straight through the physical energy barrier of the ultra-thin oxide. When the SiO_2 thickness drops below 1 nm, this direct tunnelling current becomes so massive that the transistor consumes an unacceptable amount of power even when switched "off".
By switching to a High-κ dielectric (such as Hafnium Oxide, HfO_2), engineers can achieve the same electrical capacitance as an ultra-thin SiO_2 layer while using a physically much thicker film. This metric is known as Equivalent Oxide Thickness (EOT). The physically thicker High-κ layer effectively suppresses the quantum tunnelling probability, restoring the insulating property of the gate while maintaining scaled transistor performance.

To be a viable alternative to SiO_2, a High-κ material must satisfy several stringent physical and thermodynamic requisites:

1. A significantly higher dielectric constant (κ).
2. A band offset with respect to silicon of greater than 1 eV, and an overall bandgap of greater than 4 to 5 eV to adequately block both electron and hole emission.
3. Strict thermodynamic stability when in direct physical contact with silicon at high processing temperatures.
4. Full compatibility with standard IC process flows and etching chemistries.

Today, there are several metrology tools to determine the thickness of an oxide layer. Before their advent, engineers could quickly estimate the thickness of a grown SiO_2 layer simply by looking at it. Due to thin-film optical interference, different oxide thicknesses reflect specific wavelengths of light, producing a predictable and highly vivid colour chart, which can be seen in Figure 2.6.

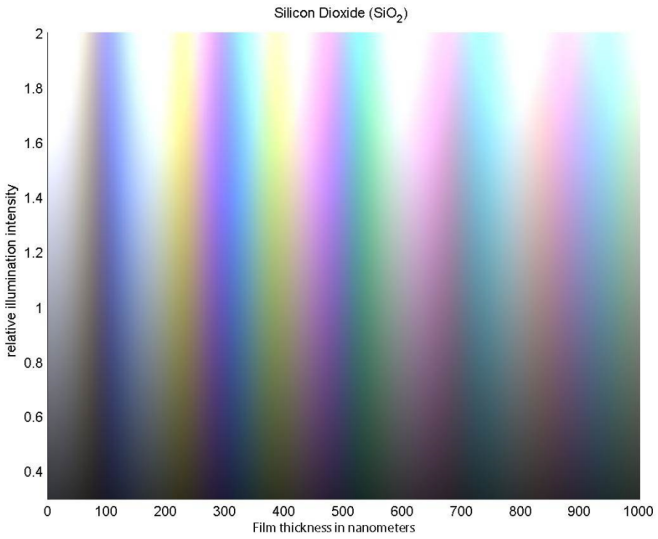


Figure 2.6: This chart shows what the light reflected from an oxide layer would look like after thin-film interference.

With the mechanisms of thermal oxidation established, we now understand how to grow the critical insulating layers that isolate devices and control transistor gates. However, as we saw with phenomena like dopant segregation and oxidation-enhanced diffusion (OED), high-temperature processing does not leave the underlying silicon untouched. The substrate is a highly dynamic environment where atoms are constantly interacting and relocating.

While oxides provide the necessary electrical isolation, a functional integrated circuit ultimately relies on precisely controlling the electrical conductivity of the silicon itself. To build the source, drain, and well regions of our transistors, we must introduce and position specific impurity atoms with nanometre precision.

In the next chapter, we will delve into **Dopant Diffusion**. We will shift our focus from the growth of surface dielectrics to the movement of impurity atoms deep within the crystal lattice, exploring the macroscopic laws and microscopic mechanisms that govern how electrical junctions are formed.

Chapter 3

Dopant Diffusion

3.1 Introduction to Doping and Scaling

While the controlled growth of silicon dioxide allows us to build insulating barriers and gate dielectrics, the actual computational power of an integrated circuit relies on the semiconductor itself. To build functional devices, such as the source, drain, and channel regions of a MOS transistor, we must introduce specific impurity atoms into the silicon lattice to precisely modulate its electrical conductivity. This process is governed by the physics of **dopant diffusion**.

3.1.1 Technology Scaling and Junction Depth

The specific shape and depth of a doping profile determine many of the critical short-channel characteristics in modern MOS devices. For instance, the electrical resistance of the source and drain regions heavily impacts the maximum drive current the transistor can deliver.

As the semiconductor industry continuously follows the International Technology Roadmap for Semiconductors (ITRS) to scale down devices, a fundamental geometric rule applies: all lateral and vertical dimensions must scale down by the same factor. Consequently, to maintain high drive currents while suppressing short-channel effects, local doping concentration levels need to progressively increase, while the vertical junction depth (x_J) must drastically decrease.

To quantify the electrical properties of these shallow junctions, engineers rely on the macroscopic concepts of resistivity (ρ) and sheet resistance (R_s). For a non-uniformly doped layer, calculating the exact sheet resistance requires integrating the dopant concentration over the depth of the junction. Historically, this was numerically integrated by Irvin for simple analytical profiles, yielding standard reference charts known as Irvin curves. Experimentally, this sheet resistance is measured using a four-point probe technique.

3.1.2 The Two-Step Doping Process

Fundamentally, diffusion is the spontaneous redistribution of atoms from regions of high concentration to regions of low concentration. While this physical process theoretically occurs at all temperatures, the diffusivity (the speed at which atoms migrate through the lattice) has an exponential dependence on temperature.

In traditional semiconductor manufacturing, establishing a precise doping profile is usually split into two distinct high-temperature steps:

1. **Pre-deposition:** An initial step designed to introduce a specific, controlled dose of dopant atoms into the very top surface of the silicon substrate. This area is highlighted in Figure 3.1.

2. **Drive-In:** A subsequent thermal annealing step (often performed without any additional dopant source) that redistributes the already-introduced dopants deeper into the bulk,

achieving the required final junction depth (x_J) and surface concentration.

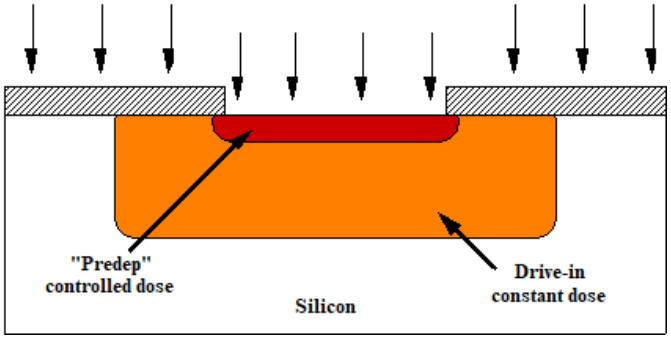


Figure 3.1: The drawing portrays a comparison of the doped areas in the two steps of the doping process.

Process Insight

The physical brilliance of dividing the doping process into a "Pre-deposition" and a "Drive-In" step lies in process control. If engineers attempted to dope the wafer in a single, prolonged high-temperature step, the total number of dopant atoms (the dose) and the depth of the junction (x_J) would be inextricably linked.
By splitting the process, the Pre-deposition step is used exclusively to set the total inventory of dopant atoms in the silicon. Once that dose is locked in, the dopant gas supply is shut off. The subsequent Drive-In step simply acts as a thermal engine to push that fixed inventory of atoms deeper into the crystal. This two-step methodology successfully decouples the total electrical charge of the junction from its physical depth, granting designers the independent control required to build complex microelectronic structures. The table in Figure 3.2 shows the advantages of this technique compared to a direct solid/gas diffusion process.

	Ion Implantation and Annealing	Solid/Gas Phase Diffusion
Advantages	Room temperature mask	No damage created by doping
	Precise dose control	Batch fabrication
	10^{11} - 10^{16} atoms cm^{-2} doses	
	Accurate depth control	
Problems	Implant damage enhances diffusion	Usually limited to solid solubility
	Dislocations caused by damage may cause junction leakage	Low surface concentration hard to achieve without a long drive-in
	Implant channeling may affect profile	Low dose predeps very difficult

Figure 3.2: This table highlights advantages and problems of the two doping techniques.

3.1.3 Solid vs. Electrical Solubility

When attempting to achieve the extraordinarily high doping levels required by modern scaling laws, we encounter fundamental thermodynamic limits. Dopants are only soluble in bulk silicon

up to a strict maximum value; if this limit is exceeded, the excess dopant atoms will precipitate out of the silicon lattice and form a completely separate physical phase. This maximum threshold is known as the **solid solubility**.

Furthermore, it is critical to distinguish between physical solid solubility and *electrical* solubility. An atom only contributes a free charge carrier if it successfully occupies a substitutional site within the silicon lattice. At very high concentrations, dopants may satisfy the solid solubility limit but fail to act as active donors or acceptors. A classic example is heavy arsenic doping: at high concentrations, arsenic tends to form an electrically inactive complex known as As_4V (four arsenic atoms clustered around a single silicon vacancy). Because these clustered atoms do not contribute to the conductivity of the material, the effective electrical solubility caps out much earlier than the physical solid solubility.

3.2 Macroscopic Diffusion: Fick's Laws and Analytical Solutions

To model the redistribution of dopants within the silicon lattice, we treat the semiconductor as a continuous medium. On a macroscopic scale, the movement of impurity atoms from regions of high concentration to regions of low concentration is governed by Fick's laws of diffusion.

3.2.1 Fick's Laws and the Diffusion Equation

The fundamental driving force for macroscopic diffusion is the concentration gradient. Fick's First Law states that the dopant flux (F) is directly proportional to the negative gradient of the dopant concentration (C):

$$\vec{F} = -D \cdot \vec{\nabla} C$$

In one dimension (depth x into the wafer), this simplifies to:

$$F = -D\frac{\partial C}{\partial x}$$

where D is the diffusivity, or diffusion coefficient, of the dopant species.

By applying the principle of mass conservation to a differential volume element, we get Fick's Second Law. According to this principle, the change in concentration over time must equal the difference between the flux entering and the flux leaving the volume, $F_{in} - F_{out}$.

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right)$$

If we assume that the diffusivity D is constant (which is valid under intrinsic doping conditions at a constant temperature), the equation simplifies to the classic linear diffusion equation:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

Steady-State Solution

In a purely steady-state scenario, the concentration profile no longer changes with time ($\frac{\partial C}{\partial t} = 0$). The diffusion equation reduces to:

$$D\frac{\partial^2 C}{\partial x^2} = 0$$

Integrating this twice yields a simple linear solution:

$$C(x) = A + Bx$$

where A and B are constants determined by the boundary conditions. However, in actual manufacturing, diffusion is an inherently transient process, requiring time-dependent solutions.

3.2.2 The Pre-deposition Step: The Error Function Solution

During the pre-deposition step, the silicon wafer is exposed to what can be modelled as a continuous, infinite supply of dopant gas. This clamps the surface concentration of the silicon at its maximum possible solid solubility limit C_s.

The mathematical boundary conditions for this "infinite source" problem are:

$$\begin{aligned} C(x,0) &= 0 \quad \text{for } x > 0 \text{ (initially un-doped substrate)} \\ C(0,t) &= C_s \quad \text{for } t > 0 \text{ (constant surface concentration)} \\ C(\infty,t) &= 0 \quad \text{for } t > 0 \text{ (deep bulk remains un-doped)} \end{aligned}$$

Solving Fick's second law under these conditions yields the complementary error function profile (which can be seen in Figure 3.3)

$$C(x,t) = C_s \text{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

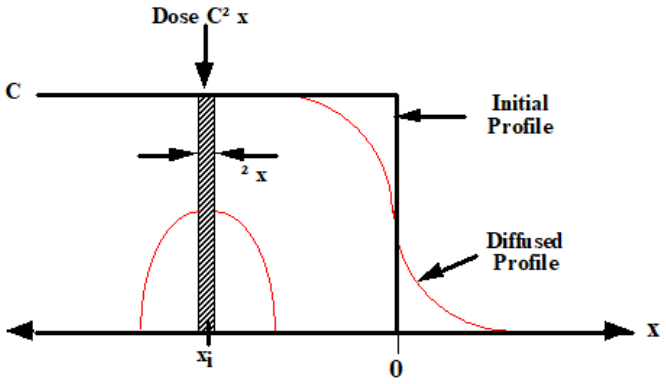


Figure 3.3: The complementary error function (also called erfc) profile. It is quite intuitive to see the step "melt" into this profile.

A critical consequence of the error function solution is that the total dose of dopants (Q, the integral of the concentration over depth) continuously increases with annealing time. The total dose injected into the wafer after a pre-deposition time t is:

$$Q(t) = \int_0^\infty C(x,t)\,dx = \frac{2C_s\sqrt{Dt}}{\sqrt{\pi}}$$

Physically, the erfc solution can be thought of as a continuous "sum" of infinitely many point-source Gaussian diffusion problems occurring over time[12].

3.2.3 The Drive-In Step: The Gaussian Solution

Following pre-deposition, the external dopant source is removed, and the wafer undergoes the drive-in anneal. The total dose Q of dopant atoms is now fixed.

If we approximate this initial pre-deposited dose as an infinitesimally thin sheet of dopants perfectly situated at the surface, we can model it mathematically as a Dirac delta function. The boundary conditions for this "constant dose" problem are:

$$\begin{aligned} \int_0^\infty C(x,t)\,dx = Q &= \text{constant} \\ C(x,0) &= Q\delta(x) \\ C(\infty,t) &= 0 \end{aligned}$$

[12]Indeed, the erfc function is defined as $\text{erfc}(z) = 1 - \text{erf}(z)$, while erf is defined as $\text{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,dt$.

Solving the diffusion equation for a fixed dose spreading out from the origin yields the Gaussian profile:

$$C(x,t) = \frac{Q}{\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

Unlike the pre-deposition step, the surface concentration here is not clamped. As the dopants spread deeper into the bulk, the surface concentration drops:

$$C_s(t) = C(0,t) = \frac{Q}{\sqrt{\pi Dt}}$$

Physical Insight

When deriving the Gaussian solution for a drive-in step, a subtle mathematical challenge arises. A standard delta-function solution assumes that the dopant spreads out symmetrically in both the $+x$ and $-x$ directions. However, the surface of the wafer ($x = 0$) is a hard boundary. To solve this, we can treat the surface at $x = 0$ as a perfectly reflecting wall. The formal solution assumes a symmetric Gaussian distribution extending from $-\infty$ to $+\infty$, but because the flux must be zero at the surface, the "negative" half of the Gaussian is effectively folded over or "mirrored" back into the positive bulk. This is why the concentration factor is $\frac{Q}{\sqrt{\pi Dt}}$ instead of the $\frac{Q}{2\sqrt{\pi Dt}}$ typically found in an unbounded 1D random walk.

3.2.4 Temperature Dependence of Diffusivity

The diffusion coefficient D is not a universal constant; it is strongly dependent on the temperature and the specific dopant atom. Under intrinsic doping conditions, the diffusivity follows a strict Arrhenius law:

$$D = D_0 \exp\left(-\frac{E_a}{k_B T}\right)$$

where D_0 is the pre-exponential factor ($\mathrm{cm}^2 \cdot \mathrm{s}^{-1}$), E_a is the activation energy (eV), k_B is the Boltzmann constant, and T is the absolute temperature in kelvin. Because of this exponential dependence, a small change in temperature produces a massive change in the diffusion rate.

3.2.5 Thermal Budget and Multiple Diffusion Steps

Real semiconductor manufacturing does not consist of a single thermal step. Wafers undergo numerous oxidations, anneals, and depositions, each causing the dopants to diffuse further.

If a dopant is diffused at temperature T_1 for time t_1, and then subsequently diffused at temperature T_2 for time t_2, the effective diffusion length can be computed by summing the individual Dt products. The total effective Dt is:

$$(Dt)_{\text{eff}} = \sum_i D_i t_i$$

Because the diffusivity D depends exponentially on temperature, lower-temperature steps contribute virtually nothing to the final profile. Only the highest temperature diffusion steps dictate the final junction depth and profile. This effectively means the final Gaussian profile can be written as:

$$C(x,t) = \frac{Q}{\sqrt{\pi (Dt)_{\text{eff}}}} \exp\left(-\frac{x^2}{4(Dt)_{\text{eff}}}\right)$$

3.3 Corrections to Fick's Law: Extrinsic Effects

The analytical solutions derived in the previous section rely on a critical mathematical assumption: the diffusivity D must be constant. This assumption holds true only under *intrinsic* conditions, where the local concentration of dopant atoms is less than the intrinsic carrier concentration (n_i) of silicon at the annealing temperature.

However, modern scaling requires incredibly high, heavily *extrinsic* doping levels. At these high concentrations, the simple laws of diffusion break down, and macroscopic profiles severely deviate from perfect Gaussian or erfc shapes. To accurately model real-world junctions, Fick's laws must be corrected to account for internal electric fields and concentration-dependent diffusivities.

3.3.1 Electric Field Enhancement

When a region of silicon is heavily doped, the dopant atoms ionise to create a massive local gradient of free charge carriers (electrons for n-type, holes for p-type). Because electrons and holes are incredibly light and possess a much higher mobility than the heavy, massive dopant ions, they tend to diffuse forward much faster than the dopants themselves.

As the fast-moving electrons or holes run ahead, they leave behind the slowly diffusing, oppositely charged donor or acceptor ions. This macroscopic charge separation sets up a strong internal electric field. Under steady-state conditions, the drift flow of electrons/holes driven by this electric field will exactly balance their outward diffusion flow.

Crucially, this internal electric field exerts a Coulombic force on the charged dopant ions. It effectively "drags" the donor or acceptor atoms forward, significantly enhancing their overall diffusivity. To account for this drift component, Fick's First Law can be rewritten as:

$$F = -hD\frac{\partial C}{\partial x}$$

where h is equal to $h = 1 + \frac{C}{\sqrt{C^2+4n_i^2}}$ and transforms the equation into a drift-diffusion one.

3.3.2 Concentration-Dependent Diffusion

Because of the electric field enhancement and the complex interactions between dopants and charged point defects, the overall diffusivity D is no longer a constant; it becomes a direct function of the local dopant concentration, $D(C)$.

Under these extrinsic conditions, Fick's Second Law still holds, but it must be written in its more general, non-linear form:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D(C)\frac{\partial C}{\partial x}\right)$$

Experimentally, it is found that the total effective diffusivity is the sum of parallel diffusion mechanisms mediated by point defects in various charge states (neutral, single negative, double negative and so on for n type and positives for p type):

$$D_{\text{eff}} = D^0 + D^-\left(\frac{n}{n_i}\right) + D^{--}\left(\frac{n}{n_i}\right)^2 + \dots$$

Each of these individual D components follows its own Arrhenius relationship ($D = D_0 e^{-\frac{E_a}{k_B T}}$), with its own specific pre-exponential factor and activation energy. Because the concentration of charged point defects depends on the Fermi level (which in turn depends on the dopant concentration C), the overall D_{eff} scales with the doping level.

Physical Insight

The fact that D depends on C completely alters the physical shape of the junction. In high-concentration regions, the diffusivity is significantly enhanced, meaning diffusion is much faster where the concentration is highest.
Physically, this means dopants in the heavily doped plateau of the junction race forward quickly, but as soon as they reach the lightly doped tail of the profile, their diffusivity plummets, and they violently slow down. This causes the dopants to "pile up" behind the advancing front. Instead of the smooth, sweeping tail of a Gaussian or erfc curve, the resulting junction assumes a distinct "box-shaped" profile: a flat, highly doped plateau followed by a precipitous, almost vertical drop-off. This abruptness is incredibly beneficial for modern transistors, as it provides highly defined, low-resistance source/drain extensions.

3.3.3 Recovering the correction to Fick's First Law

In this subchapter, we will go over the mathematical derivation of the corrected Fick's First Law. It will be, of course, very math-heavy and although interesting, possibly not that useful to many. This part can be safely skipped if one wishes to do so.

Nevertheless, as mentioned in subchapter 3.3.1, we must acknowledge that the total flux is now given by two components: a diffusion one and a drift one.

$$F_{tot} = F_{diff} + F_{drift} = -D\frac{\partial C}{\partial x} + Cv$$

where v is the drift speed induced by the electric field and it is equal to:

$$v = \mu E = \mu\left(-\frac{\partial \varphi}{\partial x}\right) = \mu\left[-\frac{\partial}{\partial x}\left(\frac{k_B T}{q} ln\frac{n}{n_i}\right)\right]$$

where φ is the potential and q the fundamental unit of charge.

We can use Einstein's relation to write the mobility as a function of diffusivity ($\mu = \frac{qD}{k_B T}$). With this, the total flux becomes:

$$F_{tot} = -D\frac{\partial C}{\partial x} - DC\frac{\partial}{\partial x} ln\frac{n}{n_i}$$

It turns out that it is possible to write the partial derivative of the logarithm in terms of the concentration. Using the neutrality condition[13] ($N_D + p = N_A + n$) and the mass action law ($np = n_i^2$):

$$\frac{\partial C}{\partial x} = \frac{\partial}{\partial x}(N_D - N_A) = \frac{\partial n}{\partial x} - \frac{\partial p}{\partial x} = \frac{\partial n}{\partial x} - n_i^2\frac{\partial}{\partial x}\left(\frac{1}{n}\right) = \frac{\partial n}{\partial x} + \frac{n_i^2}{n^2}\cdot\frac{\partial n}{\partial x} = \frac{1}{n}\cdot\frac{\partial n}{\partial x}\left(n + \frac{n_i^2}{n}\right) = \frac{\partial}{\partial x} ln(n)\cdot(n+p)$$

Since adding constants to a function does not change its derivative, we can invert the previous relation and write:

$$\frac{\partial}{\partial x} ln(n) = \frac{\partial}{\partial x} ln\left(\frac{n}{n_i}\right) = \frac{1}{n+p}\cdot\frac{\partial C}{\partial x}$$

With this, the total flux equation becomes:

$$F_{tot} = -D\frac{\partial C}{\partial x} - DC\frac{\partial}{\partial x} ln\frac{n}{n_i} = -D\frac{\partial C}{\partial x} - DC\frac{1}{n+p}\cdot\frac{\partial C}{\partial x}$$

[13] We'll assume n-doping so the concentration will be written as $N_D - N_A$. The steps are exactly the same with p-doping, just with n and p swapped.

Using again the neutrality condition and mass action law, we can also write $n + p$ in terms of C.

$$C^2 = (N_D - N_A)^2 = (n - p)^2 = n^2 - 2np + p^2 = n^2 - 2n_i^2 + p^2 = (n + p)^2 - 4n_i^2$$

which yields:

$$n + p = \sqrt{C^2 + 4n_i^2}$$

Substituting this into our total flux equation, we finally get the corrected Fick's First Law:

$$F_{tot} = -D\frac{\partial C}{\partial x} - DC\frac{1}{n+p} \cdot \frac{\partial C}{\partial x} = -D\frac{\partial C}{\partial x} - D\frac{C}{\sqrt{C^2 + 4n_i^2}}\frac{\partial C}{\partial x} = -hD\frac{\partial C}{\partial x}$$

3.3.4 Segregation and Interfacial Pile-Up

Diffusion does not occur in a vacuum; it often occurs simultaneously with other high-temperature processes, most notably thermal oxidation. As discussed in the previous chapter, different materials (such as Si and SiO_2) possess different solid solubilities for a given dopant.

This discrepancy leads to dopant segregation at the boundary. As the oxide grows or the wafer is annealed, dopant atoms redistribute at the Si/SiO_2 interface:

- **N-type dopants (e.g., phosphorus, arsenic, antimony):** Tend to be rejected by the oxide and pile up at the interface. This results in dopants accumulating in a very thin, highly concentrated, but often electrically inactive interfacial layer between the Si and SiO_2.
- **P-type dopants (e.g., boron):** Tend to be readily absorbed by the oxide, causing the silicon surface to severely deplete of boron.

These segregation phenomena severely distort the topmost region of the diffusion profile and must be carefully modelled when designing the threshold voltage implants of a transistor.

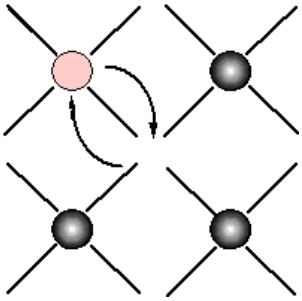
3.4 Atomic-Scale Microscopic Diffusion

Fick's laws and their drift-diffusion corrections successfully describe the macroscopic redistribution of dopants, but they are phenomenological. They do not explain *how* a solid atom actually moves through a dense, rigid crystal lattice. At the atomic scale, the diffusion of dopants is strictly related to the presence of point defects.

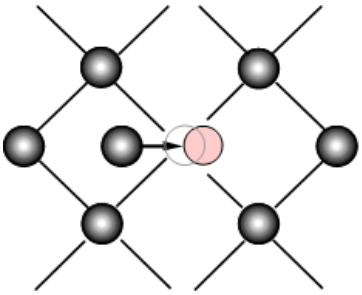
3.4.1 Vacancy and Interstitial Mechanisms

There are two primary microscopic point-defect mechanisms that facilitate dopant diffusion, shown in Figure 3.4.

- **Vacancy-Assisted Diffusion:** A substitutional dopant atom can move simply by hopping into an adjacent, empty lattice site (a vacancy) as Figure 3.4a shows.
- **Interstitial-Assisted Diffusion:** An interstitial silicon atom can collide with a substitutional dopant atom, "kicking" it out of its lattice position and forcing it into a mobile interstitial state as Figure 3.4b shows. Alternatively, an interstitial defect and a dopant atom can bind together to form an interstitial/dopant pair, which can then travel rapidly and efficiently along the bond directions of the crystal.



(a) Vacancy-Assisted Diffusion



(b) Interstitial-Assisted Diffusion

Figure 3.4: a) The picture shows how a dopant atom moves through a vacancy in the lattice. b) In this case, a dopant atom is pushed out of its lattice site by an interstitial silicon atom, thus allowing for its further diffusion.

3.4.2 Fractional Diffusivity

The overall diffusivity of a specific dopant is a weighted combination of these mechanisms. The general diffusivity can be mathematically formulated to account for the fraction of diffusion mediated by interstitial-type mechanisms (f_i) and the fraction mediated by vacancy-type mechanisms (f_v). If D^* is the thermal equilibrium diffusivity in strictly inert conditions, the effective actual diffusivity is given by scaling it with the local point-defect concentrations:

$$D^{eff} = D^* \left(f_i \frac{C_I}{C_I^*} + f_v \frac{C_V}{C_V^*} \right)$$

where C_I and C_V are the actual transient interstitial and vacancy concentrations, and C_I^* and C_V^* are their respective thermal equilibrium concentrations.

Different elements strongly prefer different diffusion mechanisms based on their atomic radius and bonding characteristics:

- **Boron and phosphorus** diffuse almost exclusively via interstitial mechanisms ($f_i = 1.0, f_v = 0$).
- **Antimony** diffuses almost entirely via the vacancy mechanism ($f_i = 0.02, f_v = 0.98$).
- **Arsenic** and **silicon self-diffusion** utilise a mixed mechanism (e.g., for As, $f_i = 0.4, f_v = 0.6$; for Si, $f_i = 0.6, f_v = 0.4$).

3.4.3 Modelling Interstitial-Assisted Diffusion

Because diffusion is fundamentally coupled to point defects, any fabrication step that alters the defect concentration will drastically alter the diffusion rate. As noted previously, the process of thermal oxidation actively injects silicon interstitials into the bulk substrate and absorbs vacancies. Consequently, oxidation strongly enhances the diffusion of interstitial-driven dopants (like boron and phosphorus) while it explicitly retards the diffusion of vacancy-driven dopants (like antimony).

To formally model interstitial-assisted diffusion, we need not look at just how the dopants move in the lattice, but how dopant-defect couples move in the lattice. To this purpose, let's consider an interaction between a substitutional dopant A (like boron) and silicon interstitial I of the type:

$$A + I \rightleftharpoons AI$$

In this framework, the AI pair is considered the highly mobile species. For this, we can write a diffusion equation similar to Fick's law:

$$f_{AI} = -d_{AI}\frac{\partial C_{AI}}{\partial x}$$

We are essentially shifting our view from a "macroscopic" perspective, in which we follow dopant diffusion, to a microscopic one, in which we follow the actual dopant-interstitial pairs. However, these quantities are not known. To go from this equation to one which we can plug into a simulator, we can make some assumptions. Let's suppose that the interaction we wrote between a dopant A and an interstitial I behaves as a chemical reaction at equilibrium[14]. Given this, we can write the concentration of the pair as:

$$C_{AI} = kC_AC_I$$

Substituting this into the diffusion equation:

$$f_{AI} = -d_{AI}(kC_I\frac{\partial C_A}{\partial x} + kC_A\frac{\partial C_I}{\partial x}) = -d_{AI}(\frac{C_{AI}}{C_A}\cdot\frac{\partial C_A}{\partial x} + \frac{C_{AI}}{C_I}\cdot\frac{\partial C_I}{\partial x})$$

This flux is then what causes a variation in concentration of the dopant species. Therefore, using this flux in Fick's second law we can write:

$$\frac{\partial C_A}{\partial t} = \frac{\partial}{\partial x}d_{AI}C_{AI}(\frac{1}{C_A}\frac{\partial C_A}{\partial x} + \frac{1}{C_I}\frac{\partial C_I}{\partial x})$$

Assuming the concentration of interstitials is spacially uniform, we can safely neglect the last term. With this, the equation is actually very similar to the macroscopic version of Fick's second law which we found:

$$\frac{\partial C_A}{\partial t} = \frac{\partial}{\partial x}D_A^{eff}\frac{\partial C_A}{\partial x}$$

where we wrote D_A^{eff} to mean the adjusted effective diffusivity of the dopant based on the presence of defects. Thus, equating them, we get:

$$\frac{\partial}{\partial x}D_A^{eff}\frac{\partial C_A}{\partial x} = \frac{\partial}{\partial x}d_{AI}C_{AI}\cdot\frac{1}{C_A}\frac{\partial C_A}{\partial x}$$

Removing derivatives and the equal terms on both sides of the equation, we find:

$$D_A^{eff} = d_{AI}C_{AI}\cdot\frac{1}{C_A}$$

Recalling the equation we wrote for D^{eff} in the previous subchapter, and the fact that we are considering a model with only interstitials, we find:

$$D^*\frac{C_I}{C_I^*} = d_{AI}C_{AI}\cdot\frac{1}{C_A} \implies d_{AI}C_{AI} = D^*\frac{C_I}{C_I^*}\cdot C_A$$

From this, we can substitute $d_{AI}C_{AI}$ in our flux equation, which yields:

$$f_{AI} = -D^*\frac{C_I}{C_I^*}\cdot C_A(\frac{1}{C_A}\frac{\partial C_A}{\partial x} + \frac{1}{C_I}\frac{\partial C_I}{\partial x})$$

We can write the derivative in a more compact form and use the properties of logarithms to transform it into:

$$f_{AI} = -D^*\frac{C_I}{C_I^*}\cdot C_A[\frac{\partial}{\partial x}ln(C_A\frac{C_I}{C_I^*})]$$

[14]This is a strong hypothesis and not entirely true, especially at low temperatures. Despite this, it works well enough at high temperatures. More refined models consider different speeds for the two directions of the "chemical reaction".

We've finally written the microscopic flux as a function of terms we can measure such as effective equilibrium diffusivity (D^*), supersaturation of point defects (C_I^*) and concentration of dopant species (C_A). Taking the negative derivative with respect to x of this (using Fick's second law) we get the continuity equation which is actually plugged into simulators[15].

Physical Insight

The mobile AI pairing mechanism described above leads to a fascinating phenomenon known as the **chemical pumping** of interstitial defects deep into the substrate. Consider a highly concentrated phosphorus diffusion at the surface of a wafer. The phosphorus atoms pair up with native interstitials to form highly mobile PI pairs. These pairs rapidly diffuse out of the highly concentrated surface region and travel deep into the bulk. When the pair eventually dissociates deeper in the crystal, it deposits the phosphorus atom and permanently releases the interstitial.
This continuous one-way transport creates a massive supersaturation of interstitials deep in the bulk, which perfectly explains two famous anomalous diffusion phenomena. First, it accounts for **enhanced tail diffusion**, where the deep "tail" of a Phosphorus profile diffuses much faster than Fick's laws predict because the local bulk area is constantly flooded by the excess interstitials, which are pumped in from the heavily doped surface. Second, it explains the **emitter push** effect. If a rapid phosphorus diffusion (such as one used to form the emitter of a bipolar transistor) is performed directly above a previously established boron layer (the base), the interstitials pumped downward by the phosphorus will flood the boron layer. Because boron is strictly an interstitial diffuser ($f_i = 1.0$), this flooding artificially and violently accelerates the diffusion of the boron, physically "pushing" the boron junction much deeper into the substrate than intended.

3.5 Metrology of Doped Profiles

Once the pre-deposition and drive-in anneals are complete, it is vital to measure the resulting doping profile to verify that the junction depth (x_J) and dopant concentrations meet the required specifications. Metrology techniques for dopant profiles are generally divided into two categories: those that measure the total *chemical* concentration of the impurity atoms, and those that measure the *electrically active* carrier concentration.

3.5.1 Secondary Ion Mass Spectrometry (SIMS)

To determine the total chemical concentration of dopants as a function of depth, Secondary Ion Mass Spectrometry (SIMS) is the industry standard.

In a SIMS measurement, the surface of the silicon sample is bombarded with a highly focused beam of primary ions (typically oxygen or caesium). The immense kinetic energy of these impinging ions physically sputters atoms away from the sample surface. A fraction of these ejected atoms become ionised. These "secondary ions" are then collected, accelerated, and passed through a mass spectrometer (typically using a magnetic sector or quadrupole mass filter) to be mass-analysed. By continuously sputtering the surface and analysing the ejected mass, SIMS provides a highly accurate, depth-resolved profile of the exact atomic concentration of the dopant.

[15] This whole derivation can be corrected by taking into account the electric field component in the first flux equation, making it $f_{AI} = -d_{AI}[\frac{\partial C_{AI}}{\partial x} + \frac{q}{k_B T} C_{AI} \frac{\partial \varphi}{\partial x}]$. This would recover a final flux of $f_{AI} = -D^* \frac{C_I}{C_I^*} \cdot C_A[\frac{\partial}{\partial x} ln(C_A \frac{C_I}{C_I^*} \frac{n}{n_i})]$.

Process Insight

A common and crucial observation in semiconductor metrology is that a SIMS profile often does not perfectly match an electrical resistivity profile, particularly at very high doping concentrations.
SIMS is a brute-force mass analysis technique; it counts every single dopant atom it sputters away, regardless of where that atom was sitting in the silicon lattice. However, as established in the discussion on solid versus electrical solubility, not all dopant atoms are electrically active. Atoms that are trapped in interstitial sites, clustered into neutral complexes (such as As_4V), or precipitated out of the lattice do not contribute free electrons or holes. Therefore, the total chemical concentration measured by SIMS will significantly exceed the electrically active carrier concentration measured by electrical profiling in heavily doped regions.

3.5.2 Electrical Profiling: SRP and SSRM

To measure the electrically active portion of the doping profile, spreading resistance techniques are employed.

Spreading Resistance Profiling (SRP) relies on mechanically magnifying the depth of the junction. The silicon sample is polished at a very shallow, precisely known bevel angle (often less than one degree). Two perfectly aligned probes are then systematically stepped down the bevelled surface. At each step, a small voltage is applied, and the spreading resistance is measured. Because the geometry of the bevel stretches a junction depth of a few hundred nanometres over several micrometres of lateral distance, SRP can extract a high-resolution 1D depth profile of the active carrier concentration. This process is represented in Figure 3.5.

Scanning Spreading Resistance Microscopy (SSRM) takes this concept to the two-dimensional nanoscale. Operating similarly to an Atomic Force Microscope (AFM), SSRM drags a highly conductive, diamond-coated tip across a cleaved cross-section of the transistor. This provides an extraordinary 2D map of the local resistance, directly reflecting the complex spatial distribution of the active dopants.

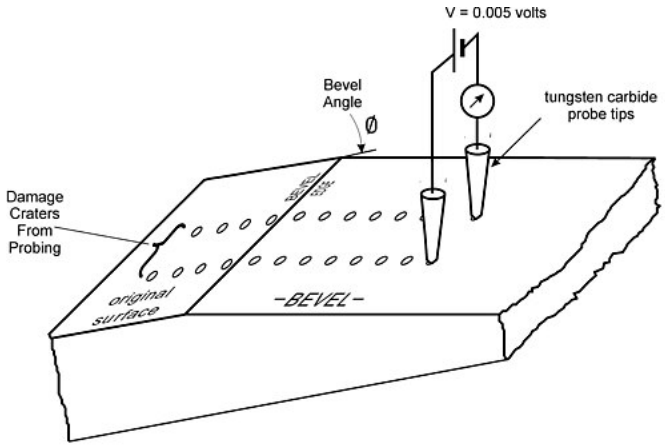


Figure 3.5: This is a schematic representation of what SRP looks like. Typically 60 to 100 or more measurements are made. Image taken from Wikipedia.

3.5.3 Electron Holography

As transistor dimensions shrink, locating the exact boundaries of the active pn-junction becomes increasingly difficult. While conventional Transmission Electron Microscopy (TEM) provides unparalleled structural imaging (revealing metallic contacts and crystalline boundaries through amplitude contrast), it cannot visually distinguish between n-doped and p-doped silicon.

To map the active junction directly, **Electron Holography** is used. In this advanced TEM technique, the coherent electron beam is split into two paths, one passing through the sample and one passing through a vacuum reference, using a Möllenstedt biprism. When the two beams are recombined, they create an interference pattern (a hologram). Figure 3.6 shows the structure of a TEM for holography.

Crucially, as the sample beam passes through the transistor, the phase of the electron wave is shifted by the local electrostatic potential of the built-in pn-junction. By computationally reconstructing the phase from the hologram, engineers can directly visualise the projected 2D potential landscape of the transistor, clearly delineating the n-doped source/drain extensions from the p-doped bulk.

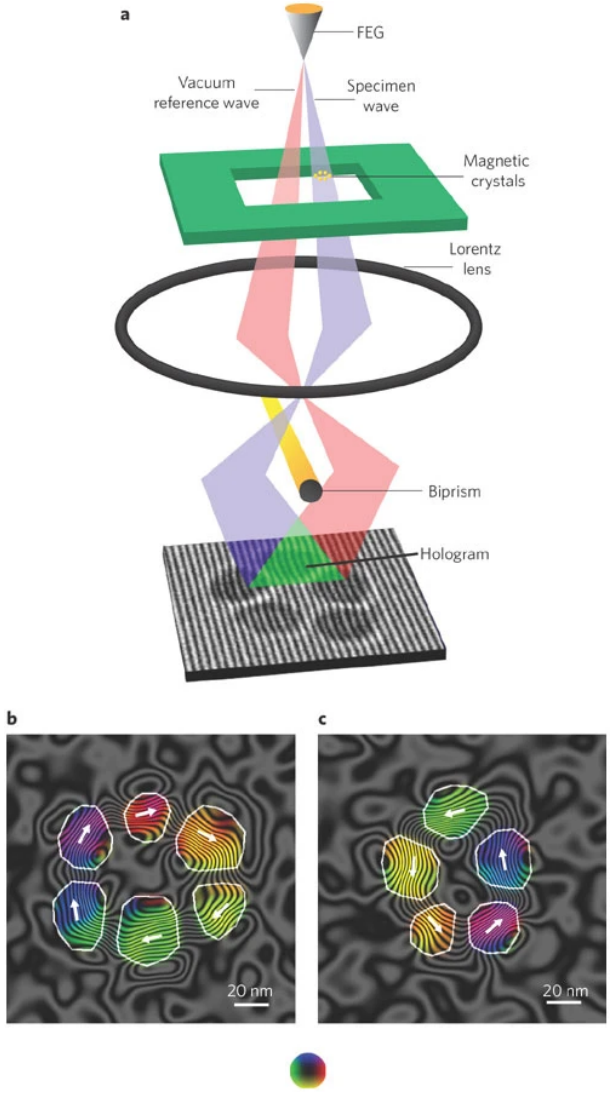


Figure 3.6: This is a TEM (a) used to create a hologram of magnetic nanocrystal rings of cobalt (b and c). In this experiment, variations in the spacing and direction of the recorded holographic interference fringes even contain information about the projected magnetic flux density. Image taken from Nature materials.

In this chapter, we have transitioned from the macroscopic continuum equations of Fick's laws to the intricate, atomic-scale mechanisms of point-defect-assisted diffusion. We have seen how high-concentration extrinsic effects, internal electric fields, and non-equilibrium point defect injection fundamentally alter the ideal Gaussian and error-function profiles. Armed with advanced metrology techniques like SIMS and SSRM, process engineers can verify and fine-tune these complex dopant distributions.

In the next chapter, we will explore **Ion Implantation**. We will shift from the gentle, thermodynamic spreading of atoms to a violent, ballistic process. We will examine how accelerating charged dopant ions directly into the crystal structure grants us unparalleled control over the doping profile, and how we must subsequently deal with the massive crystallographic damage this process leaves behind.

Chapter 4

Ion Implantation

While thermal diffusion and oxidation rely on the gentle, thermodynamically driven movement of atoms at high temperatures, modern semiconductor manufacturing demands a much more precise and forceful approach to doping. As devices scale down to the nanometre regime, relying on the solid solubility limits and isotropic spreading of thermal diffusion becomes untenable. To achieve ultra-shallow junctions and exact dopant concentrations, the industry relies on **ion implantation**.

Ion implantation is a ballistic, non-equilibrium process. Dopant atoms are ionised, accelerated to high kinetic energies, and physically smashed into the silicon lattice. This technique has become the undisputed mainstream doping method in the integrated circuit industry due to several overwhelming advantages: it provides unparalleled control over the total dopant dose (ranging seamlessly from 10^{11} to 10^{16} cm^{-2}), allows for highly accurate depth control by tuning the acceleration energy, and permits the use of simple, room-temperature masking materials such as photoresist.

4.1 The Physics and Architecture of Ion Implantation

An ion implanter is essentially a highly specialised linear particle accelerator, a scheme of which can be seen in Figure 4.1. It must generate a plasma, extract the desired ions, filter out any unwanted chemical species, accelerate the chosen ions to a precise kinetic energy, and uniformly distribute them across the surface of the wafer.

4.1.1 Ion Generation and Extraction

The process begins in the ion source chamber (Figure 4.2a), where a solid dopant sample is vaporised, or a dopant-bearing gas, such as BF_3 or AsH_3 (these are highly poisonous), is introduced. Inside this chamber, a thermionic tungsten filament (20 V, 200 A) acts as a cathode, emitting energetic electrons that are accelerated towards an anode plate. When these high-speed electrons collide with the dopant gas molecules, they strip away electrons, creating a plasma of positive ions.

To maximise the ionisation efficiency, a magnetic field is applied across the chamber. This field forces the emitted electrons into helicoidal trajectories, significantly increasing their path length and thus the probability of them colliding with gas molecules before reaching the anode. Once the plasma is formed, an extraction electrode at the exit side of the chamber (Figure 4.2b) is biased to a very large negative potential (20-80 kV) with respect to the chamber itself. This strong electric field forcefully extracts the positive ions from the plasma, while an exit slit physically shapes the newly formed ion beam.

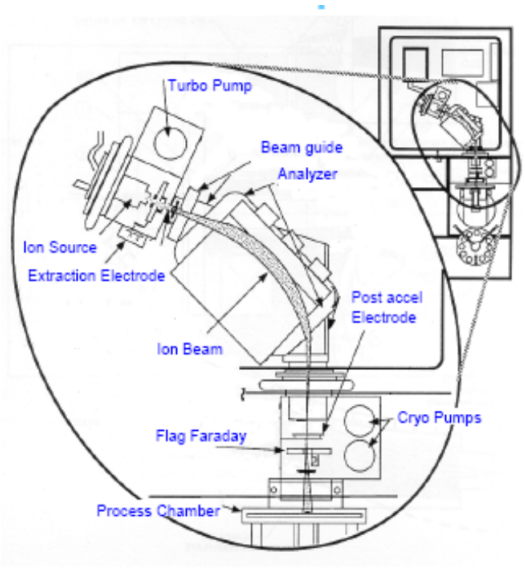


Figure 4.1: This picture shows the scheme of an ion implanter. They are roughly the size of two rooms.

4.1.2 Mass Analysis and Filtering

The beam extracted from the source is chemically impure. If BF_3 gas is used to implant boron, the plasma will contain B^+, F^+, BF^+, BF_2^+, and various isotopes like ^{10}B and ^{11}B. Implanting this raw mixture would completely destroy the electrical characteristics of the device. To purify the beam, it is passed through an analyser magnet, which is shown in Figure 4.3.

The magnetic mass analyser is based on the Lorentz force. As the ions enter the analyser region, they possess roughly the same kinetic energy, given by $E = \frac{1}{2}mv^2 = qV_{ext}$, where V_{ext} is the extraction voltage.

A magnetic field $\vec{B}$ is applied perpendicular to the beam's velocity $\vec{v}$, exerting a Lorentz force $F = qvB$ that acts as a centripetal force. By equating the Lorentz force to the centripetal force requirement ($qvB = \frac{mv^2}{r}$), the radius of curvature r for an ion's trajectory is found to be:

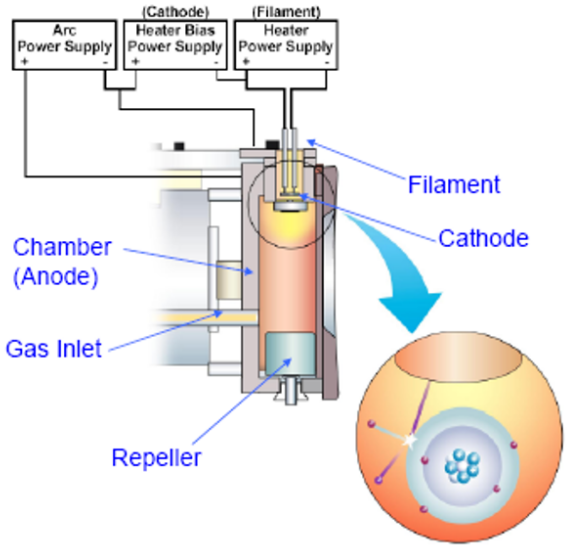
$$r = \frac{1}{B}\sqrt{\frac{2mV_{ext}}{q}}$$

Because the radius r depends strictly on the mass-to-charge ratio ($\frac{m}{q}$), ions of different masses will bend at different angles. By carefully tuning the magnetic field B, engineers can steer only the exact desired isotope (for example, strictly $^{11}B^+$) through a narrow resolving slit at the end of the analyser, while lighter or heavier fragments crash safely into the chamber walls.

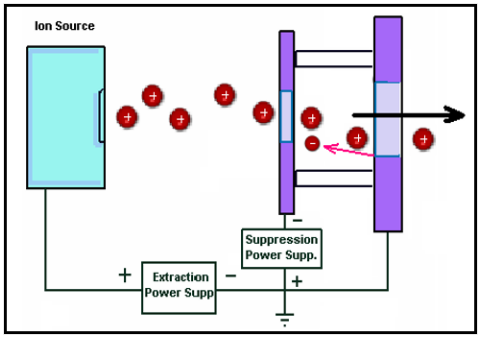
4.1.3 Acceleration and Neutral Filtering

Once the beam is isotopically pure, it enters a linear acceleration column. The whole structure is shown in Figure 4.4. Here, a series of electrodes applies strong electric fields to accelerate the ions to their final target energy, which can range from a few keV for ultra-shallow junctions up to several MeV for deep well formations.

During this flight, a small fraction of the fast-moving ions may collide with residual gas molecules in the vacuum and capture an electron, becoming neutral atoms. Because they no longer carry a charge, these neutralised atoms will not respond to any subsequent electromagnetic steering and would cause dose inaccuracies if they hit the wafer. To prevent this, the beam is passed through electrostatic deflection plates just before the process chamber. The electric field



(a) Ion source chamber



(b) Ions exiting the chamber

Figure 4.2: a) This is a scheme of an ion source chamber, also known as Arc Chamber. The tungsten filament can sometimes be covered by another material that acts as a cathode and emits secondary electrons. The potential accelerating these electrons is known as Arc voltage and is around 80V. b) The picture shows the exit of the chamber. Aside from the Front Slit shaping the beam, there is also a suppressor in front of it. It is set at a lower potential (-2 kV) so that electrons generated from the impact of ions with the front slit don't go back into the chamber causing X-rays.

bends the charged ions towards the wafer, while the neutral atoms fly straight ahead into a beam trap.

4.1.4 Dosimetry and Beam Heating

In the final process chamber, the ion beam must be uniformly distributed over the entire wafer. Depending on the tool's architecture, this is achieved either by electrostatically raster-scanning the beam across a stationary wafer, or by mechanically sweeping the wafer across a stationary ribbon beam.

To measure the exact dose being implanted, the wafer is placed inside a "Faraday cup" assembly. This apparatus captures every ion that strikes the target, allowing the system to measure a highly precise electrical current. By integrating this current over time, the implanter's computer calculates the exact number of dopant atoms injected into the silicon, ensuring perfect dose control.

Process Insight

While integrating current provides a theoretically perfect dose measurement, high-energy ion impacts routinely eject secondary electrons from the wafer. To an ammeter, a departing electron is electrically indistinguishable from an arriving positive ion. If these electrons escape, the system overestimates the ion flux, resulting in an under-doped wafer.
To prevent this, the assembly is shaped as a deep "Faraday cup" to physically trap escaping particles. Additionally, a negatively biased suppression electrode is positioned in front of the wafer, forcing electrons back into the target so the measured current strictly represents the true dopant dose.

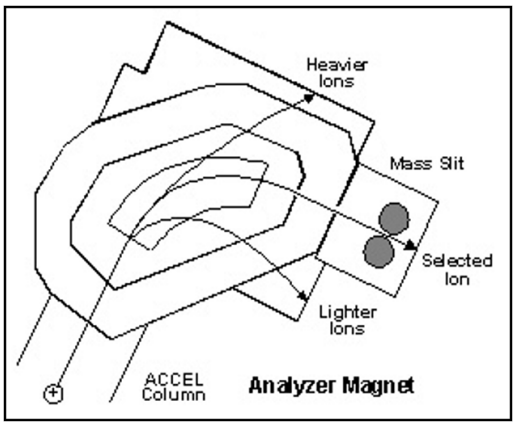


Figure 4.3: The scheme shows how a mass analyser works. Due to the Lorentz force, heavier and lighter ions have different trajectories compared to those selected.

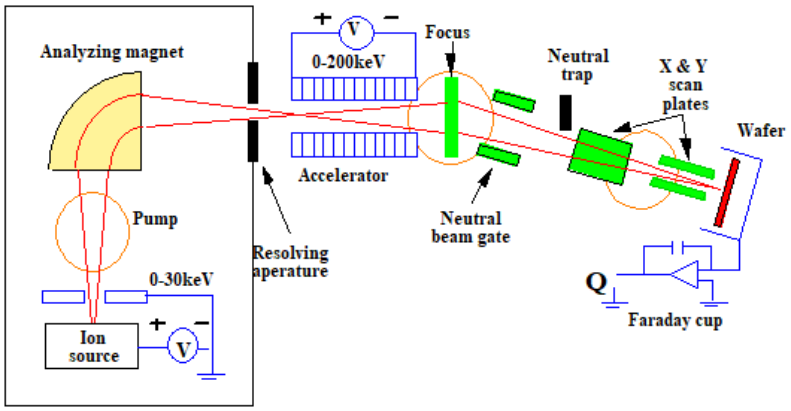


Figure 4.4: In this scheme, the rays trace the path taken by the ion beam. The neutral trap across the electrostatic lens captures the neutral atoms that are not deviated by the neutral beam gate.

However, stopping high-energy ions deposits a massive amount of energy into the substrate. For instance, a heavy dose of 5×10^{15} cm^{-2} implanted at 60 keV deposits roughly 20 kJ of thermal energy into a 200 mm wafer. This intense ion beam heating raises the temperature of the silicon and the masking materials, requiring active cooling of the wafer chuck to prevent photoresist burning and uncontrolled dynamic annealing of the crystal.

4.2 Ion Stopping Mechanisms and Profile Modelling

Once the accelerated ions strike the silicon wafer, they violently penetrate the surface and travel into the bulk. Inside, they rapidly lose their kinetic energy through a series of collisions until they come to rest.

While the exact trajectory of an individual ion is statistical, its average energy loss is governed by two distinct mechanisms: nuclear stopping and electronic stopping. The total energy loss per unit path length is defined as:

$$\frac{dE}{dx} = -N[S_n(E) + S_e(E)]$$

where N is the target atom density of the silicon, $S_n(E)$ is the nuclear stopping power, and $S_e(E)$ is the electronic stopping power.

4.2.1 Nuclear and Electronic Stopping

Nuclear Stopping occurs when the incoming ion undergoes a two-body elastic collision directly with the nucleus of a target silicon atom. This interaction is modelled as Coulomb scattering, heavily modified to account for the shielding effect of the target atom's electron cloud. Nuclear stopping dominates at low ion energies. Consequently, as an energetic ion slows down and reaches the end of its trajectory (the "end of range"), nuclear stopping becomes the primary energy-loss mechanism. Because massive momentum is transferred to the host nuclei during these collisions, nuclear stopping is responsible for almost all the crystallographic damage produced during implantation.

Physical Insight

Nuclear stopping is modelled as a classic two-body Coulomb scattering event between the incoming ion (mass M_1) and a target silicon atom (mass M_2). To account for the electron clouds shielding the nuclei, the purely repulsive Coulomb potential $V(r)$ is modified by a correction factor, Φ:

$$V(r) = \frac{Z_1 Z_2 q^2}{4\pi\epsilon_0 r}\Phi\left(\frac{r}{a}\right)$$

During the collision, the kinetic energy transferred to the target atom (E_{trans}) depends strictly on the initial energy (E_0) and the scattering angle (θ) in the centre-of-mass frame:

$$E_{trans} = \frac{4M_1M_2}{(M_1+M_2)^2}E_0\sin^2\left(\frac{\theta}{2}\right)$$

The scattering angle itself is a function of the impact parameter p and the modified potential $V(r)$, integrating from the distance of closest approach (r_{min}) to infinity:

$$\theta = \pi - 2\int_{r_{min}}^{\infty}\frac{p\,dr}{r^2\sqrt{1-\frac{V(r)}{E_c}-\frac{p^2}{r^2}}}$$

Crucially, the nuclear stopping power (S_n) is relatively small at high energies because the fast-moving ion interacts only briefly, minimising momentum transfer. As the ion slows down, the scattering cross-section increases, causing nuclear stopping to completely dominate at the "end of range". This elegantly explains why severe crystallographic damage is highly concentrated at the very end of the ion's trajectory.

Electronic Stopping occurs when the positively charged ion interacts with the sea of electrons surrounding the target atoms. This mechanism consists of two components:

- **Non-local electronic stopping:** As the charged ion moves through the semiconductor, it creates a trailing polarisation field in the electron cloud. This field "lags" behind the moving ion, creating an electrostatic drag force, much like moving through a highly viscous medium.
- **Local electronic stopping:** The direct exchange of momentum between the fast-moving ion and the orbital electrons of the target atoms.

Because both of these electronic effects depend heavily on the ion's velocity, the electronic stopping power is generally proportional to the square root of the ion's kinetic energy ($S_e \propto \sqrt{E}$). Electronic stopping dominates at high energies, but because the mass of an electron is negligible compared to a silicon nucleus, these interactions do not physically displace silicon atoms or cause structural damage.

4.2.2 The Gaussian Approximation

Because, macroscopically, the stopping process consists of a sequence of random scattering events, ion implantation is an inherently statistical process[16].

To a first-order approximation, as seen in Figure 4.5, the final depth distribution of the implanted ions can be modelled as a Gaussian distribution:

$$C(x) = C_p e^{-\frac{(x-R_p)^2}{2\Delta R_p^2}}$$

where:

- C_p is the peak concentration of the dopant.
- R_p is the **Projected Range**, which is the average depth at which the ions come to rest. It is strictly a function of the ion's mass and its incoming energy. The higher the energy, the higher R_p and ΔR_p.
- ΔR_p is the **Straggle** (or standard deviation), representing the statistical spread of the ions around the projected range.

By integrating this concentration profile over all depths, the total implanted dose (Q) can be directly related to the peak concentration and the straggle:

$$Q = \int_0^\infty C(x)dx \approx \sqrt{2\pi}\Delta R_p C_p$$

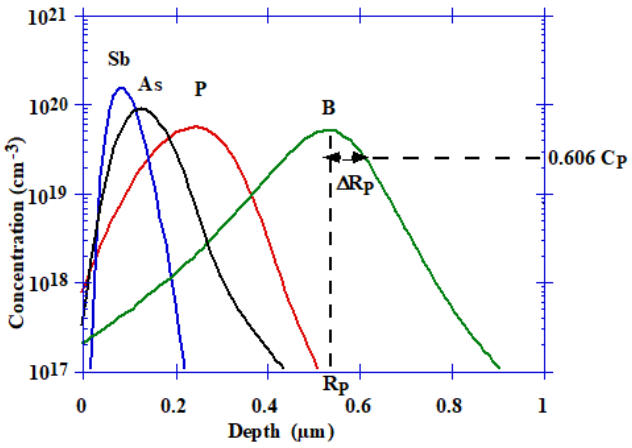


Figure 4.5: The graph shows the concentration profiles of different elements. Indeed, as a first-order approximation, the lineshapes are similar to a Gaussian around the peak.

4.2.3 Masking Rules

One of the greatest advantages of ion implantation is that it can be selectively blocked by patterned layers of photoresist, silicon dioxide, or the poly-silicon transistor gates themselves. To effectively mask a region of the underlying silicon, the masking material must be thick enough to completely stop the incoming ions.

However, the Gaussian profile dictates that the dopant concentration never truly drops to zero, not even inside the mask. To ensure efficient masking, engineers calculate the exact dose that manages to penetrate completely through a mask of thickness x_m inside the silicon. This

[16] With Monte Carlo simulations, engineers are able to find accurate and reliable predictions for concentration profiles.

penetrating dose ($Q_{penetrating}$) is given by integrating the tail of the Gaussian distribution from the back of the mask to infinity:

$$Q_{penetrating} = \int_{x_m}^{\infty} C(x)dx \approx \frac{Q}{\sqrt{2\pi}} \frac{\Delta R_p}{x_m - R_p} \exp\left[-\frac{(x_m - R_p)^2}{2\Delta R_p^2}\right]$$

Process Insight

In industrial fabrication, the formula for $Q_{penetrating}$ dictates a practical rule of thumb for designing masking layers. Since the goal is to protect the underlying active silicon, the mask thickness (x_m) is generally chosen such that the concentration of dopants exiting the bottom of the mask is strictly lower than the pre-existing background doping of the silicon substrate (see Chapter 1.3.3).
To achieve this, process engineers typically design the mask thickness to be the projected range plus several multiples of the straggle: $x_m = R_p + m\Delta R_p$. Depending on the acceptable leakage, m is usually chosen to be between 3 and 6. This ensures that the exponential decay of the Gaussian tail has dropped by several orders of magnitude before the ions ever reach the pristine silicon below.

4.2.4 Higher-Order Moments and Subsequent Diffusion

While the simple Gaussian approximation is highly useful, it is only truly accurate near the peak of the distribution. Real implanted profiles are rarely perfectly symmetrical. Lighter ions (like boron) tend to "bounce back" off heavier silicon atoms, skewing the profile towards the surface. Heavier ions (like arsenic) tend to scatter silicon atoms forward, skewing the profile deeper into the bulk.

To accurately model these asymmetrical profiles, an arbitrary distribution must be described using higher-order statistical moments, given by equation:

$$m_i = \frac{1}{Q} \int_{-\infty}^{+\infty} (x - R_p)^i \cdot C(x)dx$$

In particular, we use:

- **Skewness:** The third central moment ($i = 3$, divided by ΔR_p^3), which measures the asymmetry of the profile (Skewness = 0 for a perfect Gaussian).
- **Kurtosis:** The fourth central moment ($i = 4$, divided by ΔR_p^4), which measures the "flatness" or "peakedness" of the tail regions (Kurtosis = 3 for a perfect Gaussian).

By utilising these higher-order moments, advanced Technology Computer-Aided Design (TCAD) simulators generate highly accurate distributions using complex analytical functions, such as Pearson IV equations.

Finally, it is crucial to remember that ion implantation is almost always followed by a high-temperature activation anneal. Drawing on the mathematics developed in the previous chapter, an implanted Gaussian distribution can be treated as if it were a delta function that has already been thermally diffused by an initial "effective" $(Dt)_0$ equal to $\frac{\Delta R_p^2}{2}$.

The subsequent high-temperature anneal simply adds its own real thermal budget (Dt) to this initial spread. The final, post-anneal dopant distribution will remain a Gaussian, but with an increased, effective straggle:

$$\Delta R_{p,eff} = \sqrt{\Delta R_p^2 + 2Dt}$$

4.3 Channelling and Crystallographic Damage

The Gaussian approximation assumes that the silicon target acts as an isotropic, amorphous medium. However, silicon is a highly ordered crystal. The periodic arrangement of its atoms dramatically impacts the trajectories of the implanted ions and leads to severe physical damage.

4.3.1 The Channelling Effect

If the incident ion beam perfectly aligns with a major crystallographic symmetry direction (such as the $\langle 110 \rangle$ or $\langle 100 \rangle$ planes), the ions encounter wide, open "corridors" of empty space within the lattice. Ions travelling down these channels, like in Figure 4.6 experience virtually no nuclear stopping. As a result, they can channel effortlessly through the crystal and penetrate vastly deeper into the substrate than the standard Gaussian projected range predicts, creating a massive, unwanted "tail" in the doping profile.

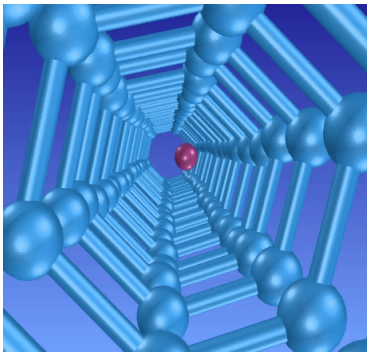


Figure 4.6: In this computer visual we pictorially understand what the channelling effect is. By squeezing through the lattice, ions are able to completely ignore the stopping power of nuclei.

Process Insight

Because channelling completely distorts the intended ultra-shallow junctions required for modern transistors, fab engineers employ several techniques to actively suppress it:

1. **Sample Tilt:** The most common industrial solution. The wafer is mechanically tilted (typically by about 7°) relative to the incoming ion beam. This forces the ions to "see" a dense, random arrangement of silicon atoms rather than open crystallographic corridors, thereby reducing the chance of channelling. Figure 4.7 shows the effects of this technique.

2. **Screen Oxides:** Growing a thin, amorphous layer of silicon dioxide on the wafer surface before implantation. As the ions pass through this amorphous oxide, they are scattered randomly, ensuring they enter the underlying silicon crystal at random, non-channelling angles.

3. **Pre-Amorphisation:** Deliberately bombarding the silicon with heavy, electrically neutral ions (like silicon or germanium) to completely amorphise the top layer of the crystal before implanting the actual dopant.

4.3.2 Crystallographic Damage

As established, nuclear stopping transfers immense momentum to the host lattice. The displacement energy required to knock a silicon atom completely off its stable lattice site and create a

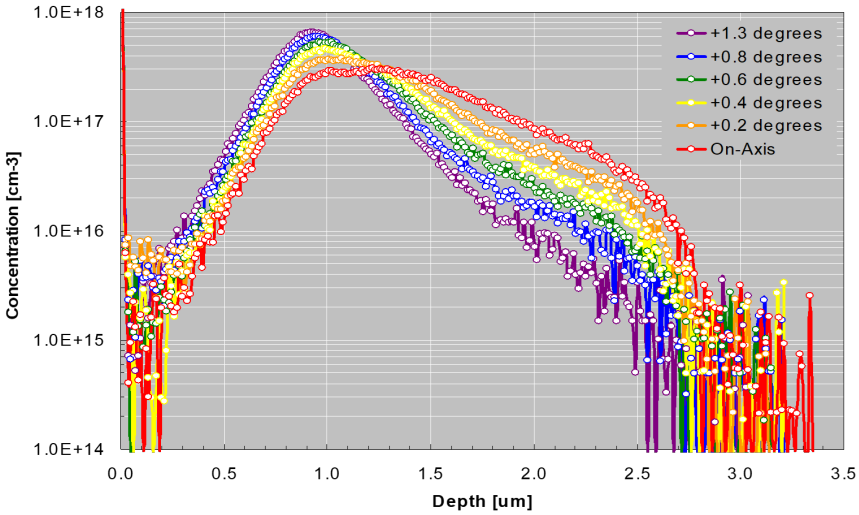


Figure 4.7: This graph shows the concentration profiles generated by different tilting angles. The usual angle is around 7°; steeper angles would cause shadowing problems (features of the wafer would block the target areas).

Frenkel pair[17] (an interstitial and a vacancy) is approximately 15 eV.

Because standard implantation energies range from tens to hundreds of keV, a single implanted ion possesses enough energy to displace a massive number (n) of silicon atoms. These primary displaced atoms often have enough kinetic energy to displace even more atoms, initiating a violent cascade of secondary collisions. While a fraction of these point defects will spontaneously recombine as the cascade cools, every new ion adds additional damage. The total defect density is a combination of the new amorphising damage (N_α), pre-existing damage from previous cascades (N), and ongoing thermal recombination (f_{rec}), which can be written as:

$$\Delta n(x) = n \cdot f_{rec} \cdot \left(1 - \frac{N}{N_\alpha}\right)$$

4.3.3 Solid Phase Epitaxy (SPE) and Activation

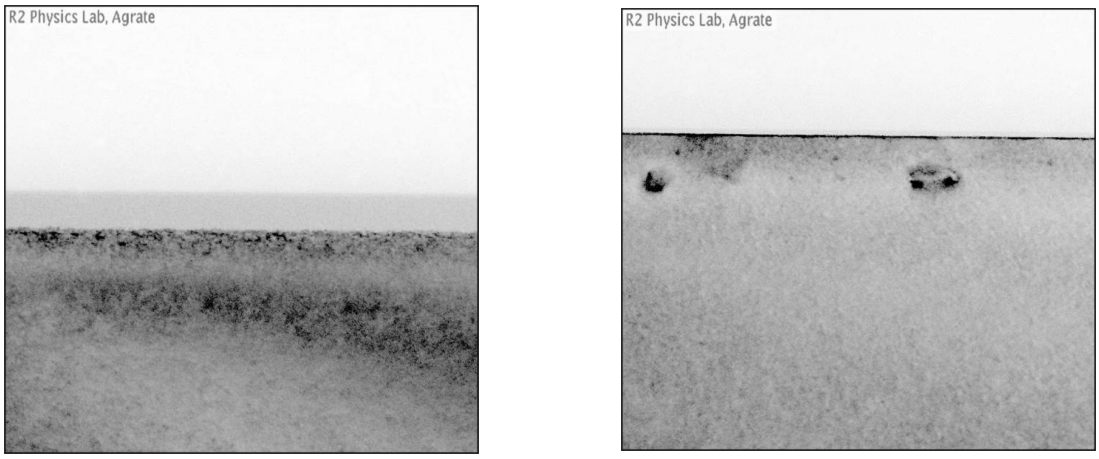
If the implantation dose is sufficiently high, the accumulated damage completely overwhelms the local crystal structure, rendering the targeted silicon layer amorphous. Alternatively, the surface could also undergo a similar process via a controlled pre-amorphisation step. To recover the mandatory crystalline lattice, the wafer is subjected to a thermal anneal to induce Solid Phase Epitaxy (SPE).

During SPE, the amorphous silicon regrows into a crystalline phase by using the underlying, undamaged substrate as a perfect atomic "seed" or template. Figure 4.8 shows the before and after of this process. This regrowth is a relatively rapid process; for instance, along the $\langle 100 \rangle$ orientation, the crystal regrows at approximately 50 $\mathrm{nm \cdot min^{-1}}$ at a temperature of 600 °C. The activation energy for this regrowth rate is 2.3 eV, which directly corresponds to the thermal energy required for Si-Si bond breaking and formation.

The SPE process is highly dependent on the local environment; it proceeds even quicker if the substrate is already doped, which facilitates the rapid annealing of defects and the activation of dopants. However, if an amorphous layer is entirely buried and the regrowth occurs simultaneously from both a top and bottom crystalline interface, the two advancing growth fronts

[17] To estimate the total number of Frenkel pairs generated by a single incident ion, the modified Kinchin-Pease (NRT) model is widely used. It predicts the number of displaced atoms as $N_d = 0.8\frac{E_v}{2E_d}$, where E_v is the portion of the ion's kinetic energy lost to nuclear collisions, and E_d is the displacement threshold energy ($\approx$ 15eV for Si).

can collide, producing severe extended line defects known as “clam-shell” defects. Furthermore, Transmission Electron Microscopy (TEM) reveals that even after successful SPE, clustered “end of range” defects typically remain visibly stranded just beyond the maximum depth of the amorphised zone.



(a) Before annealing

(b) After annealing

Figure 4.8: a) This TEM image shows a silicon substrate right after implantation. The darker spots are the amorphised areas of the lattice. b) After SPE, we can see that most of the lattice has returned to its original position. Nevertheless, some dislocation damage (end of range defects) is visible near the surface.

Dopant Activation

Merely physically embedding dopant atoms into the silicon is entirely insufficient; as we've established, to be electrically active, the dopants must occupy stable, substitutional lattice positions. The post-implantation anneal serves this crucial activation function.

For low-dose implants, or for highly destructive amorphising implants undergoing SPE, excellent dopant activation can be achieved at relatively low temperatures (e.g., 600 °C). However, achieving activation is notoriously difficult for high-damage implants (such as a 10^{15} atoms·cm^{-2} boron dose) that fall just short of full amorphisation. In these highly damaged, but technically still crystalline regions, a massive supersaturation of excess silicon interstitials competes directly with the boron atoms for the available substitutional lattice sites. This intense competition leads to the formation of inactive Interstitial-Boron (I/B) complexes, severely limiting the electrical yield of the implant.

4.4 Defect Evolution and Transient Enhanced Diffusion

While annealing after implantation is absolutely mandatory to dissolve crystal defects, regrow the amorphised silicon, and activate the dopants, applying high temperatures intrinsically causes the dopants to diffuse. Because the implanted silicon is severely damaged and flooded with non-equilibrium point defects, this diffusion does not follow the ideal, intrinsic Fickian models developed in the previous chapter.

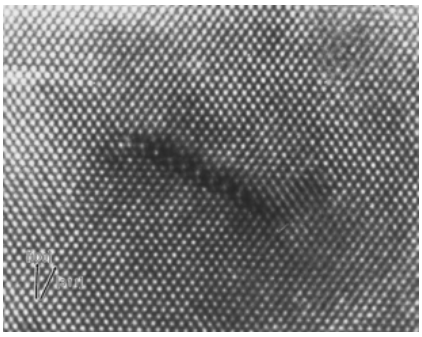
4.4.1 The “+1” Model and 311 Clusters

However, the dopant atoms themselves must occupy substitutional lattice sites to become electrically active. When a dopant atom kicks a native silicon atom out of its lattice position to take its place, an extra silicon interstitial is generated. The “+1” model posits that after the initial

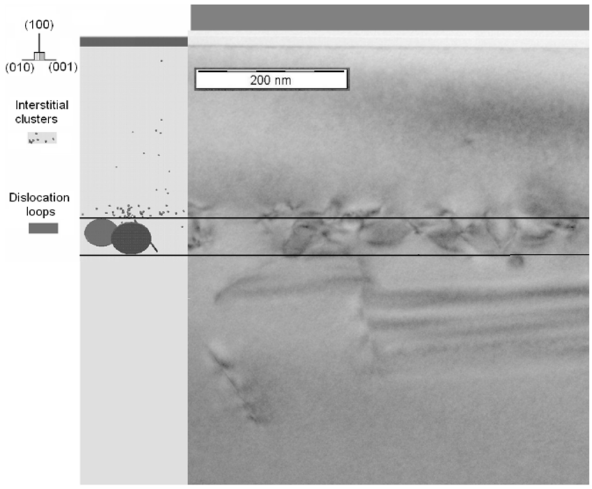
Frenkel pair recombination is complete, there remains exactly one excess silicon interstitial for every substitutional dopant atom incorporated into the lattice. While there is no direct microscopic proof for this exact one-to-one ratio, the model provides an excellent, experimentally validated estimate of the excess interstitial population.

Upon further annealing, these excess interstitials do not simply wander aimlessly. They agglomerate into small, rod-shaped clusters that form strictly on the {311} crystallographic planes, extending along the ⟨110⟩ directions. These structures are metastable and depending on the severity of the implant damage and the distance from the wafer surface, these {311} clusters can evolve in several ways:

- They can "evaporate" by releasing interstitials from their ends, which subsequently diffuse to the wafer surface and recombine.
- They can undergo a process known as *unfaulting* and collapse into stable, two-dimensional dislocation loops (frequently detectable by TEM exactly at the projected range, R_p).
- If the density is high enough, these loops can agglomerate to form massive extended dislocations, which require exceedingly high temperatures (greater than 1100 °C) to fully dissolve.



(a) A rod-like cluster of silicon interstitials



(b) Loop dislocations

Figure 4.9: a) This TEM image shows a rod cluster in a silicon substrate. Microscopically, these defects are made of chains of 5, 7 or 8 atoms, which develop along the ⟨110⟩ direction to minimise dangling bonds. You can read more about this on IEEE Xplore. b) The picture compares a simulation (on the left) with the actual damage caused by a 10^{15} boron implant at 80 keV followed by a 30-minute annealing at 900 °C. Dislocation loops are easily detectable at the predicted depth.

4.4.2 Transient Enhanced Diffusion (TED)

The formation and subsequent dissolution of these {311} clusters drive a massive anomaly in dopant redistribution known as Transient Enhanced Diffusion (TED).

TED is a phenomenon where the interstitial damage from the implant drastically enhances the diffusion rate of the dopant for a brief, transient period. Because {311} clusters form extremely rapidly during the initial temperature ramp-up, they initially trap the excess interstitials. However, as the anneal continues, these clusters become unstable and begin to shrink, slowly emitting a continuous flood of free interstitials into the surrounding silicon.

This emission creates a massive supersaturation of point defects. For example, during a 750 °C anneal, the thermal equilibrium concentration of interstitials (C_I^*) is roughly 10^8 cm^{-3}. However, the shrinking {311} clusters maintain a transient interstitial concentration (C_I) of approximately 10^{13} cm^{-3}. This means the local diffusivity is enhanced by a factor of more than 10^5 compared to standard thermal diffusion.

TED is the dominant effect determining final junction depths in shallow profiles today, and it exhibits highly counter-intuitive behaviour: profiles can actually diffuse *more* at low temperatures than at high temperatures for the same thermal budget (Dt). At very high temperatures, the {311} clusters dissolve almost instantly (in milliseconds), terminating the TED effect quickly. However, at lower temperatures, the clusters decay very slowly, sustaining the massive interstitial supersaturation (and the vastly accelerated dopant diffusion) for hours.

Physical Insight

The anomalous diffusion caused by TED is not confined to the implanted junctions themselves; the emitted interstitials travel deep into the surrounding silicon and wreak havoc on previously established profiles. A classic example is the effect of deep arsenic source/drain implants on a buried boron channel profile.
The damage from the source and drain junctions acts as a massive reservoir of interstitials. As these interstitials diffuse laterally and vertically away from the source/drain regions, they flood the transistor's active channel. Because boron diffuses exclusively via an interstitialcy mechanism, this flood violently accelerates the boron diffusion.
Crucially, the wafer surface acts as an interstitial sink. This sets up a flux that literally "pushes" the boron atoms from the deep channel up towards the surface of the wafer. Consequently, the active dopant concentration immediately beneath the gate oxide increases dramatically. This unwanted surface pile-up artificially raises the transistor's threshold voltage (V_{th}). Because this interstitial flooding is more concentrated when the source and drain are physically closer together, this threshold voltage shift becomes far more pronounced in heavily scaled devices. In semiconductor physics, this phenomenon is famously known as the Reverse Short Channel Effect (RSCE).

4.5 Advanced Doping for Ultra-Shallow Junctions

As transistor dimensions scale ever downward, the formation of ultra-shallow junctions becomes a critical bottleneck. The primary adversary in this regime is Transient Enhanced Diffusion (TED). If traditional furnace annealing is used, the slow temperature ramp-ups provide ample time for {311} clusters to form and dissolve, driving the dopants far deeper than the nanoscale geometry allows. To circumvent this, the industry has developed advanced thermal, chemical, and physical techniques to suppress TED and conformally dope modern 3D architectures.

4.5.1 Millisecond and Microsecond Annealing

Because TED is highly dominant at relatively low temperatures and long timescales, the fundamental strategy to defeat it is to apply extremely high temperatures for incredibly short durations. This maximises dopant activation while physically denying the interstitials the time required to diffuse.

- **Rapid Thermal Processing (RTP):** Operates in the timescale of seconds using banks of halogen lamps. While better than standard furnaces, it still allows a noticeable degree of TED.
- **Flash Anneal:** Operates in the millisecond regime. The wafer is pre-heated by halogen

lamps, and then an intense bank of Xenon (Xe) flash lamps delivers a massive, instantaneous burst of thermal energy specifically to the wafer surface.

- **Laser Anneal:** Operates in the microsecond regime. A highly focused laser beam is rapidly scanned across the wafer surface at speeds of approximately 300 $\text{mm} \cdot \text{s}^{-1}$.

In the most extreme cases, laser annealing is used to actively melt the topmost layer of the silicon, triggering Liquid Phase Epitaxial (LPE) regrowth. By perfectly tailoring a pre-amorphisation implant depth to match the exact melt depth of the laser, engineers can force a virtuous interaction between laser-created vacancies and the End of Range (EOR) interstitials, effectively annihilating them and completely suppressing TED.

4.5.2 Defect Engineering and Surface Treatments

Instead of just fighting the symptoms of TED thermally, engineers can actively engineer the point-defect population within the silicon lattice.

Co-implantation involves deliberately implanting non-dopant impurities that chemically compete with the dopants for interactions with point defects, thereby reducing the defect supersaturation. For example, Carbon (C) can be co-implanted to form stable C-I (Carbon-Interstitial) clusters in silicon and SiGe. By sweeping up the free interstitials, Carbon starves the Boron atoms of their primary diffusion vehicle, severely reducing Boron diffusivity. Similarly, Fluorine (F) can be implanted to form F-V (Fluorine-Vacancy) clusters, reducing vacancy-assisted diffusion.

Furthermore, the wafer surface itself acts as a massive sink for interstitial annihilation. Surface treatments can significantly modulate how dopants interact with point defects. An atomically clean silicon surface possesses a very high number of dangling bonds, which drastically increases the probability of interstitial adsorption. By removing the native oxide and performing high-vacuum annealing or ultra-low pressure NH_3 treatments, the bare surface effectively "vacuums up" excess interstitials, significantly reducing the anomalous diffusion of arsenic and Boron even in the presence of heavy pre-amorphising implants.

4.5.3 3D Doping: PLAD and Monolayer Doping

Traditional beam-line ion implantation is inherently a line-of-sight technique; ions travel in straight lines and embed themselves into planar surfaces. However, modern logic and memory devices (such as FinFETs and buried word-line DRAMs) rely on complex, highly vertical 3D architectures. Bombarding a vertical silicon fin with a directional ion beam results in heavily doped horizontal surfaces but severely under-doped vertical sidewalls.

To achieve conformal doping on 3D structures, alternative technologies must be utilised:

- **Plasma Doping (PLAD):** The wafer is immersed directly into a high-density plasma containing the dopant ions. A pulsed negative voltage is applied to the wafer chuck. During each pulse, ions are accelerated from the plasma across the electrostatic sheath region and driven into the wafer.

- **Molecular Monolayer Doping:** A highly controlled, self-limiting chemical reaction is used to coat the silicon surface with a single monolayer of boron- or phosphorus-containing molecules. These molecules covalently bond to the exposed silicon. A subsequent Rapid Thermal Anneal (RTA) breaks these bonds and drives the dopants into the substrate. The dose is finely tuned by mixing dopant-containing precursors with "blank" precursors during the deposition phase, and the depth is strictly controlled by the RTA temperature.

Process Insight

PLAD is a revolutionary tool for 3D devices like FinFETs. Because the plasma surrounds the entire 3D structure, and the accelerating sheath conforms directly to the topography of the wafer, the ions strike the silicon from multiple angles simultaneously. This intrinsically conformal delivery mechanism ensures that the vertical sidewalls of a fin receive a higher dopant dose, closer to that of the horizontal top surface. Furthermore, because PLAD operates at very low energies and delivers massive currents across the entire wafer simultaneously, it can achieve doses several times higher than conventional beam-line tools while practically eliminating unwanted electrical charging on the wafer surface.

Having explored the violent, ballistic reality of Ion Implantation, we now possess the tools to precisely modulate the electrical conductivity of our silicon substrate. We have seen how magnetic mass analysers and electrostatic accelerators grant us unprecedented control over dopant doses and depths, and how thermal and defect engineering can repair the massive crystallographic damage left behind.

In the next chapter, we will turn our attention to **Thin Film Deposition**, specifically focusing on **Chemical Vapour Deposition (CVD)**. We will shift from altering the existing silicon crystal to actively building new materials on top of it, exploring the chemical kinetics and gas-phase reactions necessary to deposit the highly conformal films that wire, isolate, and protect the billions of transistors on a modern chip.

Chapter 5

Thin Film Deposition

5.1 Introduction to Thin Film Deposition

While doping modifies the electrical properties of the existing silicon substrate, the creation of a modern integrated circuit requires building upwards. Many additional layers, ranging from insulating dielectrics to conductive metals, must be deposited above the silicon substrate to form the complex, three-dimensional architecture of the device.

When engineering these deposited layers, process engineers evaluate success based on two primary categories:

- **Film Quality:** The layer must possess the correct chemical composition, exhibit the required mechanical and electrical properties, and maintain exceedingly low defectivity and contamination levels.
- **Thickness Control:** The thickness of the film must be strictly uniform from deposition to deposition, across the entire diameter of the wafer, and crucially, as a function of the underlying wafer topology.

5.1.1 Conformality and Gap Filling

Because these films are deposited onto wafers that already feature etched trenches, protruding gates, and patterned spacers, the topography is far from flat. This introduces two critical deposition metrics:

- **Step Coverage (Conformality):** The ability of the deposition process to coat horizontal and vertical surfaces with a constant, uniform thickness. An example is shown in Figure 5.1.
- **Gap Filling:** The specific ability of the process to completely fill high-aspect-ratio structures (deep, narrow trenches) without leaving voids.

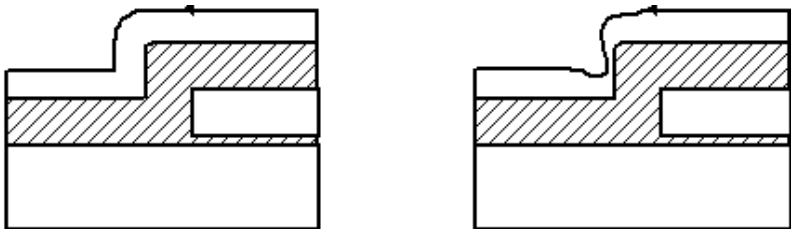


Figure 5.1: The left image shows a deposition with good conformality. In the right image, the step is covered unevenly.

Process Insight

Achieving perfect conformality is one of the most difficult challenges in semiconductor manufacturing. If a deposition process is non-conformal, the film tends to grow faster on the top horizontal surfaces and upper corners of a trench than on the vertical sidewalls or the bottom.
As the film grows thicker, the faster-growing material at the top corners will eventually pinch together and seal the top of the trench before the bottom is fully filled. This creates a permanent, empty cavity (a "void" or "keyhole") trapped inside the dielectric or metal layer. During subsequent Chemical Mechanical Polishing (CMP) or etching steps, these buried voids can be broken open, trapping chemical contaminants or causing fatal electrical shorts. Therefore, highly conformal deposition techniques are strictly required for high-aspect-ratio gap filling.

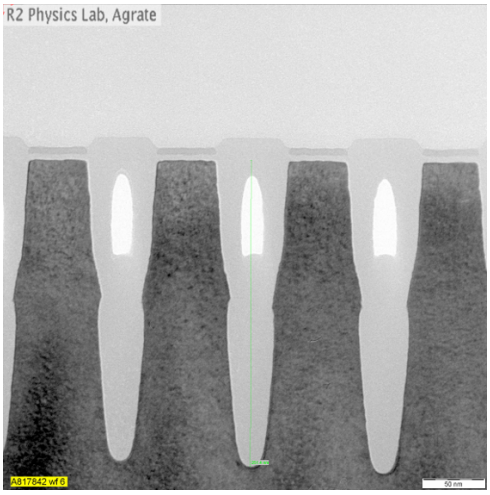


Figure 5.2: The white space in these trenches are voids covered up by a deposition with poor conformality. Thus, the deposited material created a "bridge" and prevented the complete filling of the trenches.

5.1.2 Deposition Methods: PVD vs. CVD

The industry relies on two fundamentally different methods to deposit thin films from a vapour phase:

1. **Physical Vapour Deposition (PVD):** Purely physical methods (such as thermal evaporation or ion sputtering) are used to eject constituent atoms from a solid target, which then travel through a vacuum and condense onto the wafer surface.

2. **Chemical Vapour Deposition (CVD):** Reactant gases are introduced into a heated deposition chamber. The film is deposited entirely as the result of a chemical reaction between these gases directly at the heated wafer surface.

While other techniques exist, such as liquid spin-coating (e.g., for photoresists or Spin-On Dielectrics) and electrolytic deposition (e.g., for copper interconnects), CVD remains the absolute workhorse for depositing high-quality, conformal dielectrics and semiconductor films.

5.2 CVD Kinetics and Fluid Dynamics

To understand how a CVD film grows, we must examine the sequence of physical and chemical events occurring inside the deposition chamber.

5.2.1 The Seven Steps of CVD

The overall CVD process in a traditional atmospheric chamber (APCVD) can be broken down into seven sequential steps, which can be observed in Figure 5.3.

1. Transport of the reactant gases from the main gas stream to the deposition region.
2. Transport of reactants from the main gas stream, through the boundary layer, to the wafer surface.
3. Adsorption of the reactants directly onto the wafer surface.
4. Surface reactions, which include the chemical decomposition of the gas, surface migration to attachment sites (like kinks and ledges), site incorporation, and other surface phenomena.
5. Desorption of the reaction by-products.
6. Transport of the by-products back through the boundary layer.
7. Transport of the by-products away from the deposition region via the main gas stream.

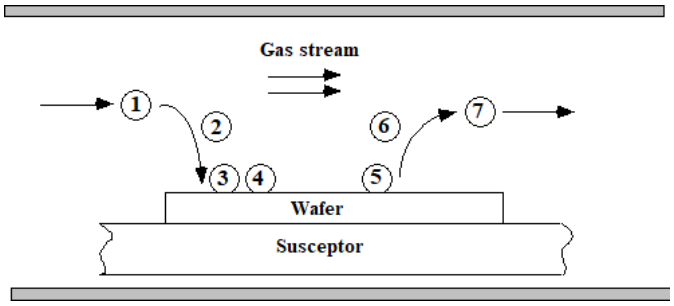


Figure 5.3: This drawing schematically shows where each step takes place inside the chamber.

In macroscopic modelling, gas-phase reactions and the intricate details of surface migration are often neglected. The overall growth rate is typically determined by whichever is slower: the mass transfer across the boundary layer (Step 2) or the chemical surface reactions (Steps 3-5), which is what we'll focus on.

5.2.2 Fluid Dynamics and the Stagnant Layer

To accurately model the mass transfer of reactants (Step 2), we must analyse the fluid dynamics of the gas inside the CVD reactor. In a traditional APCVD system, wafers are held on heated susceptors while reactant gases flow through the chamber at atmospheric pressure.

Basic Transport Mechanisms

When a fluid flows with a primary convection current in one direction (e.g., the y-direction), its velocity is rarely uniform across the transverse direction (the x-direction). Regions of fast convection drag adjacent regions of slower convection along with them. This transfer of momentum

between fluid layers can be mathematically described by a Fick-like law:

$$j_p = -\eta \frac{\partial v_y}{\partial x}$$

where j_p is the momentum flux, η is the fluid's dynamic viscosity, and v_y is the velocity of the fluid in the y-direction.

By defining a differential control volume with cross-sectional area S and thickness dx, we can evaluate the forces acting on the fluid volume. The momentum entering one side is $j_p(x)S$, and the momentum leaving the other is $j_p(x+dx)S$. Just as Fick's First Law leads to the second law of diffusion, this momentum gradient leads to the second-order differential equation:

$$\frac{\partial v_y}{\partial t} = \frac{\eta}{\rho} \frac{\partial^2 v_y}{\partial x^2}$$

where ρ is the density of the gas and η represents the viscosity.
By equating these viscous forces to the driving force of the gas, that is the pressure gradient $\frac{p_1 - p_2}{l}$ across a length l, we can solve for the exact velocity profile in stationary conditions.

- **Parallel Plates:** For a fluid moving between two parallel plates separated by a distance a, the velocity profile forms a parabola:

$$v_y = \frac{p_1 - p_2}{2\eta l}(ax - x^2)$$

- **Cylindrical Tube:** Similarly, for a tube of radius a, the velocity profile in the z-direction is:

$$v_z = \frac{p_1 - p_2}{4\eta l}(a^2 - r^2)$$

In both cases, the no-slip condition forces the velocity to drop exactly to zero at the physical boundaries ($x = 0, x = a$ or $r = a$).

The Boundary Layer

In a real CVD chamber, gas is often injected as a "plug" flow, entering with a completely uniform velocity U_∞. As it travels down the tube, the friction from the walls forces this uniform plug to gradually transform into the parabolic profiles calculated above.

To determine the nature of this flow, we calculate the Reynolds number, N_{re}. If N_{re} is large enough[18], but still strictly below the threshold that would induce chaotic, turbulent flow, broad regions of the gas stream will maintain the approximately uniform velocity U_∞. Consequently, the velocity profile can be approximated as completely constant, *except* in a thin transition region immediately above the susceptor.

This region is the **boundary or stagnant layer** (δ_s), which we've already established in the Physical Insight of Chapter 2.3.1. In this region, the macroscopic velocity of the gas effectively drops to zero.

It is important to note that the thickness of this boundary layer (δ_s) is not constant; it progressively thickens as the gas travels further along the susceptor, as seen in Figure 5.4. Because the gas inside this boundary layer is stagnant, reactant molecules must physically diffuse across it to reach the wafer surface, creating the bottleneck present in Step 2.

[18] In general, in our cases, the Reynolds number is indeed large enough.

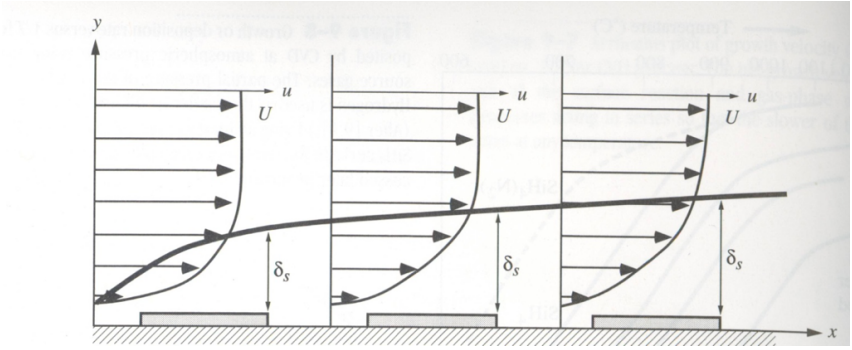


Figure 5.4: In this graph we can see what the gas profile looks like. The thick line represents the stagnant layer thickness, which gets increasingly large the further the gas travels.

5.2.3 Flux Equations for CVD

Because the physical setup of diffusion across a stagnant layer followed by a surface reaction is identical to the Deal-Grove oxidation model, we can derive a very similar set of flux equations.

The scheme for this derivation is shown in Figure 5.5. Let C_G be the reactant concentration in the bulk gas, and C_S be the reactant concentration exactly at the wafer surface. The flux of mass transfer across the stagnant layer (F_1) is:

$$F_1 = h_G(C_G - C_S)$$

where h_G is the mass transfer coefficient. The flux of the chemical surface reaction (F_2) is:

$$F_2 = k_S C_S$$

where k_S is the surface reaction rate constant.

In steady state, these two fluxes must be equal ($F_1 = F_2 = F$):

$$h_G(C_G - C_S) = k_S C_S$$

Solving for the unknown surface concentration C_S:

$$h_G C_G = C_S(k_S + h_G) \implies C_S = \frac{h_G C_G}{k_S + h_G}$$

Substituting this back into the reaction flux equation yields the total steady-state flux:

$$F = k_S \left(\frac{h_G C_G}{k_S + h_G} \right) = \frac{k_S h_G}{k_S + h_G} C_G$$

The physical growth velocity of the film (v) is simply this flux divided by the atomic density (N, the number of atoms incorporated per unit volume):

$$v = \frac{F}{N} = \frac{k_S h_G}{k_S + h_G} \frac{C_G}{N}$$

5.2.4 Mass Transfer vs. Surface Reaction Regimes

Just as with thermal oxidation, this overarching velocity equation reveals two distinct, limiting growth regimes:

- **Surface Reaction Controlled ($k_S \ll h_G$):** If mass transfer across the boundary layer is extremely fast, the growth velocity simplifies to $v \approx k_S \frac{C_G}{N}$. In this regime, the deposition is heavily limited by the chemical reaction. Because k_S follows an Arrhenius law, the growth rate is *very* temperature-sensitive.

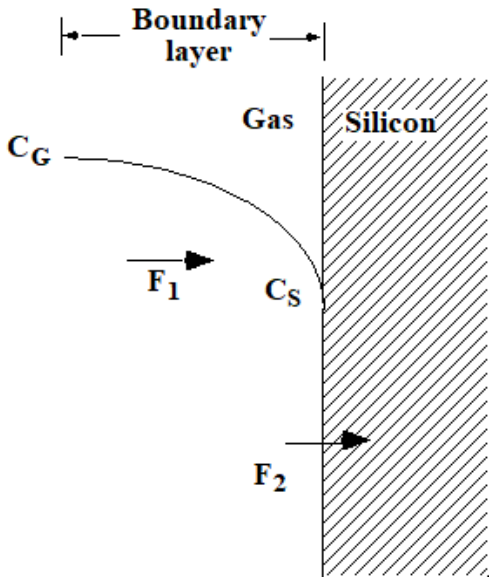


Figure 5.5: Schematic representation of the setup for diffusion in CVD.

- **Mass Transfer Controlled ($h_G \ll k_S$):** If the chemical reaction is extremely fast, the growth velocity simplifies to $v \approx h_G \frac{C_G}{N}$. The surface is in lack of reactants. Because h_G shows little dependence on temperature, the growth rate is *very* sensitive to the boundary layer geometry. In general, we don't want to work in h_G-limited regimes.

This presents an industrial challenge for APCVD systems. If the reactor operates in the mass-transport regime (like high-temperature silicon epitaxy), the deposition is extremely sensitive to the local gas flow and boundary layer thickness. To ensure every wafer sees the exact same gas flow, wafers must be placed side-by-side horizontally on a sloped susceptor, as though to "follow" the stagnant layer profile.

One theoretical solution is to lower the temperature to force the process into the surface-reaction-controlled regime, where geometry matters less. However, at atmospheric pressure, lowering the temperature results in commercially unviable, exceptionally low deposition rates and extremely poor film quality. To solve this, the industry had to rethink the reactor architecture entirely.

5.3 Low-Pressure CVD (LPCVD)

As established by the flux equations, operating a traditional APCVD chamber in the mass-transport-controlled regime creates severe throughput limitations. A brilliant, alternative method to force the system into the uniform, surface-reaction-controlled regime without dropping the temperature to unviable levels is to drastically lower the pressure. Low-Pressure Chemical Vapour Deposition (LPCVD) is now widely used in integrated circuit manufacturing.

5.3.1 Reactant Gas Diffusivity

According to the kinetic theory of gases, the diffusivity (D) of a gas is inversely proportional to its pressure. By dropping the chamber pressure from atmospheric levels down to the millitorr or torr range, the diffusivity of the reactant gases increases by several orders of magnitude.

Because the mass transfer coefficient (h_G) is directly proportional to diffusivity, this massive increase in D forces h_G to become significantly larger than the surface reaction rate constant (k_S). The inequality $k_S \ll h_G$ is strictly satisfied, planting the LPCVD process firmly in the surface-reaction-controlled regime.

5.3.2 Advantages of LPCVD

By eliminating the reliance on boundary layer geometries, LPCVD offers several monumental industrial advantages:

- **Wafer Stacking:** Because the gas diffuses so rapidly, wafers no longer need to be laid out horizontally. They can be stacked closely together in dense vertical or horizontal quartz boats, massively increasing manufacturing throughput.
- **Improved Film Quality:** Operating at lower pressures significantly reduces the likelihood of unwanted, premature gas-phase reactions (where gases react in the empty space above the wafer and rain down as solid particulate defects).
- **Less Autodoping:** The high diffusivity effectively sweeps away by-products and outgassing dopants from the substrate before they can re-incorporate into the growing film.

5.4 Plasma Physics and PE-CVD

While LPCVD solves the geometric limitations of deposition, it still fundamentally relies on thermal energy to drive the chemical reactions. However, modern microelectronic scaling demands strict thermal budgets. To deposit high-quality films at low temperatures, the industry replaces thermal energy with non-thermal energy by generating a plasma.

5.4.1 Definition of a Plasma

A plasma is formally defined as a collection of free charged particles moving in random directions that is, on average, electrically neutral. If a gas is sufficiently heated, the atomic collisions become energetic enough to decompose the atoms into freely moving electrons and positive ions.

In a plasma at true thermal equilibrium (such as the interior of a star or a fusion reactor), the electron density (n_e) is roughly equal to the ion density (n_i), and their temperatures are identical ($T_e = T_i = T$), typically reaching tens to hundreds of millions of kelvin. Industrial thermal plasmas work in the range of 4000 K to 20000 K.

5.4.2 Weakly Ionised Plasmas

In semiconductor manufacturing, we utilise **weakly ionised plasmas** (such as low-pressure glow discharges), which operate between 1 mTorr and 1 Torr. These plasmas are distinctly *not* in thermal equilibrium. They are driven electrically rather than thermally.

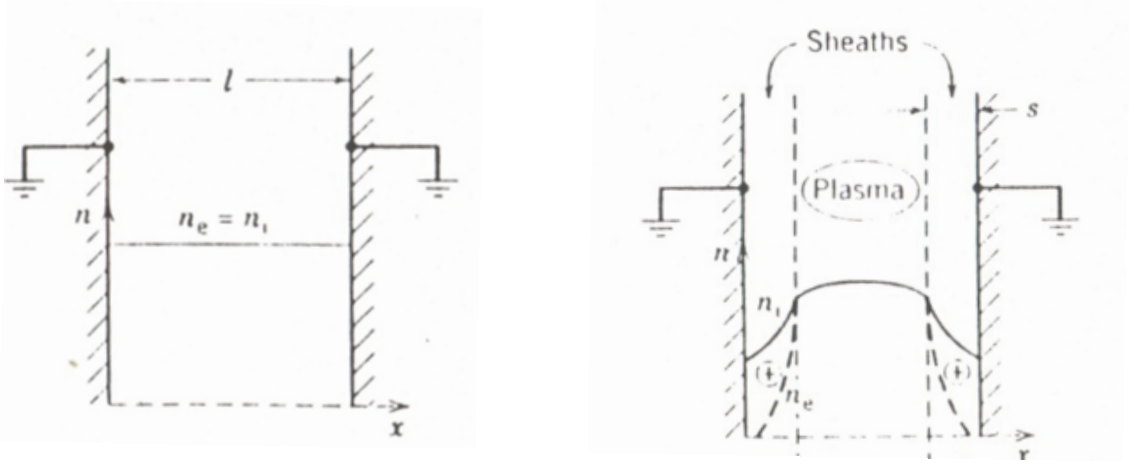
In a glow discharge, the electrons absorb energy from an applied radio-frequency (RF) electric field. Because electrons are incredibly light, they accelerate to massive velocities, reaching extremely high electron temperatures ($T_e \approx 1$ to 10 eV, which corresponds to tens of thousands of degrees kelvin). The heavy ions, however, barely respond to the high-frequency oscillating field, remaining at roughly room temperature ($T_e \gg T_i$). The plasma is sustained in a steady state by the continuous collisions between these fast electrons and neutral molecules, which causes further ionisation. Typical charge densities (n) range from 10^8 to 10^{13} cm^{-3}.

The fundamental spatial scale of a plasma is the **Debye length** (λ_{De}). It represents the typical length scale over which significant charge densities can spontaneously exist without being screened out. On any distance larger than λ_{De}, the plasma rigorously enforces macroscopic electrical neutrality. Within this neutral bulk, the electron density responds to a spatially varying electrostatic potential (Φ) according to Boltzmann's relation:

$$n_e = n_i e^{\frac{\Phi}{T_e}}$$

Physical Insight

Because the net charge density in the bulk plasma is zero, there is initially no electric field to confine the particles. However, because the electrons are vastly lighter and hotter than the ions, their thermal velocity is staggeringly high. These fast electrons collide with the chamber walls, generating a current and getting lost in a very short time scale.
To prevent the total collapse of the plasma, an electric field spontaneously builds up to oppose this imbalance between the ions and the electrons. The bulk plasma gets raised to a positive potential (V_p) with respect to the confining walls. This creates thin, positively charged boundary layers known as **sheaths** joining the plasma to the physical walls. This behaviour is shown in Figure 5.6.
In a high-voltage DC sheath, the strong electric field repels electrons so violently that the electron density drops practically to zero ($n_e \approx 0$), leaving a purely ionic, positively charged region. The simplest model (the Matrix Sheath) assumes a constant ion density, but a more accurate physical model (the Child Law Sheath) accounts for energy and flux conservation as the ions continuously accelerate toward the wall.
Crucially, for ions to enter the sheath and fall to the wall, they must cross the sheath boundary with a minimum velocity called the **Bohm velocity** ($v_B = \sqrt{\frac{k_B T_e}{M}}$). To gently accelerate the ions up to this speed, a **pre-sheath** must exist just inside the bulk plasma. In the pre-sheath, macroscopic neutrality ($n_e \approx n_i$) is maintained, but a weak electric field exists to drift the ions toward the sheath edge.



(a) A plasma chamber before sheath formation

(b) A plasma chamber after sheath formation

Figure 5.6: a) This drawing represents the electron and ion density before the formation of the sheaths. Of course, the gas is neutral and the densities are the same across the chamber. b) The picture illustrates the distribution of electron and ion densities across the chamber after the formation of the sheaths. The electron density drops due to the formation of electric fields, while the ion density drops due to conservation of flux (ions are accelerated towards the walls).

5.4.3 Plasma-Enhanced CVD (PE-CVD)

By maintaining a plasma above the wafer, Plasma-Enhanced CVD (PE-CVD) introduces massive amounts of non-thermal energy into the chemical system. The plasma environment is a chaotic mix of electrons, ionised molecules, neutrals, excited species, and free radicals.

Free radicals (such as SiO, SiH_3, and F) are electrically neutral species that possess incomplete chemical bonding, rendering them very reactive. The intense fragmentation of the

precursor gases into radicals, combined with the continuous physical bombardment of the wafer surface by ions falling through the plasma sheath, allows surface reactions and film deposition to occur at dramatically lower temperatures than standard thermal CVD. Figure 5.7 shows the structure of a chamber for PECVD.

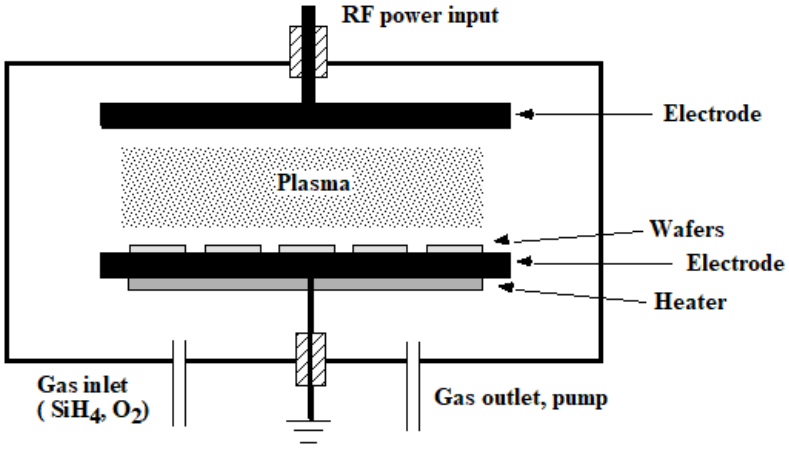


Figure 5.7: This picture shows what the chamber for PECVD looks like. The area above the wafers is occupied by the plasma instead of the regular gas stream in a standard CVD chamber.

5.4.4 High Density Plasma CVD (HDP-CVD)

For the most demanding gap-filling operations, engineers use High Density Plasma CVD (HDP-CVD). This architecture utilizes a remote, extremely dense plasma coupled with an independently controlled RF substrate bias.

By biasing the substrate independently, engineers can perfectly balance the chemical deposition of the film (usually SiO_2) with the simultaneous physical sputtering (etching) by heavy argon ions. As the oxide grows and threatens to pinch off the top of a narrow trench, the directional ion bombardment selectively sputters away the protruding upper corners. This continuous deposit-and-etch cycle guarantees exceptional planarisation and entirely void-free filling of high-aspect-ratio structures (avoids the issue shown in Figure 5.2).

5.5 Atomic Layer Deposition (ALD)

For the most extreme geometries in modern microelectronics, even advanced CVD techniques struggle to provide adequate step coverage. To achieve absolute atomic-scale thickness control and perfect conformality, the industry employs Atomic Layer Deposition (ALD).

ALD relies on a strictly controlled sequence of self-limiting surface reactions. Instead of continuously flowing a mixture of reactant gases into the chamber, ALD pulses precursor gases sequentially. A precursor is introduced in a large enough dose to completely saturate all available surface reaction sites, forming exactly one atomic monolayer[19]. Because the reaction is self-limiting, it stops abruptly once the surface is fully covered, regardless of how much extra precursor is supplied. The chamber is then purged, and the next precursor is introduced to react exclusively with the newly formed monolayer. This process is schematically summarised in Figure 5.8.

This unique layer-by-layer growth mechanism grants ALD several unparalleled advantages:

[19] Actually, one cycle only deposits a fraction of a monolayer (10% to 40%) due to the steric hindrance from bulky precursor ligands. Multiple cycles are required for exactly one monolayer.

- **Ultimate Conformality:** The self-limited growth regime guarantees perfectly uniform growth across highly structured substrates. This makes ALD the mandatory deposition technique of choice for high-κ dielectrics in heavily scaled DRAM capacitors, which frequently feature cylindrical containers with immense aspect ratios of up to 100:1.
- **Composition Control:** Because the film is grown atomic layer by atomic layer, ALD offers an intrinsically high degree of control over the final chemical composition. Dopants can be introduced seamlessly by substituting different precursors during specific cycles, and complex nano-laminate structures can be easily synthesised.
- **Low Thermal Budget:** The temperatures required for ALD are significantly lower than standard CVD processes, typically ranging between 200 °C and 350 °C. This thermal budget can be driven even lower by utilising radical-assisted or Plasma-Enhanced ALD (PE-ALD) processes.

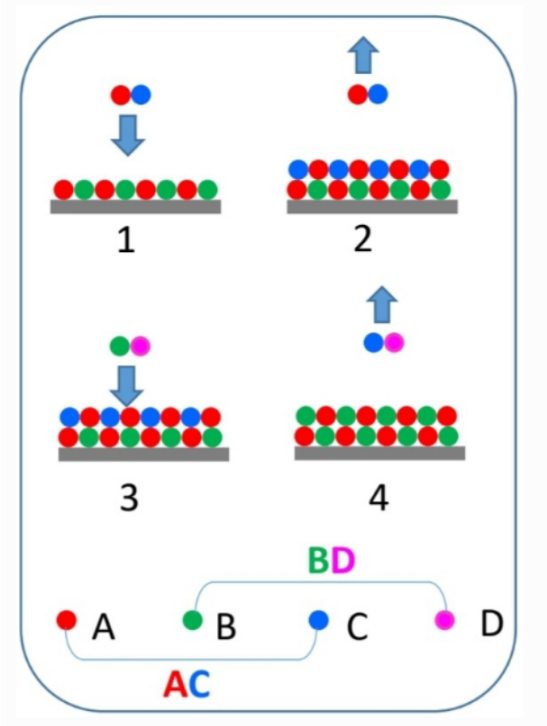


Figure 5.8: This picture is a schematic representation of an ALD process. In steps 1 and 3 the precursor gases are introduced, while steps 2 and 4 purge the by-products outside the chamber. The chemical species (A, B, C, D) must be chosen so that the A-B bond is stronger than A-A and B-B bonds, while C-D should be stronger than A-C and B-D. Image taken from the notes of the Micro and Nano Fabrication Technology course by Professor Petti at Polimi.

5.6 Common CVD Films and Reactions

Depending on the specific architectural needs of the integrated circuit, a variety of materials must be deposited using the architectures discussed above. Below is a summary of the most common CVD films, their primary equipment, and their fundamental chemical reactions:

- **Epitaxial Silicon (EPI-Si):** Deposited via APCVD or LPCVD at very high temperatures (700 °C to 1250 °C). The primary reaction uses silane ($SiH_4 \rightarrow Si + 2H_2$), though hydrogen chloride (HCl) can be added to enable selective deposition processes. Alternatively, silicon tetrachloride is used ($SiCl_4 + 2H_2 \rightarrow Si + 4HCl$).

- **Polycrystalline Silicon (Poly-Si):** Deposited via LPCVD using the same silane reactions, but at lower temperatures (575 °C to 650 °C). The final grain structure is heavily dependent on the specific deposition temperature and the doping concentration; notably, if deposited below 570 °C, the silicon forms as a purely amorphous film.
- **Silicon Nitride (Si_3N_4):** Deposited via LPCVD or PE-CVD across a wide temperature range (200 °C to 800 °C). The standard reaction combines silane with ammonia ($3SiH_4 + 4NH_3 \rightarrow Si_3N_4 + 12H_2$).
- **Silicon Dioxide (SiO_2):** The most versatile dielectric, capable of being deposited using nearly all architectures (LPCVD, PE-CVD, HDP-CVD, APCVD) from room temperature up to 800 °C. It can be formed using simple silane and oxygen ($SiH_4 + O_2 \rightarrow SiO_2 + 2H_2$) or via the decomposition of TEOS ($Si(OC_2H_5)_4$), often combined with ozone (O_3).
- **Tungsten (W):** Deposited via LPCVD at relatively low temperatures (250 °C to 500 °C) to form highly conductive metallic contact plugs. The reduction of tungsten hexafluoride is driven either by silane ($2WF_6+3SiH_4 \rightarrow 2W+3SiF_4+6H_2$) or hydrogen gas ($WF_6+3H_2 \rightarrow W + 6HF$).

5.7 Physical Vapour Deposition (PVD): Evaporation

While Chemical Vapour Deposition relies on complex, surface-driven chemical reactions, Physical Vapour Deposition (PVD) takes a much more brute-force approach. PVD techniques generate a vapour of the desired film material through purely physical means, transport it across a vacuum, and allow it to condense onto the wafer surface. The simplest and oldest of these techniques is thermal evaporation.

5.7.1 The Evaporation Mechanism and Langmuir-Knudsen Theory

In a standard evaporation system, the solid source material is placed inside a crucible and heated in a high-vacuum chamber. Heating is typically achieved either by passing a massive current through a tungsten filament or by bombarding the source with a highly focused electron beam (E-beam). Once the material absorbs enough thermal energy, it melts and evaporates (or directly sublimates) into a gas.

Because the chamber (shown in Figure 5.9) is held at an extremely low pressure, the mean free path of the evaporated atoms is significantly longer than the dimensions of the chamber. Consequently, the atoms travel from the crucible to the wafer in straight lines without undergoing any gas-phase collisions. This creates a strict line-of-sight deposition.

The fundamental rate at which atoms leave the heated source is governed by the Langmuir-Knudsen theory. The molecular evaporation rate (Φ) can be expressed as:

$$\Phi = 3.513 \times 10^{22} \frac{P_e}{\sqrt{mT}}$$

where m is the gram-molecular mass of the source material, T is the absolute temperature in kelvin, and P_e is the vapour pressure of the material at that temperature in Torr. The flux is measured in number of particles per unit area per unit time ($cm^{-2} \cdot s^{-1}$).

To achieve a commercially viable deposition rate, the vapour pressure P_e must typically reach between 1 and 10 mTorr. Because vapour pressure depends exponentially on temperature, reaching this threshold is relatively easy for metals like aluminium or gold, but exceedingly difficult for refractory metals like tungsten (W) or molybdenum (Mo), which require immense temperatures to evaporate effectively.

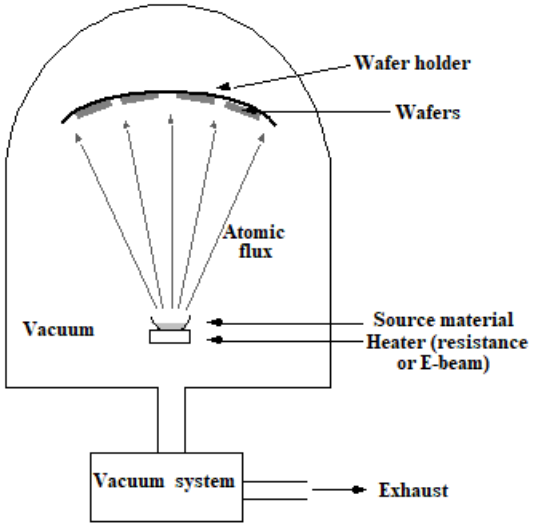
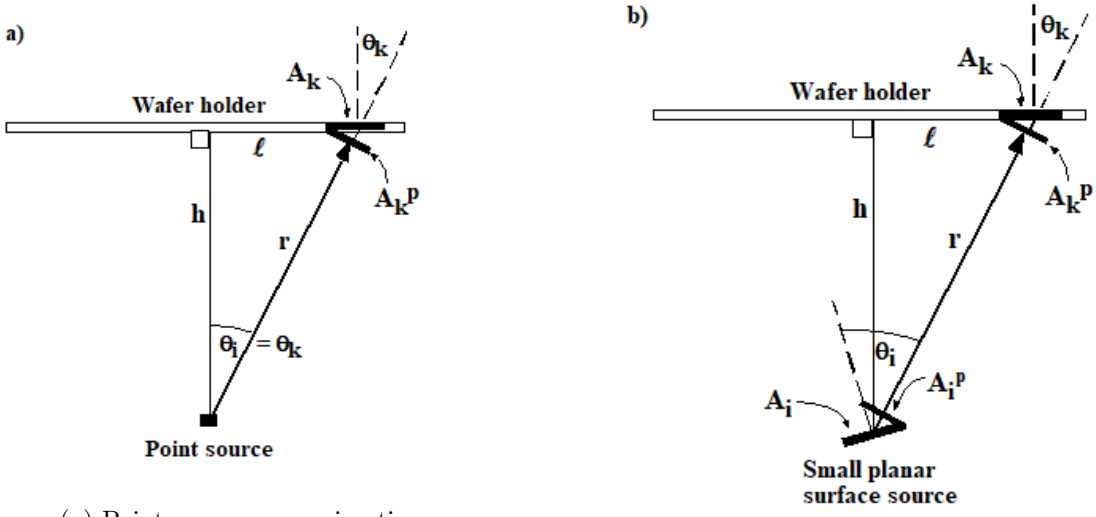


Figure 5.9: This scheme shows the chamber of a Thermal PVD process. The wafers are distributed along a curved surface to increase uniformity in the deposition, as will be discussed later.

5.7.2 Deposition Rate and Geometric Approximations

Because evaporation is a line-of-sight process, the deposition rate across the wafer is strictly a function of the source geometry and the reactor's physical layout. We can model the flux of atoms arriving at the wafer surface using two idealised source geometries.



(a) Point source approximation

(b) Surface source

Figure 5.10: a) The scheme shows the geometry in the case of a point-like source. Here, only the target surface must be projected to recover the deposition rate. b) With the proper geometry, one must account for the projected surface of the source as well.

1. The Point Source Approximation:
If the crucible is incredibly small, it can be modelled as a point source that emits atoms uniformly in all directions, as shown in Figure 5.10a. The emitted flux radiates outward across a solid angle Ω (where $\Omega = 4\pi$ for emission in all directions, or 2π if baffled to emit strictly upwards). If a wafer is positioned at a distance r from the source, and its surface is tilted at an incidence

angle θ_k relative to the incoming atomic flow, the deposition rate (v) is given by:

$$v = \frac{R_{evap}}{\Omega r^2} \frac{\cos\theta_k}{N}$$

where R_{evap} is the total evaporation rate and N is the atomic density of the deposited material.

2. The Small Planar Surface Approximation:

Real crucibles are not infinitely small points; they possess a flat surface area, as is represented in Figure 5.10b. A small planar surface does not emit atoms isotropically; rather, it emits following a cosine distribution, acting as a perfectly diffused source. To account for this, the arrival flux must include the emission angle θ_i (the angle relative to the normal of the source surface). The deposition rate is modified by a $2\cos\theta_i$ term (where the factor of 2 accounts for the normalisation of the total evaporation rate over the half-sphere):

$$v = \frac{R_{evap}}{\Omega r^2} \frac{2\cos\theta_i \cos\theta_k}{N}$$

For highly directional effusion cells, more realistic, narrow angular distributions (pictured in Figure 5.11) can be modelled using higher-power cosine functions ($\cos^n\theta_i$, where $n > 1$).

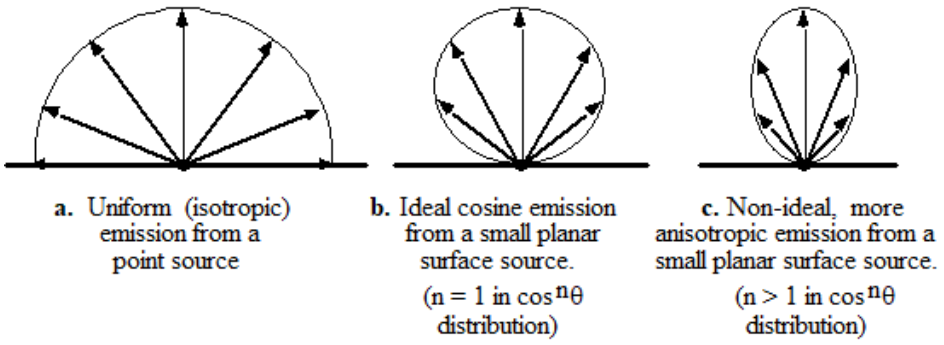


Figure 5.11: The drawings show more realistic flow shapes depending on the cells used and evaporation parameters.

Process Insight

While evaporation is conceptually simple, produces very low wafer damage, and can vaporise almost any single element, it suffers from severe physical limitations that have largely relegated it to a research technique rather than a mainstream manufacturing tool for modern ICs.

- **Non-Uniformity:** As dictated by the $\frac{1}{r^2}$ dependence and the cosine angles in the flux equations, a flat wafer placed above a crucible will grow film significantly faster in the centre than at the edges. To combat this, wafers must be mounted in spherical or hemispherical "planetary" domes that physically rotate the wafers through the vapour cloud to average out the thickness.
- **Alloy Deposition:** Evaporating an alloy (like Al-Cu) from a single crucible is a thermodynamic nightmare. Because aluminium and copper possess vastly different vapour pressures at a given temperature, the element with the higher vapour pressure will evaporate much faster, severely altering the stoichiometry of the final film. Controlling this requires multiple independent crucibles and heating systems, compounding the uniformity problems.
- **Terrible Step Coverage:** Evaporated atoms arrive with relatively low kinetic energy. Upon striking the wafer, they instantly stick exactly where they land (a sticking coefficient, $S_c \approx 1$) with virtually no surface mobility or re-sputtering. Combined with the strict line-of-sight arrival, this can lead to catastrophic "shadowing" effects and extremely poor step coverage over highly topographical features.

5.7.3 Molecular Beam Epitaxy (MBE)

It is worth noting one highly specialised variant of evaporation: Molecular Beam Epitaxy (MBE). Operating under ultra-high vacuum conditions, MBE uses precisely controlled "effusion cells", known as Knudsen Effusion Cells, to evaporate highly pure elemental beams.

If the silicon substrate is heated to provide thermal energy, the arriving atoms have enough surface mobility to align with the underlying crystal lattice, allowing for the growth of perfect epitaxial layers. Because the temperature of the effusion cells flawlessly controls the relative evaporation rates, MBE can build complex semiconductor alloys literally atomic layer by atomic layer. However, because the deposition rate is extraordinarily slow (approximately 1 atomic layer per second), MBE remains widely used in university research and optoelectronics, but is generally far too slow for high-volume CMOS manufacturing.

5.8 Sputter Deposition

To overcome the severe thermodynamic limitations of evaporation, specifically the inability to deposit alloys uniformly and the awful step coverage, the semiconductor industry relies on Sputter Deposition. Instead of using heat to gently boil atoms off a source, sputtering uses a high-energy plasma to physically blast atoms off a solid target using brute-force momentum transfer.

Sputtering operates at significantly higher pressures than evaporation, typically ranging from 1 to 100 mTorr. In a standard sputtering chamber, shown in Figure 5.12, a plasma is ignited in an inert gas, which is almost universally argon. The heavy argon ions (Ar^+) are accelerated across a high-voltage sheath and slammed into the solid source material, known as the "target". The immense kinetic impact physically dislodges (sputters) the target atoms, sending them flying across the chamber to condense on the wafer.

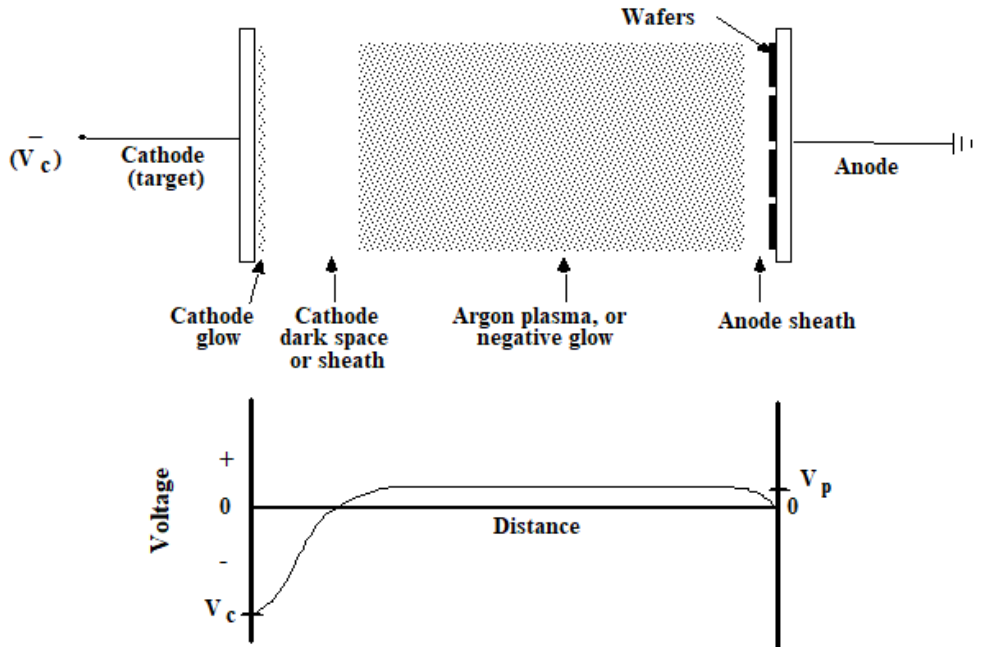


Figure 5.12: The chamber for a sputtering deposition process looks just like a PECVD chamber. An advantage of this setup is that polarisation can be inverted to instead do a process called "ion bombardment" on the wafer, thus cleaning it.

5.8.1 DC Sputtering and Arrival Dynamics

In a basic Direct Current (DC) sputtering system, a net DC current flows through the circuit. The target acts as the cathode, where the current is carried by the incoming positive ion flux. The wafer acts as the anode, where the current is mainly carried by electrons; due to the Maxwellian distribution of the electron energies, only a very small drop in the plasma potential is required to sustain this anode current.

When an energetic Ar^+ ion strikes the target (for example, an aluminium target), multiple physical processes occur simultaneously:

- The target atom (Al) is sputtered and migrates to the wafer surface. Upon arriving, it can stay adsorbed, migrate to another surface site, or be re-emitted.
- The incoming Ar^+ ion may capture an electron, become neutralized, and inadvertently become incorporated into the growing metallic film as an impurity.
- The kinetic impact forces the target to emit secondary electrons. As these electrons are accelerated away from the cathode, they collide with neutral Argon atoms to sustain the plasma, or they may ionise other impurities travelling to the wafer.

Unlike an idealised point source, sputtering targets are generally massive plates that provide a very wide range of arrival angles for the sputtered atoms. This broad arrival angle distribution is typically modelled mathematically by a $\cos^n \theta$ function. Crucially, just as with evaporation, it is the normal component of this incident flux striking the wafer surface that ultimately determines the local deposition or growth rate. The exact shape and spread of this angular distribution are highly dependent on the physical size and type of the source, the overall geometry of the vacuum system, and the frequency of gas-phase collisions the atoms experience during their transit.

5.8.2 Sputtering Yield and Alloy Deposition

The efficiency of the sputtering process is quantified by the Sputtering Yield (Y), which is defined as the number of sputtered target atoms per incident Argon ion. Y typically ranges between 0.1

and 3, and depends strictly on the target material, the kinetic energy of the incident ion, and the ion's arrival angle.

Crucially, the physical momentum-transfer nature of the sputtering yield provides an elegant solution to the alloy deposition problem that plagues evaporation. If we attempt to sputter an alloy target (e.g., Al-Cu), the two materials will inevitably possess different sputtering yields. At the very beginning of the process, the material with the higher yield is sputtered away much faster. However, this rapid removal leaves the target surface depleted of the high-yield element and enriched with the low-yield element.

Because more of the low-yield atoms are now exposed at the surface, their overall emission rate catches up. In a very short time, the system reaches a steady state where the elements leave the target surface in the exact same ratio as the bulk composition of the alloy. The differing individual yields are perfectly compensated by the altered surface concentrations, allowing alloys to be deposited effortlessly and accurately. This process is called **pre-sputtering**.

Physical Insight

While DC sputtering works well for metals, it fails for dielectrics, which, of course, are poor conductors. Incoming positive argon ions accumulate on the non-conductive target, quickly cancelling the applied cathode voltage and extinguishing the plasma.
Radio Frequency (RF) Sputtering solves this by applying a high-frequency AC voltage. During the positive half-cycle, highly mobile electrons flood the target to build a massive negative charge. The heavy positive ions cannot move fast enough to neutralise this during the negative half-cycle, naturally creating a permanent, negative DC self-bias that sustains continuous Argon bombardment.
However, to prevent symmetrical RF sheaths from violently sputtering the wafer alongside the target, the system must be designed asymmetrically. The voltage drop (V) across a sheath is inversely related to its electrode's surface area (A):

$$\frac{V_1}{V_2} = \left(\frac{A_2}{A_1}\right)^m$$

where m is 4 theoretically, but 1 to 2 experimentally. By connecting the wafer holder to the massive chamber walls, its effective area (A_2) becomes vastly larger than the target (A_1). Consequently, the wafer's voltage drop (V_2) falls to nearly zero, leaving it safely in a low-energy deposition zone.

5.8.3 Advanced Sputtering Processes

To address the increasingly difficult geometries of modern high-aspect-ratio contacts and vias, the basic sputtering architecture has been heavily modified, leading to several advanced PVD techniques:

- **Reactive Sputter Deposition:** A reactive gas is intentionally added to the Argon plasma. The sputtered metal atoms chemically react with this gas on the fly or at the wafer surface. For example, sputtering a pure titanium (Ti) target in a mixed Ar/N_2 plasma effortlessly deposits a titanium nitride (TiN) film.
- **Bias Sputtering:** The wafer chuck is intentionally biased with an independent RF power supply. This creates a controlled sheath above the wafer, allowing a slight, simultaneous sputter-etch during deposition to clean the surface or improve planarisation.
- **Magnetron Sputtering:** Powerful magnets are placed behind the target. The magnetic field traps the secondary electrons, forcing them into tight, spiral trajectories near the

target surface. This massive increase in the electron path length drastically improves the argon ionisation rate, sustaining a much denser plasma at far lower operating pressures.

- **Collimated Sputtering:** To improve gap-filling in very deep, narrow trenches, plates with circular or hexagonal holes (a collimator) are placed directly between the target and the wafer. This physically blocks atoms arriving at oblique angles, filtering the flux into a highly directional, narrow arrival angle distribution.
- **Ionised Sputtering:** An RF coil is wrapped around the plasma region between the target and the wafer. As neutral sputtered atoms pass through this coil, they are stripped of an electron and become positive ions. As they approach the wafer, the vertical electric field of the wafer sheath accelerates these heavy ions straight down, providing an extremely narrow, perfectly vertical arrival angle that easily coats the bottoms of very deep contact holes.

5.9 Topographical Modelling of Deposition

To accurately predict the final geometry of a deposited film on complex 3D structures, process engineers rely on advanced computer simulations. Instead of calculating the computationally impossible trajectory of every single gas species from the source through the entire reactor, these programs simplify the physics. They calculate the reactant flow at each point on the wafer surface and mathematically describe every local physical process.

If we were to write out the total flux, considering all phenomena that contribute to it (which are schematically shown in Figure 5.13), we would get:

$$\vec{F}^i_{net} = \vec{F}^i_{direct(neutrals)} + \vec{F}^i_{direct(ions)} + \vec{F}^i_{redep.} + \vec{F}^i_{diff.in} + \vec{F}^i_{emitted} + \vec{F}^i_{sputtered} + \vec{F}^i_{diff.out}$$

where:

- $\vec{F}^i_{direct(neutrals)}$ is the flux given by the direct deposition of neutral species.
- $\vec{F}^i_{direct(ions)}$ is the flux given by the direct deposition of ionic species.
- $\vec{F}^i_{redep.}$ is the flux given by the redeposition effect.
- $\vec{F}^i_{diff.in}$ is the flux given by species migrating in via diffusion on the surface.
- $\vec{F}^i_{emitted}$ is the flux given by emitted species (desorbed) from the surface.
- $\vec{F}^i_{sputtered}$ is the flux given by sputtered atoms.
- $\vec{F}^i_{diff.out}$ is the flux given by species migrating out via diffusion on the surface.

Obviously, the last three fluxes bring a negative contribution to the total flux compared to the other ones. The superscript i refers to a specific local site on the surface.

5.9.1 Arrival Flux and Sticking Coefficients

Rather than tracking full trajectories, simulators account for the angular distribution of the flux arriving at the wafer surface. Because film growth requires atoms to physically bond to the surface, the simulation is primarily interested in the component of the flux normal to the local surface orientation ($\hat{n}$):

$$F_{normal} = \vec{F}_{incident} \cdot \hat{n}$$

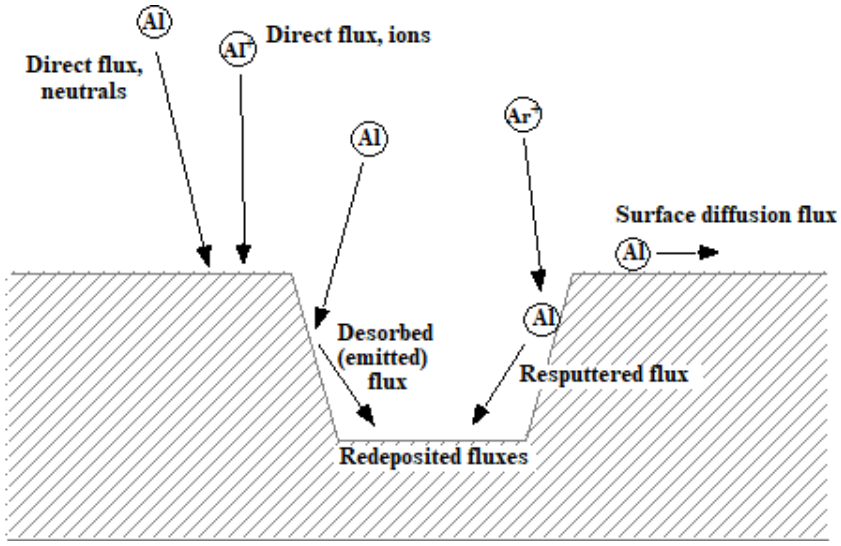


Figure 5.13: In this picture we can see all fluxes contributing to the deposition process. All these phenomena happen at the same time and at different intensities.

where $\vec{F}_{incident} = \vec{F}_{direct(neutrals)} + \vec{F}_{direct(ions)}$.

A critical parameter governing how this flux behaves upon impact is the sticking coefficient (S_c), given by $S_c = \frac{F_{reacted}}{F_{incident}}$. This means the fraction of emitted particles is $F^i_{emitted} = (1 - S_c)F^i_{incident}$. Depending on the sticking coefficient, we get different behaviours:

- **High Sticking Coefficient ($S_c = 1$):** Typical of PVD processes. Atoms stick exactly where they land on their first impact. This leads to heavy shadowing effects and poor step coverage.
- **Low Sticking Coefficient ($S_c < 1$):** Typical of CVD processes. Atoms bounce off the surface, allowing reactant molecules to ricochet deep into narrow trenches, or they bind to a precursor state in which they are free to diffuse on the surface.

5.9.2 Redeposition Mechanics

When $S_c < 1$, or when atoms are physically knocked off the surface during sputtering, simulators must calculate the redeposition flow. The redeposition flow arriving at a specific point i is calculated by taking into account the emission flow from all other points k inside the trench. This is governed by a geometrical factor (g_{ik}) that mathematically accounts for the distance and relative orientation between the points:

$$F^i_{redep} = \sum_k F_k g_{ik}$$

In these models, the initial emission flux from a surface point is generally considered to be isotropic, following a standard cosine distribution.

5.9.3 Diffusion mechanics

If high temperatures are used during the deposition, the arriving atoms possess significant lateral mobility, and surface diffusion must be taken into account. The simulator mathematically defines the physical evolution of the surface using a differential equation governed by specific thermodynamic variables:

$$v_{surface} \propto D_S \gamma_s \Omega \nu \frac{\partial^2 K}{\partial S^2}$$

where:

- D_S is the surface diffusivity.
- γ_s is the per area surface energy.
- Ω is the atomic or molecular volume.
- ν is the number of atoms or molecules per unit area.
- K is the local physical curvature.
- S is the unit length across the surface.

This surface diffusion physically smooths out sharp corners, directly aiding in the planarisation of the deposited film.

5.9.4 Process-Specific Modelling

By utilising the equation for the total flux, simulators can accurately model specific deposition technologies by keeping only the flux terms that correspond to the dominant physics of the chosen process.

1. LPCVD Modelling:

In Low-Pressure CVD, the deposition is driven purely by the thermal chemical reactions of neutral gas species. Because there is no plasma, the flux from ions and the effects of sputtering are entirely absent. Assuming a low-temperature process with negligible surface diffusion, the master equation simplifies to:

$$F^i_{net} = F^i_{direct(neutrals)} + F^i_{redep.} - F^i_{emitted}$$

We can rewrite $F^i_{emitted}$ using the sticking coefficient:

$$F^i_{net} = F^i_{direct(neutrals)} + F^i_{redep.} - (1 - S_c)\left(F^i_{direct(neutrals)} + F^i_{redep.}\right)$$

By using the equation we wrote for $F^i_{redep.}$ in Chapter 5.9.2:

$$F^i_{net} = S_c\left(F^i_{direct(neutrals)} + F^i_{redep.}\right) = S_c\left(F^i_{direct(neutrals)} + g^{ik}(1 - S_c)F^k\right) = S_cF_D$$

Which means:

$$rate = \frac{S_cF_D}{N}$$

where F_D is the direct incident flux of neutral precursor species arriving on the wafer and N is the deposited material's atomic/molecular density.

2. PECVD Modelling:

Plasma-Enhanced CVD introduces both chemical radicals and energetic ions, meaning both neutral and ionic fluxes must be accounted for. To accurately model the dual nature of the plasma, we must account for the ion-induced reactions flux. The model is formulated as:

$$F^i_{net} = F^i_{direct(neutrals)} + F^i_{redep.} - F^i_{emitted} + F^i_{ion-induced}$$

By using a similar approach as before, and introducing K_D and K_i as rate constants for the non-ion-enhanced and ion-enhanced terms, we get:

$$rate = \frac{K_DS_cF_D + K_iF_i}{N}$$

where F_i is the ionic counterpart to F_D.

3. Standard PVD Modelling:

In standard physical sputtering, arriving ions do not play a significant chemical role at the wafer surface, meaning the basic mathematical framework is essentially the same as in LPCVD systems (though it utilizes a highly directional angular flux distribution and a high sticking coefficient, S_c). Meaning:

$$rate = \frac{S_c F_D}{N}$$

However, if high temperatures (T) are used during the deposition, surface diffusion is not negligible, thus yielding:

$$rate = \frac{S_c F_D + \frac{D_s}{k_B T}\gamma_s \Omega \nu \frac{\partial^2 K}{\partial s^2}}{N}$$

4. Ionised-PVD Modelling:

In Ionised-PVD, an RF coil deliberately ionizes the sputtered metal atoms before they reach the wafer, creating a massive influx of charged particles. The direct ion flux term is reactivated, and because these heavy ions are accelerated vertically by the wafer sheath, they may cause simultaneous sputter-etching upon impact. The resulting equation is:

$$F^i_{net} = F^i_{direct(neutrals)} + F^i_{direct(ions)} + F^i_{redep.} - F^i_{emitted} - F^i_{sputtered}$$

In this case the rate is:

$$rate = \frac{S_c F_d + F_i - K_{sp} Y F_i + K_{rd} F_{rd}}{N}$$

where K_{sp} is the sputtering rate constant, Y is the sputtering yield, K_{rd} is the redeposition rate constant and F_{rd} is the redeposition flux.

Having explored the diverse techniques of Thin Film Deposition, we now possess the methods to build the critical structural layers of an integrated circuit. By manipulating temperature, pressure, and fluid dynamics, Chemical Vapour Deposition (CVD) allows us to grow highly conformal dielectrics and semiconductors perfectly tailored to intricate 3D topographies. Conversely, Physical Vapour Deposition (PVD) harnesses the violent kinetic energy of plasmas to sputter complex metal alloys, ensuring the robust conductive wiring required to tie billions of transistors together.

However, whether deposited chemically or physically, these thin films are initially laid down as continuous, unbroken blankets across the entire wafer surface. To transform these uniform layers into functional, isolated components, such as distinct transistor gates, individual contact plugs, and discrete copper wires, we must carve them into precise nanoscale geometries.

The next chapter will delve into the absolute cornerstone of semiconductor scaling: **Photolithography and Pattern Definition**. We will explore the complex optical systems, precision masks, and photosensitive chemistries used to project nanoscale blueprints onto the wafer, granting us the ability to selectively etch and sculpt the films we have so carefully deposited.

Chapter 6

Photolithography

6.1 Introduction to Lithography

If thin film deposition and ion implantation provide the raw materials of an integrated circuit, lithography provides the blueprints. Photolithography is the fundamental process used to define the incredibly precise, nanoscale geometries of modern semiconductor devices.

The lithographic cycle is the most frequently repeated sequence in the fab. A typical process flow involves depositing a structural film (conductive or insulating), coating it with a photosensitive polymer (photoresist), and using lithography to pattern the resist. This resist pattern acts as a protective mask for the subsequent etching or ion implantation step, after which the remaining resist is stripped away so the cycle can begin anew.

There exist different lithographic techniques, each with different sources and operation methods. Optical exposure methods may use UV, DUV, EUV or X-ray radiation or charged particles, such as in Electron Beam Lithography (EBL) or Focused Ion Beam lithography (FIB). These are all paired with resists. Resistless methods also exist, such as using probes (Atomic Force Microscopes, Scanning Tunneling Microscopes, Scanning Near-field Optical Microscopes). Nevertheless, optical lithography is the most used in standard IC manufacturing.

At its core, optical lithography (Figure 6.1) is remarkably similar to traditional photographic printing. A light source is projected through a master template (the mask), and the image is captured in a photosensitive layer coating a prepared silicon wafer. The discipline is broadly divided into two main fields:

- **Optical Imaging:** The physics of manipulating light to create an intensity modulation (an aerial image) at the wafer level.
- **Chemical Processes:** The material science of how the photoresist absorbs that radiation and undergoes chemical changes to become soluble or insoluble.

6.2 Illumination Sources and Photomasks

To resolve ever-smaller features, the industry has relentlessly pursued shorter and shorter optical wavelengths, governed by the fundamental limits of diffraction. The reason for this can be found in the resolution equations for the employed optical systems; resolution is always directly proportional to the wavelength used in the lithographic process. The smaller the wavelength, the smaller the features can be printed on the wafer, the more circuits can be fit on a chip, the more Moore's law can be followed.

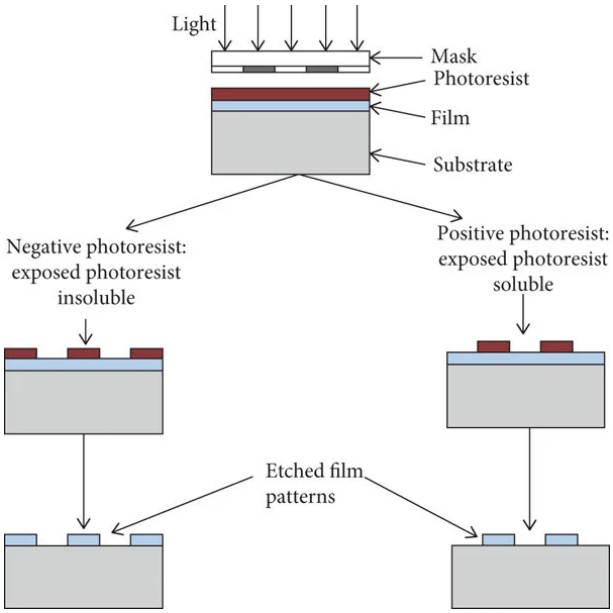


Figure 6.1: This picture shows one cycle of the photolithographic process. Positive and negative refer to the tone of photoresists; this will be expanded upon in Chapter 6.4. After etching the pattern in the film (light blue layer), a process known as "resist stripping" removes the excess resist that remains on the patterned features. Image taken from ResearchGate.

6.2.1 Illumination Sources

Early lithography systems relied on the discharge of high-pressure mercury (Hg) vapour arc lamps. By applying several kilovolts across electrodes, the mercury gas ionises, emitting a spectrum with strong peaks at specific ultraviolet wavelengths. Systems typically filter out unwanted emission to isolate a single spectral line, most notably the **g-line (435 nm)** and the **i-line (365 nm)** highlighted in Figure 6.2.

However, below 300 nm, mercury lamps no longer provide adequate intensity. To drive modern deep-ultraviolet (DUV) lithography, the industry shifted to Excimer Lasers due to their massive power output and extreme spectral purity (bandwidths of approximately 0.2 pm). "Excimer" stands for "Excited Dimer", that is a molecule formed by two elements (like a noble gas and a halogen) that only bond when excited. When the molecule naturally breaks apart and returns to its ground state, it emits a high-energy photon. By pulsing these lasers at several kilohertz, fabs generate the standard DUV wavelengths: **KrF (248 nm)**, **ArF (193 nm)**, and theoretically $\mathbf{F_2}$ **(157 nm)**.

6.2.2 Photomasks (Reticles)

The photomask, or reticle, is the master blueprint of the circuit layer. It consists of a highly polished, incredibly flat piece of quartz glass (flatness $\approx 0.5\,\mu m$) coated on one side with a thin, patterned layer of opaque chrome. An example of one is shown in Figure 6.3. The quartz provides 100% transmission to DUV light, while the chrome provides 0% transmission, creating a high-contrast binary image. The image is a negative of the circuit to be printed.

Modern masks are completely defect-free reduction reticles, meaning the pattern on the mask is artificially enlarged, typically by a factor of 5X for older i-line systems, and 4X for current DUV systems. Manufacturing these reticles is exceptionally difficult and mirrors the semiconductor process flow itself, requiring specialised e-beam or laser direct-writing tools to precisely etch the chrome layer.

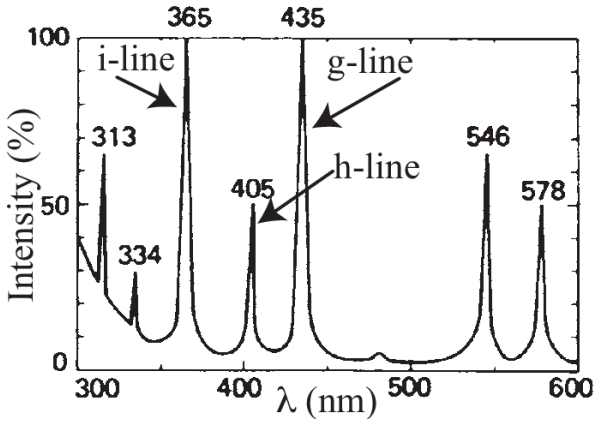


Figure 6.2: This picture shows the emission spectrum of a mercury lamp. The i-line and g-line are the most intense wavelengths. Image taken from ResearchGate.

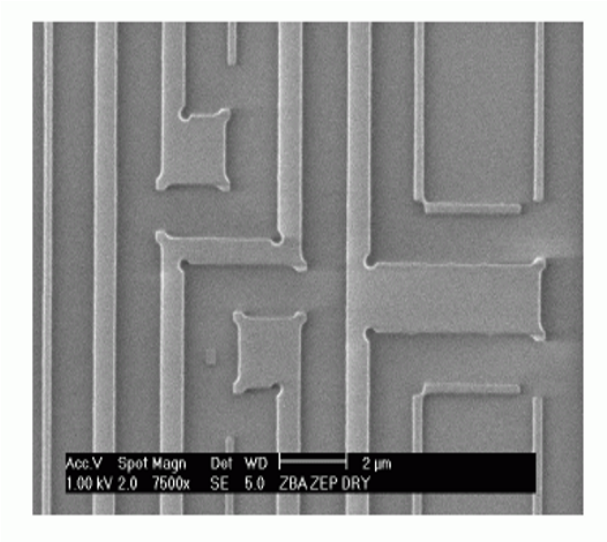


Figure 6.3: The image shows a top-down SEM picture of a photomask from the Cr side.

6.3 Wafer Exposure Systems

Once the light is generated and passed through the reticle, it must be projected onto the wafer. The architecture of the exposure tool dictates both the ultimate resolution and the defectivity of the process. The three main techniques are Contact Printing, Proximity Printing and Projection Printing. A scheme for each is shown in Figure 6.4.

6.3.1 Contact and Proximity Printing

The oldest and simplest exposure tools are 1:1 (lens-less) exposure systems:

- **Contact Printing:** The mask is brought into direct physical contact with the photoresist layer on the wafer. Because there is no gap, diffraction effects are practically eliminated, yielding extraordinarily high resolution (well below 100 nm). However, the hard physical contact inevitably damages both the mask and the resist, generating catastrophic defect densities that render the technique useless for high-volume manufacturing.
- **Proximity Printing:** To eliminate contact defects, a small gap (g) of 5 to 25 µm is introduced between the mask and the wafer. While this saves the mask, the separation

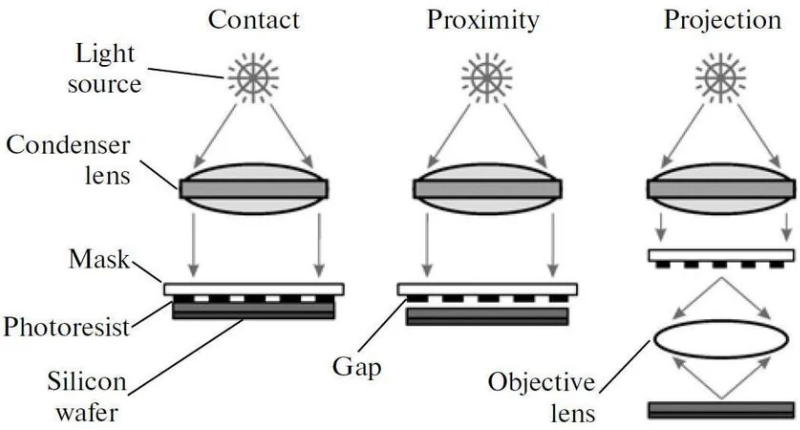


Figure 6.4: This picture shows the three structures for the main optical lithography techniques. In contact printing, as the name suggests, there is no gap between the mask and the resist, whereas proximity printing and projection printing have some distance from the target. This means phenomena such as diffraction are able to occur. Image taken from ResearchGate.

thrusts the imaging into the Near-Field (Fresnel) diffraction regime, severely degrading resolution. The minimum resolvable feature (R_{min}) is approximated by:

$$R_{min} \approx k\sqrt{\lambda g}$$

where λ is the wavelength and k is a process constant (typically ≈ 1). For an i-line source ($\lambda = 365$ nm) and a 20 µm gap, the resolution limit is roughly 2.7 µm, which is far too large for modern transistors. Furthermore, both contact and proximity systems require 1X masks, which are incredibly difficult to manufacture without defects.

6.3.2 Projection Printing

To achieve high resolution without the defectivity of contact printing, the industry universally adopted Projection Printing. In these systems, a complex objective lens (often containing tens of distinct optical elements) is placed between the mask and the wafer to capture the Far-Field (Fraunhofer) diffraction pattern and focus the demagnified 4X or 5X image onto the resist. A comparison of the intensity profiles between the three techniques can be seen in Figure 6.5.

The fundamental drawback of this architecture is that the lens has a finite physical size; it cannot capture the diffraction orders that scatter at very wide angles (Figure 6.6). Because higher spatial frequencies scatter wider, critical high-frequency information is lost, creating the absolute resolution limit of the optical system.

To quantify the information lost, we can simply use Rayleigh's criterion for resolution (which will be seen in more detail in Chapter 6.9.4). The optical system will have a certain Numerical Aperture (NA). With it, the resolution limit is given by:

$$R = k\frac{\lambda}{NA}$$

where k is again a process constant (which in this case can assume values < 1), while $NA = n \cdot \sin\alpha$ contains information on the refractive index of the medium (n) and the maximum angular span of the lens (the physical meaning behind this is explained in Chapter 6.7.3).

6.3.3 Steppers and Scanners

Because it is impossible to engineer a single lens capable of projecting a perfect, distortion-free image across the entire 300 mm diameter of a modern wafer, projection systems expose the wafer in small segments (rectangular fields in which it is divided, kind of like a grid).

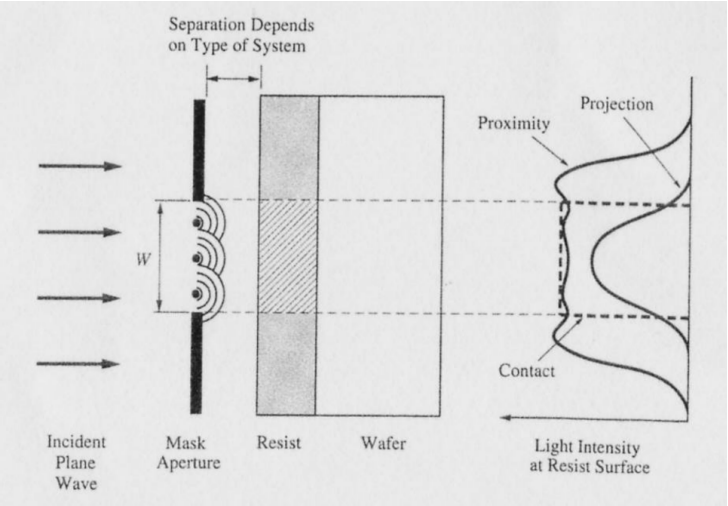


Figure 6.5: The effects of diffraction can be seen in the intensity at the resist surface for the three techniques. In contact printing, the profile is, of course, rectangular. In proximity and projection printing the wave effects of light become evident.

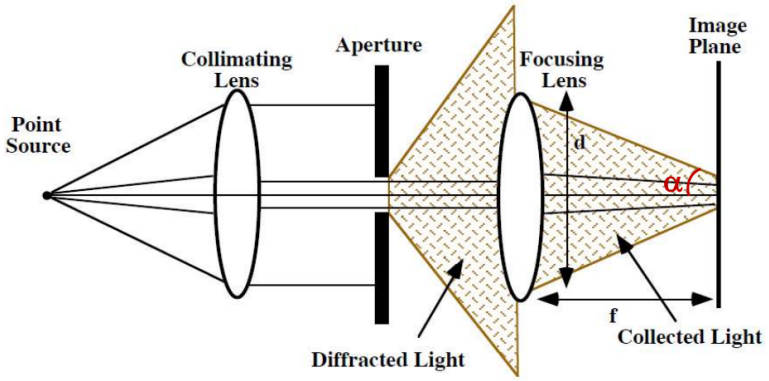


Figure 6.6: The picture traces the path light takes in an optical system like the ones used in Projection Printing. The focusing lens can't "catch" all incoming diffracted light due to being of finite size, thus it intrinsically loses information. Image taken from Shanghaitech.

- **Step and Repeat (Stepper):** The optical system exposes a full reticle field (e.g., 21 × 21 mm) in a single flash. The wafer stage then physically moves ("steps") to the next adjacent field, and the flash is repeated.
- **Step and Scan (Scanner):** The modern evolution of the stepper. Instead of illuminating the entire mask at once, a scanner illuminates only a narrow rectangular slit (e.g., 26 × 8 mm). To expose the full field, the reticle stage and the wafer stage physically move in perfect synchronisation to sweep the slit across the entire pattern.

Process Insight

Moving from a Stepper to a Scanner was an optical necessity. By only using a narrow slit in the absolute centre of the objective lens, engineers eliminated the severe optical aberrations that plague the outer edges of a lens, resulting in vastly superior dimensional control. It also allowed for physically wider exposure fields (e.g., 26×33 mm), dramatically increasing throughput.
However, this requires staggering mechanical precision. Because the optics demagnify the image by 4X, the reticle must physically move four times faster than the wafer to keep the projected image stationary on the resist. State-of-the-art wafer stages move at incredibly high velocities (250 to 500 $\text{mm} \cdot s^{-1}$) and accelerate at 0.5G (4.9 $\text{m} \cdot s^{-2}$). Consequently, the reticle stage above it must move at blistering speeds of 1000 to 2000 $\text{mm} \cdot s^{-1}$ while maintaining nanometre-level synchronisation.

6.3.4 The Pellicle

Because projection printing relies on absolute focal precision, any dust particle landing on the reticle would be projected onto the wafer, destroying the circuit. To prevent this, reticles are protected by a **pellicle**, which is a remarkably thin, highly transparent membrane (often made of nitrocellulose or Mylar) stretched tightly over a metal frame mounted 5 to 10 mm above the chrome side of the mask. By sealing the volume immediately above the mask, any airborne dust particle is forced to land on the pellicle rather than the reticle itself. Because the pellicle is positioned completely outside the optical depth of focus, the shadow of the particle is severely blurred and does not print on the wafer.

6.4 The Photoresist Chemistry

While the optical system projects the aerial image, it is the photoresist that must capture it. Photoresist materials are complex, photosensitive chemical formulations designed to dramatically change their physical properties upon exposure to incident photons.

Resists are fundamentally classified into two categories based on their response to light:

- **Positive Resist:** The regions exposed to light undergo a chemical change that makes them *soluble* in the developer solution[20]. The unexposed regions remain insoluble.
- **Negative Resist:** The regions exposed to light undergo a chemical cross-linking that makes them *insoluble* in the developer. The unexposed regions are washed away.

Regardless of tone, a production-worthy resist must possess three main characteristics: high sensitivity to incident energy (measured in $\text{mJ} \cdot \text{cm}^{-2}$, to be able to induce the chemical reactions needed), excellent pattern fidelity and resolution, and robust physical resistance to subsequent destructive processes like etching and ion implantation.

Photoresists are made of three main components:

1. **Inactive resin:** a hydrocarbon polymer matrix that provides the physical structure and etch resistance of the resist.
2. **Casting solvent:** a liquid vehicle used to adjust the viscosity of the resist. This allows it to be spun onto the wafer: the resist is deposited on the wafer, which is in turn spun at high rpm to evenly distribute the resist via the centrifugal force.

[20]The developer solution is what is used to "wash away" the resist after it has been exposed. For positive resists solvents are usually inorganic, while for negative resists they are organic.

3. **Photo-Active Compound (PAC):** the component which actually undergoes the chemical reactions. In the case of positive resists, this induces Wolff rearrangement (like in DNQ, see Chapter 6.4.1) or chain scission (like in PMMA) in the matrix, while in the case of negative resists it induces a cross-linking process (the long polymer chains form a 3D molecular network) which renders them insoluble.

After spin coating the wafer, the resist contains between 20% and 40% of casting solvent as well as built-in stress. A **Prebake** at $75 - 100\,^{\circ}\mathrm{C}$ is performed to remove the casting solvent, eliminate mechanical stress and promote adhesion to the wafer.

Usually, Prebake happens on a hot plate, which is a fast and easily controllable technique. For higher throughput, heating can also happen in a convection oven or with IR radiation.

6.4.1 g-line and i-line Chemistry (DQN)

For older g-line and i-line lithography (436 nm and 365 nm), the industry standard is the DQN positive resist system. In this system, **Novolac (N)** provides the polymer matrix. On its own, Novolac is naturally soluble in aqueous basic developers. As for the photo-active compound, **Diazoquinone (DQ)** is used. When mixed with Novolac, its hydrophobic behaviour severely retards the resin's natural dissolution rate.

When the DQN resist is exposed to ultraviolet light, the Diazoquinone molecules undergo a chemical transformation known as the **Wolff Rearrangement**. The incident photon provides enough energy to break the weakly bonded N_2 molecule off the PAC. To stabilise itself, the molecule rearranges its carbon ring and bonds with ambient water to form a carboxylic acid. Because this newly formed carboxylic acid is highly soluble in basic developer, the previously inhibited Novolac resin can rapidly dissolve in the exposed areas.

6.4.2 Deep UV (DUV) Chemistry and Chemical Amplification

As the industry shifted to Deep UV wavelengths (248 nm and below) to improve resolution, traditional DQN resists completely failed for two critical reasons:

- **Resist Absorption:** DQN molecules strongly absorb photons below 365 nm. This means the top of the resist acts as a completely opaque blanket, preventing the DUV light from penetrating through the full thickness of the film.
- **Resist Sensitivity:** The intensity of early DUV light sources was very low compared to mercury arc lamps. Because the Wolff Rearrangement requires exactly one photon to convert exactly one PAC molecule, DQN resists were far too insensitive, requiring commercially unviable exposure times.

To solve this, chemists invented **Chemically Amplified Resists (CAR)**. In a positive CAR, the Novolac resin is replaced by a polymer that is deliberately rendered insoluble by "protecting groups" bonded to its side chains. The PAC is replaced by a **Photo-Acid Generator (PAG)**.

When struck by a DUV photon, the PAG molecule does not directly make the resist soluble; instead, it generates a single molecule of acid. During a mandatory subsequent heating step called the **Post Exposure Bake (PEB)**, thermal energy mobilises this acid. The acid attacks the protecting groups on the polymer, cleaving them off to render the polymer soluble. Crucially, this reaction is *catalytic*, meaning the acid molecule is completely regenerated after the cleavage and is free to attack tens or hundreds of other protecting groups.

Because one photon can initiate hundreds of deprotection reactions, CAR sensitivity is massively amplified. While standard i-line resists require doses of 100 to 150 $\mathrm{mJ \cdot cm^{-2}}$, DUV CAR resists require a mere 15 to 40 $\mathrm{mJ \cdot cm^{-2}}$. Because the chemical amplification relies entirely on

thermal mobility, PEB temperature and time are arguably the most crucial control parameters in modern lithography.

Process Insight

Because the entire DUV imaging process relies on trace amounts of catalytically generated H^+ acid, Chemically Amplified Resists are incredibly susceptible to chemical poisoning. Airborne alkaline contaminants in the cleanroom, specifically amines, can easily diffuse into the top few nanometres of the resist film. These basic amines instantly neutralise the photo-generated acid molecules near the surface. During the PEB, the polymer deep in the bulk of the film is successfully deprotected, but the neutralised surface layer remains stubbornly insoluble. Upon development, this produces a horrific, flared profile known as "T-topping" (or in cases of high contamination, completely un-cleared spaces). To combat this, modern DUV tracks employ extensive chemical filtration systems to scrub amines from the ambient air down to the parts-per-billion level.

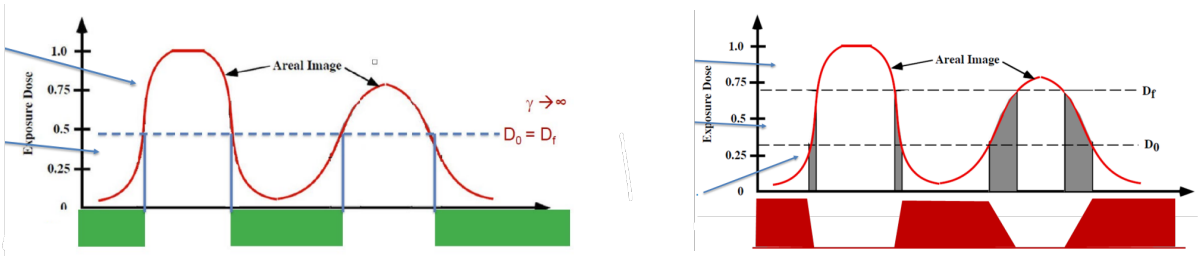
6.4.3 Resist Contrast

The performance of a resist is mathematically quantified by its contrast (γ), which measures the resist's ability to sharply distinguish between the bright and dark areas of an aerial image. This is experimentally determined by plotting the fraction of resist remaining after development against the logarithm of the exposure dose. Contrast is defined as:

$$\gamma = \frac{1}{\log \frac{D_f}{D_0}}$$

where D_0 is the dose at which the exposure first begins to have an effect (thickness begins to drop), while D_f is the dose at which the exposure is complete (the resist clears entirely).

The ideal response is given by $\gamma \to \infty$, which means $D_f = D_0$. In this case, the resist is able to reproduce sharp, vertical walls (as can be seen in Figure 6.7a). Of course, in the real world, $D_f \neq D_0$. Therefore, the exposure will start to have an effect even as the dose is rising, creating slanted walls (as can be seen in Figure 6.7b). Thus, the higher the contrast, the better.



(a) Ideal contrast response

(b) Non ideal contrast response

Figure 6.7: a) This graph shows what the exposure response looks like in the resist in the ideal case (dose vs spatial coordinate). Since $D_f = D_0$, the chemical reactions immediately stop when the dose is below a certain threshold. The green boxes at the bottom represent the features dug in the resist. b) Here, since $D_f \neq D_0$, the chemical reactions gradually stop as the dose goes from D_f to D_0. This means the resist in that interval is only partially cleared, as shown by the red shapes at the bottom. Images taken from the notes of the Micro and Nano Fabrication Technology course by Professor Petti at Polimi.

How the resist responds to "grey areas" (the blurred edges of an aerial image caused by diffraction) is fundamental. Standard g-line and i-line resists have a contrast of approximately 2 to 3. Advanced DUV CAR resists boast extremely high contrasts of 5 to 10. This exceptionally

steep chemical response acts as a threshold, allowing even slightly blurry, intermediate-quality aerial images to produce nearly perfectly vertical, high-quality resist profiles.

6.5 The Lithographic Process Flow

In a modern fab, the lithography cell consists of two massive, integrated tools: the Exposure Tool (the Stepper or Scanner) and the Track. The Track handles all the chemical and thermal preparation of the resist before exposure, and the subsequent development and baking after exposure.

6.5.1 The Track Processing Steps

The standard photolithographic process flow on a silicon wafer proceeds through seven strictly controlled steps:

1. **Dehydration Bake and HMDS Application:** The bare wafer is baked at approximately 150 °C for 60 to 90 seconds to completely drive off any ambient water vapour. Immediately after, Hexamethyldisilazane (HMDS) is applied in vapour form. HMDS acts as an adhesion promoter; one end of the molecule bonds firmly to the inorganic substrate, while the other end bonds chemically with the organic photoresist.

2. **Resist Application (Spin Coating):** The liquid resist (containing the polymer and solvent) is dispensed onto the centre of the wafer. The wafer is then spun at high speeds, typically between 1500 and 5000 RPM for about 30 seconds. The final thickness heavily depends on the spin speed and the resist's viscosity, ranging from 0.1 µm to 5 µm, with an incredibly precise uniformity of $\approx 20 - 100$ Å across the wafer.

3. **Pre-Exposure Bake (Soft Bake - SB):** The wafer is placed on a hot plate at $\approx 100\,^{\circ}\text{C}$ for 60 seconds. This enhances the evaporation of the remaining solvent within the resist and further strengthens the adhesion bonds between the resist, the HMDS, and the substrate.

4. **Alignment and Exposure:** The wafer moves into the Stepper/Scanner. As previously discussed, typical exposure times are 60 to 90 seconds per wafer, utilising doses ranging from $100 - 150\text{mJ} \cdot \text{cm}^{-2}$ for i-line resists down to $15 - 50\text{mJ} \cdot \text{cm}^{-2}$ for DUV resists.

5. **Post-Exposure Bake (PEB)[21]:** The wafer is baked again at $100 - 150\,^{\circ}\text{C}$ for 60 to 90 seconds. As established for Chemically Amplified Resists (CAR), this step is absolutely crucial, as it provides the thermal energy necessary to drive the chemical reaction rates and diffusion of the photo-generated acid.

6. **Development:** The latent image inside the resist is transformed into a physical topography. A "puddle" of basic developer (almost universally Tetra Methyl Ammonium Hydroxide), is placed on the wafer. The developer dissolves the exposed areas (for positive resist), and the process is abruptly stopped by rinsing the wafer with deionised water (H_2O).

7. **Final Bake (Hard Bake):** A final thermal curing step is performed at higher temperatures ($\approx 150\,^{\circ}\text{C}$) to completely harden the resist, drastically improving its mechanical and chemical resistance to the subsequent dry etching or high-energy ion implantation processes.

[21] PEB is often needed for negative resists, is optional for positive resists and it is mandatory for CAR. It helps rearrange overexposed and underexposed areas thanks to the thermal movement of the molecules. More importantly, it reduces the standing wave effect (it smoothens out feature walls which might end up in a wavy pattern due to reflection at the resist/wafer interface of light during lithography).

6.5.2 Thermal Reflow and Optical Challenges

Engineers frequently use the Final Bake to actively modify the final dimensions of the printed features. Using a technique called **Thermal Reflow**, the resist is intentionally heated to its glass transition temperature, causing the polymer to slightly melt and flow. This can successfully shrink critical dimensions, such as reducing 120 nm contact holes down to 100 nm (a 20 nm reduction).

However, the resist must also combat severe optical reflections from the underlying substrate:

- **Notching:** When printing over highly topographical substrates (e.g., pre-existing metal lines), oblique reflections can bounce off the angled topography and inadvertently expose dark, non-exposed regions of the resist.
- **Standing Waves:** Because the resist acts as a thin optical film ($n_{resist} \approx 1.6$ to 1.7), light reflecting off a highly reflective flat substrate interferes with the incoming light. This sets up a vertical standing wave of intensity within the resist thickness, causing the sidewalls of the developed resist to look "scalloped", as can be seen in Figure 6.8. This interference makes it so the final printed linewidth oscillates severely depending on microscopic variations in resist thickness.

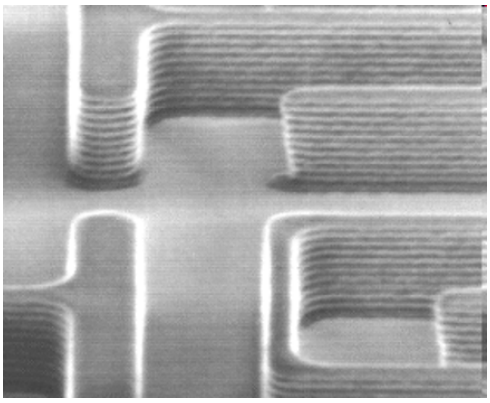


Figure 6.8: This SEM image highlights the wavy pattern of resist walls after an exposure without an ARC.

To suppress these effects, fabs employ **Anti-Reflective Coatings (ARC)** directly beneath the resist. The ARC's thickness and refractive index are finely calibrated to match optical parameters and absorb incident light, thereby eliminating bottom reflectivity and preserving vertical resist profiles.

6.6 Registration and Overlay

A photolithographic process is fundamentally judged by three metrics: Resolution, Throughput, and **Registration** (also known as Overlay).

Because an integrated circuit is built layer by layer, aligning the current photomask to the patterns already physically etched into the wafer from a previous step is vital. For example, the source and drain contact vias must align perfectly with the active area, and the transistor gate must be perfectly centred. As features shrink, overlay tolerances become brutal; for a 32 nm technology node, the required 3σ overlay accuracy is a mere 10.5 nm.

6.6.1 Alignment Marks and Measurement

To achieve this, special structures called alignment marks are generated on the wafer during previous masking steps. The scanner's alignment system recognizes these marks and calculates their

absolute coordinates. The wafer and reticle stages are then manoeuvred using highly complex laser interferometers that control movement across six degrees of freedom (X, Y, Z, R_x, R_y, R_z) with an accuracy and repeatability of 10 to 20 nm.

After the wafer is developed, metrology tools inspect dedicated measurement structures to evaluate the mutual alignment error (misregistration in X and Y) between the two layers.

6.6.2 Overlay Models

Steppers and scanners can produce several distinct mechanical and optical "fingerprints" during exposure that cause overlay errors. To maintain tight registration, these errors are mathematically modelled and fed back into the scanner's software for automatic correction. These models divide the geometrical errors into two distinct categories:

1. Intrafield Errors (Within the Exposure Field):
These errors originate from the projection optics and the reticle stage. They apply relative to the centre of the individual scanner field:

- Field Translation (T_x, T_y)
- Field Magnification (M)
- Field Rotation (R)
- Field Skew/Orthogonality

2. Interfield Errors (Field to Field over the Wafer):
These errors originate from the gross positioning of the wafer stage. They apply globally across the entire substrate:

- Wafer Translation (t_x, t_y)
- Wafer Scale/Magnification (m)
- Wafer Rotation (R_{wafer})

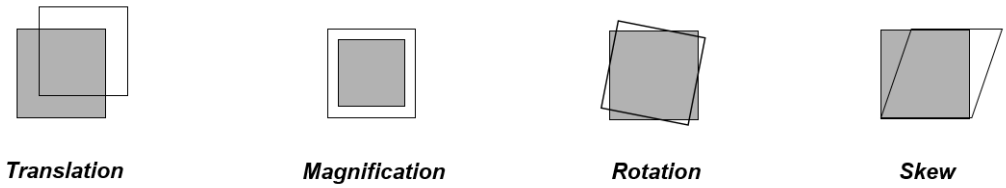
To properly classify and correct these overlay errors, metrology tools measure several points across the wafer to generate a large system of equations. A least-squares method model is then applied to extract the specific translation, scaling, and rotation coefficients ($\{T_x, T_y, M, \dots\}$).

6.7 The Sub-Wavelength Gap and Diffraction Physics

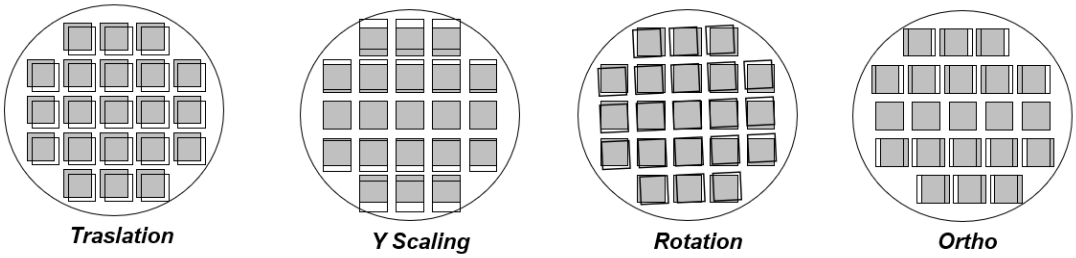
For the first few decades of semiconductor manufacturing, the wavelength of the exposure light was significantly larger than the minimum feature size being printed on the wafer. During this "Above Wavelength" era, optical projection was relatively straightforward. However, by the late 1990s, the aggressive scaling demanded by Moore's Law outpaced the development of shorter-wavelength light sources, forcing the industry into the "Sub-Wavelength" gap. Today, fabs routinely print features smaller than 50 nm using 193 nm light. Printing features so much smaller than the wavelength itself requires a profound understanding of diffraction physics and the limits of optical projection.

6.7.1 Diffraction at the Reticle

Why is there a fundamental resolution limit in projection lithography? The answer lies in the wave nature of light. When the dimensions of the features on the photomask become comparable to the wavelength of the source light, the mask no longer acts as a simple stencil; it acts as a diffraction grating. An example of this effect can be seen in Figure 6.10.



(a) Intrafield errors



(b) Interfield errors

Figure 6.9: a) A list of what intrafield errors look like. b) A list of what interfield errors look like. The grey represents the substrate while the square frame represents the current mask. If you can't see the frames try zooming in.

As light passes through the dense arrays of lines and spaces on the mask, it is diffracted into multiple discrete beams (orders). This creates a Far-Field (Fraunhofer) diffraction pattern at the entrance pupil of the objective lens. The angle (ϕ) at which these diffraction peaks are dispersed is dictated by the fundamental grating equation:

$$\sin\phi = \frac{n\lambda}{p}$$

where λ is the source wavelength, p is the periodicity (pitch) of the mask structures, and n is an integer representing the diffraction order ($\pm 1, \pm 2, \dots$).

This equation reveals that pitch is fundamental for lithography. Under the same wavelength, as the feature pitch gets smaller (representing a higher spatial frequency), the diffraction peaks are dispersed at progressively wider angles (a comparison is shown in Figure 6.11). The diffraction pattern theoretically extends to infinity, with the higher spatial frequency information, which is responsible for defining the sharp edges and fine details of the mask, carried by the peaks positioned furthest away from the optical centre.

6.7.2 The Objective Lens and Pattern Fidelity

The function of the objective lens is to collect this expanding diffraction pattern and reconstruct the image at the wafer plane. However, the lens possesses a finite physical diameter. It can only collect the portion of the diffracted light that falls within its capture angle.

- **Large Pitch:** The diffraction peaks are grouped tightly together at small angles. Several diffraction peaks (e.g., $0^{\text{th}}, \pm 1^{\text{st}}, \pm 2^{\text{nd}}, \pm 3^{\text{rd}}$) successfully enter the lens, resulting in a sharply reconstructed aerial image with excellent modulation at the wafer.
- **Small Pitch:** The diffraction peaks are dispersed at much wider angles. Only a few peaks (e.g., $0^{\text{th}}, \pm 1^{\text{st}}$) manage to enter the lens, causing the image resolution to worsen and the features to blur.

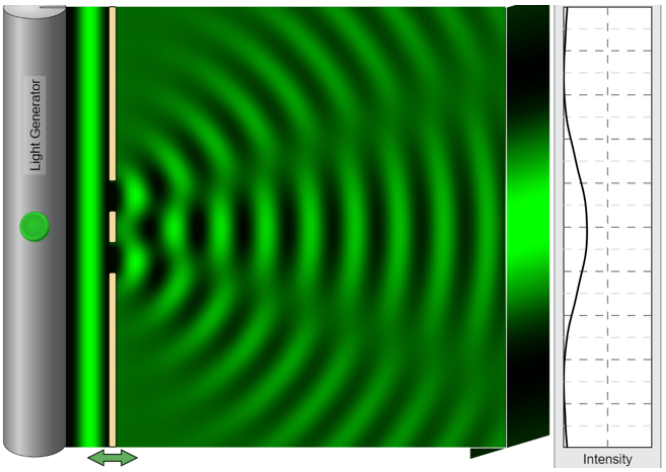


Figure 6.10: The scheme shows the diffraction pattern generated by two close gaps in the mask. The interference causes an uneven distribution of the intensity at the lens plane, which in turn may create artifacts at the wafer plane.

(a) Large pitch

(b) Small pitch

Figure 6.11: a) Diffraction pattern generated by a large pitch. b) Diffraction pattern generated by a small pitch. Here the diffraction peaks are more spread out.

- **The Resolution Limit:** If the pitch is excessively small, the diffraction angle becomes so extreme that the $\pm 1^{\text{st}}$ orders completely miss the lens. Only the central, undiffracted 0^{th} order peak enters the lens. In this scenario, there is absolutely no intensity modulation at the wafer level; the resist simply sees a uniform, featureless grey blur.

Physical Insight

To understand why capturing only the 0^{th} order results in a complete loss of resolution, we must view the optical system through the lens of Fourier analysis.
The 0^{th} order beam travels straight through the mask without diffracting. In Fourier terms, it represents the "DC component" of the signal, that is the average background light intensity. It contains absolutely no spatial information about the mask's geometry. The higher diffraction orders ($\pm 1^{\text{st}}, \pm 2^{\text{nd}}$, etc.) represent the "AC components," which carry the alternating spatial frequencies (the actual lines and spaces).
To create an image with intensity modulation (light and dark areas), at least two coherent beams must interact and interfere at the wafer surface. If the lens is too small and only the DC component (0^{th} order) arrives at the wafer, there is nothing for it to interfere with. Therefore, capturing at least the $\pm 1^{\text{st}}$ diffraction orders is the strict physical prerequisite for transferring any geometric information from the mask to the resist.

6.7.3 Optical Parameters: NA and σ

To quantify the capability of an exposure tool to capture these diffracted orders, engineers rely on two primary optical parameters:

1. Numerical Aperture (NA):

The Numerical Aperture determines the maximum number of diffraction orders that can be captured by the projection lens, directly dictating the quality and fidelity of the reconstructed image. It is defined as:

$$NA = n \cdot \sin \alpha$$

where α is the maximum semi-angle of the cone of light that can enter the lens, and n is the refractive index of the medium between the lens and the wafer (for air, $n = 1$). Increasing the NA of the optical system allows it to capture wider diffraction angles, leading to vastly improved pattern fidelity (transforming a blurred circular dot into a sharp, well-defined square).

2. Partial Coherence (σ):

While NA defines the objective lens, the partial coherence factor (σ) defines the illumination of the mask itself. It is a measure of the spatial extension of the light source and dictates the range of angles striking the reticle. It is mathematically defined as the ratio of the illumination numerical aperture to the objective numerical aperture:

$$\sigma = \frac{\sin \alpha_0}{NA}$$

where α_0 is the semi-angle of the illumination cone (which is the cone made by the solid angle through which the mask "sees" the source). The parameter ranges from $0 < \sigma < 1$, where $\sigma = 0$ represents a perfectly coherent point source (laser-like, unidirectional illumination) and $\sigma = 1$ represents highly incoherent, multi-directional illumination. Modulating σ is a powerful tool used to optimise image contrast for specific feature pitches.

6.8 Fourier Optics and Image Formation

To rigorously quantify the resolution limits and image quality of a projection lithography system, the optical sequence must be mathematically analysed using Fourier optics. By treating the propagation of light as a series of spatial frequency transformations, engineers can precisely model how a physical mask pattern is transferred into an aerial image at the wafer level.

6.8.1 The Mask Spectrum

The first step in this optical sequence occurs when incident light passes through the reticle. Consider a simple mask structure consisting of alternating lines of transparent quartz and opaque chrome, with a spatial periodicity, or pitch, of p, and an opening width of $\frac{p}{2}$, as shown in Figure 6.12.

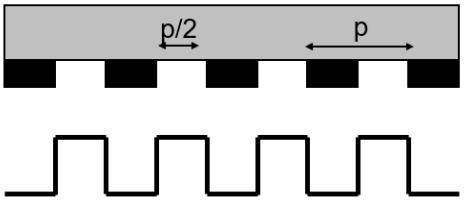


Figure 6.12: The picture shows a mask with an evenly alternating pattern. The line at the bottom represents the transmission function of this mask.

This physical structure is mathematically described by the mask transmission function, $O(x, y)$. For a binary mask, $O(x, y)$ operates as a square wave, yielding a value of 1 where light transmits fully through the quartz, and 0 where it is completely blocked by the chrome.

Applying diffraction scalar theory, the electric field immediately after the mask can be calculated using the Huygens-Fresnel principle in the Fraunhofer (far-field) approximation. Mathematically, this dictates that the electric field arriving at the diffraction plane is proportional to the integral of the mask transmission function modulated by a complex phase factor:

$$E(x', y') \propto \iint O(x, y) e^{-i\frac{2\pi}{\lambda z}(x'x+y'y)} dxdy$$

where z is the distance from the mask to the diffraction plane, while x' and y' are the physical coordinates in the diffraction plane.

Crucially, this Fraunhofer diffraction integral is mathematically identical to a spatial Fourier transform. By defining the spatial frequencies of the diffraction pattern in terms of the physical coordinates:

$$f_x = \frac{x'}{\lambda z} \qquad \text{and} \qquad f_y = \frac{y'}{\lambda z}$$

we can observe that the resulting electric field at the pupil is exactly the Fourier transform of the mask transmission function. Therefore, the diffracted light propagating away from the reticle natively encodes the spatial frequencies of the mask pattern.

6.8.2 The Objective Lens (Coherent Imaging $\sigma = 0$)

Following the creation of the mask spectrum in the diffraction plane, the diffracted light propagates towards the objective lens. For a coherent imaging system (where the partial coherence factor $\sigma = 0$), the fundamental function of the objective lens is to reconstruct this diffraction pattern and focus it onto the wafer.

Since the mask spectrum arriving at the lens is the spatial Fourier transform of the mask pattern, the objective lens needs to perform an inverse Fourier transform to recreate the spatial image. Physically, this is exactly what spherical lenses do.

However, a physical lens is bounded by its finite diameter; it can only perform this inverse transform on the specific portion of the diffracted light that successfully enters its aperture. Because higher spatial frequencies diffract at progressively wider angles, they naturally fall outside the lens aperture. Consequently, the objective lens inherently acts as a diffraction-limited system, behaving as a strict low-frequency pass filter.

For circularly symmetric optical systems, this spatial filtering behaviour is described mathematically by the Lens Pupil Function, $P(f_x, f_y)$. It acts as an ideal binary filter:

$$P(f_x, f_y) = \begin{cases} 1 & \text{if } \sqrt{f_x^2 + f_y^2} \leq \frac{NA}{\lambda} \\ 0 & \text{otherwise} \end{cases}$$

This filtering threshold dictates the Critical Cut-off Frequency (ν_c), which represents the absolute maximum spatial frequency the lens can capture:

$$\nu_c = \frac{NA}{\lambda}$$

By applying this pupil function to the incoming spectrum, the lens strips away all higher-order frequencies. The final Electric Field at the Wafer level, $E(x, y)$, is then generated by evaluating the inverse Fourier transform of this filtered spectrum. Written explicitly as a two-dimensional integral, this takes the form:

$$E(x, y) = \iint_{-\infty}^{+\infty} \tilde{E}(f_x, f_y) P(f_x, f_y) e^{i2\pi(f_x x + f_y y)} df_x df_y$$

where $\tilde{E}(f_x, f_y)$ is the incident electric field spectrum arriving from the reticle.

6.8.3 Aerial Image Formation

To determine the actual physical light pattern projected onto the wafer, we must apply the theoretical filtering limits of the objective lens to the mathematics of the electric field.

By explicitly substituting the definition of the circular pupil function into the previous equation, we can redefine the limits of integration.

The resulting Electric Field at the Wafer level, $E(x, y)$, is therefore given by the restricted integral:

$$E(x,y) = \iint_{\sqrt{f_x^2+f_y^2} \leq \frac{NA}{\lambda}} \tilde{E}(f_x, f_y) e^{i2\pi(f_x x + f_y y)} df_x df_y$$

This equation mathematically demonstrates how the finite numerical aperture (NA) cuts off the higher-order spatial frequencies, preventing them from contributing to the reconstructed spatial image.

While this yields the precise amplitude and phase of the light arriving at the substrate, the photosensitive chemical resist does not respond directly to the electric field vector; rather, it reacts to the physical energy deposited by the photons. Therefore, the final Aerial Image Intensity at the wafer level, $I(x, y)$, is calculated as the squared magnitude of this reconstructed electric field:

$$I(x,y) = |E(x,y)|^2$$

This intensity distribution, $I(x, y)$, represents the final contrast pattern of light and dark regions that will selectively expose the photoresist to define the integrated circuit features. A summary of this transfer can be found in Figure 6.13.

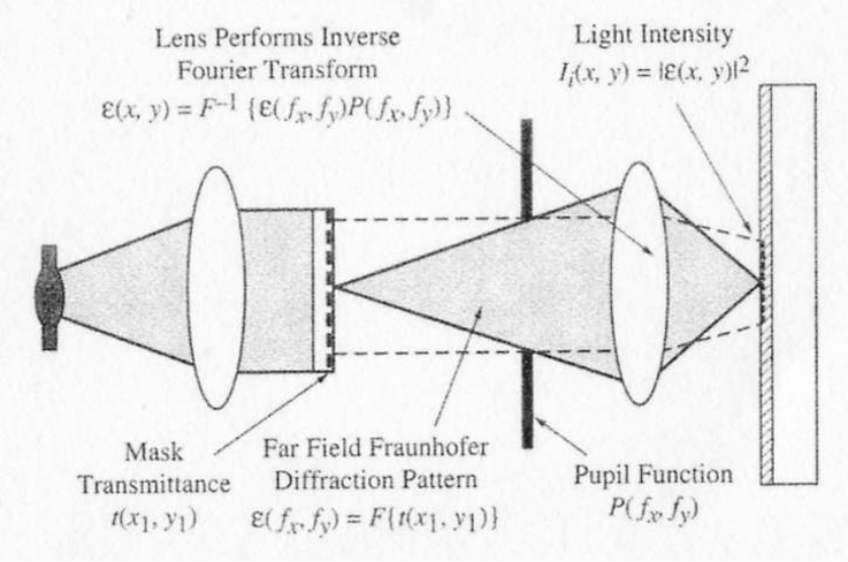


Figure 6.13: This picture schematically shows each Fourier transform and anti-transform that happens during image formation. Here, only the mathematical shape of the field is followed. To make it physically correct, the electric field in the Far Field approximation after the mask would be $\varepsilon(f_x, f_y) = \frac{E_0 e^{ikf}}{i\lambda f} \mathcal{F}\{t(x_1, y_1)\}_{f_x = \frac{x}{\lambda f}, f_y = \frac{y}{\lambda f}}$.

6.8.4 Resolution Limits in 1D Coherent Imaging

To simplify the mathematical analysis of the resolution limit in coherent imaging ($\sigma = 0$), it is convenient to rewrite the aerial image formation equations using normalised coordinates

(indicated by a "hat" symbol) and considering just one spatial direction, meaning $x = \hat{x}\frac{\lambda}{NA}$ and $f = \hat{f}\frac{NA}{\lambda}$.

The image is synthesised from normalised spatial frequencies ranging strictly from -1 (where $f = -\frac{NA}{\lambda}$) to $+1$ (where $f = \frac{NA}{\lambda}$). Only the components of the spectrum passing through this low-pass pupil filter are synthesised to form the final image.

The complete 1D mathematical sequence for this aerial image formation is given by:

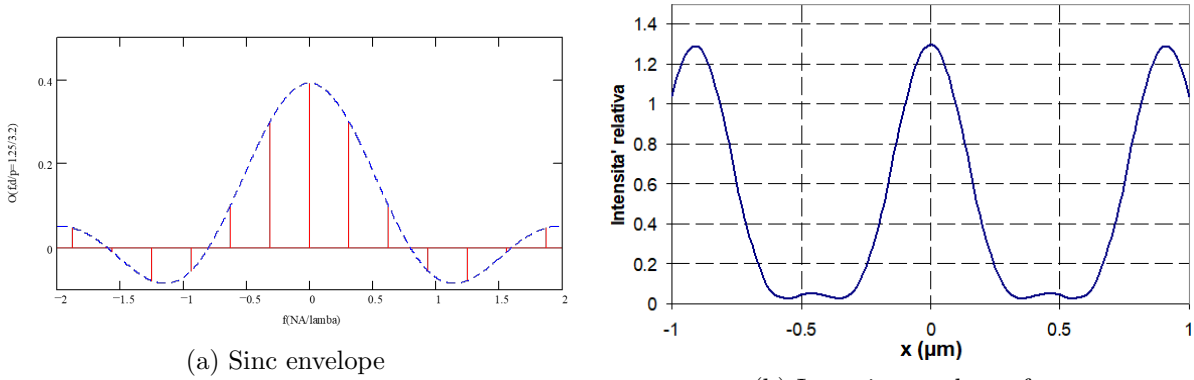
$$\begin{aligned} E_m(\hat{x}) &= E_0(\hat{x})\hat{O}(\hat{x}) \\ \hat{O}_m(f_x) &= \mathcal{F}\{E_m(\hat{x})\} \\ \hat{O}_l(f_x) &= \hat{O}_m(f_x)P(f_x) \\ E_w(\hat{x}) &= \mathcal{F}^{-1}\{\hat{O}_l(f_x)\} \end{aligned}$$

where $E_m(x)$ is the transmitted electric field, $O_m(f_x)$ is the mask spectrum generated by the Fourier transform, $O_l(f_x)$ is the pupil-filtered spectrum, and $E_w(x)$ is the final electrical field at the wafer level.

Now, consider a repeating pattern on the reticle characterised by a structural periodicity (pitch) p and an opening width d, such that $p = \hat{p}\frac{\lambda}{NA}$ and $d = \hat{d}\frac{\lambda}{NA}$. Because the transmission pattern is periodic, its Fourier transform, which is the mask spectrum, is discrete rather than continuous. This discrete spectrum takes the mathematical form of Dirac delta functions bounded within a Sinc envelope (as shown in Figure 6.14a). Crucially, these frequencies are spaced with exactly a $\frac{1}{\hat{p}}$ separation.

With this, the final intensity (Figure 6.14b) becomes:

$$\hat{I}(\hat{x}) = \left(\frac{\hat{d}}{\hat{p}}\right)^2 \left| 1 + \sum_{m=1}^{m_0} 2\text{sinc}\left(\frac{m\hat{d}}{\hat{p}}\right) \cos\left(\frac{2\pi m\hat{x}}{\hat{p}}\right) \right|^2$$



(a) Sinc envelope

(b) Intensity on the wafer

Figure 6.14: a) Distribution of the Dirac deltas enveloped by a Sinc function. b) An example of an intensity profile at the wafer level, generated by 5 diffraction orders.

6.8.5 Partially Coherent Imaging and Off-Axis Illumination

In physical manufacturing systems, the mask is illuminated by light travelling in various directions, rather than a perfectly perpendicular, coherent beam. The final image is calculated by adding the images arising from all these incident light sources.

This behaviour is quantified by the partial coherence factor ($0 < \sigma < 1$), which, as mentioned in Chapter 6.7.3, is mathematically defined by the ratio of the illumination numerical aperture to the objective numerical aperture:

$$\sigma = \frac{\sin\alpha_0}{NA}$$

where α_0 is the semi-angle of the illumination source.

To account for this factor, the resolution equation can be rewritten as:

$$R = \frac{1}{2}\frac{\lambda}{NA(1+\sigma)}$$

Engineers exploit this partial coherence using Off-Axis Illumination (OAI). The effect of Off-Axis Illumination is to shift the mask spectrum on the pupil plane, which increases the effective resolution (the setup is presented in Figure 6.15). This frequency shift is mathematically defined by:

$$\frac{\sin\phi}{\lambda} = \sigma\frac{NA}{\lambda} = \sigma\nu_c$$

By shifting the spectrum relative to the entrance pupil of the projection optics (which is bounded by the critical cut-off frequencies $\pm\nu_c = NA/\lambda$), higher diffraction orders that would have otherwise fallen outside the lens are successfully captured.

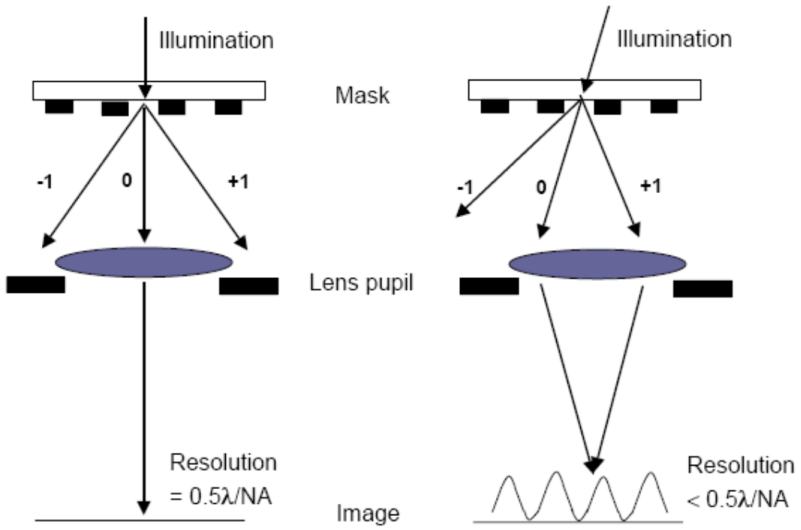


Figure 6.15: The graph compares the same optical system used with and without OAI. The setup on the right is able to capture the first diffraction order as well, despite it falling outside the lens in the setup on the left.

Off-Axis Illumination can be considered a Resolution Enhancement Technique (see Chapter 6.10).

6.9 Quantification of Image Quality and the Rayleigh Criteria

6.9.1 Modulation Transfer Function and Image Contrast

To formally determine the theoretical resolution limit of the optical imaging system and quantify the resulting image quality, engineers use the Modulation Transfer Function (MTF). The MTF is a critical metric used to calculate the precise fraction of energy transmitted by the optical system at any particular spatial frequency.

As established in the analysis of optical pupils, the illumination coherence defines the maximum transmittable frequency:

- For purely coherent imaging ($\sigma = 0$), the optical system can capture frequencies up to $\frac{NA}{\lambda}$.
- For partially coherent imaging ($\sigma \leq 1$), the shifted spectrum enables the system to resolve higher spatial frequencies, transmitting energy up to $(1+\sigma)\frac{NA}{\lambda}$.

Ultimately, the quality of this transferred image is evaluated by measuring the Image Contrast of the aerial intensity at the wafer level. Given the maximum light intensity (I_{max}) and the minimum light intensity (I_{min}) of the projected pattern, the physical contrast C is formulated as:

$$C = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

A higher contrast means sharper transitions between light and dark regions, which is strictly required to perfectly expose the photoresist without blurring the feature edges.

6.9.2 Depth of Focus (DoF)

While maintaining high image contrast in the exact focal plane is crucial, physical semiconductor wafers are never perfectly flat. Therefore, a lithographic system must maintain a usable image over a certain vertical range. This is defined as the Depth of Focus (DoF), which is shown in Figure 6.16. It is the maximum amount of focus change z that can be tolerated before the printed pattern size falls outside the strict manufacturing specifications[22].

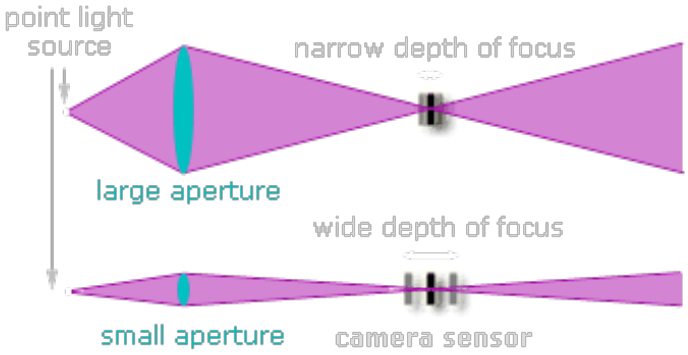


Figure 6.16: The picture shows a comparison between an optical system with a long DoF and one with a short DoF, denoted by the grey rectangles. This parameter must be tuned in accordance to feature sizes: high aspect ratio features require a longer DoF to correctly expose the resist. Image taken from Cambridge in colour.

If the wafer sits at a negative defocus or positive defocus relative to the ideal focal plane, the aerial image blurs, drastically reducing the contrast and altering the critical dimensions of the printed features.

6.9.3 The Rayleigh Criterion for Depth of Focus

To mathematically derive the Depth of Focus, we must consider the optical interference of image formation. Consider the image formed by exactly two rays: one originating from the centre of the optical pupil, and the other arriving from the extreme edge of the aperture at an angle θ.

When the system is perfectly focused, these rays interfere to form a perfectly sharp image at the ideal focus level z_0. However, at a defocused focal plane located a vertical distance z away, there is a relative phase change between the two rays. Rayleigh states that the depth of focus is the specific distance z such that the Optical Path Difference (OPD) between the central ray and the edge ray is exactly a quarter of the wavelength:

$$\text{OPD} = \frac{\lambda}{4}$$

[22]Depth of Focus and Depth of Field are two different quantities. The Depth of Field is the distance between the nearest and farthest objects in a scene that appear sharp and clear, while the Depth of Focus is the maximum distance the image (e.g., image sensor, wafer, etc.) can move while it stays in focus.

By mapping the geometry of the rays across a small angle θ, the optical path difference limits the maximum tolerable defocus before destructive interference permanently degrades the image.

6.9.4 The Rayleigh Equations: The Fundamental Trade-off

The optical physics governing the aerial image can be distilled into two supreme equations, known as the Rayleigh criteria, which dictate the capability of all optical imaging systems.

1. **Resolution (R):** The minimum resolvable feature size, given by:

$$R = k_1 \frac{\lambda}{NA}$$

where:

- λ is the source wavelength.
- NA is the Numerical Aperture of the objective lens.
- k_1 is the measure of "lithography aggressiveness," a process coefficient dependent on the specific resist, the lens quality, and the exposure tool. The absolute theoretical limit for k_1 is 0.25.

2. **Depth of Focus (DoF):** The available depth of focus is given by:

$$DoF = k_2 \frac{\lambda}{NA^2}$$

where k_2 is a process coefficient strictly dependent on the feature shape and configuration, the partial coherence of the illumination, and the optical aberrations in the lens.

Physical Insight

The Rayleigh equations highlight the most fundamental engineering trade-off in photolithography. The best lithographic process demands the highest possible resolution (the smallest R) and the biggest, most forgiving Depth of Focus (DoF).
To improve resolution, an engineer can either decrease the wavelength λ or increase the Numerical Aperture NA. However, looking at the DoF equation, the Numerical Aperture is squared in the denominator. Therefore, while increasing the size of the lens linearly improves the resolution, it quadratically destroys the Depth of Focus. As NA approaches 1.0, the depth of focus shrinks to a few hundred nanometres, meaning the silicon wafer must be polished to near-perfect mathematical flatness to keep the image in focus.

6.9.5 Numerical Aperture and Wavelength Evolution

To consistently scale transistors according to Moore's Law without completely sacrificing Depth of Focus, the industry has historically relied on a delicate balance of increasing NA and decreasing λ.

Since the 1980s, the evolution of exposure systems has followed this path:

- **1980s:** g-line Hg lamps ($\lambda = 436$ nm) with an NA of approximately 0.50.
- **Early 1990s:** i-line Hg lamps ($\lambda = 365$ nm).
- **1995:** KrF Excimer Lasers ($\lambda = 248$ nm) driving NA up to 0.63.
- **2001:** ArF Excimer Lasers ($\lambda = 193$ nm) driving NA to 0.75 and eventually 0.93.

- **2004 onwards:** The introduction of Immersion Lithography, allowing the NA to physically exceed 1.0.
- **Future Generation (2019+):** Extreme Ultraviolet Lithography (EUVL) at $\lambda = 13.4$ nm.

Despite these massive leaps in hardware, the physical wavelength limits are continually reached, forcing process engineers to rely on extensive software and optical tricks, such as Resolution Enhancement Techniques (RET), to artificially lower the k_1 factor and print features deep within the sub-wavelength regime.

6.10 Resolution Enhancement Techniques (RET)

As the physical limitations of exposure wavelengths and objective lenses are reached, engineers can no longer rely solely on hardware improvements to scale down feature sizes. Recalling Rayleigh's criterion for resolution:

$$R = k_1 \frac{\lambda}{NA}$$

In order to achieve higher resolution (lower R) and to continue printing smaller features, the lithography process can aim to decrease the k_1 factor.

This is achieved through Resolution Enhancement Techniques (RET), which apply a variety of deliberate manipulations to the optical wavefront. By doing so, additional contrast is obtained in the high spatial frequency components, resulting in a significant enhancement of the practical resolution.

RET corresponds to the meticulous control of the three major components of the optical wavefront:

1. **Wavefront Direction:** Manipulated via Off-Axis Illumination (OAI).
2. **Wavefront Phase:** Manipulated via Phase Shift Masks (PSM).
3. **Wavefront Amplitude:** Manipulated via Optical Proximity Corrections (OPC).

6.10.1 Wavefront Direction: Off-Axis Illumination (OAI)

As established, illuminating the mask from an off-axis angle shifts the diffraction spectrum across the objective lens pupil. OAI focuses on carefully filtering the spatial frequencies useful to reconstruct the image at the wafer level, which assures maximum contrast by minimising the zero-order "DC" background light.

To achieve this, the illumination shape must be heavily customised depending on the critical features[23] to be printed and the specific device application (e.g., MPU, Flash memory, DRAM). Standard OAI schemes include **Annular**, **Dipole**, and **Cross-Pole** (Quadrupole) illumination (illustrated in Figure 6.17). For example, by measuring the intensity distribution of a Dipole illumination source, engineers can perfectly optimise the light angles to capture the critical $\pm 1^{\text{st}}$ diffraction orders for highly periodic, unidirectional lines and spaces.

6.10.2 Wavefront Phase: Attenuated Phase Shift Masks (Att-PSM)

While conventional binary masks use opaque chrome to completely block light (0% transmission) and bare quartz to transmit it (0° phase reference), this sharp transition blurs into a grey area at the wafer level due to diffraction.

[23] OAI systems are optimised for predetermined pitches. At specific feature spacings, image contrast drops sharply, causing issues in the transfer of the pattern. These specific spacings are called "forbidden pitch".

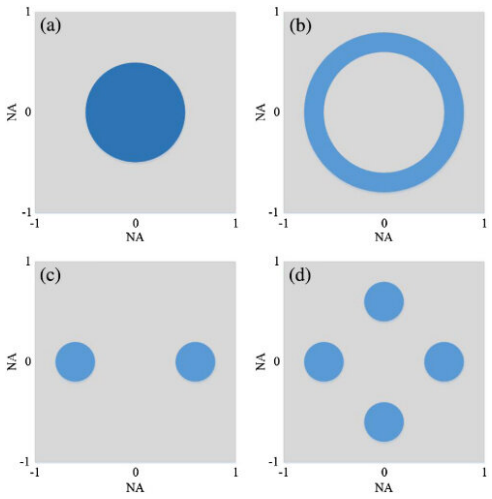


Figure 6.17: This picture shows the common patterns adopted in OAI: conventional (a), annular (b), dipole (c) and quadrupole (d). Specifically, these are maps of the illumination pupil, that is the cross-sectional shape of the light source as it passes through the condenser lens before reaching the photomask. Image taken from ResearchGate.

Attenuated Phase Shift Masks (Att-PSM) replace the opaque chrome with a semi-transparent "shifter" material that allows a small amount of light (typically a 6% transmission) to pass through[24]. Crucially, the thickness of this shifter material is engineered to shift the phase of the transmitted light by exactly 180° relative to the light travelling through the bare quartz. When the light from the quartz and the light from the shifter meet at the feature edges, destructive interference occurs, artificially sharpening the aerial image contrast to a level unattainable by a conventional binary mask, as shown in Figure 6.18.

When analysing the Att-PSM mask spectrum in the frequency domain, introducing this 6% background transmission (T) with a 180° phase shift drastically alters the diffraction orders. Compared to a conventional mask, the amplitude of the 0^{th} order peak strictly decreases, while the amplitudes of all non-zero orders $(\pm 1, \pm 2, \dots)$ increase by a factor of $(1 + t)$, where t is the amplitude transmittance ($t = \sqrt{T}$). This suppresses the unwanted background light and amplifies the crucial high-frequency alternating components. More specifically, we get:

$$\tilde{\hat{O}}_x(\hat{f}) = \left[(1+t)\frac{\hat{d}}{\hat{p}} \cdot \operatorname{sinc}(\hat{f}\hat{d}) \sum_{n=-\infty}^{n=\infty} \delta\left(\hat{f} - \frac{n}{\hat{p}}\right)\right] - t\delta(\hat{f})$$

6.10.3 Wavefront Amplitude: Optical Proximity Corrections (OPC)

When pushing lithography to the absolute resolution limits of the current wavelength and feature sizes, severe distorting effects naturally occur. Because higher spatial frequencies (which define sharp corners) are lost outside the lens pupil, a perfectly rectangular mask feature will print on the wafer with severely rounded corners and shorter lines (a phenomenon known as line-end shortening).

[24]There are different kinds of masks with a principle similar to Attenuated Phase Shift Masks. There are Alternating Phase Shift Masks (Alt-PSM), which introduce a 180° shift by etching part of the quartz on the photomask, so that light travels different paths inside the quartz and comes out out of phase, ready to interfere destructively in a similar way to Att-PSM. The fabrication cost of these masks is 10 times higher than regular (binary) photomasks since you need two patterning steps with higher alignment. There are also Pixelated Masks, introduced by Intel in 2007, which are an extreme case of a phase shift mask. Here, there are no opaque parts and the light always interferes on the whole wafer. The mask is made so the total interference yields exactly the desired pattern.

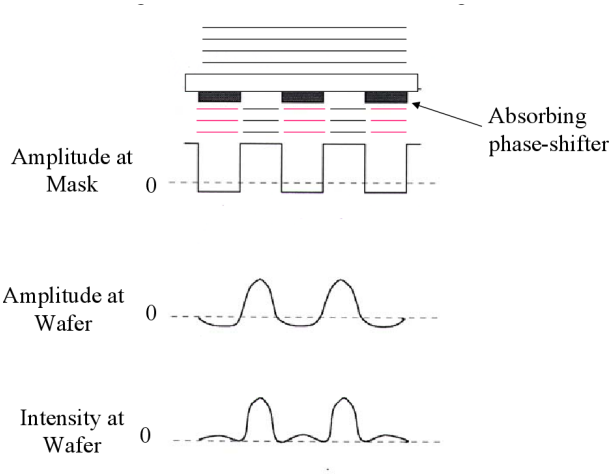


Figure 6.18: This picture shows the interference given by an Attenuated Phase Shift Mask. Due to the phase shift, there is destructive interference in the area that is supposed to be unexposed. This way, the intensity is much lower and only the correct features are exposed. Image taken from Semantic Scholar.

To counter these optical proximity effects, engineers use Optical Proximity Corrections (OPC). OPC involves making deliberate, sub-resolution changes to the physical shape of the pattern on the mask itself. By adding extra geometry to the mask (such as in Figure 6.19) that will not print directly, such as "hammerheads" on the ends of lines to make them print longer, or "serifs" on the outside of corners to make them print squarer, the resulting aerial image perfectly matches the original intended layout. Modern reticles are completely saturated with these complex OPC geometries to compensate for optical proximity near the resolution limit.

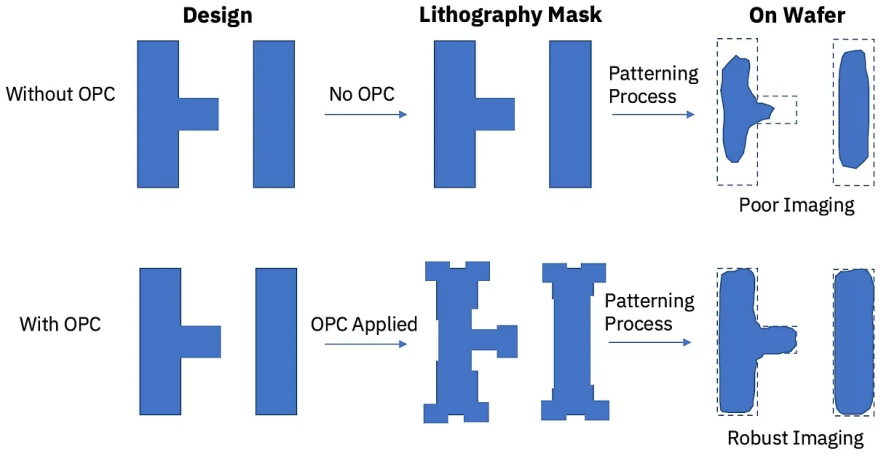


Figure 6.19: This picture shows the result of the exposure of the same design on a mask with no OPC and one with OPC. Due to diffraction, the image on the wafer has little to no resemblance to the original design. With OPC, the added serifs on the corners and hammerheads account for diffraction, projecting the desired pattern. Image taken from IBM.

6.11 Immersion Lithography

While RET successfully lowered the k_1 factor, scaling eventually required a physical increase in the Numerical Aperture (NA) of the optical system. Because $NA = n \cdot \sin\alpha$, a conventional

"dry" lithography system (where the medium between the lens and the wafer is air, $n \approx 1$) is physically capped at a theoretical maximum NA of 1.0, and a practical maximum of about 0.93.

To break the $NA = 1.0$ barrier, the semiconductor industry turned to an old idea originally pioneered by G. B. Amici (1786–1868) for microscopy: **Immersion Lithography**. This concept has been successfully applied to the most advanced scanner systems by replacing the air gap between the final objective lens and the wafer with an immersion fluid (as shown in Figure 6.20).

The idea is very simple, and indeed oil immersion has been used for over a century in optical microscopy. However, for lithography, the immersion fluid must have specific characteristics to be compatible with the process. The medium between the lens and the wafer must:

- Have a refractive index > 1. Thankfully, that is the case for all substances.
- Have low optical absorption at 193 nm, to be able to provide enough dose to the resist.
- Be compatible with the photoresist and the lens material.
- Be uniform and non-contaminating.

Surprisingly, ultra-pure water meets all these requirements.

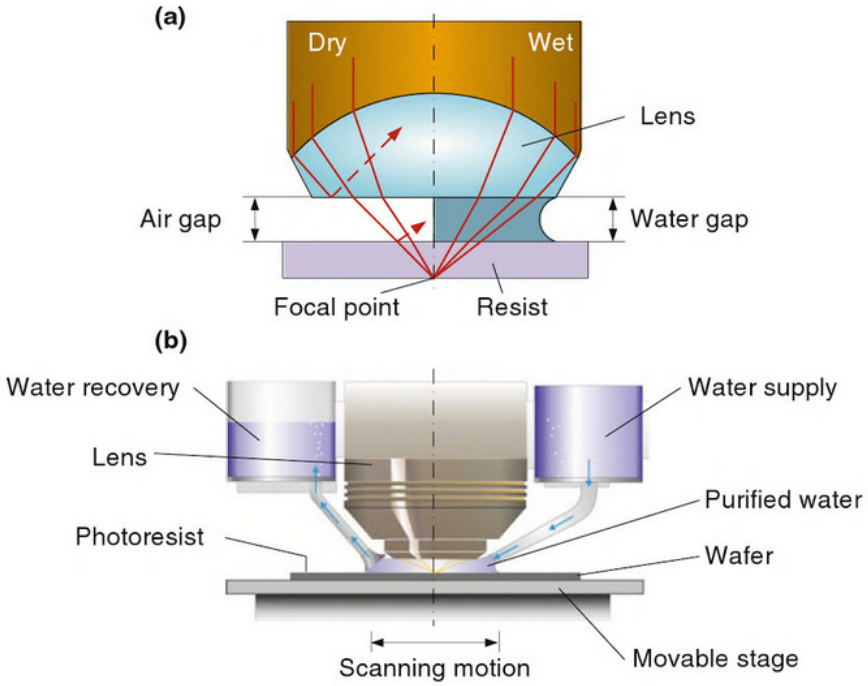


Figure 6.20: a) Ray-tracing scheme showing the advantage of immersion lithography. Some rays get reflected at the interface in the dry case due to being past the critical angle of incidence. The water allows the resist to receive more diffraction orders, meaning more resolution. b) This picture shows what the actual system looks like. A water supply inserts a water "puddle" between the resist and the lens. Image taken from Micro and Nano Fabrication Tools and Processes by H. Gatzen, Volker Saile & Jürg Leuthold.

6.11.1 The Resolution Advantage

When ultra-pure water is used as the immersion medium, the refractive index (n) in the focal region increases to 1.34. Because n is generally larger in liquid than in air, vacuum, or gas, the numerical aperture can be significantly higher than that of a conventional dry system.

This physically scales down the wavelength of the light interacting with the resist. The effective wavelength (λ_{eff}) of the system can be intended as:

$$R = k_1 \frac{\lambda}{NA} = k_1 \frac{\lambda}{n \sin\theta} = k_1 \frac{\frac{\lambda}{n}}{\sin\theta} = k_1 \frac{\lambda_{eff}}{\sin\theta}$$

For a 193 nm ArF laser firing into water ($n = 1.44$), the effective wavelength drops to roughly 144 nm, allowing the system to achieve an NA of up to 1.35 and resolve vastly smaller features.

6.11.2 The Depth of Focus Advantage

Counterintuitively, immersion lithography also drastically improves the Depth of Focus (DoF), which normally degrades as NA increases.

For a given diffraction order (and thus a given lateral spatial frequency), the physical angle of the light travelling inside the high-index immersion fluid will be strictly less than if the medium were air. Because the light rays are travelling at smaller, steeper angles, the Optical Path Difference (OPD) between the rays changes much more slowly as the wafer moves out of focus. Smaller angles mathematically equate to a smaller optical path difference, and therefore a smaller degradation of the image for a given amount of defocus.

Process Insight

While immersion lithography provides a brilliant physical advantage, implementing it into a Step-and-Scan system moving at 500 $\text{mm} \cdot \text{s}^{-1}$ is an engineering nightmare. The scanner must dynamically maintain a perfect "puddle" of ultra-pure water strictly trapped beneath the lens while the wafer flies back and forth beneath it. If microscopic air bubbles form in the water, they act as tiny lenses, destroying the aerial image. Furthermore, because the refractive index of water is highly temperature-dependent, the fluid must be thermally controlled to within milli-kelvins to prevent optical distortion during exposure.

6.12 Pushing the Limits: Multiple Patterning

For decades, the semiconductor industry has continually braced for the physical end of optical scaling. The so-called "Sturtevant Plot" (Figure 6.21) humorously illustrates that the impending "end of optical lithography" has been confidently predicted to be exactly seven years away every single year since 1976.

The absolute theoretical resolution limit of a single optical exposure is dictated by the Rayleigh criterion, which states that the process coefficient k_1 cannot drop below 0.25. However, by the time the industry reached the 22 nm and 14 nm nodes, the required pitches demanded a k_1 factor significantly below this hard physical limit. To break the barrier and continue scaling without moving to shorter wavelengths, the industry adopted **Multiple Patterning**, the technique of splitting a single dense layout across two or more separate masks.

6.12.1 Double Exposure: Litho-Etch-Litho-Etch (LELE)

For complex layouts (such as logic circuits), the technique of choice is the Litho-Etch-Litho-Etch (LELE) approach, commonly referred to as Double Exposure. By taking a dense layout and splitting adjacent lines onto two separate masks, the pitch on each individual mask is effectively doubled, relaxing the optical requirements back into a printable regime.

The LELE process flow (shown in Figure 6.22) requires a complex stack of sacrificial layers:

1. **Stack Deposition:** A double hard mask is deposited over the substrate, typically consisting of amorphous PECVD carbon topped with a dielectric layer (e.g., silicon nitride or silicon oxide).

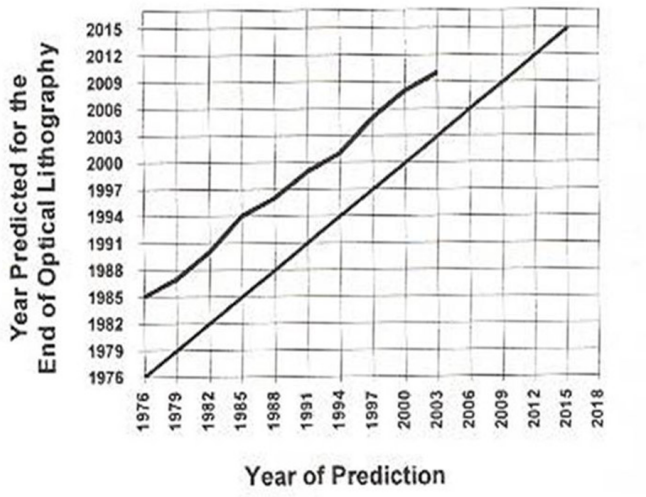


Figure 6.21: This is the Sturtevant plot. It is a famous running joke in the semiconductor industry.

2. **First Patterning:** The first mask (containing half the features) is exposed and developed. The pattern is etched into the top silicon nitride layer.

3. **Second Patterning:** The wafer is recoated with photoresist, and the second mask (containing the remaining interleaved features) is exposed and developed.

4. **Final Transfer:** The second pattern is etched into the silicon oxide, and both patterns are then simultaneously transferred down through the carbon hard mask and into the underlying substrate.

Because the second exposure is entirely physically distinct from the first, it is not self-aligned. Consequently, the absolute main issue with the LELE approach is overlay control; any slight misalignment between the two exposures will catastrophically shift the distances between adjacent lines.

6.12.2 Self-Aligned Double Patterning (SADP)

To circumvent the brutal overlay requirements of LELE, fabs utilise Self-Aligned Double Patterning (SADP). This technique is uniquely suited for highly repetitive, periodic patterns, such as dense memory arrays. Rather than relying on two optical exposures, SADP uses a single exposure followed by a conformal thin-film deposition to effectively halve the pitch.

Consider a target design requiring a 40 nm pitch. The SADP physical sequence (shown in Figure 6.22) proceeds as follows:

1. **Core Exposure:** Standard optical lithography is used to print a relaxed 80 nm pitch pattern (e.g., 40 nm resist lines separated by 40 nm spaces).

2. **Core Trimming:** The photoresist lines are intentionally subjected to a chemical trim, shrinking their width down to exactly 20 nm.

3. **Conformal Deposition:** A 20 nm thick, highly conformal dielectric layer (e.g., silicon oxide) is deposited over the trimmed resist cores.

4. **Spacer Etch:** A highly anisotropic plasma etch is applied to remove the dielectric from the horizontal surfaces, leaving it exclusively on the vertical sidewalls of the core. These remaining sidewalls are called “spacers.”

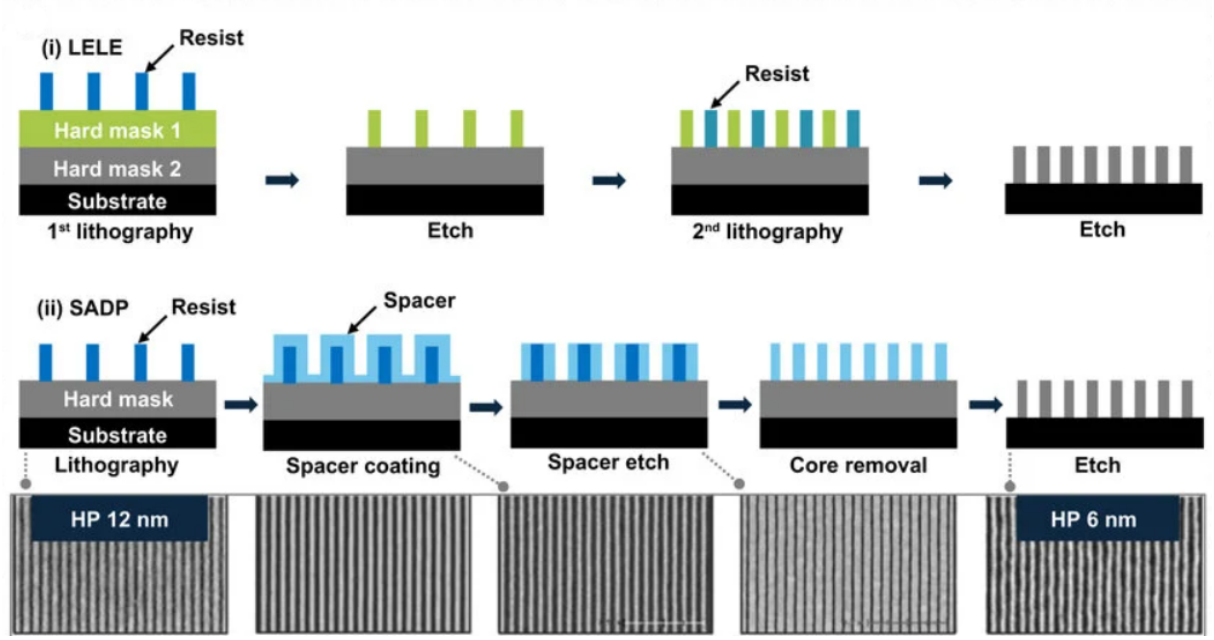


Figure 6.22: i) This scheme follows the LELE process. After the resist is exposed and patterned on the first silicon nitride layer, another mask is used to expose features between the already existing ones. Both are then transferred onto the substrate. ii) This scheme follows the SADP process. The spacer is effectively able to halve the pitch. After an etching to expose the underlying resist, a lift-off is performed to remove it, thus leaving only the pattern on the silicon oxide. Image taken from ResearchGate.

5. **Core Strip:** The original photoresist core is stripped away (e.g., using an O_2 plasma ash), leaving only the freestanding spacers.

6. **Final Transfer:** Because there is one spacer on each side of the original core, the number of lines has doubled. These spacers act as the final hard mask, successfully yielding a 40 nm pitch (20 nm lines and 20 nm spaces).

This technique is incredibly scalable. By repeating the exact same spacer sequence on the newly formed lines, engineers can achieve **Self-Aligned Quadruple Patterning (SAQP)**, yielding structures at exactly one-quarter of the original lithographic pitch.

Process Insight

While the 2D cross-section of SADP appears elegant, implementing it in 3D topography introduces severe industrial challenges.

- **Thermal Limits:** The high temperatures required to deposit a robust conformal spacer are frequently incompatible with organic photoresist. Therefore, the resist pattern must first be transferred into a temporary "dummy core" (often amorphous carbon) before the spacer is deposited.
- **The Ring Effect and Chop Masks:** Because the spacer deposits conformally around the entire perimeter of the rectangular core, it naturally forms a continuous, unbroken ring. To separate this ring into two distinct, parallel lines, a secondary Chop Mask must be exposed to physically etch away the looped ends of the spacers.
- **Pad Masks:** Spacers inherently strictly dictate a single uniform line dimension. If a larger feature is required (e.g., a landing pad for a via), an additional **Pad Mask** must be exposed to print these larger features.
- **Even/Odd Variability:** Dimensional control in SADP is complex. The width of a line is determined purely by the spacer deposition thickness. However, the spacing between the lines alternates in origin: one space is defined by the width of the trimmed core, while the adjacent space is defined by the gap between the original cores. This combination of lithographic variability and deposition variability dictates that "even" spaces and "odd" spaces often suffer from differing structural variability.

6.13 Extreme Ultraviolet (EUV) Lithography

As the industry exhausted the physical limits of 193 nm immersion lithography and multiple patterning techniques became economically unsustainable, a massive leap in wavelength was required[25]. Extreme Ultraviolet (EUV) lithography operates at a wavelength of 13.5 nm (historically cited in early roadmaps as 13.4 nm).

In many theoretical respects, EUV lithography is a direct continuation of legacy optical systems (Figure 6.23 shows a comparison with DUV technology). It builds heavily on the established optical lithography experience base: it employs Step-and-Scan printing mechanisms, uses 4X reduction optics, theoretically supports optical extension tricks (such as Off-Axis Illumination, Phase Shift Masks, and Optical Proximity Corrections), and its resolution and Depth of Focus still scale strictly according to the Rayleigh equations (NA and λ). However, the physical reality of manipulating 13.5 nm light requires a complete paradigm shift in tool architecture.

[25] You may wonder why go straight to EUV instead of the next excimer line at 157 nm. It's because the return on investment is very small. First of all, fused silica (photomask material for DUV) and atmospheric oxygen become absorptive at 157 nm. Furthermore, resists would be expensive fluoropolymers, while immersion fluids compatible are exotic and dangerous hydrocarbons. Moving to the 157 nm line would require a complete material and substance overhaul for only a $\approx 20\%$ improvement in resolution. The 14x reduction in wavelength provided by EUV offered "simpler" single-patterning and guaranteed several generations of scaling.

ArFi (193nm)	EUV (13.5nm)
Transmission optics (lenses)	Reflection optics (Bragg mirrors)
Excimer laser source	Laser produced plasma source (LPP)
Immersion (NA_{water}=1.33)	Vacuum (NA= 0.33)

Figure 6.23: The picture compares a DUV system using an ArF laser and an EUV system using a complex plasma source. The ASML logo can be noticed on the EUV system. Indeed, as you may know, this Dutch company is the only company in the world able to create these machines, although there have been reports of China having manufactured a similar one in December of 2025. The laser system takes up to 20 square metres out of $\approx$ 80 square metres of the floor space required for a single machine.

Process Insight

Generating 13.5 nm photons is an extraordinarily violent process. This wavelength can only be obtained by exciting the relatively inner electrons of an atom, which requires the ionisation of the atom to form a hot, dense plasma.
In a modern Laser-Produced Plasma (LPP) source, microscopic droplets of liquid tin (Sn) are fired into a vacuum chamber. A tremendously powerful CO_2 laser, operating at 40 to 60 kW, is applied to bombard the droplets. The laser vaporises the tin into a dense plasma at temperatures exceeding those on the surface of the sun, triggering the EUV radiation. Collecting this light and directing it into the scanner without the tin debris destroying the first collector mirror is one of the greatest engineering feats in semiconductor history. To get an overview on this, I highly recommend this video by Veritasium, in which they explore the EUVL process in collaboration with ASML.

6.13.1 The Paradigm Shift: Reflective Optics

The fundamental problem with 13.5 nm radiation is that it is strongly absorbed by all known materials, including atmospheric air and the ultra-pure quartz glass used in DUV lenses. Consequently, an EUV system must operate in a high vacuum.

Because transmissive lenses are impossible, the entire optical train is replaced by reflective optics. The projection system relies on highly complex mirrors coated with quarter-wave Bragg reflectors (shown in Figure 6.24), which typically are alternating nanometre-scale layers of molybdenum and silicon, designed to reflect the EUV light via constructive interference. Similarly, the reticle can no longer be a transmissive piece of quartz; it must be a reflective blank coated with a patterned absorber material to define the circuit layout.

6.13.2 Primary Challenges of EUV

Implementing EUV for high-volume manufacturing presented a triad of immense challenges that delayed its adoption for over a decade:

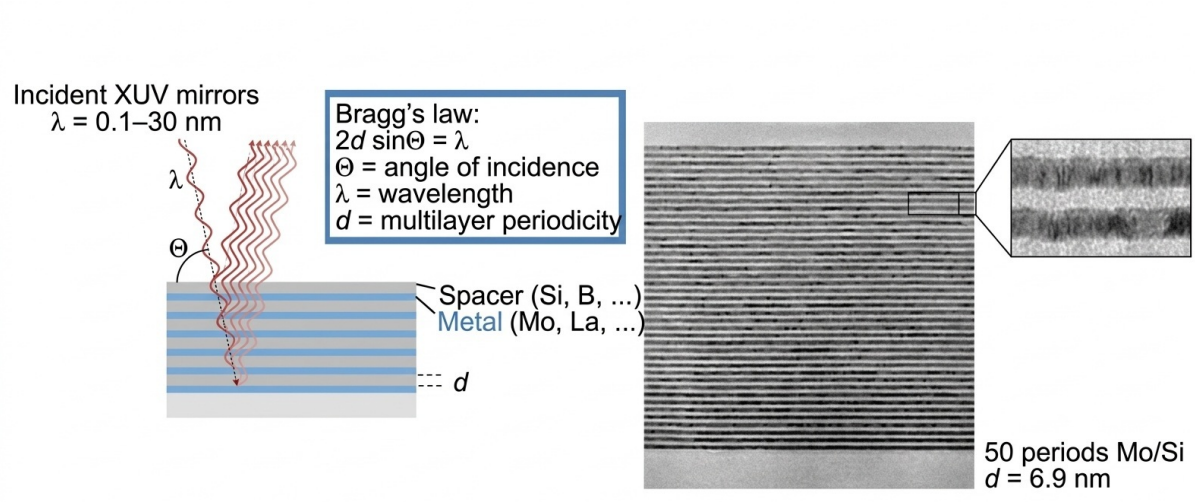


Figure 6.24: The picture shows the structure of a Bragg reflector. The scheme on the left illustrates the working principle: via carefully alternated layers of Mo and Si the light gets partially reflected and is able to undergo constructive interference. To achieve this, the acceptable surface roughness is about 0.2 nm (RMS), corresponding to a phase shift error of 10°. This way, the "mirror" is able to reach $\approx 70\%$ reflectance. The TEM picture on the right highlights the alternating layers. Image adapted from ForterTech.

- **Tools:** The sheer output power required to achieve acceptable wafer throughput was historically difficult to reach. Furthermore, manufacturing high-NA reflective optics and ensuring the lifetime and reliability of the illuminator mirrors under constant plasma bombardment remain critical challenges.
- **Resists:** EUV photoresists suffer from a strict physical trade-off between Resolution, Sensitivity (Exposure Dose), and Line Edge Roughness[26] (LER). Reducing the LER to ensure smooth transistor gates without requiring unfeasibly high exposure doses is a major chemical hurdle.
- **Masks:** Producing EUV mask blanks requires absolute physical flatness and completely defect-free multilayer coatings (Figure 6.25). Because the masks are reflective, inspecting them for sub-nanometre phase defects and repairing those defects is vastly more difficult than with transmissive optical masks.

6.14 Alternative Next-Generation Lithography (NGL)

While EUV currently dominates the leading edge, several alternative Next-Generation Lithography (NGL) technologies have been researched in an attempt to bypass the diffraction limits of photons entirely.

6.14.1 Electron Beam Lithography (EBL)

Rather than using photons, Electron Beam Lithography (EBL) uses a focused beam of electrons to expose the resist. To create the electron beam there are different options:

[26]Line Edge Roughness is the unintentional deviation of a feature's edge from its smooth, ideal straight line. It is mainly caused by Shot Noise, that is variance in the frequency of impinging photons (think of listening to raindrops; in heavy rain the sound is continuous, while in very light rain the droplets make sounds at uneven intervals). Since EUV photons are very energetic, fewer are required to fully expose resists, thus leading to bigger Shot Noise and consequently imprecise feature exposure (LER).

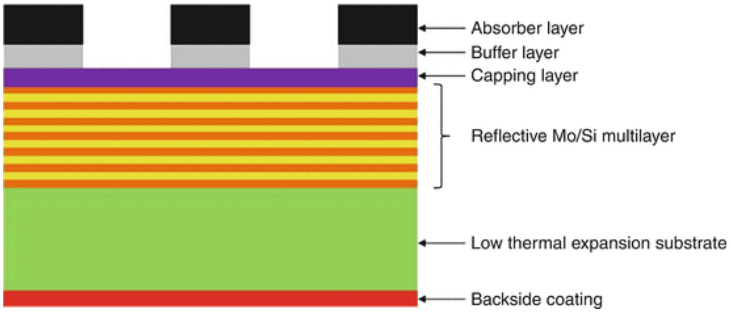


Figure 6.25: The picture shows the structure of an EUVL mask. The low thermal expansion substrate prevents thermal distortion during exposure. The alternating layers form the Bragg Reflector. The capping layer (Si) prevents oxidation or damage during cleaning. The absorber (usually TaN) and buffer (Ru or SiO_2) layers are what create the features. Image taken from Encyclopedia of Nanotechnology.

- **Thermionic Emission:** Tungsten filaments or a LaB_6 crystal tip are heated and electrons are emitted via thermionic effect.
- **Field Emission Gun:** A very high electric field is applied to a tungsten tip, which allows electrons to tunnel through the tip and be emitted.
- **Schottky emitters:** It's halfway between the two previous techniques; a ZrO_2 reservoir is heated, causing zirconium to diffuse to the tungsten tip. This forms a dipole layer that drastically lowers the work function of the tungsten surface, allowing an applied electric field to easily pull electrons from the tungsten over the lowered barrier,

Because the effective wavelength of these high-energy electrons is extremely low (≈ 0.1 nm), diffraction effects are practically negligible, offering near-perfect resolution. Furthermore, the technology is already deeply understood, as it is the standard method used to manufacture the photomasks themselves. Indeed, EBL is "maskless", saving massive reticle costs.

However, EBL suffers from severe physical and economic disadvantages:

- **Throughput:** Because the wafer is exposed pixel-by-pixel, throughput is catastrophically low (approximately 5 wafers/hr, compared to over 60 wafers/hr for optical scanners).
- **Optical Aberrations:** During travel, many effects take place which hinder the resolution: imperfections in the optics (electro and magnetostatic lenses) cause spherical aberrations, different electron energies cause chromatic aberrations, non-uniform electric and magnetic fields cause astigmatism. On top of this, since electrons are charged particles, they repel during travel, which naturally blurs the beam.
- **High Proximity Effect:** When the electrons strike the resist[27] and the underlying substrate, they scatter violently (Figure 6.26). These electrons can travel great distances ($< 1\,\mu$m), as well as backscatter inside the resist before coming to rest. This inadvertently exposes the dark regions of the pattern.

[27] EBL resists are different from Optical Lithography resists, since using electrons instead of photons means no light-induced reactions can occur. PMMA is usually the resist of choice: it has great adhesion, low sensitivity and a positive tone. Upon exposure, the molecular weight is reduced due to secondary electrons that break chains. This causes the exposed area to become more soluble.

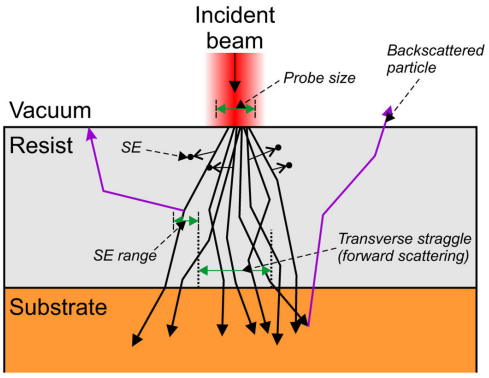


Figure 6.26: This picture illustrates the scattering events that can happen inside the resist during EBL. The picture is actually for an ion beam instead of an electron beam (Focused Ion Beam Lithography). The concept is the same as EBL, except ions are heavier and positively charged. As a consequence, some effects are mitigated (for example they sway less during travel). Nevertheless, they scatter violently inside the resist, which causes a spread in the beam's profile. Some electrons penetrate more, some scatter to the sides while some are entirely backscattered. Image adapted from AIP Publishing.

6.14.2 Nano Imprint Lithography (NIL)

Nano Imprint Lithography abandons optics entirely in favour of mechanical contact printing. A physical template containing the 3D relief of the circuit pattern is pressed directly into a liquid resist on the wafer (Figure 6.27), physically moulding the material. Standard template manufacturing technology is identical to the production of phase-shift masks, but because it relies on 1:1 contact, it requires a 1:1 template dimension. Consequently, dimensional control is incredibly demanding, and contact defectivity remains the dominant hurdle for volume fabrication.

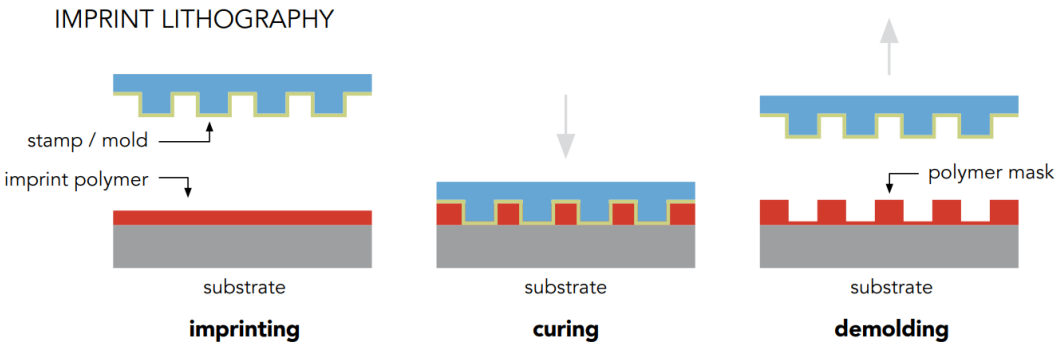


Figure 6.27: This picture shows the process of NIL, more specifically thermal NIL. Here, heat is used to slightly melt the resist so the stamp can transfer the pattern. An alternative method is UV-curing, where UV light is used to harden a slightly more liquid resist. Image taken from TUM, courtesy of T. Häberle PhD thesis.

6.14.3 Directed Self-Assembly (DSA)

Directed Self-Assembly (DSA) leverages the inherent thermodynamic properties of block copolymers. Under specific thermal conditions, these polymer chains naturally phase-separate and organise themselves into highly regular, dense nano-structures (such as tightly packed cylinders or lamellae). By printing a relatively coarse "guide" pattern using standard optical lithography, the polymers can be "directed" to assemble within those boundaries (Figure 6.28), effectively multiplying the resolution of the optical pre-pattern chemically rather than optically.

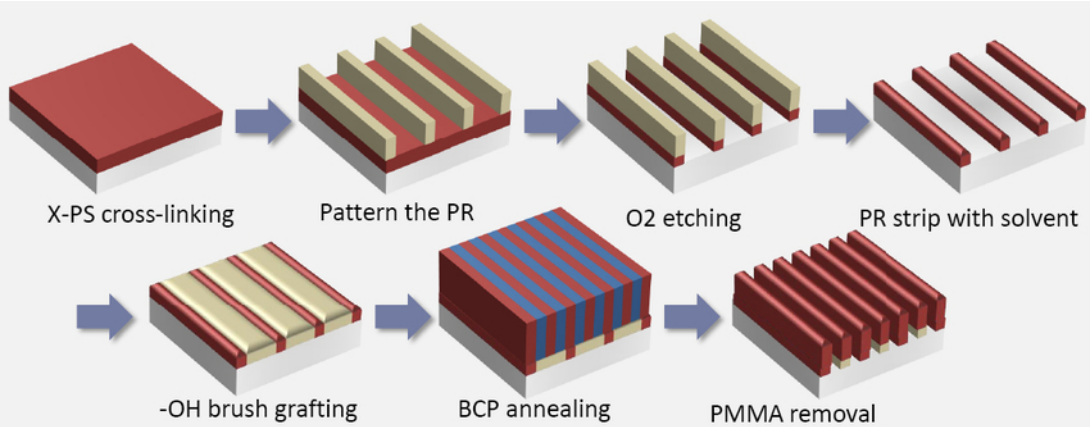


Figure 6.28: The picture follows the steps of a DSA process flow. A sparse chemical pre-pattern is used to guide the self-assembly of a block copolymer (BCP) film into aligned domains. Selective removal of one polymer block (PMMA) then yields a dense, high-resolution nanoscale line pattern. Image taken from ResearchGate, courtesy of Chi-Chun Liu, Adam Welander and Paul Nealey, Univ. of Wisconsin.

The journey through photolithography has revealed the extraordinary lengths to which the semiconductor industry goes to cast nanoscale shadows. From the precision of excimer lasers and the optical gymnastics of immersion lenses to the catalytic reactions within chemically amplified resists, we have seen how light is harnessed to draw impossibly fine blueprints. We now possess the capability to align and print billions of features across a silicon substrate with nanometre-level accuracy.

Yet, for all its optical brilliance, the immediate output of the lithography cell is fundamentally a sacrificial polymer layer resting atop the wafer. To turn these transient chemical blueprints into the permanent, electrically active structures of a microchip, the underlying layers of metal, silicon, and dielectric must be physically removed wherever the resist has left them exposed.

This critical translation from a polymeric map to physical reality brings us to the next vital stage of fabrication: **Etching and Pattern Transfer**. In the forthcoming chapter, we will examine how highly energetic plasmas, volatile chemical reactions, and directed ion bombardment are utilised to bite into the substrate. We will explore how process engineers achieve the extreme selectivity and perfect anisotropy required to carve deep trenches and sheer transistor sidewalls, ultimately sculpting the definitive three-dimensional architecture of the integrated circuit.

Chapter 7

Etching and Pattern Transfer

7.1 Introduction to Etch Metrics

After thin films are deposited onto a wafer, they are usually removed from the surface to form the desired geometric pattern. Sometimes, the underlying silicon substrate itself is patterned as well. In addition to pattern transfer, etching processes are heavily utilised to strip away photoresist and hard masks from the entire wafer once they have served their purpose. These subtractive processes can be performed in either a wet chemical environment or a dry plasma environment[28].

To characterise and evaluate the success of any etching process, engineers rely on four fundamental parameters:

- **Etch Rate:** The amount of material removed per unit of time. A high etch rate is generally desired to maintain acceptable wafer throughput in the fab.
- **Selectivity:** The ratio of the etch rates of different materials. High selectivity is strictly required to ensure that the protective mask (photoresist or hard mask) and the materials immediately underneath the target film are preserved during the etch.
- **Anisotropy:** It is defined as $A = 1 - \frac{R_{lat}}{R_{ver}}$ where R_{lat} is the lateral etch rate and R_{ver} is the vertical etch rate. A high degree of anisotropy is often needed, as the microscopic features patterned on the photoresist must be faithfully reproduced into the underlying film without lateral undercutting (Figure 7.1). The final profile shape of the structures must be accurately controlled.
- **Uniformity:** The consistency of the etch rate across the entire diameter of the wafer.

7.2 Wet Chemical Etching

Wet etching relies on immersing wafers into baths or spraying them directly with liquid chemical etchants. While it was the first pattern transfer method used in microelectronics, its application for defining critical features has drastically diminished.

Today, wet etching is widely used primarily for non-critical steps, such as removing hard masks, stripping photoresist, and cleaning away the stubborn polymeric by-products left behind by dry etching processes.

[28]The terms wet and dry refer to the presence of chemical liquid-phase etchants. There exist dry etching techniques that do not involve a plasma as well, but they are less common (mostly used for MEMS, advanced 3D-NAND memory and Thermal Atomic Layer Etching).

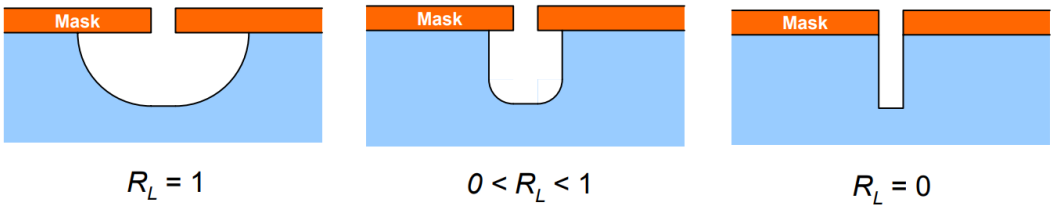


Figure 7.1: This picture shows the possible etching profiles depending on anisotropy. R_L is the lateral etch rate over the vertical etch rate. In the first case, etching is completely isotropic, extending in all directions at the same time. In the last one there is no lateral etching and the process is completely anisotropic. Picture adapted from "Introduction to Nanofabrication" by Erli Cheng at Harvard.

- **Pros:** Wet chemistry offers very high selectivity due to being driven by chemical reactions. Furthermore, it provides excellent etch rate control by being able to tune reactants/products transport from the surface as well as the temperature to influence reaction rate. Also, it has exceptionally high uniformity across the wafer and thanks to a batch system can yield high throughput.
- **Cons:** Wet etching suffers from a complete lack of anisotropy. The etch proceeds at the exact same rate in all directions (isotropic), which destroys the critical dimensions of nanoscale features. It is best used with large features when sidewall slope does not matter or to undercut the mask for easy lift-off. The only notable exception of anisotropic wet etching is crystalline orientation-dependent etching of the silicon substrate (usually used for AFM tips). In addition, it has poor process control and excessive particulate contamination.

Process Insight

When a wafer is removed from a traditional aqueous chemical bath, the liquid trapped between dense, high-aspect-ratio features begins to evaporate. As the liquid recedes, the immense surface tension of the water creates a strong capillary force that pulls the adjacent structures together, causing catastrophic pattern collapse.
To prevent these fragile nanoscale pillars from snapping or sticking together, the industry must avoid standard liquid drying. Advanced techniques bypass surface tension entirely by using HF vapour, plasma-created-salts and sublimation, cryogenic nitrogen (N_2) aerosols, or Super-critical CO_2, which inherently exhibits zero surface tension.

7.2.1 Wet Etching Examples

Here are a few chemical examples used in the industry. You don't need to memorise them but they are reported for completeness.

Etching SiO_2

The reaction used is:

$$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$$

With this, the etching rate is up to $1\,\mu m \cdot min^{-1}$ and the selectivity is high (> 100). To prevent HF depletion and consequently an etch rate drop, the HF is buffered with NH_4F:

$$NH_4F \rightarrow NH_3 \uparrow + HF$$

Etching Si_3N_4

For this etching process, hot phosphoric acid is used:

$$3Si_3N_4 + 4H_3PO_4 + 27H_2O \rightarrow 4(NH_4)_3PO_4 + 9H_2SiO_3$$

Etching Al

To etch aluminium, an oxidation is induced by a nitric acid, then a dissolution by a phosphoric acid. This cycle is then repeated multiple times. The reaction is:

$$Al + 3HNO_3 + 3H_3PO_4 \rightarrow Al(H_2PO_4)_3 + 3NO_2 \uparrow + 3H_2O$$

Etching Si

The following reaction yields an anisotropic wet etch of silicon. This is because KOH etches the (100) and (110) planes much faster than (111), creating characteristic V-shaped grooves. Tetramethylammonium Hydroxide (TMAH) and Ethylene Diamine Pyrocatechol (EDP) can also wet etch silicon.

$$Si + 2KOH + 2H_2O \rightarrow K_2SiO_2(OH)_2 + 2H_2 \uparrow$$

7.2.2 Advanced Wet Processes

As device dimensions shrink into the deep nanoscale, simply dipping a wafer into a chemical bath is no longer sufficient. Achieving high particle removal efficiency without destroying fragile transistor structures is a major engineering hurdle.

To safely dislodge nanoscale particles, fabs employ foam and bubble cleaning (using CO_2 or Nitrogen), nozzle spray cleaning, and laser-generated bubbles. Furthermore, engineers have begun researching non-Newtonian fluids to achieve zero shear stress on the wafer surface, alongside the use of micro-machined nano-tweezers for defect repair in mask fabrication.

7.3 Dry Plasma Etching Mechanisms

To overcome the catastrophic isotropic limitations of wet chemistry, the semiconductor industry transitioned to dry plasma etching for defining critical nanoscale features. A plasma is used to etch the desired material, relying on a delicate balance where both chemical reactions and physical ionic bombardment contribute to film removal.

- **Pros:** The primary advantages are high anisotropy and excellent profile shape controllability. It also removes the need for handling dangerous acids and solvents (only needing small, although still quite toxic, amounts of chemicals) and allows for better process control.
- **Cons:** The disadvantages include plasma-induced radiation damage, metallic or polymeric contamination, loading and memory effects, and the strict necessity to remove persistent etch by-products.

7.3.1 Plasma Physics and Chamber Parameters

Typical RF-powered plasma etch systems are architecturally very similar to the vacuum chambers used for PECVD or sputtering (Figure 7.2). Within the glowing discharge, both highly reactive chemical species and very directional ionic species are generated and play a role in the etch.

Physically, the plasma potential (V_P) rests at a positive value to equalise the highly disparate mobility of the electron and ion fluxes. Furthermore, to maintain current continuity across the chamber, the smaller electrode naturally develops higher electric fields, resulting in a higher RF current density. In standard plasma systems, the plasma density is closely coupled to the ion

energy[29], which is determined by the sheath voltage; consequently, increasing the RF power increases both simultaneously.

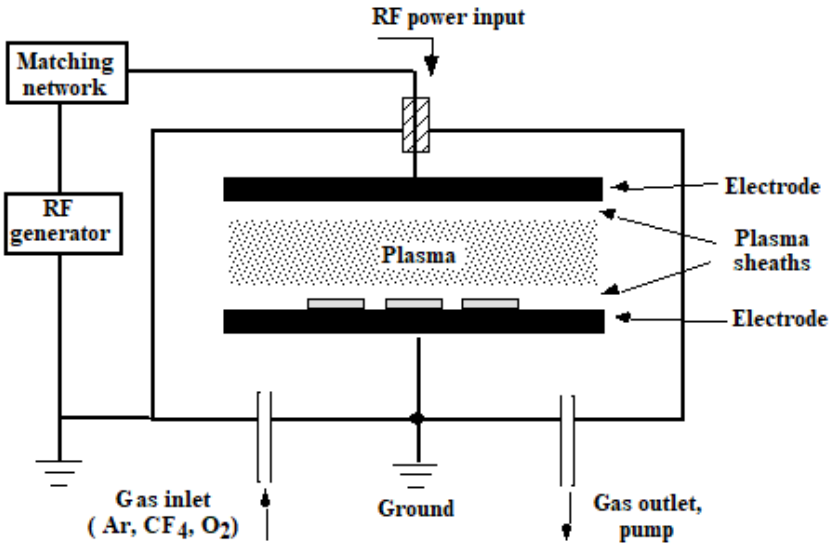


Figure 7.2: An example of a plasma chamber. The same gas system as PECVD is used. Here, the RF generator is responsible for both the ionisation and the acceleration of the ions. Nowadays, this device is used only on single wafers.

The chamber typically operates at a pressure ranging from 1 mTorr to 1 Torr. This environment contains approximately 10^{15} cm^{-3} neutral species, of which 1% to 10% may be highly reactive free radicals, alongside a much smaller concentration of 10^8 to 10^{12} cm^{-3} ions and free electrons.

Etching gases generally consist of halide-containing species such as CF_4, SF_6, Cl_2, and HBr, frequently combined with additives such as O_2, H_2, and Ar. Conversely, O_2 by itself is the standard gas used to etch (ash) organic photoresist.

7.3.2 The Three Etching Regimes

Depending on the specific chamber parameters, material removal is driven by three distinct mechanisms:

1. Purely Chemical Etching:

Gas phase etchants (the very reactive radicals) are formed in the plasma by electron collisions. For example, the dissociation of CF_4 creates reactive fluorine radicals:

$$e^- + CF_4 \rightarrow CF_3 + F + e^-$$

These radicals diffuse to the wafer and chemically react with the solid film (as shown in Figure 7.3) to form a volatile by-product, which is subsequently pumped away. For silicon, the reaction is:

$$4F + Si_{(s)} \rightarrow SiF_4 \uparrow$$

Because radicals are electrically neutral, their arrival angle distribution is uniform, mathematically modelled as $\cos^n \theta$ where $n = 1$. They also exhibit a very low sticking coefficient (S_c), meaning they easily bounce around the trench. Consequently, while purely chemical etching yields excellent selectivity, it is severely isotropic.

[29]This is because, like the system in Figure 7.2, one RF generator is used both to ionise the plasma and accelerate the ions. In modern systems these tasks are performed by different devices, allowing for more precise control.

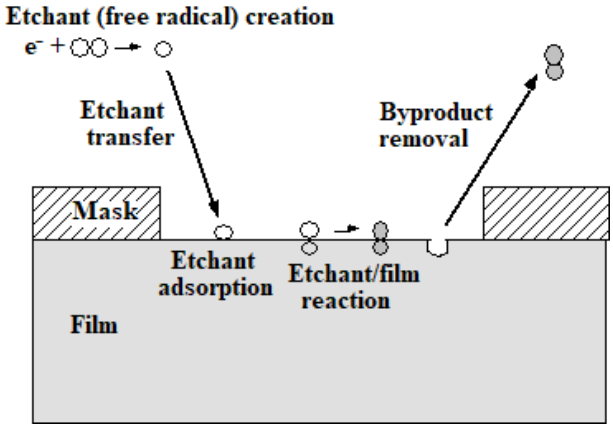


Figure 7.3: This picture illustrates the steps of the chemical part of plasma etching. Once the radicals reach the surface, they react with it, picking up atoms that are then removed as a newly-formed molecule.

2. Purely Physical Etching (Sputtering):
Alternatively, material can be removed through violent physical collisions. Because ions are accelerated vertically across the plasma sheath, they have a highly directional arrival angle distribution with a sticking coefficient of $S_c = 1$. This leads to highly anisotropic etching. However, sputtering yields are usually very low, effectively removing only one atom per incident ion. Furthermore, the sputtering yield heavily depends on the arrival angle, a physical behaviour that frequently leads to corner faceting. Note that some chemical reactions can also occur during physical ion bombardment if the ionised species themselves are reactive (e.g., Cl^+ rather than Ar^+).

3. Ion-Enhanced Etching (RIE):
Modern etching relies on a phenomenon where ion bombardment and chemical etching work in a synergetic way. When combined, the resulting etch rate is significantly higher than the mere sum of the two single, independent etch rates. Because the reaction is driven by the vertical flux of ions, the resulting anisotropy is very high. The exact physical mechanisms driving this synergetic enhancement are complex, but they primarily involve chemical reaction enhancement (where the ion impact provides the energy required for bond breaking) and active inhibitor removal.

Process Insight

To achieve perfectly vertical sidewalls, process engineers deliberately encourage the formation of an inhibitor or "passivation" layer (usually a fluorocarbon polymer) during the etch.
As the plasma generates reactive radicals and polymer precursors simultaneously, the inhibitor actively deposits on all surfaces of the trench. However, the highly directional, vertical ion bombardment constantly blasts this inhibitor layer away from the bottom of the trench, allowing the chemical etchants to continue removing the underlying film. Because the vertical ions do not strike the vertical sidewalls, the inhibitor layer stays perfectly intact on the sidewalls, acting as a microscopic shield that completely halts lateral, isotropic etching. Figure 7.4 shows the working principle of this technique.

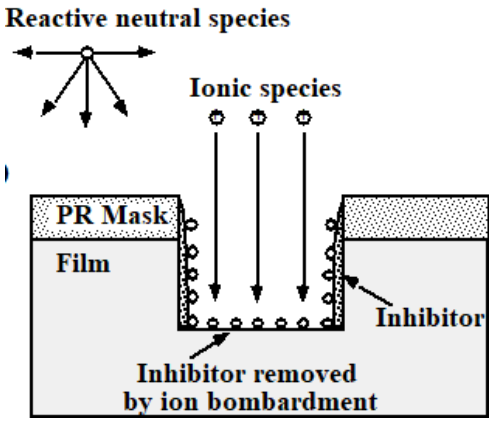


Figure 7.4: This scheme shows how the inhibitor layer can protect lateral sidewalls. The highly directional ions physically etch the passivation layer and allow chemical etching of the film. The inhibitor on the walls prevents the isotropic chemical etching from digging underneath the mask.

7.4 Etching Tools and Metrology

The physical hardware required to generate reactive plasmas has evolved significantly to meet the stringent demands of nanoscale anisotropy. It is important to note that, unlike early batch-processing furnaces, all etching systems in modern integrated circuit manufacturing are strictly single wafer tools.

7.4.1 Evolution of Etching Tools

The architecture of the etching chamber dictates the balance between chemical radical generation and physical ion bombardment. The primary tool architectures include:

1. The Barrel Etcher:

In early barrel etchers (Figure 7.5), the active plasma is kept physically separated from the wafers. Because there is no electric field directing ions toward the substrate, there is absolutely no ion bombardment. Instead, the reactive species diffuse through the chamber to react with the wafers. While this provides high chemical selectivity, it yields no anisotropy, rendering it useless for modern IC pattern transfer. However, the "downstream plasma" concept pioneered by these tools is still widely used today specifically for resist ashing.

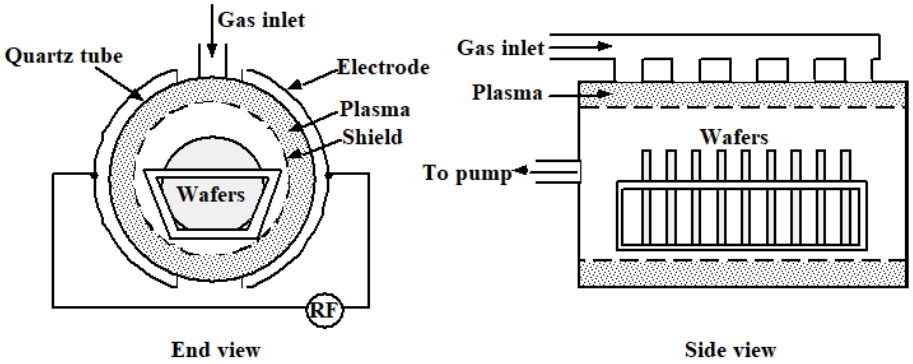


Figure 7.5: The picture shows the front and side view of a barrel etcher. The shield prevents the ions from physically impinging on the wafers. This makes it so this technique operates primarily on the chemical etching part of plasma etching.

2. The Parallel Plate Etcher:

Architecturally very similar to the PE-CVD chambers previously described, these systems sand-

wich the wafer between two electrodes (an example was shown in Figure 7.2). They can be operated in two distinct modes:

- **Plasma Mode:** Utilises symmetrical electrodes, which leads to a very small sheath voltage drop at the wafer surface and consequently poor ion bombardment.
- **Reactive Ion Etching (RIE) Mode:** Utilises strongly asymmetric electrodes; one plate is significantly smaller than the other. This asymmetry guarantees a large sheath voltage drop, driving the enhanced ion bombardment strictly required for anisotropic etching.

3. High-Density Plasma (HDP) Systems:
HDP systems represent the modern standard for critical etch steps. Their defining advantage is that plasma density and ion energy control are completely "decoupled". In these chambers (shown in Figure 7.6), the plasma is ignited in the bulk gas using Electron Cyclotron Resonance (ECR) or Inductively Coupled Plasma (ICP) systems, often utilising a planar inductive coil driven by a 13.56 MHz source power supply. Completely independently, the bottom electrode (an electrostatic chuck holding the wafer) is controlled by its own dedicated 13.56 MHz RF bias power. Operating in the very low 1 to 10 mTorr pressure range, HDP systems can simultaneously achieve exceptionally high ion density (for rapid chemical etching) and relatively low, perfectly controlled ion bombardment (to minimise substrate damage).

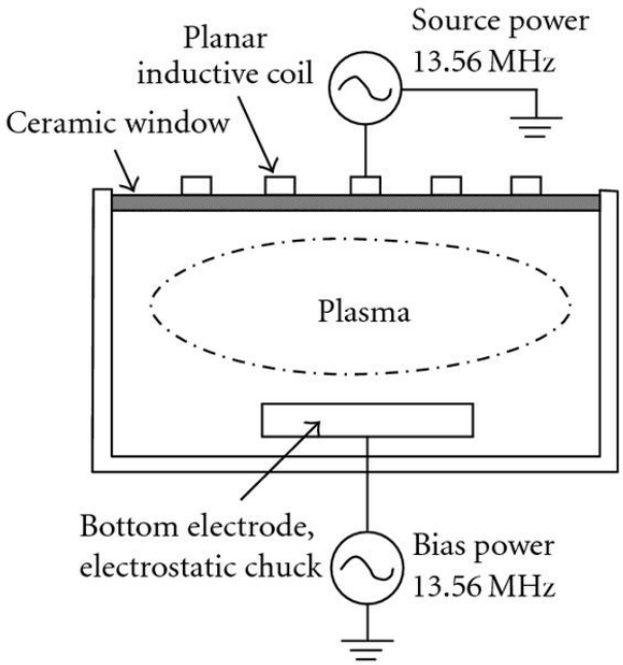


Figure 7.6: The image shows the chamber for an HDP system. Since ionisation is much more efficient, it can operate at a lower pressure. This leads to fewer collisions and thus a more anisotropic profile. Ionisation is improved by moving electrons in long circular paths, leading to higher ionisation probability.

4. Sputter Etching and Ion Milling:
In these highly specialised tools, only the physical sputtering component is exploited. A confined plasma is used to generate heavy ions (e.g., Ar^+). Charged grids are then used to accelerate these ions toward the target, while a stream of electrons is eventually used to neutralise the beam to prevent charging. Since this technique is mainly physical, it achieves rates smaller than chemical and physical plasma etching. This technique is mainly used when RIE is not

possible due to needing to avoid volatile species formation (etching Cu, Ni, Au, superconducting materials containing metals etc.)

Figure 7.7 shows a useful summary of the various techniques.

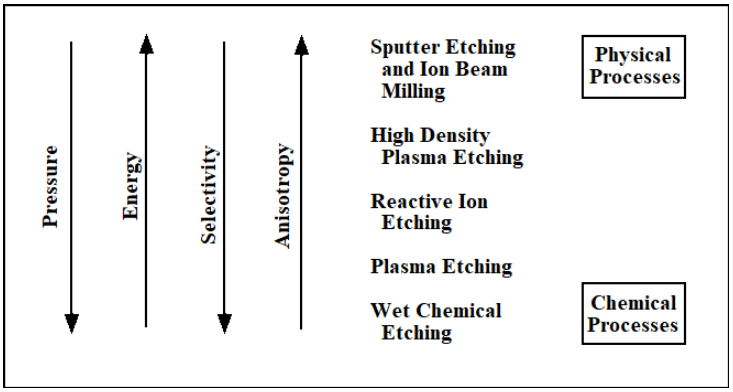


Figure 7.7: This infographic is a summary of the etching techniques shown thus far, comparing them in terms of their figures of merit and how primarily chemical/physical they are.

7.4.2 End Point Detection

Because dry etching removes material so rapidly, precisely controlling when to stop the etch is a critical metrology challenge. Ideally, it should be exactly as the target film clears but before the underlying substrate is destroyed.

Engineers achieve this using Optical Emission Spectroscopy (OES). Within the glowing plasma, various gaseous species are constantly excited and emit photons at highly specific optical wavelengths. By using OES sensors mounted to the chamber window, the system tracks the intensity of a specific wavelength relative to either a reactant gas or a reaction by-product.

For example, if the system tracks an etch by-product, the optical signal will remain high and steady as the film is actively etched. The moment the target film is completely removed, the by-product generation abruptly stops, causing a sharp drop in the tracked optical signal. This precise drop triggers the system's "End Point," instantly cutting the RF power and safely terminating the process.

7.5 Process "Knobs" and Integration Issues

7.5.1 Plasma Etching Process Parameters

Developing a successful dry etch recipe requires meticulously balancing several macroscopic chamber parameters, colloquially known as process "knobs". Because the chemical and physical mechanisms are so tightly coupled, the dependence of the etch rate and profile on these knobs is quite complicated and heavily reliant on the specific tool design.

The primary control parameters include:

- **Power:** Increasing the RF power generally increases the overall etch rate. In standard RIE systems, power ranges from 0.1 to 5 $\mathrm{W \cdot cm^{-2}}$, yielding ion energies between 10 and 700 eV. In HDP systems, power ranges from 0.1 to 3 $\mathrm{W \cdot cm^{-2}}$, producing ion energies between 10 and 500 eV. Crucially, in HDP systems, the ion bombardment can be completely independently controlled via the bias power, allowing engineers to precisely tune the etch profile.

- **Pressure:** RIE tools operate at 10 to 100 mTorr, whereas HDP tools operate at much lower pressures of 1 to 10 mTorr. Increasing chamber pressure predictably decreases the ion

density (lowering physical sputtering) but increases the chemical flux (enhancing chemical etching). Because of these competing mechanisms, the etch rate dependence on pressure is not easily predictable.

- **Gas Species and Flow Rate:** The choice of etchant gases dictates the fundamental chemistry, allowing engineers to control by-product formation and passivation layer deposition to sculpt the structural profile. The gas flow rate itself has a relatively minor impact on the etch rate; while a higher flow rate supplies more reactant species, it can also decrease the residence time of the gases in the chamber too severely.
- **Temperature:** Both the wafer chuck temperature and the chamber wall temperature must be strictly controlled, as they dictate the volatility of the etch by-products and the thermodynamic reaction rates.

7.5.2 Micro-Loading and Topographical Issues

As the plasma interacts with complex 3D topography, several integration issues inherently arise:

Loading and Micro-Loading Effects: The overall etch rate is dependent on the total exposed area, a phenomenon known as the macro-loading effect. At the microscopic scale, **micro-loading** occurs when structures with different mask opening widths or different aspect ratios etch at completely different rates (the effect is shown in Figure 7.8) due to uneven plasma density. This severe non-uniformity is caused by three primary factors:

1. The physical shadowing effects that block chemical flux.
2. The severe depletion or trapping of reactant species deep inside narrow trenches.
3. The distortion of incoming ion paths due to localised charging effects within the trench.

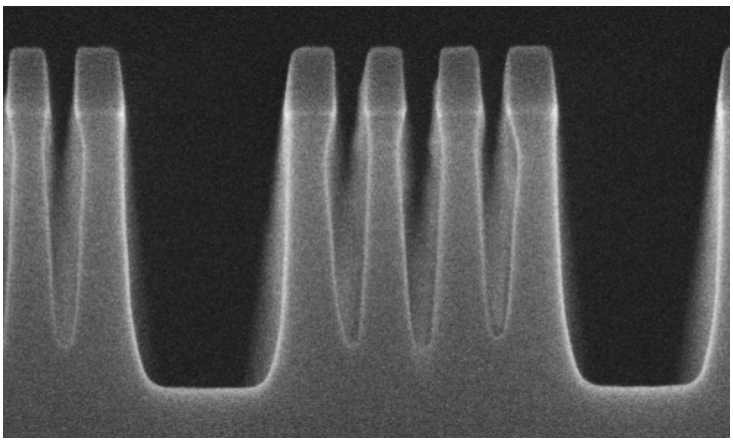


Figure 7.8: This SEM picture shows trench depths where Micro-Loading effects altered the desired shapes. The larger trenches on the sides were etched deeper due to higher plasma concentration, while the narrower trenches in the centre end up with shallower depths. The distance between these is ≈ 52 nm.

Physical Insight

Micro-loading, frequently referred to in the industry as **Aspect Ratio Dependent Etching** (ARDE), occurs when features with varying mask opening widths etch at completely different rates despite being exposed to the exact same macroscopic plasma. This severe non-uniformity is driven by a combination of restricted reactant transport, geometric shadowing, and localised electrical effects.

As a trench etches deeper and narrower, the transport of neutral chemical radicals transitions from standard viscous flow into the regime of molecular flow, known as **Knudsen diffusion**. Within these confined geometries, reactive species increasingly collide with the trench sidewalls rather than with each other, causing them to be trapped or chemically consumed before they can physically reach the bottom of high-aspect-ratio structures. This reactant depletion is further exacerbated by geometric shadowing. While isotropic neutral radicals arrive at the wafer from a wide range of angles, the physical solid angle of the open plasma "sky" visible from the bottom of a deep, narrow trench is significantly smaller than that of a wide trench, geometrically restricting the chemical flux arriving at the etch front.

Finally, the etch rate in narrow trenches is severely hindered by localised differential charging. In a plasma, highly mobile, isotropic electrons easily accumulate at the top corners of the insulating photoresist mask, charging it negatively. Conversely, the heavy, highly directional positive ions travel deep into the trench, charging the bottom positively. This separation of charge creates a severe, localised micro-electric field within the trench that actively repels and distorts the trajectories of subsequent incoming positive ions, drastically reducing the physical ion bombardment in narrow openings.

Profile Distortions: Even if the etch rate is controlled, the physical trajectory of the ions can cause highly localised defects (all shown in Figure 7.9).

- **Micro-Trenching:** When vertical ions graze the steep, passivated sidewalls of a structure, they physically deflect downwards, concentrating their bombardment at the extreme corners of the trench bottom. This physical deflection causes the corners to etch faster than the centre, leaving deep, unwanted micro-trenches at the base of the feature.
- **Resist Re-Sputtering:** The energetic ion bombardment can physically degrade the protective photoresist, sputtering its polymeric material into the trench and distorting the ion trajectories.
- **Ion trajectory distortion:** The repulsion between ions can also cause etching to spread out in between features. This contributes to the difficulty in achieving straight, vertical walls.

Figure 7.10 summarises the complex behaviour of plasma and its effects during etching.

7.5.3 Radiation Damage

Because the wafer is directly exposed to a violent plasma environment, it can suffer from severe physical and electrical degradation. The bare wafer surface can suffer from violently displaced substrate atoms, as well as the unintentional implantation of high-energy plasma species, such as hydrogen (H) and carbon (C).

Furthermore, as the plasma deposits an electrical charge on the wafer surface, large conductive structures (like metal interconnects) can act as antennas, funnelling massive amounts of charge down into the delicate gate oxides. This phenomenon, known as the **antenna effect**, causes active oxide plasma damage and gate breakdown, the severity of which can also depend on the specific substrate doping.

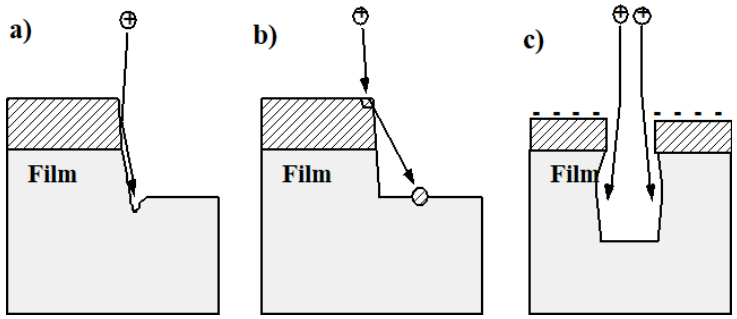


Figure 7.9: These drawings show the different localised defects that can occur during etching. a) In Micro-Trenching the edges of the trenches get etched faster due to ions "reflecting" on the walls. b) The polymer molecules constituting the resist can get dislodged and redeposit inside trenches. c) Ions can repel each other, causing etching in a more isotropic way.

7.6 Specific Etch Chemistries

The choice of reactive gas is heavily dependent on the physical and chemical properties of the target film. Process engineers carefully select gas mixtures to balance volatility, selectivity, and polymer deposition.

7.6.1 Silicon Oxide (SiO_2)

Silicon dioxide is predominantly etched using fluorine-based chemistries. Gases such as NF_3, SF_6, and CF_4 dissociate in the plasma to create highly reactive fluorine (F) radicals. An example reaction would be:

$$CF_4 + SiO_2 \rightarrow SiF_4 + CO_2$$

However, these fluorine-rich gases generally result in very isotropic etching and suffer from extremely poor selectivity to the underlying silicon.

To modify the chemistry, engineers can add O_2 to a CF_4 plasma, which increases the density of F radicals by reducing the recombination of carbon and fluorine. Conversely, to reduce isotropy and enforce vertical sidewalls, process engineers employ fluorine-poor chemistries, which are gases with a high carbon to fluorine ratio, such as CHF_3, C_3F_8, or C_2F_6. Increasing the carbon to fluorine ratio actively promotes the deposition of a fluorocarbon polymer, which serves as a robust sidewall inhibitor, granting excellent profile control.

7.6.2 Silicon and Poly-Silicon

Because the chemical by-product SiF_4 is highly volatile, fluorine-based chemistries can technically be used to etch silicon as well. However, this approach poses severe problems with SiO_2 selectivity and isotropy. Interestingly, adding O_2 to CF_4 can actually increase selectivity because the silicon etch rate gets drastically enhanced, much more than the SiO_2 etch rate.

Despite this, the industry standard for silicon and poly-silicon etching relies on halogen gases, specifically chlorine (Cl_2) and hydrogen bromide (HBr). These halogens are used to increase both selectivity and anisotropy by relying on a weaker purely chemical component. Adding small amounts of O_2 to the halogen mixture can actively trigger inhibitor deposition, allowing for precise profile control of the poly-silicon gate.

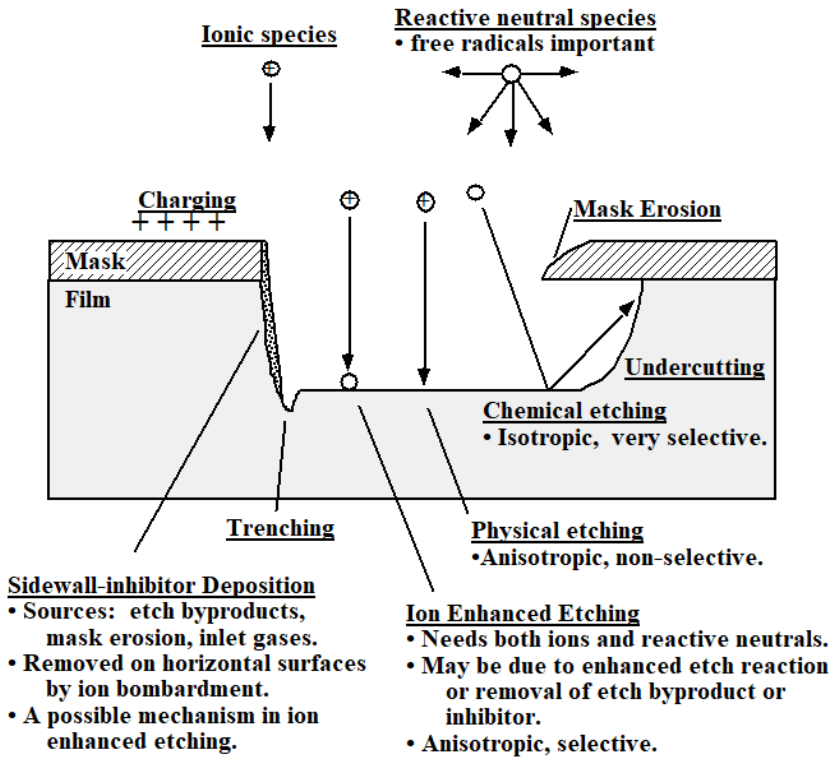


Figure 7.10: This drawing includes most outcomes that plasma ions can induce during plasma etching.

7.6.3 Aluminium (Al)

Etching aluminium interconnects presents a unique set of challenges. Initially, heavy physical ion bombardment (sputtering) is required simply to break through and remove the tough, native Al_2O_3 layer that naturally forms on the metal surface.

Once the native oxide is breached, chlorine-based chemistries are strictly used to increase the volatility of the etch by-products. Strong inhibitor usage is absolutely vital during this step to avoid severe lateral erosion of the metal lines. Furthermore, because aluminium is frequently alloyed with trace amounts of copper (Cu) to prevent electromigration, the chamber must be operated at much higher temperatures to ensure the evaporation of the stubborn copper-based by-products (copper can't be dry-etched).

Finally, aluminium etching requires immediate and stringent post-etch treatments. If any residual chlorine remains on the wafer, it will readily react with atmospheric water vapour (H_2O) to form highly corrosive hydrochloric acid (HCl), which will rapidly destroy the patterned aluminium lines.

7.7 Etching Models

Because the fundamental plasma physics and chemical interactions occurring at the atomic level are incredibly complex and not completely understood, engineers rely on simple empirical models to simulate dry etching.

In these simulations, the arrival angle distribution is typically modelled as $\cos^n \theta$, where $n = 1$ for neutral chemical species, and $n = 10$ to 80 for highly directional ions. Furthermore, the sticking coefficient (S_c) is assumed to be 1 for ionic species (which strike and embed), and very low for neutral chemicals (which bounce around the trench).

7.7.1 The Simple Linear Model

The simplest analytical approach is to imagine that physical sputtering and chemical reactions act entirely independently. The overall etch rate is mathematically defined by combining a chemical reaction rate constant (K_f), an effective sputtering yield (K_i) that depends on both the incident angle and the ion energy, the incident fluxes (F), and the physical density of the target film (N):

$$R_{etch} = \frac{(S_c K_f F_c + K_i F_i)}{N}$$

The first term of the numerator refers to the chemical contribution, whereas the second refers to the ionic one. While basic, this linear model is capable of simulating macroscopic phenomena such as micro-loading effects.

7.7.2 The Ion-Enhanced Etching Model

A far more accurate approach models the synergetic reality of Reactive Ion Etching. This model dictates that neutral species must first be adsorbed onto the wafer surface to chemically react with the film, after which the resulting by-products must be physically removed by ion bombardment.

- The number of species adsorbed is strictly proportional to the number of available sites, which are the sites on the surface that are currently free from by-products ($S_c(1-\theta)F_c$).
- The number of by-products successfully removed is strictly proportional to the incoming ion flux ($k_i \theta F_i$)

Assuming a steady-state condition:

$$S_c(1-\theta)F_c = k_i \theta F_i \implies \theta = \frac{1}{1 + \frac{k_i F_i}{S_c F_c}}$$

So the etch rate is:

$$R_{etch} = \frac{1}{N} \cdot \frac{1}{\frac{1}{k_i F_i} + \frac{1}{S_c F_c}}$$

This synergetic model dictates that if either the chemical flux or the ion flux drops to zero, the overall etch rate is zero, since both components are absolutely required to etch the material. Furthermore, the total etch rate will eventually saturate when one component becomes too large relative to the other, as the entire operation is ultimately limited by the slower of the two series processes.

While this approach boasts broad applicability for simulating plasma profiles, it is imperfect; it notably fails to mathematically account for the mechanism of independently formed inhibitor layers, nor does it perfectly model excess inhibitor formation during highly polymerising etches.

The journey through plasma etching has demonstrated how transient lithographic blueprints are violently and precisely carved into the permanent architecture of a microchip. By harnessing the synergetic interplay of reactive chemical radicals and highly directional ion bombardment, process engineers can bite into silicon, dielectrics, and metals with astounding anisotropy and selectivity.

Yet, for all this intricate patterning and physical sculpting, a wafer full of isolated transistors is fundamentally useless. These individual silicon switches cannot perform logic operations, store memory, or process information unless they are wired together into a massive, heavily coordinated electrical network. As transistor dimensions have relentlessly shrunk, the true bottleneck of semiconductor performance has shifted away from the silicon substrate itself and moved upwards, into the complex web of wiring required to tie billions of devices together without crippling signal delays.

This architectural challenge brings us to the next vital stage of fabrication: **Interconnects**. In the forthcoming chapter, we will explore how fabs construct the staggering, multi-level microscopic highways that route power and data across the chip.

Chapter 8

Interconnects and Metallisation

8.1 Introduction to the Back End of the Line (BEOL)

Once the intricate array of millions of individual transistors has been successfully fabricated within the silicon substrate, the devices remain isolated and functionally useless. The Back End of the Line (BEOL) encompasses the sequence of manufacturing steps dedicated to wiring these active devices into a specific, functional circuit configuration (like the one shown in Figure 8.1). This complex, three-dimensional architecture is constructed from local interconnects, intermediate and global metal lines, vertical vias, and the insulating dielectric layers that prevent the entire network from short-circuiting.

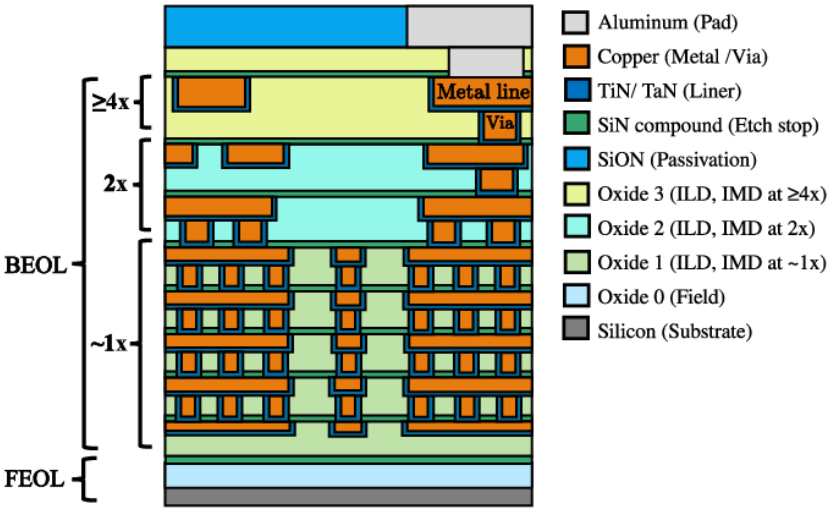


Figure 8.1: This image shows the end result of the BEOL. All circuit components created on the substrate during the Front End of the Line (FEOL) are wired together exploiting the vertical direction. Image taken from IEEE Xplore.

To build this complex, three-dimensional architecture, the BEOL relies on a distinct hierarchy of structural elements:

- **Local Interconnects:** Silicided transistor gates and silicided active areas (source and drain regions) that provide the absolute lowest level of routing.
- **Contacts:** Vertical conductive plugs responsible for connecting these local interconnects and active devices directly to the first metal level.
- **Intermediate and Global Interconnects:** The primary horizontal metal lines that wire the active elements into functional circuits.

- **Dielectrics:** Insulating materials required to electrically isolate the different metal levels and adjacent lines from one another.
- **Vias:** Vertical conductive plugs used specifically for connecting different metal lines across separated dielectric layers.

8.1.1 Local Interconnects and Silicides

The lowest level of this wiring hierarchy consists of local interconnects, which physically reside below the lowest global metal level. They are primarily used to strap source and drain regions together to allow for extremely low-resistance contacts, and to reduce the sheet resistance of polysilicon gates while leaving the critical poly-silicon dioxide interface completely unchanged.

To achieve this, the industry relies on a specific class of metal-silicon alloys known as silicides. Silicides exhibit highly desirable properties for local routing, including very low resistivity (typically 10 to 50 $\mu\Omega\cdot$cm), excellent high-temperature stability, good compatibility with both silicon and subsequent deposited materials, and an ease of plasma etching[30]. Crucially, unlike the aluminium or copper used in higher metal layers, silicides do not suffer from electromigration (a topic which will be treated in Chapter 8.4).

8.1.2 Silicide Formation: The Salicide Process

Silicides can technically be formed through direct deposition via Physical Vapour Deposition or Chemical Vapour Deposition, though CVD is practically only used for tungsten silicide (WSi2). This technique is not really used any more.

Alternatively, they can be formed through a standard reaction process where metal and amorphous silicon are sequentially deposited, patterned, and annealed to force a chemical reaction.

However, modern IC fabrication overwhelmingly relies on the elegantly efficient **Self-Aligned Silicide (Salicide)** process. In this approach, a metal is sputtered directly over the entire wafer, completely blanketing the exposed polysilicon gates, the silicon source/drain regions, and the surrounding isolation oxides. Upon annealing, the metal chemically reacts exclusively with the exposed silicon. Because no reaction occurs over the insulating field oxide zones, the unreacted metal remains pure and can be selectively stripped away, leaving precisely self-aligned conductive caps only where needed.

[30] Actually, this was the case when using tungsten or titanium silicide. Nowadays, as we'll see later, the industry uses cobalt and nickel silicides, whose by-products after standard plasma etching are non-volatile, thus creating an unwanted metallic film.

Process Insight

To illustrate the precise thermal and chemical control required in the BEOL, consider the classic Self-Aligned Titanium Silicide process:

1. **Preparation:** The native (maybe because it was in contact with the atmosphere) or screen silicon dioxide is meticulously removed using an argon sputter etch[31] and a wet HF etch[32].
2. **Deposition:** Titanium is deposited via sputtering to a thickness of roughly 50 to 100 nm.
3. **First Anneal:** The wafer undergoes a 15 to 60 second nitrogen (N2) anneal at 600 to 700°C. The titanium reacts with the underlying silicon to form TiSi2, while simultaneously reacting with the nitrogen ambient at its top surface to form a titanium nitride (TiN) barrier. This relatively low temperature intentionally prevents excessive lateral growth, but it structurally yields TiSi2 in its higher-resistance C49 crystalline phase (60 to 70 µΩ·cm).
4. **Selective Strip:** The TiN and any unreacted titanium over the oxide regions are chemically stripped off using an SC1 wet etch (NH4OH:H2O2:H2O).
5. **Phase Transformation (Second Anneal):** A secondary, higher-temperature anneal at 800°C is performed to structurally convert the TiSi2 into its highly conductive C54 phase (13 to 16 µΩ·cm). Both the temperature and the duration of this final step must be strictly minimised to prevent the physical agglomeration of the film and the unwanted diffusion of substrate dopants into the metal layer.

8.1.3 Evolution of Silicide Materials

Historically, various transition metals were explored for salicidation. While TiSi2 and cobalt silicide (CoSi2) (a comparison of which is shown in Figure 8.2) dominated early nodes due to their excellent 14 to 20 µΩ·cm resistivities, advanced scaling exposed their physical limitations.

As the International Technology Roadmap for Semiconductors (ITRS) predicted, narrow line salicidation and maintaining physical compatibility with ultra-shallow junctions became severe integration bottlenecks. Consequently, nickel silicide (NiSi), frequently alloyed with platinum (Pt), has become the current mainstream material for advanced integrated circuits. NiSi consumes significantly less silicon during its reaction than Ti or Co, making it the superior choice for preserving the integrity of incredibly shallow transistor junctions.

8.2 Contacts and Plugs

Contacts provide the critical, low-resistance vertical connections between the active semiconductor devices and the first global metal layer. To ensure optimal device performance, the materials used for these connections must guarantee an Ohmic electrical contact, exhibit low resistivity, and maintain strict thermal stability throughout the remainder of the process flow.

[31] The sputtering can be done in the same chamber as the titanium deposition; the plasma pulse can be controlled to sputter either the substrate or the target

[32] Usually after a wet etch there is always a rinse, which tends to remove all hydrogen from the surface. This leaves silicon's dangling bonds prone to react again to form another oxide layer. That's why an HF-last process is usually done, where the chemical etch is the last step and the rinse is not performed. This way, the hydrogen from the HF can bond to silicon to stop it from reacting again.

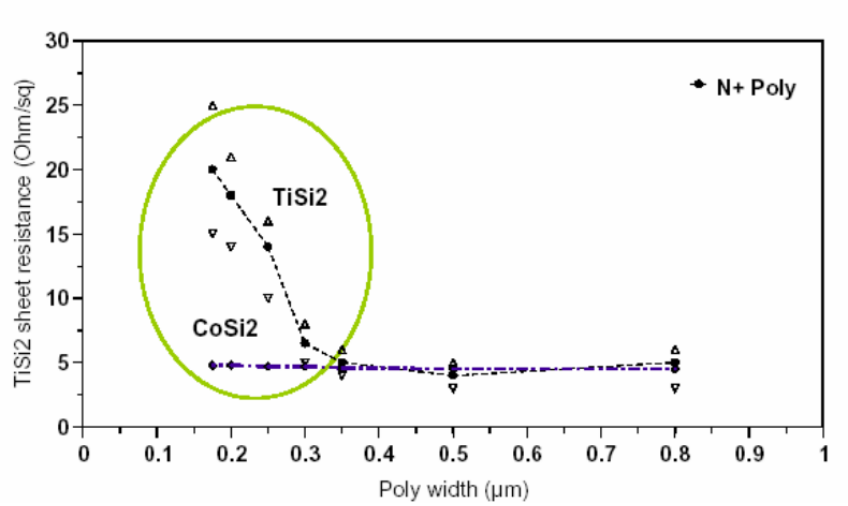


Figure 8.2: This graph shows a comparison between cobalt and titanium silicide in terms of sheet resistance vs width. In theory, the resistance should be independent, as the $CoSi_2$ behaves. However, the resistance of $TiSi_2$ increases at low dimensions. This is because in narrow lines, the number of grain boundaries drastically decreases, which in turn reduces the number of nucleation sites. With few sites, annealing doesn't allow the titanium silicide to reconstruct into a stable, low-resistance configuration.

8.2.1 Metal-Semiconductor Junction Physics

When a metal is deposited directly onto a semiconductor, the resulting electrical interface can behave either as a rectifying (Schottky) contact or as an Ohmic contact (shown in Figure 8.3). In principle, the physical barrier height (Φ_b) is strongly dependent on the metal's work function; however, in physical devices, interface states tend to pin the Fermi level deep within the semiconductor band gap. This typically results in a high barrier height to n-type silicon and a relatively low barrier to p-type silicon.

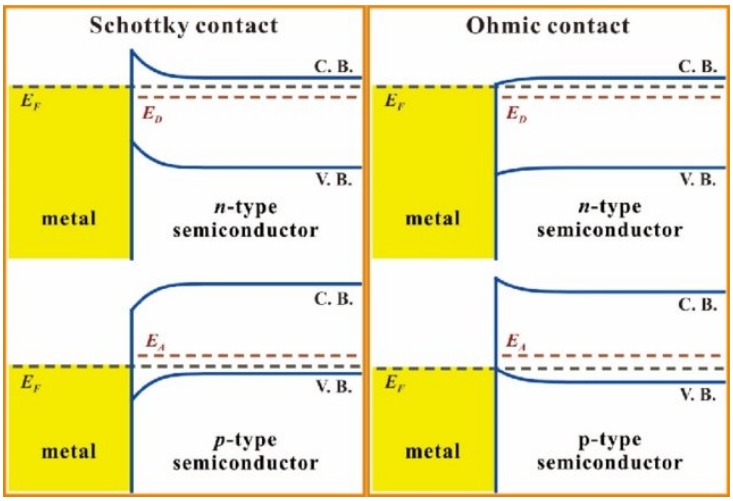


Figure 8.3: These graphs show the ideal behaviour of the bands in a semiconductor at a junction with a metal. The bands bend depending also on whether the semiconductor is p-doped or n-doped. Together with the fact that the junction can present a Schottky barrier or be Ohmic, this creates four possible outcomes. Image taken from ResearchGate.

Physical Insight

When a metal and a semiconductor are physically brought into contact, thermodynamic laws mandate that their Fermi levels must perfectly align to achieve thermal equilibrium. In an ideal, defect-free scenario, this alignment forces electrons to flow from the material with the higher Fermi energy to the lower, creating a localised region of depleted charge in the semiconductor. This depletion causes the semiconductor's energy bands to bend, establishing a physical potential barrier, Φ_b, that electrons must overcome to cross the junction. Ideally, according to the Schottky-Mott rule, the height of this barrier is determined strictly by the difference between the metal's work function (Φ_m) and the semiconductor's electron affinity (χ_s), mathematically expressed as:

$$\Phi_b = \Phi_m - \chi_s$$

However, physical reality dictates that a bare silicon crystal lattice is abruptly truncated at the surface, leaving millions of dangling atomic bonds and structural defects. These surface disruptions create a massive density of allowed electronic states that exist directly within the normally forbidden band gap, known as surface states.
When the metal contacts the semiconductor, these surface states act as a massive, localised reservoir of charge. The charge transfer required to align the two Fermi levels is almost entirely absorbed by filling or emptying these surface states, rather than stripping electrons from the bulk semiconductor itself. Consequently, the bulk energy bands remain largely unaffected, and the Fermi level becomes rigidly anchored, or "pinned," exactly at the energy level of the interface states. Because this pinning effect completely dominates the electrostatics of the junction, the final barrier height is dictated by the defective surface chemistry of the silicon, rendering it practically independent of whichever metal was actually deposited.

The standard figure of merit for evaluating the interface is the specific contact resistivity (ρ_c), mathematically defined as:

$$\rho_c = \left(\frac{\partial V}{\partial J}\right)_{V=0}$$

For a contact whose charge transport is dominated by standard thermionic emission current:

$$J_s = A^* T^2 e^{\frac{-q\Phi_b}{k_B T}} \cdot \left(e^{\frac{qV}{k_B T}} - 1 \right)$$

where A^* is the Richardson-Dushman constant $A^* = \frac{qm^* k_B^2}{2\pi^2 \hbar^3}$ and m^* is the effective mass of the charge carrier. In this case, the specific contact resistivity depends strictly on the barrier height:

$$\rho_c = \frac{k_B}{qA^* T} e^{\frac{q\Phi_b}{kT}}$$

Because this barrier is usually too high for most common metals, standard thermionic junctions exhibit unwanted rectifying behaviour. To achieve true Ohmic behaviour, the contact must operate in the quantum tunnelling regime. In this specific regime, the specific contact resistivity becomes heavily dependent on the surface doping concentration (N_d):

$$\rho_c \propto \exp\left(\frac{2\sqrt{m^* \epsilon_s}}{\hbar} \frac{\Phi_b}{\sqrt{N_d}} \right)$$

This equation mathematically dictates that increasing the surface doping (N_d) significantly lowers the specific contact resistivity, as the higher dopant concentration physically shrinks the width of the barrier, thus making it more likely to tunnel through it. Therefore, heavily doped silicon substrates are strictly required to obtain a functional Ohmic contact.

8.2.2 Contact Resistance and Scaling Issues

The actual, measurable contact resistance (R_c) of the physical structure can be roughly estimated by dividing the specific contact resistivity by the physical contact area (A):

$$R_c = \frac{\rho_c}{A}$$

Because of this inverse relationship, contact resistance remains a major technology scaling issue; as feature sizes shrink and the area A decreases for every new generation, the physical resistance inherently skyrockets. Furthermore, accurate physical models must also account for secondary electrical effects, such as current crowding at the edges of the contact interface.

8.2.3 The Evolution of the Contact: Aluminium Spiking

In early semiconductor devices, the contact and the first interconnect level were formed simultaneously by depositing aluminium directly in contact with the silicon substrate. Aluminium possesses good adhesion to silicon and effectively reduces native SiO_2 during a hydrogen anneal at 450°C.

However, silicon is highly soluble in aluminium, reaching roughly 1% solubility at 500 °C. During high-temperature processing steps, the silicon substrate would actively dissolve into the overlying metal. To fill the resulting voids in the substrate, the aluminium would migrate downwards, a phenomenon known as “aluminium spiking” (shown in Figure 8.4). This spiking destroys the ultra-shallow junction depths, completely short-circuiting the underlying transistor and strictly necessitating the introduction of robust metallic barrier layers like Ti and TiN.

8.2.4 Modern Tungsten (W) Plugs

To permanently solve the spiking and planarity issues of legacy aluminium, modern contacts are fabricated as completely isolated tungsten (W) plugs. The modern process flow (pictured in Figure 8.5) proceeds as follows:

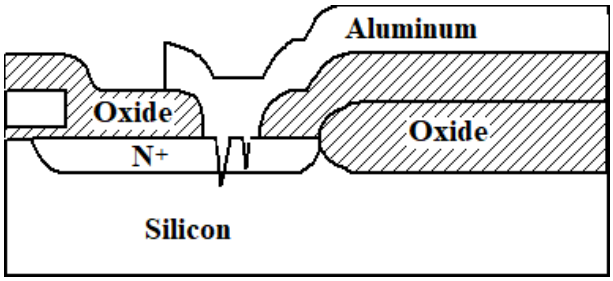


Figure 8.4: This picture shows the issues with aluminium interconnects. Due to the high solubility, aluminium is able to penetrate the n-doped silicon substrate, creating unwanted short-circuits. Moreover, planarity is also an issue, since aluminium was just deposited over the nanostructures. If it weren't for these issues aluminium would be an amazing material for interconnects.

1. **Dielectric Deposition:** A Pre-Metal Dielectric (PMD) is deposited over the transistors and subsequently planarised.
2. **Contact Etch:** The physical contact plug hole is etched vertically through the dielectric, stopping directly on the silicided active area.
3. **Barrier Deposition:** A thin layer of titanium (Ti) is deposited to improve structural adhesion, immediately followed by a layer of titanium nitride (TiN). The TiN acts as an absolute barrier to prevent WF_6 precursor gas from diffusing into the silicon during the subsequent tungsten deposition step.
4. **Tungsten Fill:** The bulk tungsten is deposited using CVD techniques to completely fill the vertical plug.
5. **Planarisation:** A W etch-back or Chemical Mechanical Planarisation (CMP) is applied to remove the excess tungsten from the wafer surface, guaranteeing an exceptionally high degree of planarity before the first global metal level is deposited.

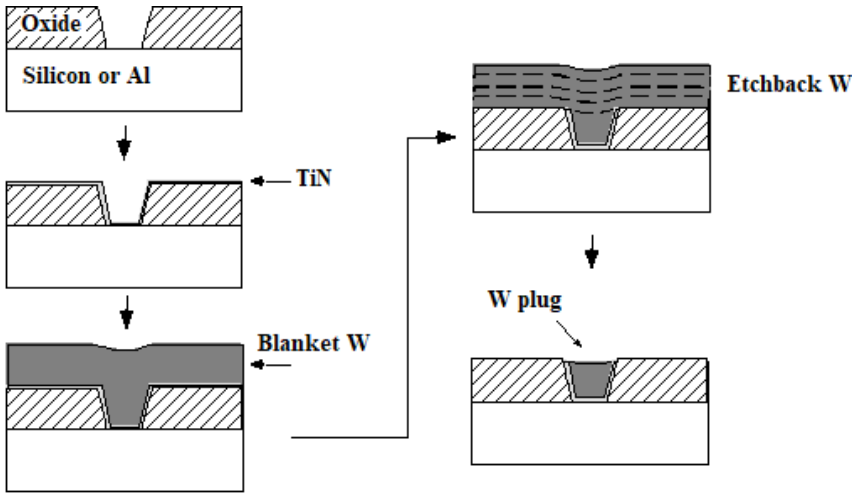


Figure 8.5: The scheme highlights the main steps in the creation of a tungsten contact. The titanium layer is like the chocolate in cannoli. At the end, a tungsten "plug" is formed, perfectly connecting the silicon layer through the oxide with the above structures.

8.3 Global Interconnects and the Aluminium Era

As the manufacturing sequence moves above the contact level, the focus shifts to routing power and data across the vast expanse of the chip. The metals chosen for these intermediate and global interconnects must strictly adhere to several core requirements:

- Extremely low electrical resistivity to minimise signal delay.
- Good physical adhesion to the underlying dielectric films.
- High thermal and mechanical stability during subsequent processing steps and continuous circuit operation.
- The physical ability to be easily deposited and accurately patterned via plasma etching.

While modern interconnects rely heavily on copper (1.7 to 2.0 µΩ·cm), the historical mainstream material for global routing was aluminium (2.7 to 3.0 µΩ·cm). Aluminium became the legacy standard because it provided a highly favourable balance of relatively low resistivity, excellent adhesion to silicon dioxide, and relatively easy deposition and plasma etching capabilities.

8.3.1 Thermal Mismatch: Hillocks and Voids

Despite its processing advantages, pure aluminium possesses a fatal physical flaw when integrated onto a silicon substrate: a severe thermal expansion mismatch. The thermal expansion coefficient for aluminium is extremely high (23×10^{-6} K^{-1}), whereas the coefficient for the underlying silicon substrate is comparatively rigid (2.6×10^{-6} K^{-1}).

Because the metal expands and contracts nearly ten times as much as the substrate, despite not touching, massive mechanical stress builds up within the aluminium film during the thermal fluctuations of the BEOL process flow.

- **Hillocks:** When the wafer is heated, the expanding aluminium is severely constrained by the substrate, placing the film under massive compressive stress. To relieve this stress, the metal extrudes upwards, forming microscopic spikes known as hillocks (shown in Figure 8.6) that can puncture overlying dielectrics and cause short circuits.
- **Voids:** Conversely, when the wafer cools back down, the aluminium attempts to shrink but is held firmly in place by the silicon, placing the film under severe tension. This tensile stress forces the metal to pull apart, creating empty voids that can completely sever the electrical connection.

To mitigate these catastrophic stress-induced deformations, engineers discovered that alloying the aluminium with a small amount of copper (a few percent by weight) effectively stabilises the metal's grain boundaries and drastically minimises the formation of hillocks.

8.4 Electromigration and Reliability

While thermal mismatch poses a significant thermo-mechanical challenge during fabrication, the most severe threat to the long-term reliability of an integrated circuit occurs entirely during its operational lifetime. Scaling forces enormous currents through microscopically thin wires, leading to a catastrophic failure mechanism known as electromigration.

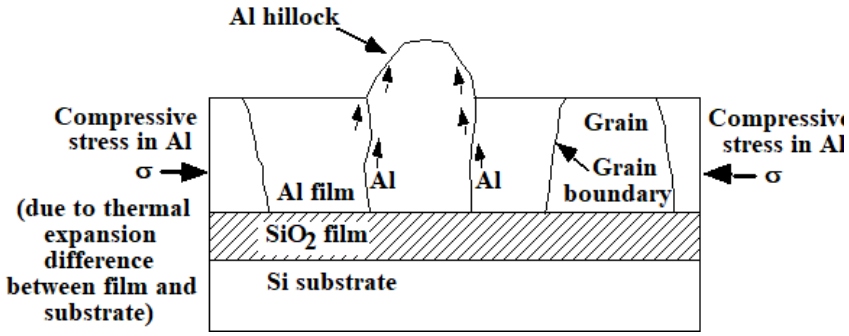


Figure 8.6: This picture shows the effect of compression of an aluminium interconnect. Due to its expansion, stress builds up inside the film, causing hillocks to form. Depending on the thickness of the film and heating length, hillocks can grow from several nanometres to a few micrometres.

8.4.1 The Electron Wind and Flux Divergence

When a metal interconnect operates at a very high current density (typically in the range of 0.1 to 0.5 MA $\cdot$ cm^{-2}), the massive flux of moving electrons violently collides with the aluminium lattice. The momentum transferred from this relentless "electron wind" physically forces the aluminium atoms to migrate in the direction of the electron flow.

Because the microcrystalline structure of a metal line is not perfectly uniform, this atomic flow experiences flux divergence. At specific points in the line, atoms migrate away faster than they are replaced, creating a severe depletion of material; conversely, at other points, atoms accumulate because the local outward flux is restricted. The effect of this migration is exactly the same as that of thermal stress: the creation of voids, which act as open circuits, and hillocks, which act as short circuits (shown in Figure 8.7).

- **Hillocks:** At regions of atomic accumulation, the excess material extrudes outward, forming hillocks. If these extrusions puncture the surrounding dielectric, they will contact an adjacent metal line and cause unwanted short circuits.
- **Voids:** At regions of atomic depletion, the loss of material eventually creates microscopic voids. As these voids grow, they can completely sever the wire, leading to an open-circuit failure.

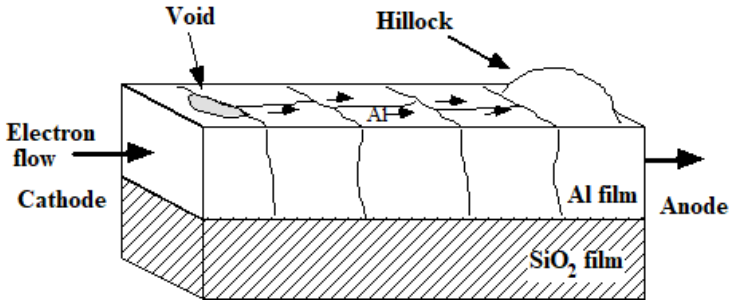


Figure 8.7: This picture shows the effects of electromigration. Atoms are dragged from the cathode side, creating voids upstream and hillocks downstream. This effect is actually synergistic with that of thermal stress, as moving electrons heat up the surrounding by Joule effect, thus augmenting the creation of voids and hillocks.

To predict the operational lifespan of a chip, the failure of interconnect lines is measured statistically. The cumulative probability of failure is typically plotted against time on a lognormal plot (Figure 8.8). From this data, engineers extract the Median Time to Failure (MTTF), a

metric that is fundamentally dependent on both the applied current density and the operating temperature via Black's equation:

$$MTTF = AJ^{-n} \cdot e^{\frac{E_a}{k_B T}}$$

where n is a dimensionless scaling factor, which is closer to 1 if the formation of voids is dominated by atomic drift and closer to 2 if Joule heating significantly accelerates the damage, while E_a represents the energy required for an aluminium atom to jump to a neighbouring vacancy.

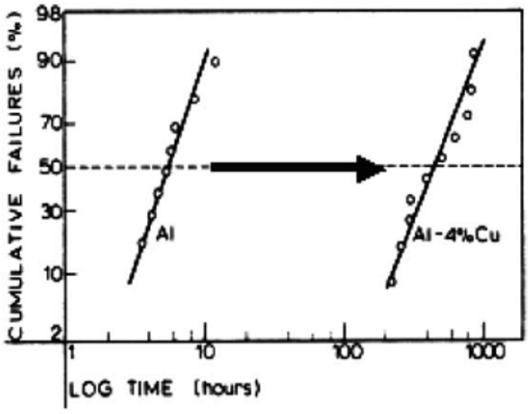


Figure 8.8: The picture shows a lognormal plot comparing cumulative failures percentage versus time. The dashed horizontal line indicates the Median Time to Failure. From this graph we can see that simply alloying aluminium with 4% copper drastically increases the MTTF.

8.4.2 Microstructure and the Bamboo Effect

Experimental data reveals that MTTF does not depend monotonically on the ratio of line width to grain size ($\frac{w}{d}$). As an aluminium line is made narrower (decreasing $\frac{w}{d}$), the MTTF initially drops, indicating faster failure. But as the line width becomes extremely narrow the MTTF abruptly increases, leading to a significantly more robust interconnect. This behaviour is shown in Figure 8.9.

This phenomenon occurs because extremely narrow lines force the metal grains to span the entire width of the wire, creating a "bamboo" structure. Because electromigration heavily exploits grain boundaries as rapid diffusion pathways, the spanning grains of a bamboo structure physically block the flow of atoms, as there are no continuous grain boundaries running parallel to the direction of current flow. Consequently, very little to no diffusion happens across these spanning grains. On the other hand, wide lines allow for an intricate network of pathways atoms can take to diffuse. This makes it so atoms seldom stop and accumulate. This means there is a middle-ground structure where atoms can diffuse for short paths parallel to the direction of flow before stopping. At these "barriers", atoms can't progress their diffusion and thus create hillocks.

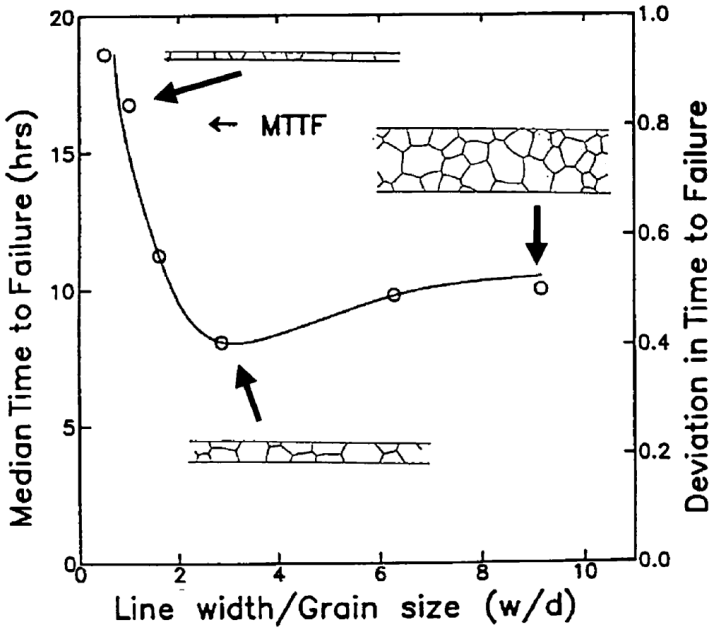


Figure 8.9: The graph shows the MTTF as a function of line width over grain size ($\frac{w}{d}$). For small $\frac{w}{d}$, the structure takes the name of "bamboo" and has high MTTF. The function has a minimum for the poly-granular structure. From there, increasing $\frac{w}{d}$ further increases the MTTF. Image adapted from Science Direct.

Physical Insight

When an interconnect is patterned into a bamboo structure, electromigration forces the aluminium atoms to deplete at one spanning grain (acting as the cathode) and accumulate at the next spanning grain (the anode).
Because the metal is rigidly confined by the surrounding dielectric, this forced redistribution of mass builds up an immense, localised stress field. The accumulation point is placed under compressive stress, while the depletion point experiences severe tensile stress. This mechanical stress gradient pushes back against the electron wind, forcing the aluminium atoms to flow backwards.
At steady state, a delicate equilibrium is reached where the forward electromigration flux and the backward stress-induced flux perfectly balance out, halting further degradation. However, if the built-up stress overcomes a specific critical level before this steady state is reached, the line will undergo irreversible deformation, physically tearing open (voids) or bursting outward (hillocks) to relieve the pressure.

8.4.3 A Mathematical Model of Electromigration Stress

To rigorously understand the physical interplay between the electron wind and the mechanical stress within a confined interconnect, we can construct a mathematical model of the atomic flux. The fundamental driving force behind electromigration is the momentum transfer from the electron wind to the metal lattice. The physical force (F) exerted on the aluminium atoms is proportional to the effective charge number (Z^*), the elementary charge (q), the metal's electrical resistivity (ρ), and the applied current density (J):

$$F = Z^* q \rho J$$

This continuous force drives a forward flux of metal atoms through the interconnect. As these migrating atoms encounter a blocking boundary (such as a spanning bamboo grain) they

begin to accumulate at the anode and deplete at the cathode. Because the interconnect is rigidly confined by the surrounding dielectric materials, this mass redistribution builds up a local mechanical stress field (σ). This resulting stress gradient ($\frac{\partial \sigma}{\partial x}$) across the length of the atomic cluster generates a secondary, reverse flux that actively pushes back against the electron wind. The total net flux (φ) of atoms is mathematically expressed by combining the thermally driven electromigration flux with the opposing stress-induced flux:

$$\varphi = \frac{DC}{k_B T}\left(Z^* q \rho J + \omega \frac{\partial \sigma}{\partial x}\right)$$

In this equation, D is the diffusion coefficient, C is the atomic concentration, and ω represents the atomic volume.

As the mechanical stress continues to build at the blocking boundaries, the system eventually approaches a stationary, steady-state condition where the forward electron wind and the backward mechanical stress perfectly balance each other out. In this stationary condition, the net atomic flux (φ) becomes precisely zero. By setting the flux equation to zero, we can isolate the stress gradient, which equates to a negative constant value ($-G$):

$$\frac{\partial \sigma}{\partial x} = -\frac{Z^* q \rho J}{\omega} = -G$$

By integrating this constant stress gradient over the physical length of the poly-granular cluster (L_p), we can determine the exact stress distribution along the line. Assuming the stress is zero exactly at the centre of the cluster, the stress equation takes the linear form $\sigma = G(\frac{L_p}{2} - x)$. This linear relationship reveals that the absolute maximum stress (σ_{max}) occurs at the extreme ends of the cluster (where $x = 0$ or $x = L_p$). Evaluating the equation at these boundaries yields the critical maximum stress value:

$$\sigma_{max} = \pm \frac{Z^* q \rho J}{\omega} \frac{L_p}{2}$$

This final equation provides profound physical intuition regarding interconnect reliability. It dictates that for a fixed operating current density, the absolute maximum stress the metal experiences is directly proportional to the physical length of the poly-granular cluster (L_p). Therefore, the longer the average poly-granular cluster, the higher the maximum stress, which inherently leads to a shorter Median Time to Failure (MTTF) as the metal is more likely to critically rupture or extrude.

This mathematical reality perfectly explains the counter-intuitive relationship between line width and failure time under fixed aluminium deposition conditions. In extremely large, wide lines, there are practically no spanning grains to block the flow of atoms; furthermore, the large metal pads at the ends of these lines act as virtually infinite sources and sinks for aluminium atoms, meaning no flux divergence occurs and the lines generally only fail at the extreme ends or vias. For intermediate line widths, the time to failure is heavily dependent on the cluster length (L_p). As these intermediate lines are scaled down to be slightly narrower, the average L_p physically shrinks, thereby reducing the maximum stress and actually leading to a longer, more favourable MTTF. Finally, in very narrow lines, the entire structure consists solely of spanning grains, creating a highly robust bamboo structure that physically prevents long poly-granular clusters from forming in the first place.

8.4.4 Mitigation Strategies in Aluminium Interconnects

To inhibit electromigration in legacy aluminium interconnects, process engineers employ two primary material strategies:

1. **Alloying:** Pure aluminium is exceptionally vulnerable to electromigration. By adding 0.5 to 4 wt% Copper (Cu) and 1 to 2 wt% Silicon (Si), the copper atoms precipitate into the grain boundaries, physically blocking the diffusion pathways and drastically increasing the MTTF.

2. **Multi-Layer Stacks:** Instead of depositing a single monolithic slab of aluminium, the interconnect is fabricated as a complex multi-layer stack, typically structured as Ti / TiN / Al(Cu) / TiN.

    - The bottom titanium's grain texture is purposely engineered so as to induce a specific grain texture on the overlying aluminium, which reduces electromigration effects.
    - The robust titanium nitride layers act as structural rigidifiers, effectively containing the stress-induced deformation of the aluminium line.
    - Most importantly, if a void does form within the Al(Cu) layer, the conductive Ti and TiN layers act as redundant electrical pathways, successfully shunting the current around the void to keep the circuit operational. Furthermore, TiN serves as a barrier for WF_6 during subsequent via processing.

8.5 Back-End Dielectrics and the RC Bottleneck

To physically support and electrically isolate the complex web of metal wiring, the interconnects are entirely encased in inter-metal and pre-metal dielectrics. The structural and electrical demands placed on these back-end insulators are exceptionally high. Process engineers require materials that exhibit excellent electrical isolation, a high breakdown electric field strictly greater than 5 $MV \cdot cm^{-1}$, good adhesion to adjacent metallic films, and strong structural density entirely free of contaminants. Furthermore, these layers must remain physically permeable to hydrogen, allowing it to diffuse down to the substrate to passivate dangling bonds at the silicon interface during final annealing steps.

Building a multi-level interconnect network also necessitates perfectly flat surfaces to prevent depth-of-focus failures during lithography. Therefore, rigorous dielectric planarisation is strictly needed before the patterning of any subsequent contacts, metal lines, or vias. While early generations achieved this through the thermal reflow of doped glasses (like PSG or BPSG) or sacrificial etch-back processes, Chemical Mechanical Planarisation (CMP) is now the definitive mainstream approach for all advanced nodes.

8.5.1 Technology Scaling and the RC Delay Crisis

For decades, the semiconductor industry operated under the assumption that shrinking transistors inherently made chips faster. However, as devices scaled deep into the sub-micron regime, a severe physical limitation emerged in the Back End of the Line: the Resistance-Capacitance (RC) bottleneck.

The time delay (τ) of an electrical signal transmitted over a metal line can be mathematically estimated as[33]:

$$\tau = 0.89RC$$

To understand why scaling is so problematic, we must express this delay as a direct function of the back-end geometrical features (shown in Figure 8.10).

[33]The 0.89 coefficient comes from the fact that we consider the rise time as delay (time it takes for the signal to go from 10% to 90%). Since interconnects can't be simply considered as a resistor and a capacitor due to their length, the actual signal wave should be treated through its Fourier components. By taking the 0th term as it's the one which decays the slowest (dominant pole approximation) and finding the rise time, we get a coefficient of $\frac{4 \ln 9}{\pi^2} \approx 0.89$.

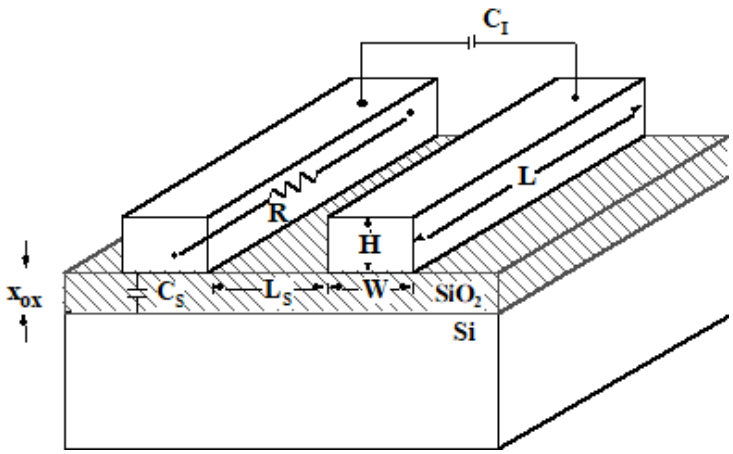


Figure 8.10: This picture shows an example of the capacitances afflicting the interconnects. In this case, C_I refers to the capacitance due to the dielectric standing directly between the two lines, not represented in the drawing.

The physical resistance (R) of the metal line is defined by its dimensions and its inherent material resistivity:

$$R = \rho \frac{L}{WH}$$

where ρ is the metal resistivity, L is the length of the line, W is its physical width, and H is its height or thickness.

Simultaneously, the total parasitic capacitance (C) is not merely a single parallel plate; it is the sum of two distinct physical components: the substrate capacitance (C_S) coupling vertically down to the underlying layers, and the inter-line capacitance (C_I) coupling laterally to adjacent metal lines on the same level:

$$C = C_S + C_I = \epsilon_{ox}\epsilon_0 \frac{WL}{x_{ox}} + \epsilon_{ox}\epsilon_0 \frac{HL}{L_s}$$

In this expression, ϵ_{ox} is the relative dielectric constant of the insulating oxide, x_{ox} is the physical thickness of the underlying dielectric layer, and L_s is the physical spacing between adjacent lines.

By substituting these expressions for R and C into the delay equation, multiplying the terms, and factoring out the common variables, we arrive at the full analytical equation for interconnect delay:

$$\tau = 0.89 K_i \epsilon_{ox}\epsilon_0 \rho L^2 \left(\frac{1}{H x_{ox}} + \frac{1}{W L_s} \right)$$

Notice that an empirical geometric multiplier, $K_i \approx 2$, is introduced into this final equation to mathematically account for the complex 3D fringing electric fields and additional capacitive coupling with other metal layers not captured by the simple parallel-plate approximation.

This final expanded equation reveals exactly why geometric scaling created an existential crisis for chipmakers. As technology nodes scale down, the critical cross-sectional geometries all scale down strictly proportionally with the minimum feature size (F_{min}). However, to accommodate increasingly complex architectures, the overall chip area (A) usually increases with scaling. Because the maximum length of a global interconnect (L_{max}) spans across this chip, it scales roughly with the square root of the area ($L_{max} \approx \frac{\sqrt{A}}{2}$).

When we substitute these scaling rules ($L^2 \propto A$ and the cross-sectional dimensions $\propto F_{min}$) back into the delay equation, we uncover the ultimate scaling trend for the RC bottleneck:

$$\tau \approx 0.89 K_{ox} \epsilon_0 \rho \frac{A}{(F_{min})^2}$$

The physical reality demonstrated by this formula is undeniable: as the minimum feature size shrinks and the chip area grows, the signal delay inherently skyrockets. The lateral capacitance dominates, and the highly resistive narrow wires choke the current. Because geometric scaling alone is mathematically disastrous for signal propagation, the industry was forced to fundamentally act on the material properties themselves to keep τ under control.

8.5.2 The Transition to Low-k Dielectrics

To address the ϵ_{ox} variable in the numerator of the delay equation, the industry transitioned away from legacy inorganic dielectrics. Traditional CVD-deposited Silicon Dioxide (SiO_2) possesses a relatively high relative dielectric constant (also written as k) of 3.9 to 5.0, while silicon nitride (Si_3N_4), which is often used in multi-layer structures, exhibits an even higher k-value of 5.8 to 6.1.

Modern fabrication relies on advanced Low-k materials to actively lower capacitance:

- **Moderate Low-k:** The initial phase of capacitance reduction involved doping silicon dioxide with fluorine to create Fluorinated Silicate Glass (FSG), or utilising organic Polyimides ($k = 2.9$ to 3.9).
- **Advanced Low-k:** Further scaling necessitated hybrid materials, primarily Si-O-C polymers. These can be deposited via Spin-On Dielectric (SOD) or CVD techniques to achieve excellent dielectric constants ranging from 2.0 to 3.8.
- **Ultra Low-k:** The absolute physical limits of dielectric scaling involve the deliberate introduction of microscopic air pockets into the film. Porous SiO_2 aerogels can drive the dielectric constant down to between 1.2 and 1.8. The theoretical limit of this progression is the "air bridge" architecture, which relies on the vacuum of empty space to achieve an absolute minimum k-value of 1.0 to 1.2.

8.6 The Copper Damascene Revolution

To combat the RC delay bottleneck, the semiconductor industry fundamentally replaced legacy aluminium interconnects with copper (Cu), which boasts a significantly lower resistivity. Today, copper is heavily integrated alongside low-k dielectrics in modern ICs, ASICs, and DRAMs.

However, integrating copper presented some major physical issues. First of all, it is a fast diffuser in almost everything; once copper gets on the substrate it will go everywhere, causing short circuits and creating traps. Also, unlike aluminium, copper cannot be dry etched because its chemical etch by-products are not volatile. If a plasma reacts with copper, the by-products simply sit on the surface rather than evaporating away, completely halting the etch. To overcome this, process engineers had to completely invert the traditional subtractive metallisation process, adopting an additive inlay method known as the Damascene approach (named after the ancient Syrian art of metal inlay).

8.6.1 Phase 1: Line 1 Single Damascene

The very first copper interconnect level (Line 1 or M1) connects directly to the underlying tungsten contacts using a Single Damascene process.

1. **Dielectric Deposition (Figure 8.11):** After standard contact formation, a thin protective layer of silicon nitride or low-k SiCN is deposited via PE-CVD or CVD to act as a strict etch-stopping layer. A thick layer of low-k dielectric (such as FSG or SiOC) is then deposited over this stop layer.

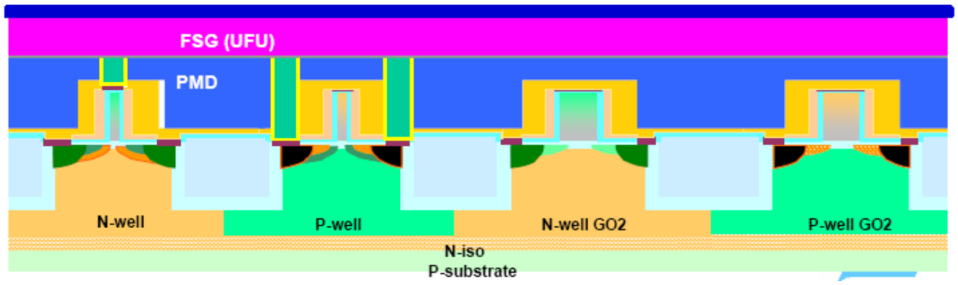


Figure 8.11: The uniform layer of FSG (pink) is covering the whole structure. Under it, the very thin protective layer (dark green) separates it from the PMD and contacts.

2. **Line 1 Lithography and Etching:** The wafer is patterned, and a plasma etches the shallow trenches for Line 1 into the low-k dielectric, halting precisely on the underlying stop layer.
3. **Reactive Pre-Clean:** Before metallisation, a reactive sputter etch is performed to physically remove any native oxides at the bottom of the trenches, ensuring a pristine electrical contact.
4. **Barrier and Seed Deposition:** To prevent copper diffusion, a robust barrier layer of tantalum nitride / tantalum (TaN/Ta) is deposited over the entire trench structure. Because the trenches will be filled using electroplating, a highly conductive Cu seed layer is immediately deposited over the Ta/TaN barrier.
5. **Electroplating and CMP (Figure 8.12):** The wafer is submerged in an electrolytic bath, and bulk copper is electroplated until it completely overfills the trenches. Finally, Cu-CMP is heavily utilised to grind away the massive metal over-burden, stopping precisely on the top of the dielectric layer to leave perfectly isolated copper wires.

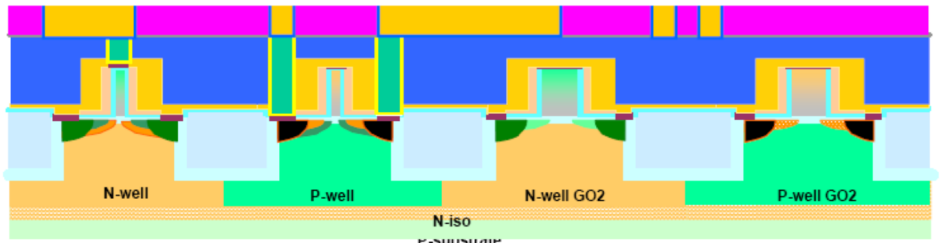


Figure 8.12: After the FSG is etched, the TaN/Ta barrier is deposited (blue) in the patterned features. The copper (orange) is deposited via electroplating, a technique which uses an ion current to move the copper atoms to the substrate. It is then made flat via CMP.

8.6.2 Phase 2: Vias and Line 2 Dual Damascene

For all subsequent metal layers, fabricating the vertical vias and horizontal lines in two completely separate Damascene loops would be prohibitively expensive. Instead, fabs use a "Dual Damascene" approach, which forms both the Vias and Line 2 (M2) simultaneously.

1. **Dielectric Stack (Figure 8.13):** A new SiCN stopping layer and a thick low-k dielectric layer are deposited directly over the planarised Line 1.
2. **Via Patterning (First Etch):** The wafer is patterned for the vias first. An anisotropic plasma etches the deep, vertical via holes completely through the new low-k dielectric, stopping on the SiCN layer just above Line 1.

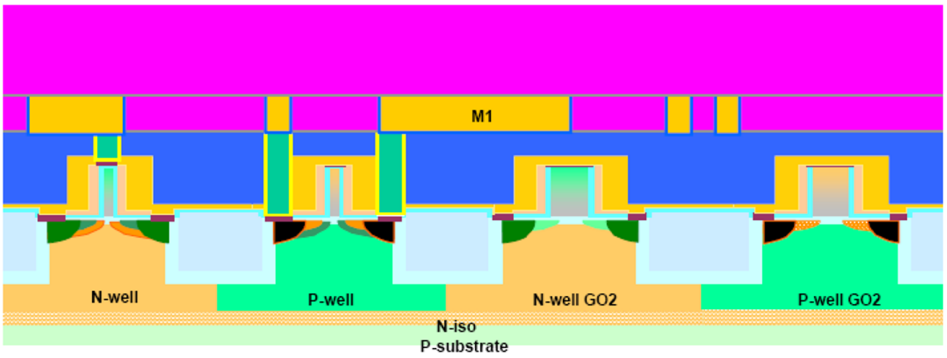


Figure 8.13: Another protective layer (dark green) and low-k dielectric layer (pink) are deposited over the substrate.

3. **Line 2 Lithography and BARC (Figure 8.14):** The wafer must now be patterned a second time to define the horizontal Line 2 trenches. To physically protect the bottoms of the already-etched vias from being destroyed during this second etch, a Bottom Anti-Reflective Coating (BARC) is spun onto the wafer, intentionally pooling inside and filling the deep vias.

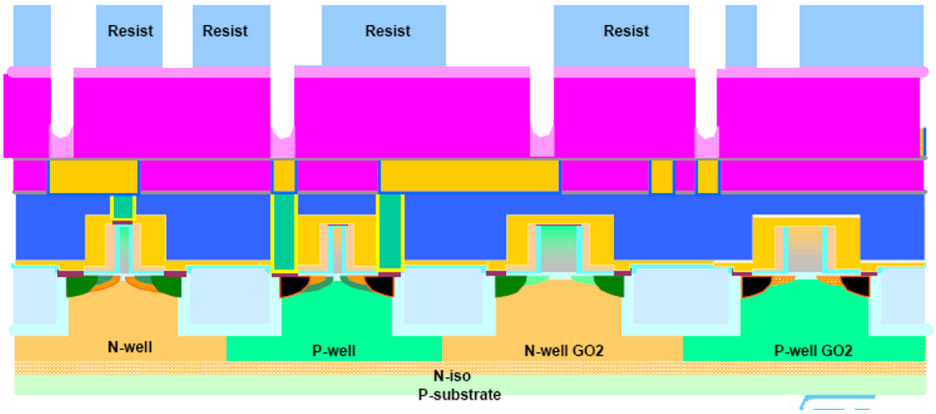


Figure 8.14: After the vertical holes have been etched on the low-k dielectric layer (pink), a BARC (light pink) is spun to protect the bottom of these trenches. A second lithography prepares the structure for the shallower etching.

4. **Line 2 Etching (Second Etch):** The plasma etches the shallower Line 2 trenches into the upper portion of the dielectric. The BARC safely protects the deep vias during this step.

5. **Simultaneous Metallisation (Figure 8.15):** Once the BARC is cleared, the TaN/Ta barrier and Cu seed are deposited. Copper is then electroplated, seamlessly filling both the deep vias and the shallow Line 2 trenches in a single step. A final Cu-CMP planarises the surface, completing the Dual Damascene loop.

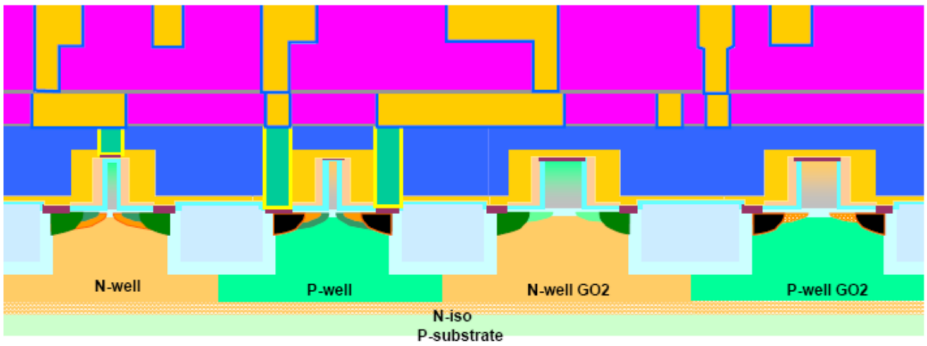


Figure 8.15: Once again, a TaN/Ta barrier (blue) and copper (orange) are finally deposited inside the trenches, completing the desired interconnects.

8.6.3 Nanoscale Copper Challenges

While copper solved the immediate RC crisis, the relentless scaling has exposed severe physical limitations in the Damascene process. As line widths shrink deep into the nanoscale regime (below 100 nm), the physical resistivity of copper increases, completely deviating from its ideal bulk resistivity.

This catastrophic resistivity spike is driven by two phenomena:

- **Grain Boundary Scattering:** In extremely narrow trenches, the copper crystal is forced to form very small, highly constrained grains. As electrons flow through the wire, they frequently collide with these dense grain boundaries, severely impeding their momentum.
- **Surface/Sidewall Scattering:** As the wire narrows, the proportion of electrons travelling adjacent to the physical edges of the wire increases. If the etched trench possesses rough sidewalls, electrons violently scatter off these irregular surfaces, further degrading the electrical current.

To mitigate this rising resistance, process engineers try to maximise the physical Cu grain size and meticulously minimise the sidewall roughness during the dielectric etch. Furthermore, as the trenches become unimaginably narrow, simply fitting the required Ta/TaN barrier and Cu seed layers into the opening without completely choking off the remaining volume for the bulk copper has become a critical scaling bottleneck.

8.7 Future Perspectives and Conflict Minerals

As the semiconductor industry pushes further into the single-digit nanometre nodes, the fundamental physical limits of copper damascene are becoming a hard wall. According to predictions by the International Technology Roadmap for Semiconductors (ITRS), in the near future, more than 80% of the total power in an integrated circuit will be dissipated solely by the interconnect network. The rising resistance of ultra-narrow wires generates immense amounts of Joule

heating, meaning entirely radical, non-traditional solutions will be required to overcome this crippling architectural bottleneck.

8.7.1 Beyond Copper: Radical Interconnect Technologies

To bypass the limitations of traditional bulk metals, researchers are currently exploring entirely new physical mechanisms for routing data across a chip:

- **Carbon Nanotubes (CNTs) and Graphene:** To circumvent the severe grain boundary scattering of nanoscale copper, researchers are investigating carbon-based nanomaterials. Carbon nanotubes (rolled cylinders of graphene) and planar graphene possess exceptionally long mean free paths for electron transport. Because electrons can travel almost ballistically (without scattering) over relatively long distances in these pure carbon lattices, they exhibit incredibly high thermal and electrical conductivity. Furthermore, they are virtually immune to classical electromigration, allowing them to theoretically handle massive current densities without physical degradation.

- **Optical Interconnects (Silicon Photonics):** Rather than fighting the physical Resistance-Capacitance (RC) delay of metal wires, the ultimate solution is to eliminate electrons entirely for long-range routing. Silicon photonics aims to integrate microscopic lasers, modulators, and optical waveguides directly onto the semiconductor chip. By converting electrical data into pulses of light, information can travel across the die at the speed of light with virtually zero RC delay, completely bypassing the resistive heating that plagues modern copper wiring.

- **Spintronics:** Traditional electronics rely on physically moving the electrical charge of an electron through a lattice, a process that inherently requires current and generates heat. Spintronics (spin transport electronics) proposes transmitting information using the intrinsic quantum mechanical spin state of the electron (up or down). Because flipping a spin state requires exponentially less energy than physically dragging an electron from point A to point B, spintronic interconnects and logic devices could facilitate ultra-low-power data transfer and persistent memory (such as MRAM) that does not succumb to extreme resistive losses.

8.7.2 The Human Cost of Metallisation

While device engineers battle the physical limits of nanoscale physics, the true cost of semiconductor scaling extends far beyond the walls of the cleanroom. The highly advanced barrier layers that prevent copper from destroying the silicon substrate rely heavily on the transition metal tantalum (Ta). Tantalum is primarily extracted from an ore known as columbite-tantalite, colloquially referred to in the industry as *coltan*.

The relentless global demand for coltan has fueled a dramatic and incredibly bloody geopolitical crisis, primarily centred in the Democratic Republic of the Congo. The extraction of this vital mineral, alongside other critical conflict minerals like cassiterite (the primary source of tin), wolframite (tungsten), and gold, is frequently overseen by brutal armed militias. These groups systematically prey on miners, using the immense profits from the illicit mineral trade to directly fund deadly civil violence and solidify regional control.

The socio-economic reality of the BEOL supply chain is harrowing. Journalistic investigations and human rights reports continually document the horrific use of forced labour, systemic slavery, and child miners operating in extreme, heavily unregulated conditions.

As the semiconductor industry looks toward the future, overcoming the physical RC delay must be coupled with strict, enforceable international legislation (such as modern EU conflict mineral laws) to clean up the global trade. The ultimate technological challenge is ensuring that

the next generation of integrated circuits is built on a foundation of scientific innovation, rather than human exploitation.

The journey through the Back End of the Line has illustrated the monumental engineering required to breathe life into isolated silicon devices. By meticulously balancing the material properties of ultra-low-k dielectrics and copper metallisation, process engineers have constructed staggering, multi-level highways that route power and data across the microscopic landscape.

When a metropolis runs out of usable land to expand horizontally, its architects are forced to build upward. The semiconductor industry now faces this exact architectural crossroads. If we can no longer shrink individual transistors without confronting catastrophic quantum tunnelling, and we can no longer simply pack more infinitely thin copper wires into a planar layer without choking the processor in its own heat and signal delay, the only remaining dimension for Moore's Law to conquer is the vertical one.

This brings us to our final structural frontier: **3D Integration and Advanced Packaging**. In the forthcoming and final chapter, we will explore how the industry is fundamentally abandoning the strict confines of the single two-dimensional die.

Chapter 9

3D Integration and Advanced Packaging

9.1 The Purpose and Evolution of Packaging

Once a silicon wafer has completed the complex Back End of the Line (BEOL) processing, it is rigorously tested and mechanically diced into individual chips. At this stage, the bare silicon die is entirely non-functional on its own; it must be integrated into a macroscopic electronic system. The semiconductor package encompasses the physical components surrounding the die that enable its active use within external circuitry.

Historically viewed as a mundane final step in manufacturing, packaging has evolved into a critical frontier of semiconductor engineering. A modern package must simultaneously fulfil five fundamental physical and electrical roles:

1. **Electrical Interfacing:** The package acts as a vital spatial adapter. Die-to-package connections are fabricated on the micrometre scale, whereas package-to-board connections exist on the millimetre scale. A robust package bridges this dimensional gap while improving the overall quality of signal transmission.

2. **Physical Protection:** Thinly diced silicon dies are exceptionally fragile and highly susceptible to mechanical stresses and environmental contamination. The package provides necessary structural integrity, typically achieved through rigid plastic injection moulding (encapsulation) and the application of hermetically sealed metal lids.

3. **Heat Dissipation:** Complex integrated circuits generate massive amounts of Joule heating, which can catastrophically burn up a die if not properly managed. Packages incorporate thermally conductive fillers and large metal heat spreaders to efficiently conduct heat away from the silicon.

4. **Functional Expansion:** Advanced packaging allows engineers to vertically stack multiple dies to multiply available memory, or to create hybrid packages that seamlessly integrate distinct logic and memory layers within a single footprint.

5. **Customer Ease of Use:** Packages provide an industry-standardised layout of interconnects (such as specific socket dimensions), allowing downstream customers to quickly attach, replace, or upgrade products on their motherboards without altering the underlying chip design.

9.1.1 The Evolution of Interconnects: Wire Bonding to Flip-Chip

To physically connect the micrometre-scale pads on the silicon die to the external package substrate, the industry relies on two primary methodologies.

The traditional approach is **Wire Bonding** (shown in Figure 9.1). In this process, chemical and laser etchings expose electrical connections on the top surface of the die. Exceptionally thin metallic wires (formerly gold, now mostly copper) are then physically bonded from these top-side pads down to metal traces on the underlying interposer or substrate. Because it is a highly adaptable, mature, and relatively inexpensive fabrication process, over 80% of all integrated circuits still utilise wire bonding today. However, wire bonding inherently requires a much larger overall package footprint to accommodate the sweeping wire loops. More critically, these long physical wires act as parasitic inductors and resistors, resulting in greater parasitic voltage loss and severely impeding high-speed signal transmission. Wire Bonding can be **direct**, when the wires are used to connect each die to the assembly substrate, or **cascading**, when the wires cascade from one die to the next, ending at the substrate.

To overcome these severe parasitic limitations, high-performance processors transitioned to the **Flip-Chip (Bumped Package)** architecture (also shown in Figure 9.1). Instead of wiring from the top down, the die is physically "flipped". Microscopic solder ball "bumps" are deposited directly onto the die's surface, establishing a direct, face-to-face electrical connection with the metal pads on the package substrate. While this requires a substantially more complex and expensive fabrication process, where any change to the chip's dimensions necessitates a complete redesign of the bonding pattern, the benefits are immense. The drastically shortened electrical connections result in far lower parasitic voltage losses, enabling extraordinarily high-density interconnects and significantly shrinking the overall volume of the package.

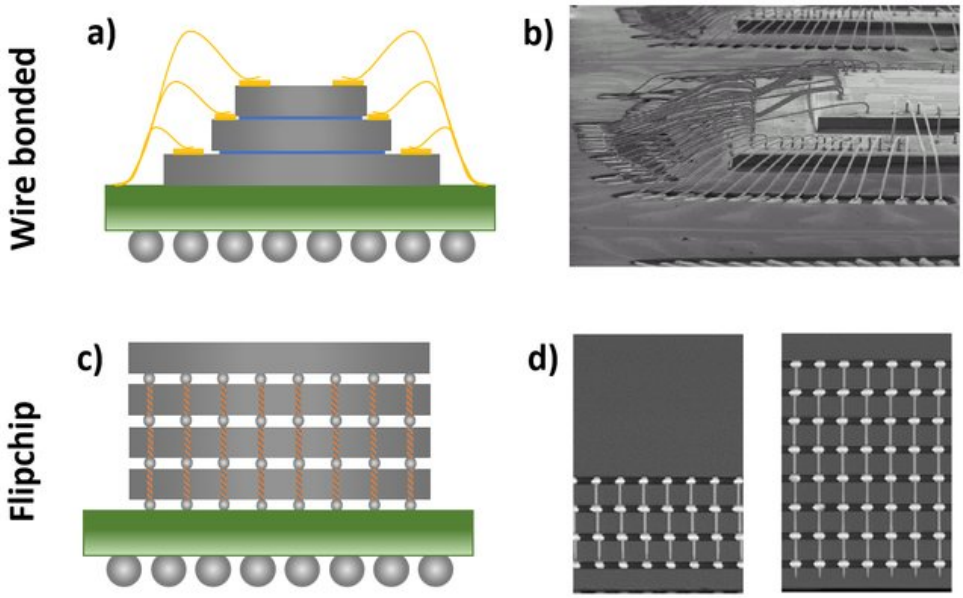


Figure 9.1: a) and b) show the structure of wire-bonds. The intricate system of gold filaments is carefully deposited using advanced software for alignment and a special tool to manipulate the wire. You can see an animation of this by Semitracks Inc. on YouTube. c) and d) show the structure of a flip-chip package. The wafers have to be handled with extreme care, but this technique provides several advantages, such as shorter electrical paths and higher connection density. Images taken from ResearchGate.

9.1.2 The Drive for Interconnect Density

As Moore's Law relentlessly forced transistor counts higher, packaging technology had to scale in proportion to provide sufficient Input/Output (I/O) density. The industry progressed from legacy Dual Inline Packages (DIP) and Flat Packages (FP) to dense Ball Grid Arrays (BGA) and Chip Scale Packages (CSP).

However, because bumped flip-chips are not easily stacked directly on top of one another, engineers developed **Package-on-Package (PoP)** architectures (Figure 9.2) to expand memory density vertically. By stacking entire pre-assembled packages and connecting them via inter-package vias, manufacturers could blend different bonding types (such as placing a wire-bonded memory package directly atop a bumped logic package). This drive for vertical density ultimately set the stage for the modern era of System in Package (SiP) and High Bandwidth Memory (HBM) architectures.

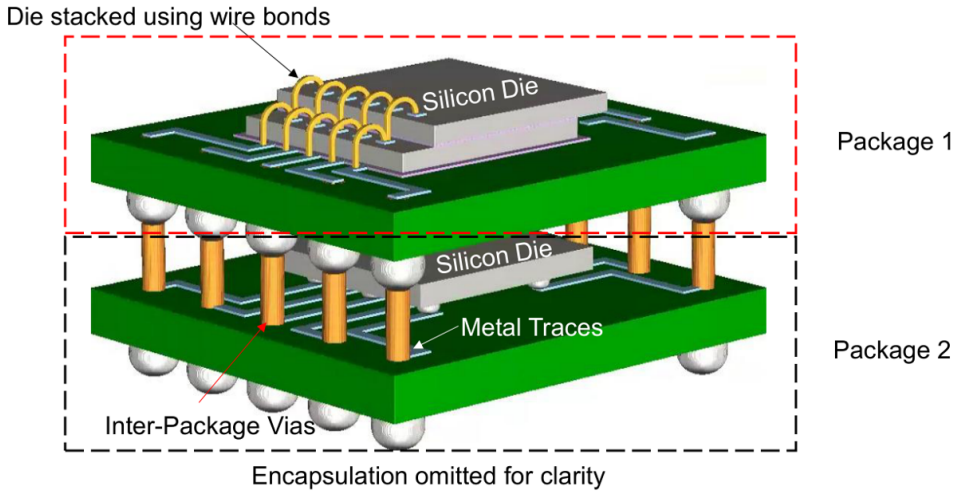


Figure 9.2: The picture shows what the Package-on-Package architecture looks like. The idea is to stack multiple packages on top of each other by connecting them with special vias and bumps. This also allows different packaging techniques to be combined. Note that the encapsulation was omitted; the whole structure would normally be encapsulated in plastic or epoxy as mentioned before.

9.2 Through-Silicon Vias (TSVs) and HBM Architecture

To achieve true vertical integration and bypass the spatial limitations of wire bonding, engineers developed the Through-Silicon Via (TSV) (Figure 9.3). Unlike traditional packaging methods that route signals around the exterior edge of a chip, a TSV is an advanced fabrication technique that etches deep vertical holes completely through the bulk thickness of the silicon die itself. These high-aspect-ratio holes are filled with conductive metal to form solid pillars, allowing electrical signals to pass directly from the active front side of a die to its back side. By creating the absolute shortest possible electrical path between vertically stacked chips, TSVs drastically reduce parasitic resistance and capacitance while enabling massive parallel I/O bandwidth.

9.2.1 TSV Integration Schemes and Scaling

Integrating TSVs into the manufacturing flow can be achieved at different stages of the chip's lifespan. Depending on the architecture, the vertical pillars can be formed between FEOL and BEOL (Via-Middle), or entirely after standard wafer fabrication is complete (Via-Last). In Via-Last processes, the TSVs can be etched from the front side or from the back side after the wafer has been temporarily bonded to a carrier and physically thinned.

As the demand for 3D interconnect density escalates, the physical dimensions of TSVs are rapidly scaling down. The industry standard metric for this scaling is the TSV pitch, which is typically defined as roughly twice the via diameter.

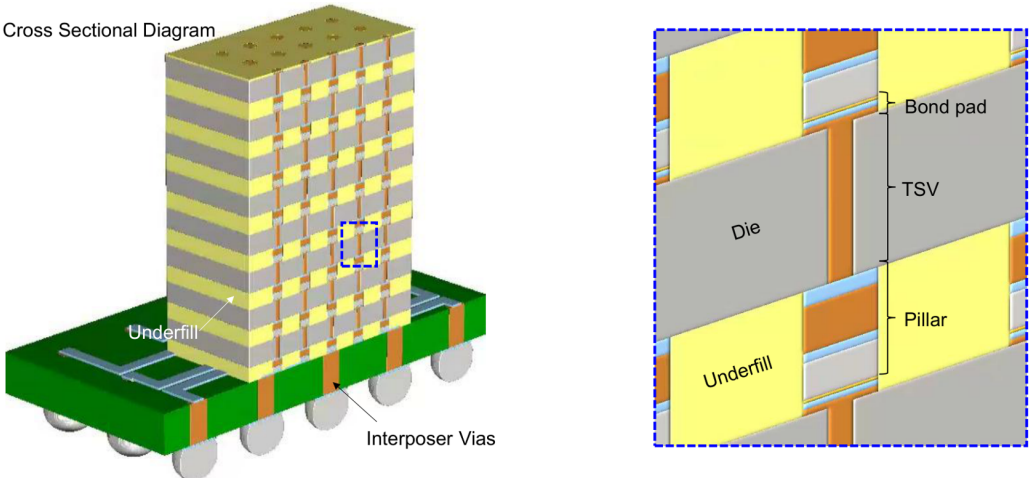


Figure 9.3: This picture shows how to package using TSVs. In the highlight in the dotted blue frame illustrates the different components of this architecture.

- **Standard TSVs:** Early generations used for standard die-level handling required relatively large dimensions, possessing pitches well above 10 µm.
- **Micro-TSVs (μTSV):** Advanced processing has enabled the transition to micro-TSVs, measuring roughly 2 µm in diameter and 5 µm in depth, significantly increasing the available interconnect density.
- **Nano-TSVs (nTSV):** The future roadmap relies on extreme wafer-to-wafer (W2W) bonding to facilitate nano-TSVs, which shrink the interconnect pitch deep into the sub-micrometre regime.

9.2.2 Case Study: High Bandwidth Memory (HBM)

The most prominent commercial success of TSV technology is the High Bandwidth Memory (HBM) architecture, designed explicitly to feed data-hungry computing units. An HBM module is not a single, monolithic chip; rather, it is a complex 3D "cube" composed of a vertical stack of multiple DRAM memory dies (typically 4 or 8 core dies) sitting atop a foundational Logic die, also referred to as the Interface die.

These distinct layers are intricately interconnected using thousands of TSVs that plunge directly through the stacked DRAM layers. At the interface between each layer, microscopic solder bumps electrically link the TSV pillars together. The bottom Interface die is uniquely responsible for managing these channels, each providing access to a discrete pool of memory, and routing the massive data flow to the external system.

9.2.3 2.5D Integration and the Silicon Interposer

While TSVs allow the internal HBM cube to function as a unified memory block, it must still communicate rapidly with the primary processor, such as a Graphics Processing Unit (GPU). Because the extremely dense micro-bumps at the base of the HBM cube and the GPU are vastly too small to interface directly with a standard, macroscopic printed circuit board, the industry utilises a "2.5D" integration approach.

Instead of stacking the GPU and memory directly on top of one another, both the central GPU and the adjacent HBM memory cubes are mounted side-by-side onto a Silicon Interposer via high-density micro-bumps (Figure 9.4). This interposer is a dedicated, passive piece of silicon possessing its own horizontal metal traces to bridge the communication between the GPU and

the memory. Furthermore, the interposer contains its own larger TSVs, which route the electrical signals from the top-side micro-bumps down to larger C4 solder bumps on the underlying package substrate. This architecture expertly steps up the physical scale of the connections, seamlessly bridging the microscopic world of the silicon die with the macroscopic world of the customer motherboard.

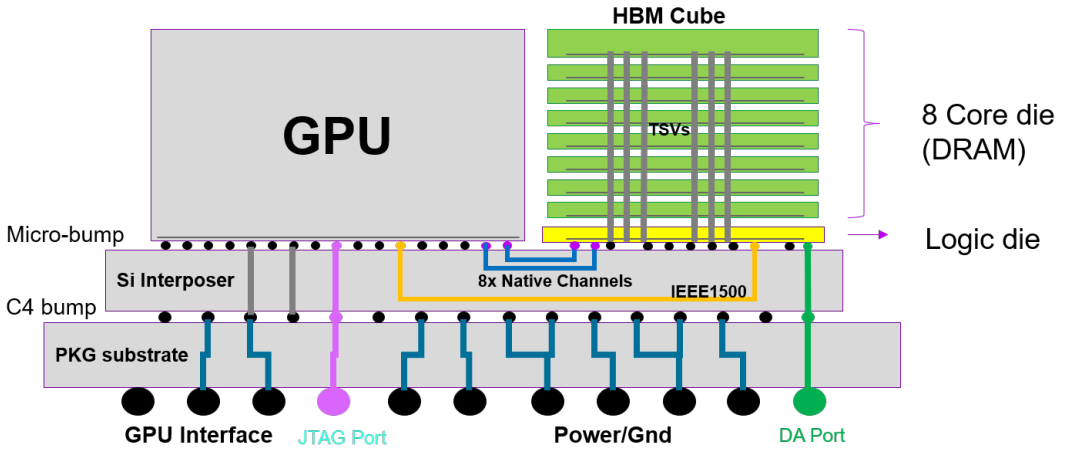


Figure 9.4: This picture shows the HBM architecture. These "cubes" are multiple and placed right next to the GPU on the chip. This shortens the physical distance data must travel, providing greater power efficiency and a massive bandwidth in a small footprint.

9.3 Advanced Bonding Techniques

While micro-bumps and Through-Silicon Vias (TSVs) enabled the first generation of 3D integration, they still rely on physical solder joints to make electrical connections. As interconnect pitches scale down toward the single-micrometre regime, solder bumps become a severe limiting factor; they are prone to bridging (short-circuiting adjacent pads) and suffer heavily from electromigration at reduced dimensions. To achieve ultra-high-density Heterogeneous Integration, the industry is shifting from solder-based connections to direct Wafer-to-Wafer (W2W) and Die-to-Wafer (D2W) bonding techniques.

9.3.1 Fusion Bonding

Fusion bonding is a direct bonding technique used to permanently join two silicon or dielectric surfaces without any intermediate adhesives or solder. The process relies on preparing ultra-clean, hydrophilic (water-attracting) surfaces.

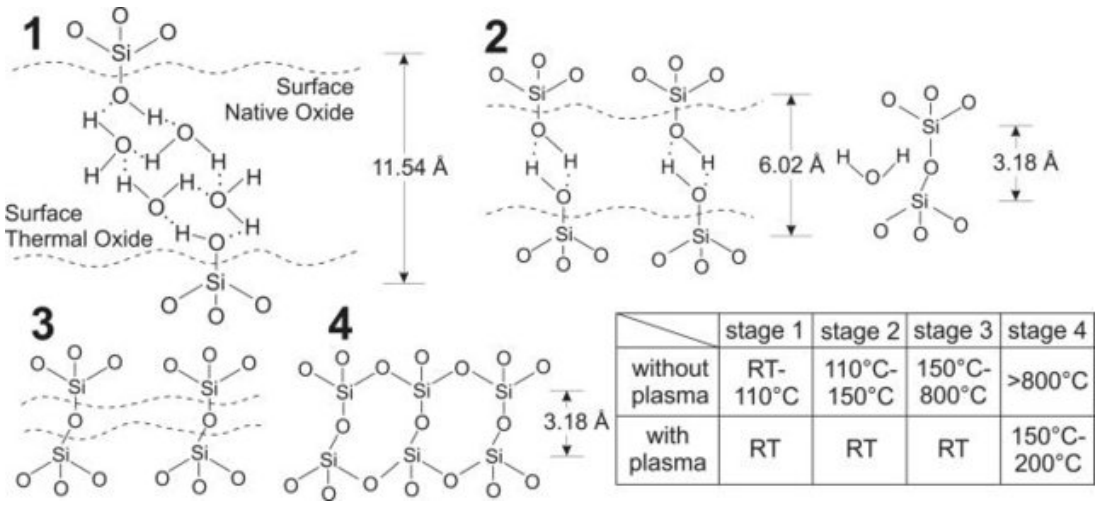
The process flow begins with low-temperature plasma activation of the wafer surfaces, followed by an optional single-wafer clean to remove particles. To prevent contamination, the pre-bonding alignment must be performed immediately after this surface preparation. At this stage, the surfaces are terminated with silanol groups (Si-OH) and covered by a few monolayers of adsorbed water molecules. The chemical steps are shown in Figure 9.5.

The physical bonding mechanism evolves through distinct temperature-dependent stages:

1. **Initial Contact (Room Temperature to 110 °C):** When the wafers are brought together at room temperature, weak Van der Waals forces and hydrogen bonds form between the opposing water molecules and silanol groups. The bond strength is initially very low (approximately 0.1 $\mathrm{J \cdot m^{-2}}$), but slightly increases as water molecules rearrange to maximise contact area.

2. **Covalent Bond Formation (110 °C to 150 °C):** As the temperature increases, the trapped water molecules diffuse away along the interface and into the bulk silicon. The opposing silanol groups react, shedding water to form robust, permanent covalent siloxane bonds (Si-O-Si). This transition triggers a massive increase in bond strength.

3. **High-Temperature Maturation (150 °C to >800 °C):** By 150 °C, most of the covalent bonds have formed, but the overall strength is physically limited by the microscopic surface roughness of the wafers, leaving tiny unbonded gaps. If the process allows for extreme temperatures above 800 °C, the silicon dioxide begins to soften and undergo viscous flow, completely closing the remaining gaps and achieving maximum bond strength (2.5 $\mathrm{J \cdot m^{-2}}$).



	stage 1	stage 2	stage 3	stage 4
without plasma	RT-110°C	110°C-150°C	150°C-800°C	>800°C
with plasma	RT	RT	RT	150°C-200°C

Figure 9.5: This picture shows the 4 stages of the fusion bonding process during the anneal. As seen from the table, plasma can enhance the process by allowing it to be carried out mostly at room temperature. Image taken from ResearchGate.

9.3.2 Hybrid Bonding

While fusion bonding is excellent for joining bare silicon or bulk dielectrics, functional integrated circuits require electrical connections. Hybrid bonding is an advanced evolution of this process that seamlessly combines direct dielectric-to-dielectric fusion bonding with metallic Copper-to-Copper (Cu-Cu) bonding in a single, simultaneous step. The process is shown in Figure 9.6.

To prepare for hybrid bonding, both the top and bottom wafers must undergo rigorous Back End of the Line (BEOL) finishing. The planarising oxide layer must be polished to an ultra-flat, highly smooth finish. Simultaneously, the copper damascene pads are polished so that they possess a very specific, minimal recess relative to the surrounding dielectric surface.

Once prepared, the wafers are aligned with extreme precision and brought together at room temperature. Because the copper pads are slightly recessed, they do not touch. Instead, the ultra-smooth dielectric surfaces adhere immediately via strong Van der Waals forces, leaving a microscopic, narrow gap between the opposing copper pads.

Finally, a high-temperature anneal is performed. This anneal serves a dual purpose: it permanently strengthens the dielectric fusion bond via covalent linkage, and it actively realises the electrical Cu-Cu contact across the microscopic gap. This solderless connection is achieved by exploiting the significantly higher coefficient of thermal expansion copper possesses compared to the surrounding silicon dioxide. As the heat increases, the fused bulk dielectric structurally locks the two wafers together, but the copper metal expands much faster and further than the surrounding oxide cavity. This massive thermal expansion forces the copper to bridge the recess

and contact the opposing pad under high compressive stress. The thermal pressure forces the copper atoms to interdiffuse across the interface, creating a solid, permanent metallic bond. Upon cooling, the copper attempts to contract, placing the newly formed joint under tensile stress and ensuring an exceptionally robust, low-resistance electrical connection.

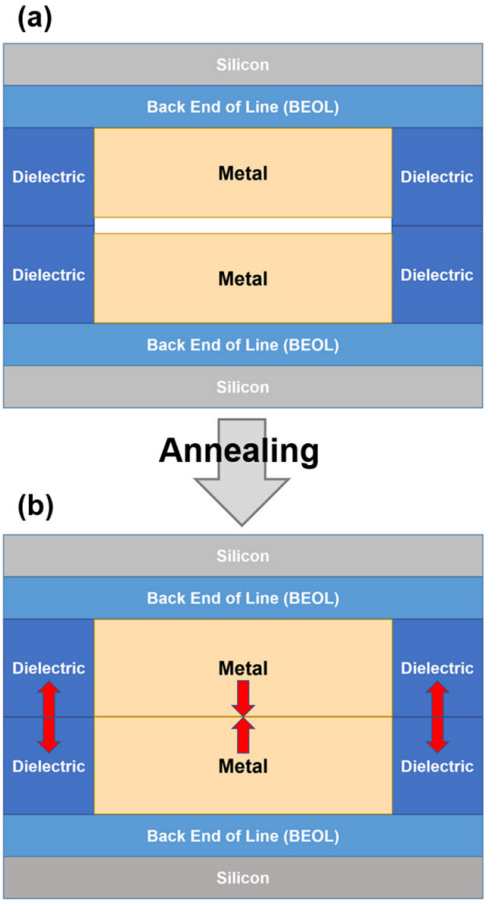


Figure 9.6: This picture explains the idea of hybrid bonding. After planarisation, copper is not at the same level as the dielectric. This means that when joining the wafers, a small gap is present between the two metals. During heating, this extra room allows the copper to expand and bond to the rest of the copper from the other wafer. Image taken from ResearchGate.

9.4 Current Applications of 3D Integration

The transition from traditional two-dimensional packaging to advanced 3D bonding has fundamentally reshaped device architectures across multiple sectors of the semiconductor industry. Direct fusion bonding and hybrid copper bonding have successfully transitioned from laboratory novelties to high-volume manufacturing standards, enabling unprecedented functional density.

9.4.1 Fusion Bonding in SOI and MEMS

Pure dielectric-to-dielectric fusion bonding has been an industrial mainstay for years, primarily driven by two specific applications:

- **Silicon-on-Insulator (SOI) Wafers:** To create high-performance SOI substrates, fabs utilise a process flow (shown in Figure 9.7) that begins with the thermal oxidation and subsequent ionic implantation of an initial donor wafer. This donor wafer is thoroughly cleaned and fusion-bonded to a blank handler wafer. The implanted layer creates a plane

of physical weakness, allowing the bulk of the donor wafer to be cleanly split off by simply baking the wafers. A final high-temperature anneal and CMP touch-polish yields a pristine, ultra-thin silicon layer completely electrically isolated by the buried oxide.

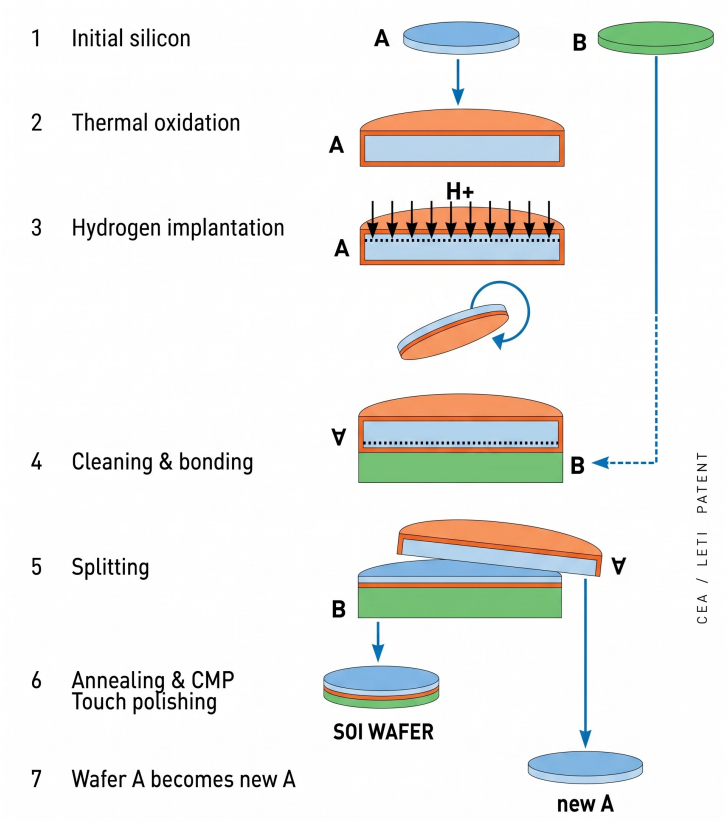


Figure 9.7: This picture shows the process flow used in SOI fabrication. The goal is to create a flawless layer of silicon sitting on top of an insulator. This prevents current leakage, allowing for faster and more power-efficient chips. The implanted hydrogen helps in splitting the wafer by forming bubbles during heating. These bubbles, trying to escape, split the wafer in two. Image adapted from an illustration of the Smart Cut process, courtesy of CEA-Leti and Soitec.

- **Micro-Electromechanical Systems (MEMS):** Fusion and anodic bonding are critical for creating three-dimensional mechanical structures, such as pressure sensors. A typical MEMS pressure sensor is fabricated by bonding a micromachined silicon wafer featuring an etched cavity and diffused piezoresistive sense elements directly to a rigid Pyrex or silicon substrate. The bonding seam creates a hermetically sealed cavity beneath a thin silicon diaphragm, allowing the piezoresistors to measure mechanical deflection caused by external pressure.

9.4.2 Hybrid Bonding in High-Density Silicon

Because it allows for simultaneous structural adhesion and ultra-dense electrical routing, Cu-Cu hybrid bonding has unlocked entirely new architectural paradigms for memory and logic.

- **Stacked CMOS Image Sensors (CIS):** Modern smartphone cameras rely almost exclusively on stacked sensor architectures enabled by hybrid bonding. Instead of placing the light-gathering Backside Illuminated (BSI) pixels and the processing logic on the same crowded plane, the array of pixels is fabricated on a dedicated top wafer, while the Image Signal Processor (ISP) logic is fabricated on a separate bottom wafer. The two wafers are

inverted and hybrid-bonded face-to-face (SiO_2-SiO_2 and Cu-Cu), drastically increasing the light-capturing area of the top surface while burying the complex signal routing below.

- **3D NAND Flash Memory:** To perpetually increase storage density without ballooning the die size, NAND manufacturers use wafer-to-wafer hybrid bonding to physically separate the memory array from its peripheral control circuitry. By bonding the massive, towering NAND memory array directly on top of the underlying CMOS logic region (typically linking the upper and lower copper metal layers), engineers can heavily shrink the overall footprint of the final memory chip.

9.4.3 The Future of Memory: VCT DRAM and 16H HBM

The most cutting edge applications of hybrid bonding are currently targeting the immense bandwidth and capacity bottlenecks of dynamic memory.

For next-generation standard memory, the industry is pioneering **3D DRAM** architectures. Using wafer-to-wafer hybrid bonding, manufacturers are implementing Cell-on-Peripheral (Core/PERI) designs, where the wordlines, bitlines, and capacitors are stacked vertically above the peripheral logic. Furthermore, the introduction of the Vertical-Cell-Transistor (VCT) structurally reorients the DRAM cell, achieving an ultra-dense $4F^2$ footprint compared to the conventional $6F^2$ layout[34].

Simultaneously, the High Bandwidth Memory (HBM) modules that power modern GPUs are reaching the physical limits of traditional packaging. While earlier HBM generations successfully used conventional Thermo-Compression Bonding (TCB) with solder micro-bumps to stack up to 8 core dies, the gap-fill requirements and bump dimensions are becoming prohibitive. To achieve extreme **16-High (16H) HBM stacks**, manufacturers such as Samsung are looking to transition to gap-less Hybrid Cu Bonding (HCB). By eliminating the micro-bumps entirely, the 16 stacked DRAM dies can fit within the exact same vertical package height previously required for smaller stacks, ensuring vastly superior interconnect density and thermal performance.

9.5 The 3D Interconnect Roadmap

As the industry exhausts traditional two-dimensional node scaling, foundries and research institutes are defining aggressive roadmaps to perpetually increase interconnect density. The ultimate goal is to seamlessly partition complex Systems-on-Chip (SoCs) into specialised functional chiplets, manufacture each on its most cost-effective node, and restack them into a unified, ultra-dense 3D device.

9.5.1 TSMC SoIC and Heterogeneous Stacking

Taiwan Semiconductor Manufacturing Company (TSMC) has heavily advanced this chiplet approach through its System-on-Integrated-Chips (SoIC) platform. Previously, high-performance packaging relied almost entirely on 2.5D CoWoS (Chip-on-Wafer-on-Substrate) architectures, where a monolithic SoC and High Bandwidth Memory (HBM) modules were placed side-by-side on a passive silicon interposer.

The SoIC architecture takes integration truly into the third dimension. Using advanced Chip-on-Wafer (CoW) and Wafer-on-Wafer (WoW) hybrid bonding, engineers can stack multiple distinct SoC logic dies directly on top of one another to create a single, exceptionally dense 3D logic block. This completely integrated SoIC logic stack can then be mounted onto a traditional

[34]The formula nF^2 refers to how the area a memory cell occupies in a DRAM. F is the minimum feature size. A $4F^2$ cell means it is $2F$ wide and $2F$ long. These cells have 2 bitlines and 2 wordlines (compared to the 3 bitlines and 2 wordlines of the $6F^2$ layout).

2.5D CoWoS interposer alongside the HBM. Because the vertical bonding is so dense, the system treats the entire vertical stack as if it were a single, extraordinarily powerful monolithic die.

9.5.2 The IMEC Interconnect Density Roadmap

The defining metric for the future of 3D integration is the 3D interconnect pitch, that is the physical distance between adjacent vertical connections. According to IMEC's semiconductor roadmap, as this pitch shrinks from the micrometre scale down to the nanometre scale, the interconnect density increases exponentially, fundamentally changing how engineers can partition a processor architecture:

- **Micro-bump Stacking (10–40 µm Pitch):** This current-generation standard is primarily used for stacking high-level chiplets and partitioning large, third-level (L3) processor caches.
- **Die-to-Wafer Hybrid Bonding (≈1 µm Pitch):** Eliminating solder allows the pitch to shrink drastically. At this density, engineers can partition much closer to the logic cores, successfully stacking fast L2 and L1 caches directly over the processing units, or stacking high-resolution image sensor pixels (IM-Pixel) directly over logic processors.
- **Wafer-to-Wafer Hybrid Bonding (100–500 nm Pitch):** At these extreme submicron pitches, the vertical connections become as dense as the intermediate Back End of the Line (BEOL) metal layers.
- **Dielectric Bonding and nTSVs (≈10 nm Pitch):** The ultimate limit of the roadmap involves direct dielectric bonding combined with nano-TSVs (nTSV). This provides such massive density that engineers can stack individual 6-Transistor (6T) SRAM cells, or even partition logic at the fundamental transistor level. This paves the way for 3D Complementary FETs (CFETs), where NMOS and PMOS devices are stacked vertically on top of one another.

9.5.3 Backside Power Delivery Networks (BSPDN)

To support the colossal interconnect density of next-generation 3D stacks, researchers must fundamentally redesign the silicon wafer itself. In a traditional chip, both the power delivery network (PDN) and the signal routing network compete for space within the front-side BEOL layers. As transistors shrink, the thick power wires choke the signal routing, causing severe physical congestion and electrical voltage drops.

The ultimate architectural solution is the Backside Power Delivery Network (BSPDN). In this new paradigm, the front-side BEOL is dedicated entirely to signal routing and high-density Cu-Cu hybrid bonding pads. The thick, bulky power cables are completely relocated to the inactive backside of the silicon substrate. Power is routed through the backside silicon using ultra-fine nano-TSVs, which connect directly to Buried Power Rails (BPR) located at the very bottom of the front-side transistor layer. By physically decoupling power and signal, BSPDN clears the congestion required to achieve the absolute limits of 3D logic stacking.

The exploration of 3D integration highlights a fundamental shift in semiconductor architecture. By breaking free from the two-dimensional plane and leveraging technologies like Through-Silicon Vias and hybrid copper bonding, the industry has successfully blurred the line between the bare silicon die and its surrounding package. These vertical stacking techniques ensure that performance, bandwidth, and functional density can continue to scale even as traditional transistor shrinking approaches its physical limits.

While this chapter concludes our structural analysis of the chip's physical fabrication and packaging, the industrial manufacturing flow continues beyond the cleanroom. Following assembly, the packaged devices must undergo rigorous Final Test and Burn-In procedures. During this critical phase, parts are subjected to extreme electrical and thermal stresses under harsh conditions to verify functionality and ensure long-term reliability. Surviving components then proceed to Module Assembly or System/SSD Testing, where packaged parts are integrated into macroscopic memory modules or composite drives for ultimate performance validation. Only after passing these final systemic gauntlets are the devices ready to be delivered to the customer.

Final Thoughts

This brings an end to our journey in the semiconductor manufacturing process. The notes ended up being longer than I anticipated, but at least I hope they helped your studies.

Remember that, although interesting and certainly useful, it is fine not to remember every single equation, chemical or temperature. What matters is that you understand the how and why of every step, and how these come together to create a working device.

On a personal note, whether you liked it or not, I appreciate you having read this work. Never stop being curious and keep studying what you like; good things will come regardless of the rest. And if you didn't like this course, congrats on pushing through it anyway; that is still a worthy achievement.

Best of luck in your future studies.